

Day1: Fundamentals of Cloud



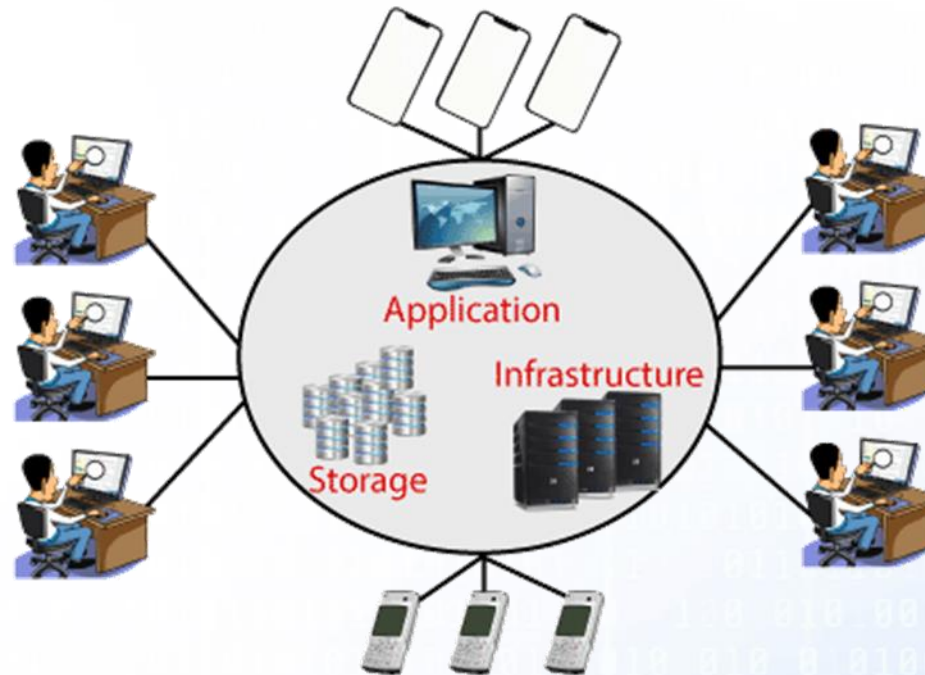
- Life Before Cloud Computing
- Introduction to Cloud Computing
- Cloud Providers
- Cloud Deployment Models
- Cloud service Offerings
- Introduction Azure Cloud and Services



- ❖ Traditional Business organizations are always costly and difficult to manage.
- ❖ The organization requires huge amount and different types of software and hardware to run the organization is challenging.
- ❖ The organization requires a lot of experts to install, check and update the software.
- ❖ on-premises datacentre, we have to manage everything, such as purchasing and installing hardware, virtualization, installing the operating system, and any other required applications, setting up the network, configuring the firewall, and setting up storage for data.



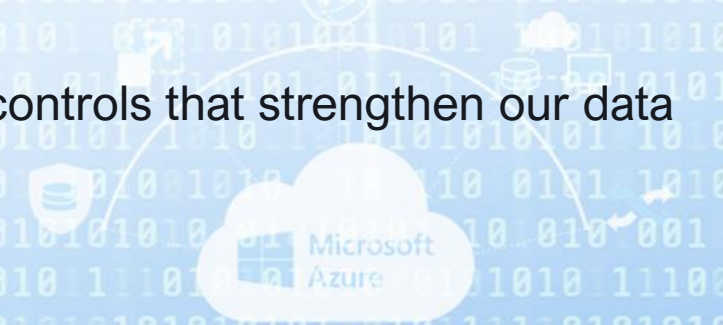
- ❖ Cloud Computing is the delivery of computing services such as servers, storage, databases, networking, software, analytics, intelligence, and more, over the Cloud (Internet).

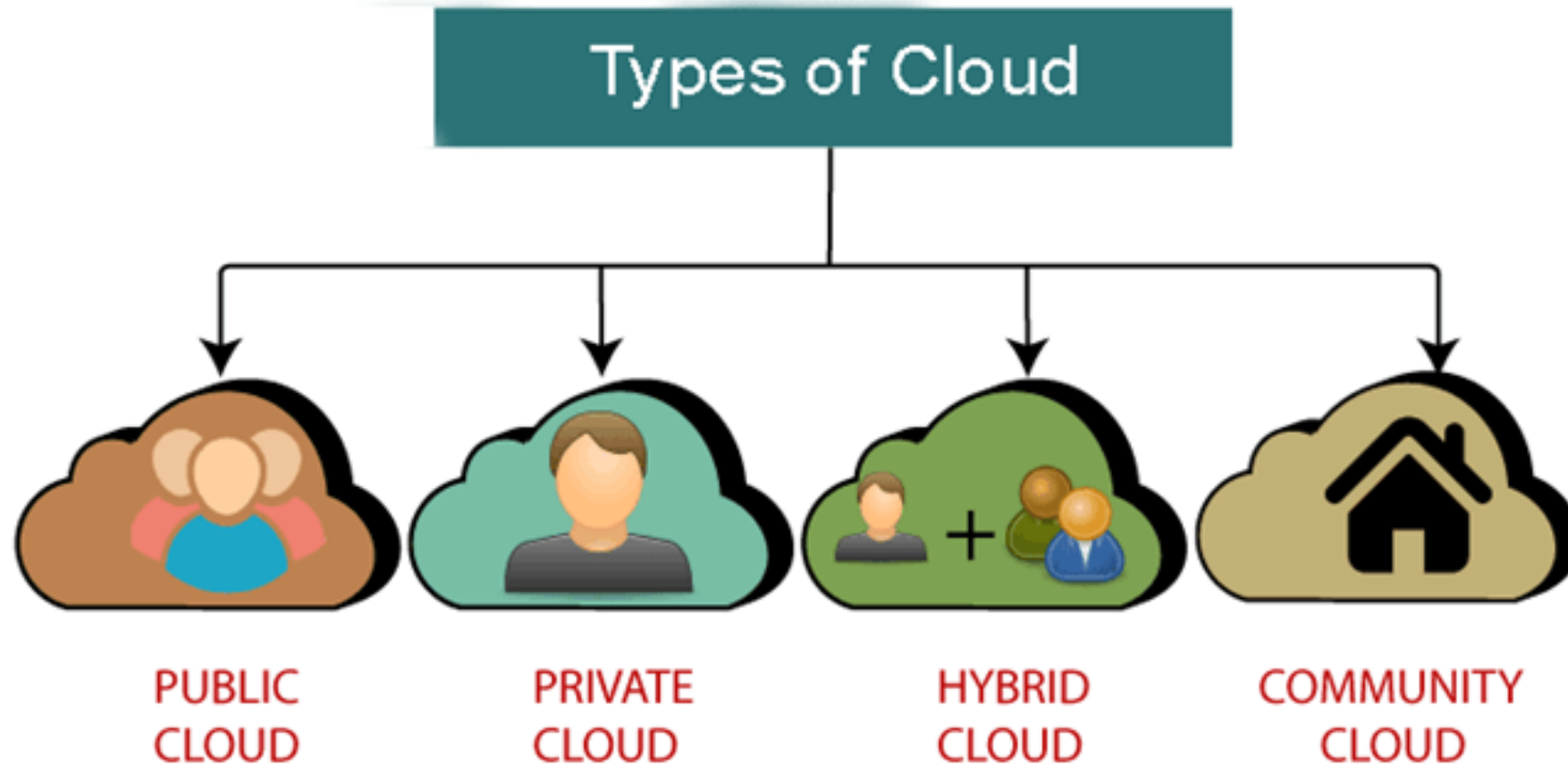


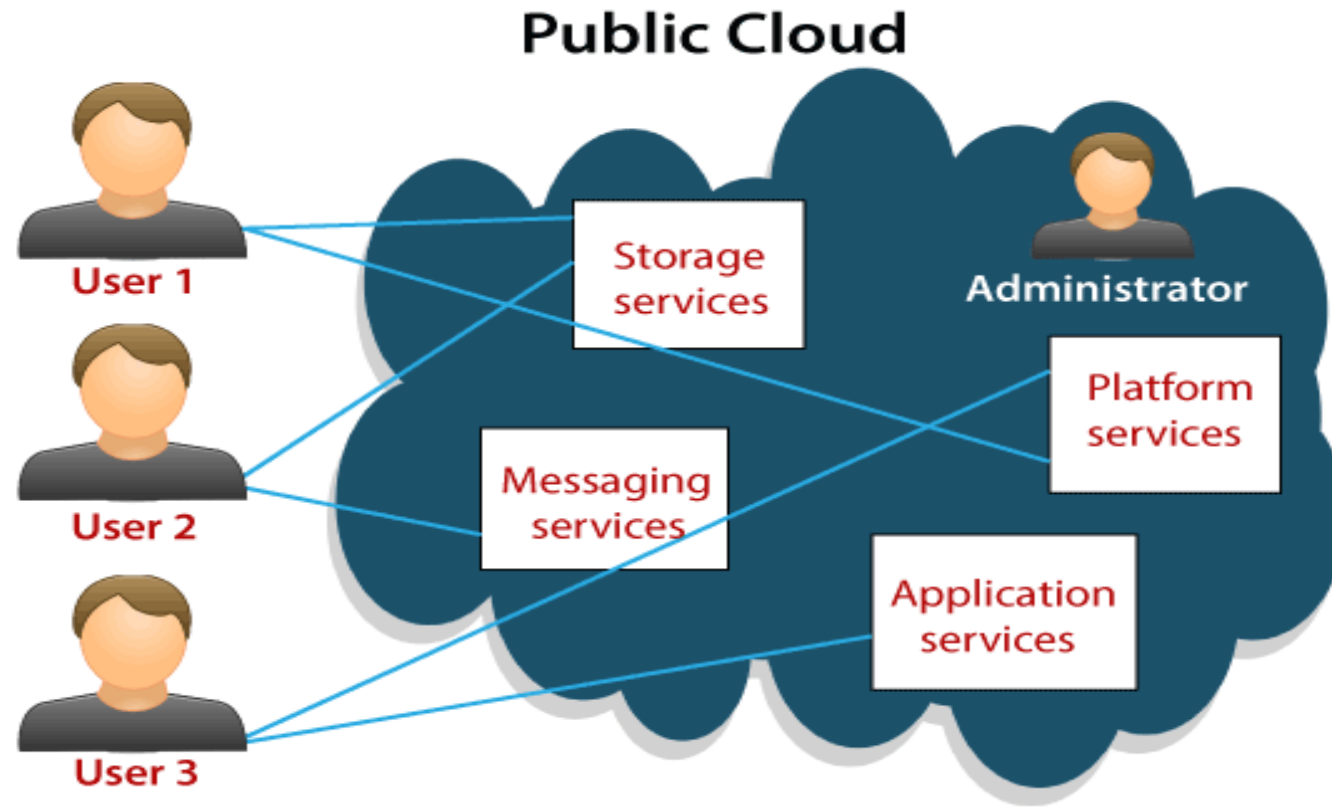
- ❖ Cloud Computing, a cloud vendor is responsible for the hardware purchase and maintenance. They also provide a wide variety of software and platform as a service. We can take any required services on rent.
- ❖ The cloud computing services will be charged based on usage.



- ❖ **Cost:** It reduces the huge capital costs of buying hardware and software.
- ❖ **Speed:** Resources can be accessed in minutes, typically within a few clicks.
- ❖ **Scalability:** We can increase or decrease the requirement of resources according to the business requirements.
- ❖ **Productivity:** While using cloud computing, we put less operational effort. We do not need to apply patching, as well as no need to maintain hardware and software. So, in this way, the IT team can be more productive and focus on achieving business goals.
- ❖ **Reliability:** Backup and recovery of data are less expensive and very fast for business continuity.
- ❖ **Security:** Many cloud vendors offer a broad set of policies, technologies, and controls that strengthen our data security.
- ❖ **High Availability**



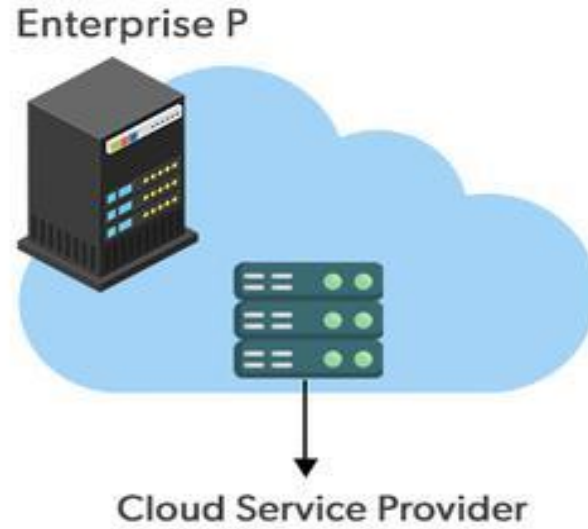




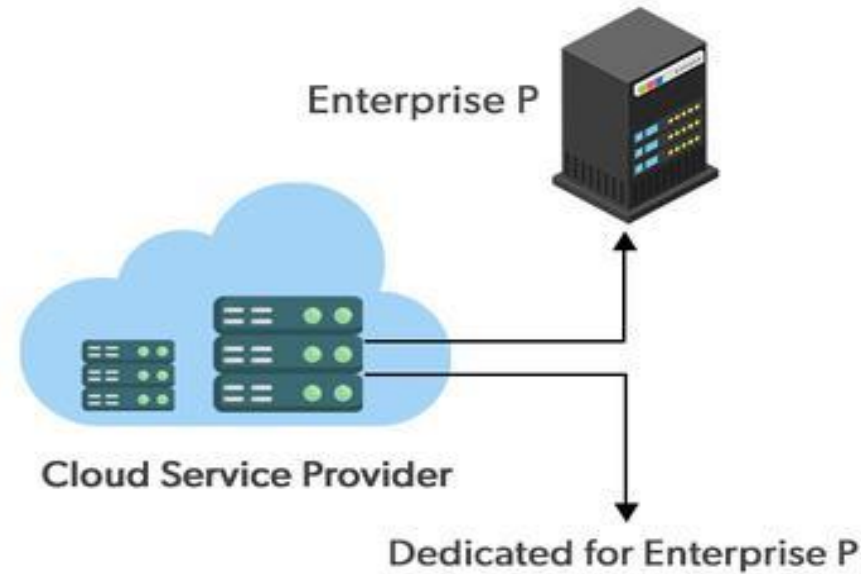
Public Cloud: The cloud resources that are owned and operated by a third-party cloud service provider are termed as public clouds. It delivers computing resources such as servers, software, and storage over the internet



On premise Private cloud



Externally hosted Private cloud

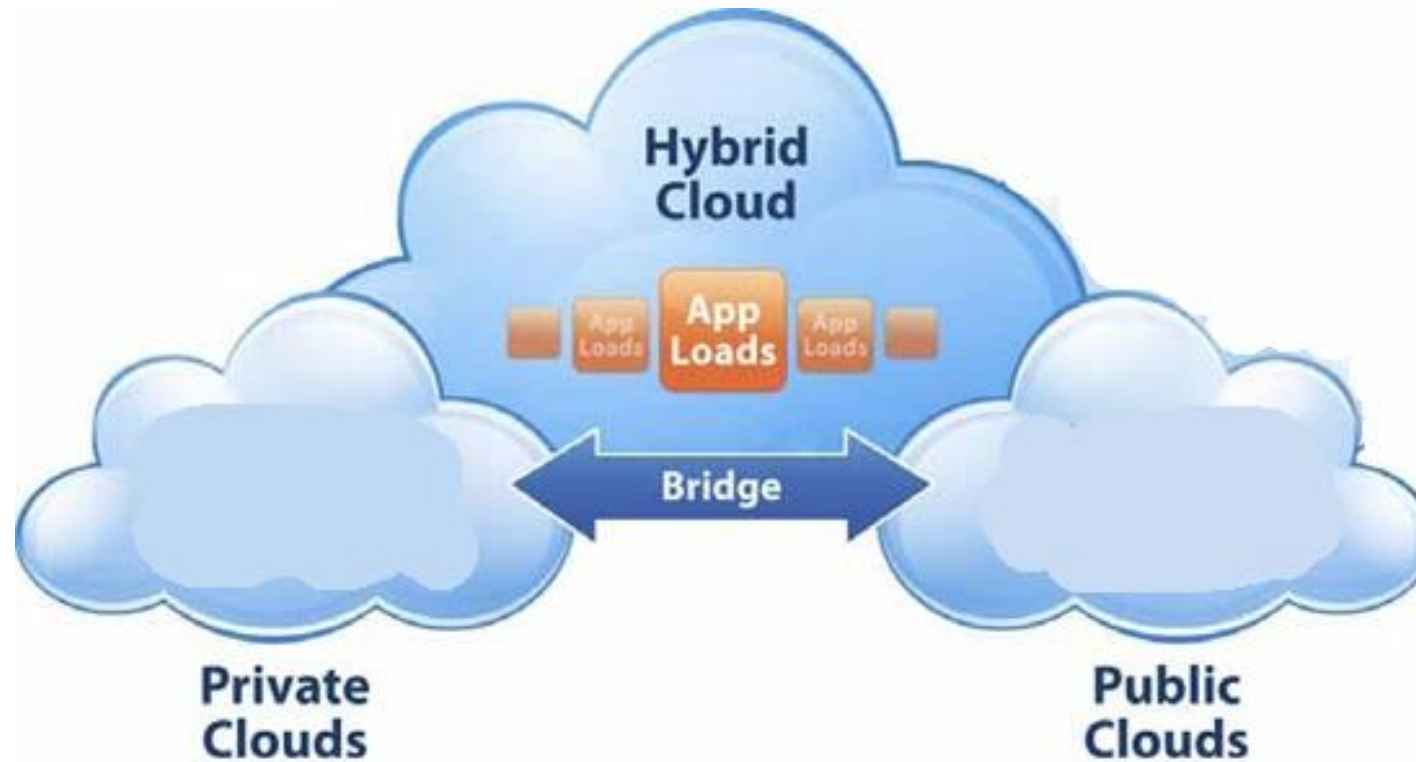


Private Cloud: The cloud computing resources that are exclusively used inside a single business or organization are termed as a private cloud.

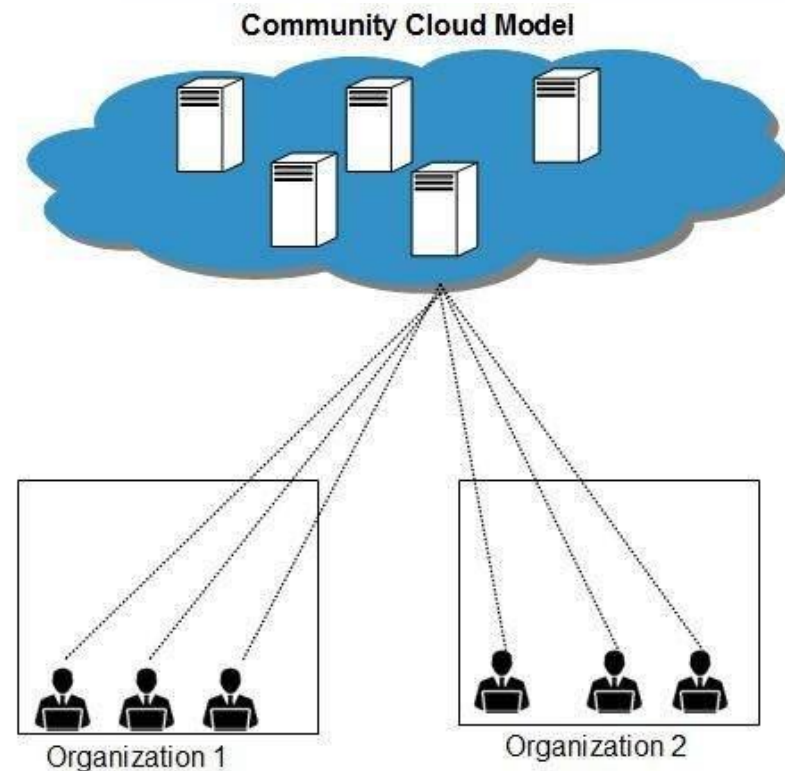
A private cloud may physically be located on the company's on-site datacentre or hosted by a third-party service provider.

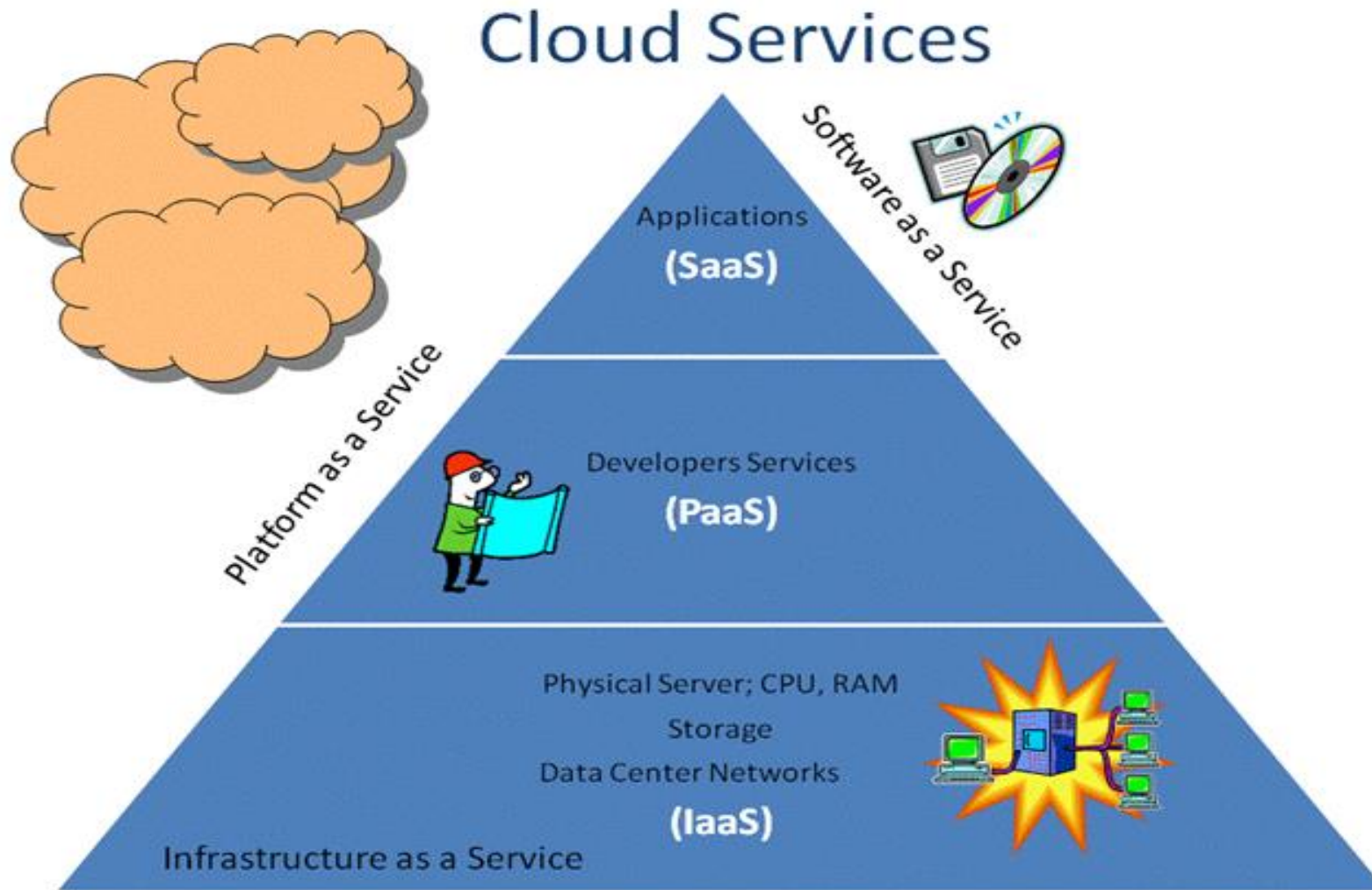


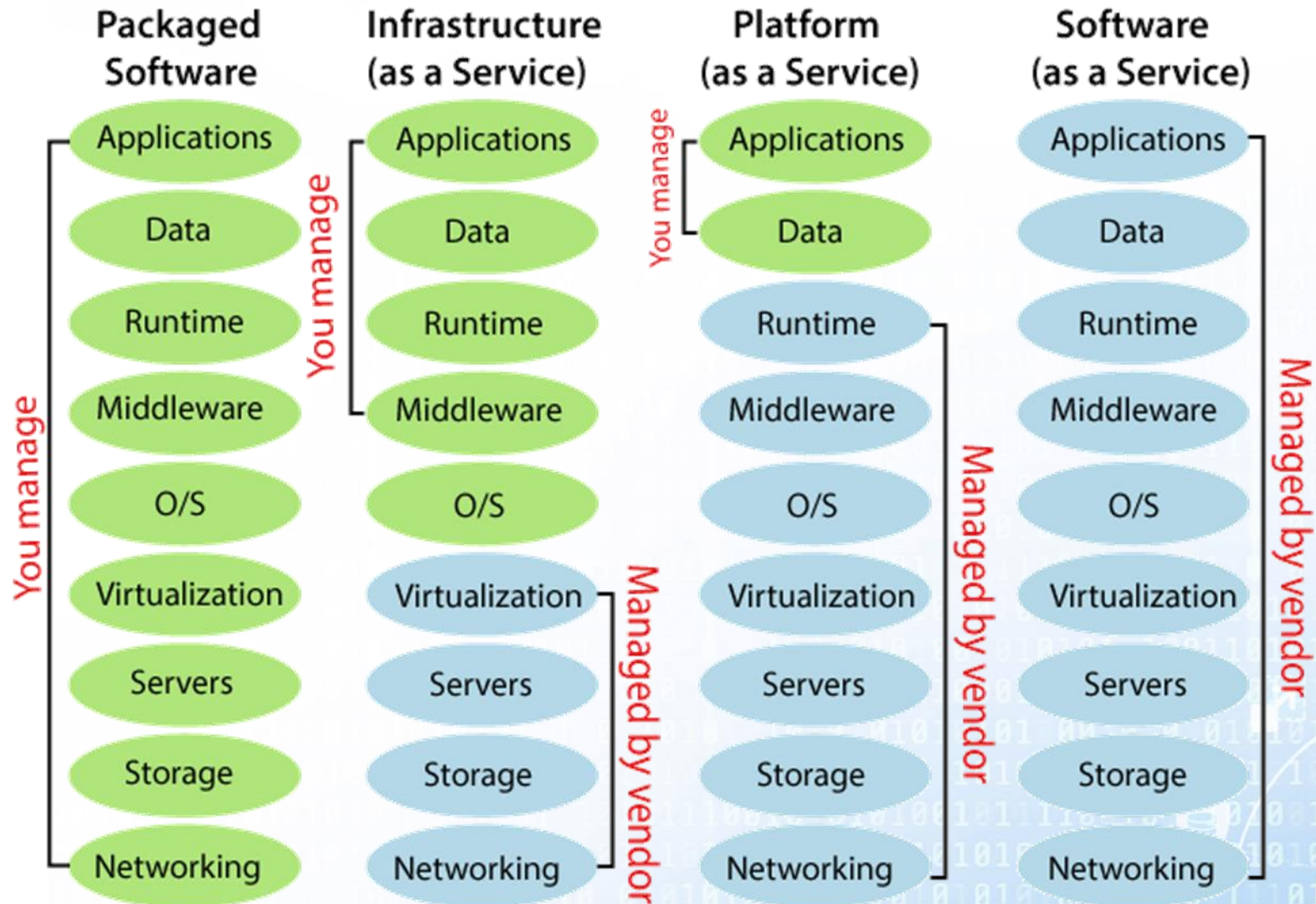
Hybrid Cloud: It is the combination of public and private clouds, which is bounded together by technology that allows data applications to be shared between them. Hybrid cloud provides flexibility and more deployment options to the business.

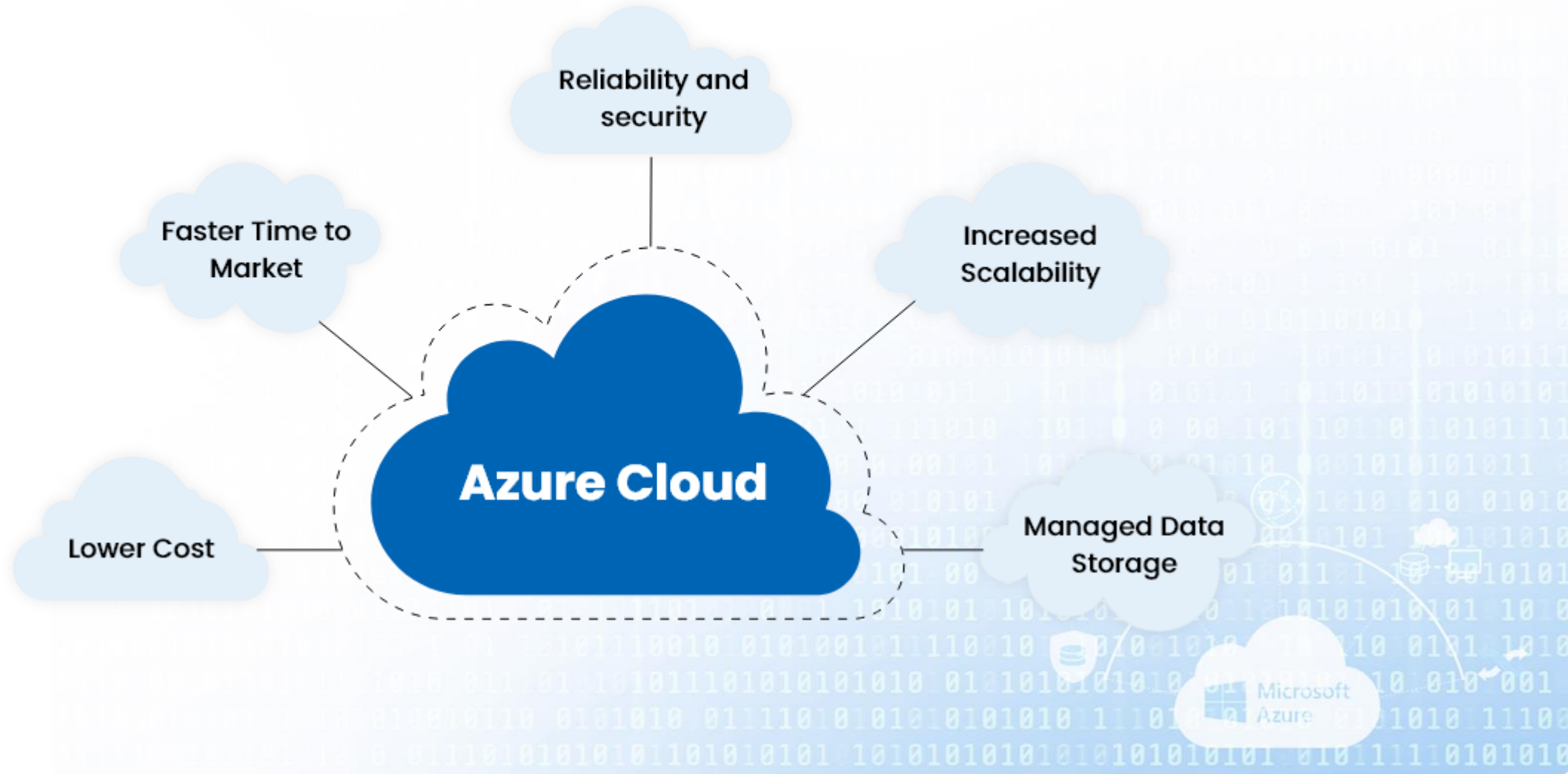


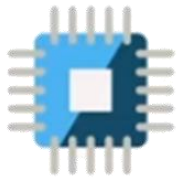
Community cloud allows systems and services to be accessible by a group of several organizations to share the information between the organization and a specific community. It is owned, managed, and operated by one or more organizations in the community, a third party, or a combination of them.







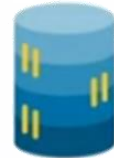




Compute



Storage



Database



Migration



Networking and
content delivery



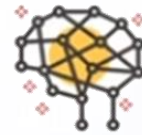
Developer tools



Management
Tools



Media Services



Machine
Learning



Analytics



Security, Identity &
Compliance



Mobile services



Application
Integration



AR and VR



Customer
Engagement



Business
Productivity



Desktop and
app
streaming



IOT
Microsoft
Azure



Virtual Machines



Virtual Machine Scale Sets



Batch



Cloud Services



Service Fabric



Container Service



Container Registry

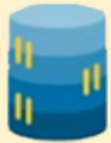


Functions



Web Apps





Azure database domain service offers cost efficient, highly secure and scalable database instances in the cloud



SQL
Database



Azure
Cosmos DB



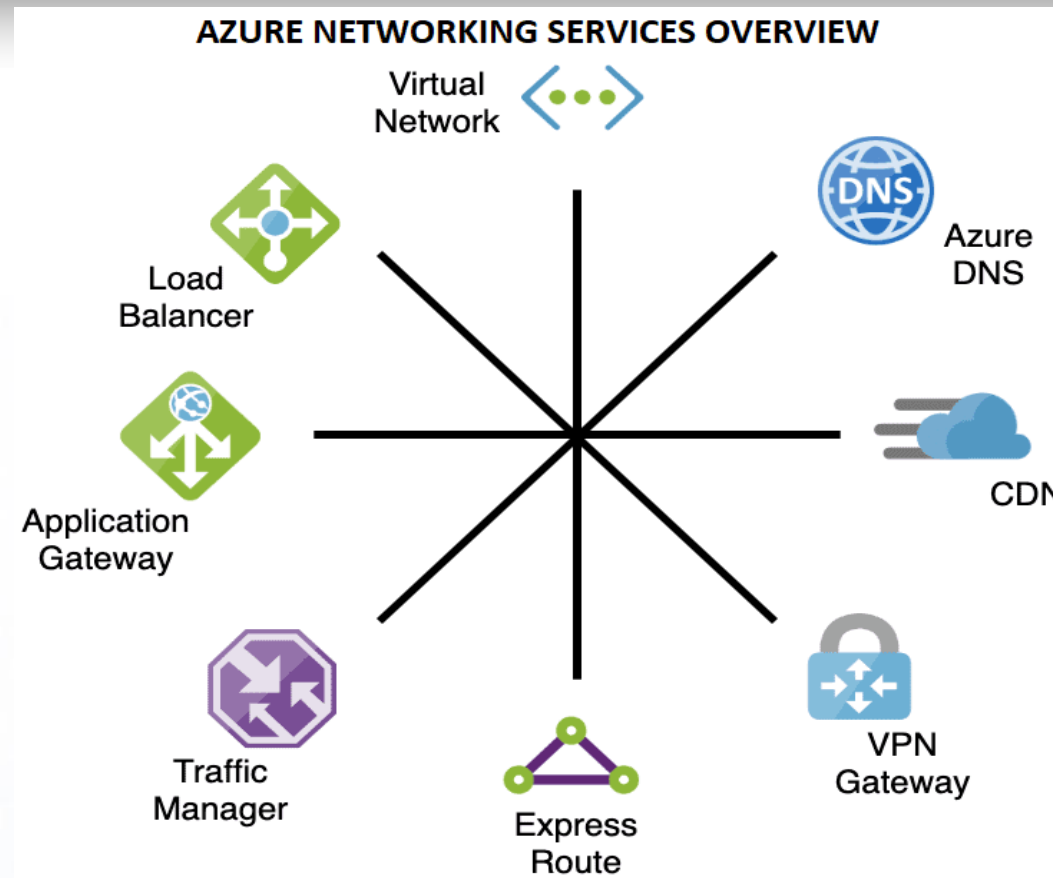
SQL
Data Warehouse

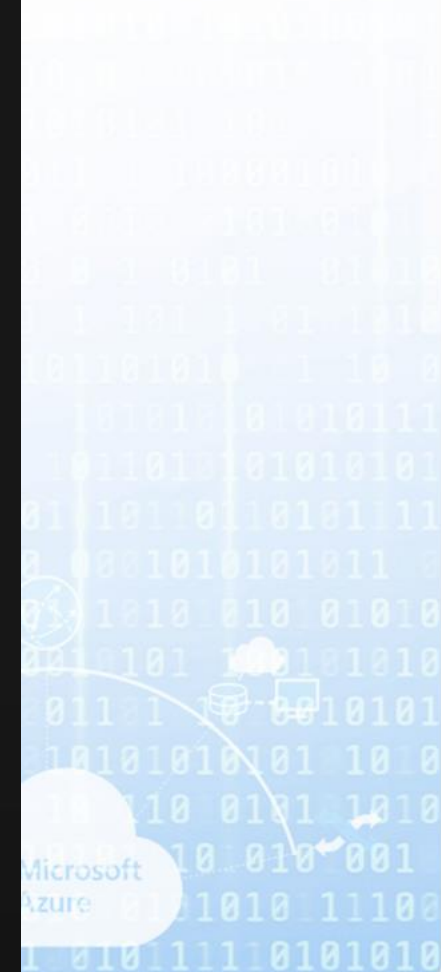
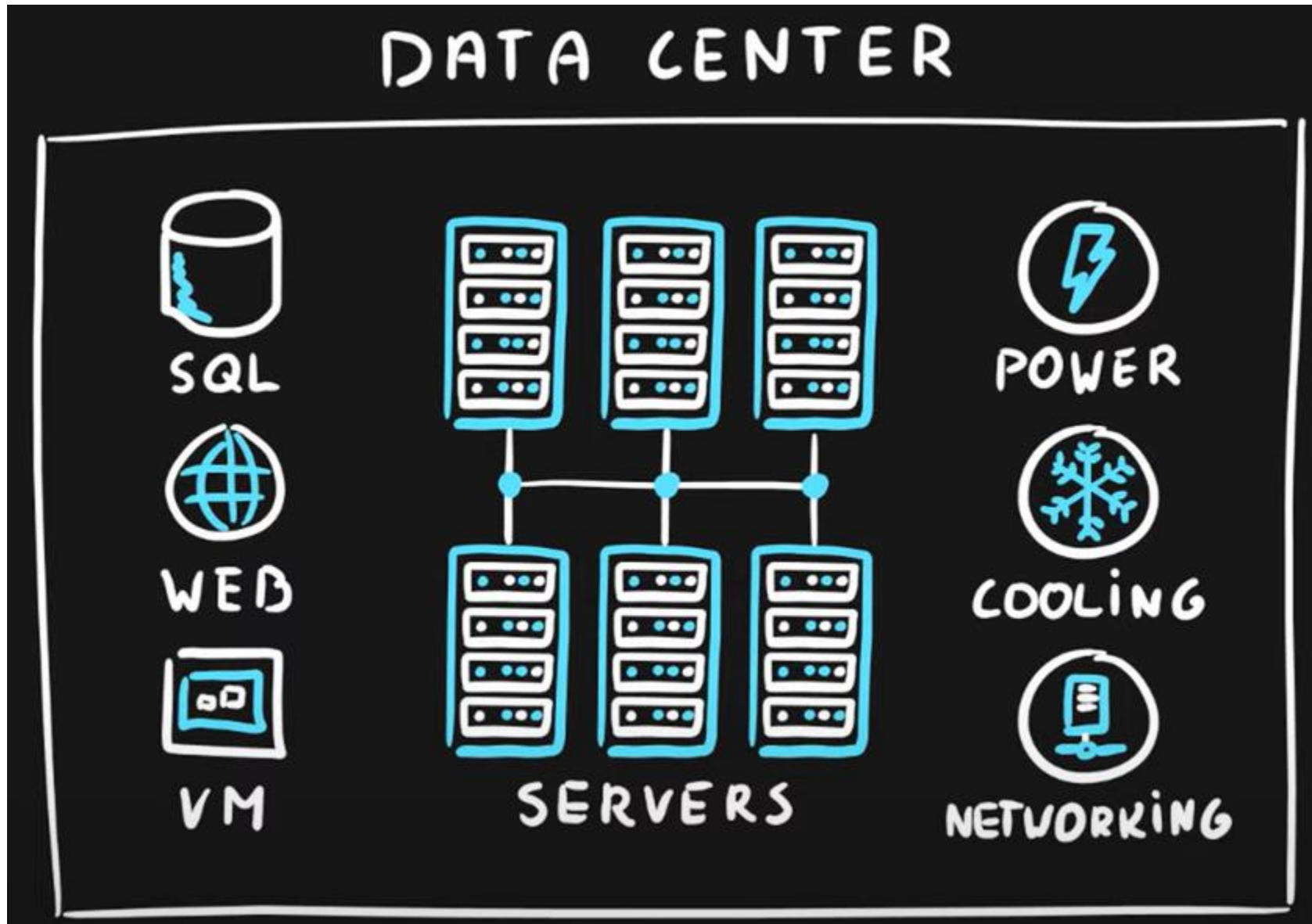


Redis
Cache

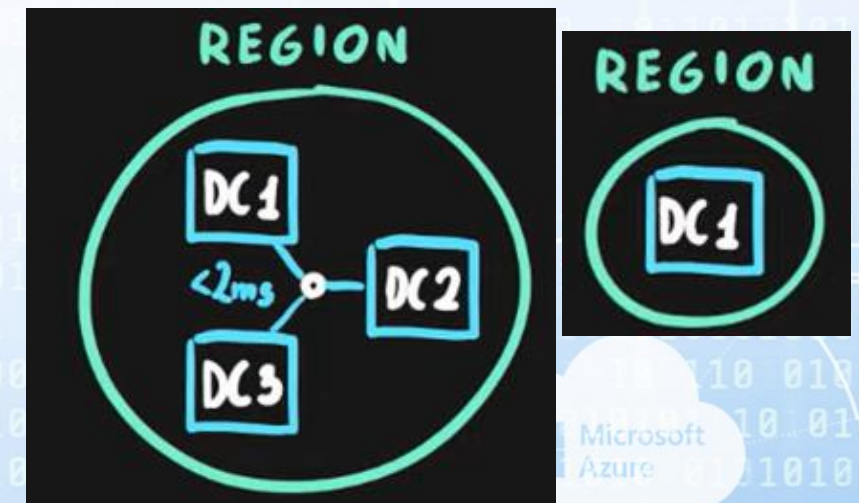
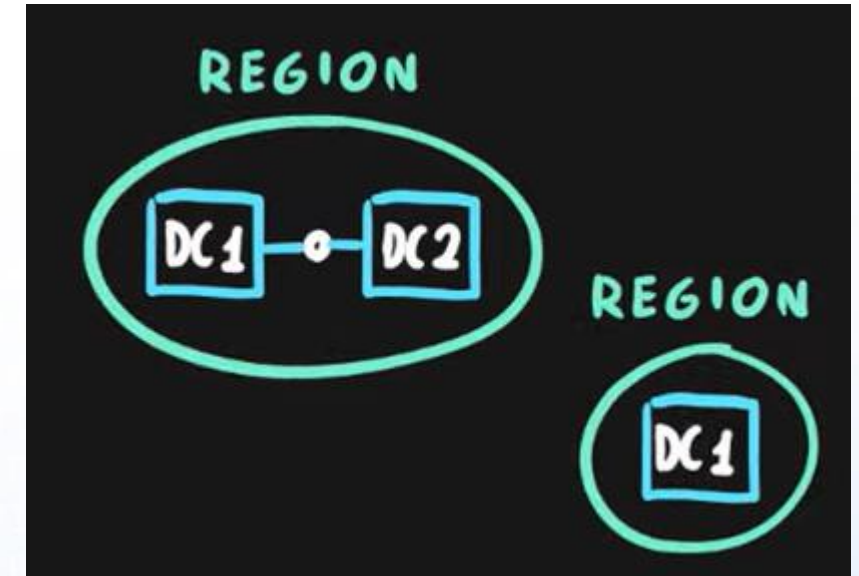


Networking and Content Delivery : It offers a highly secure cloud platform and helps in connecting your physical network to your private Virtual network with high transfer speed

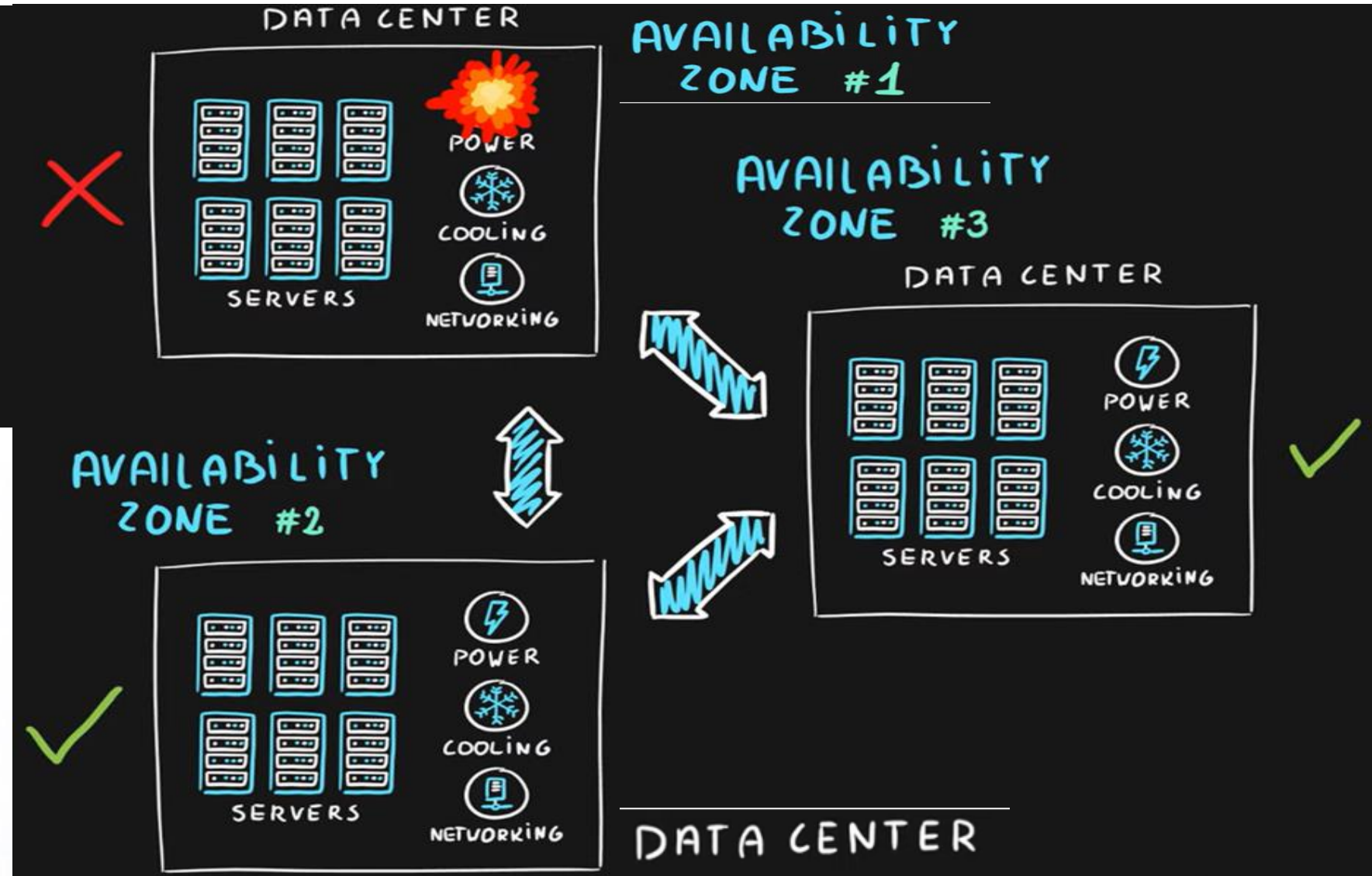
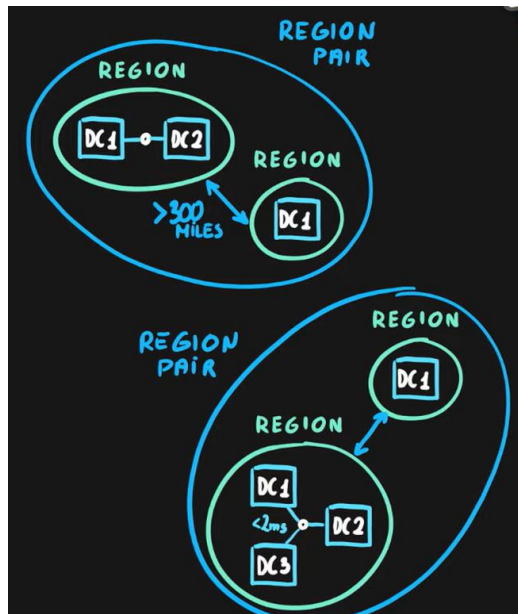


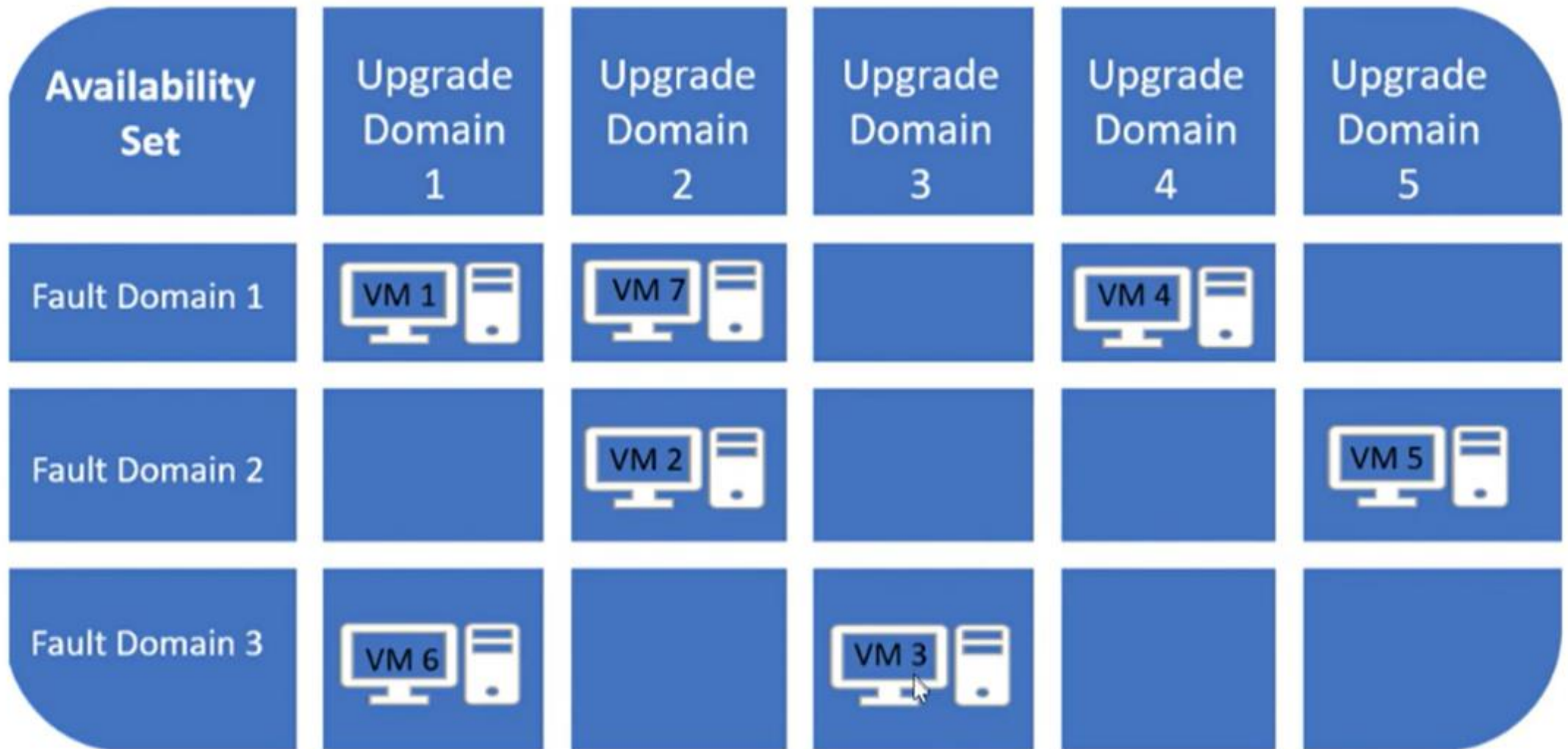


Regions



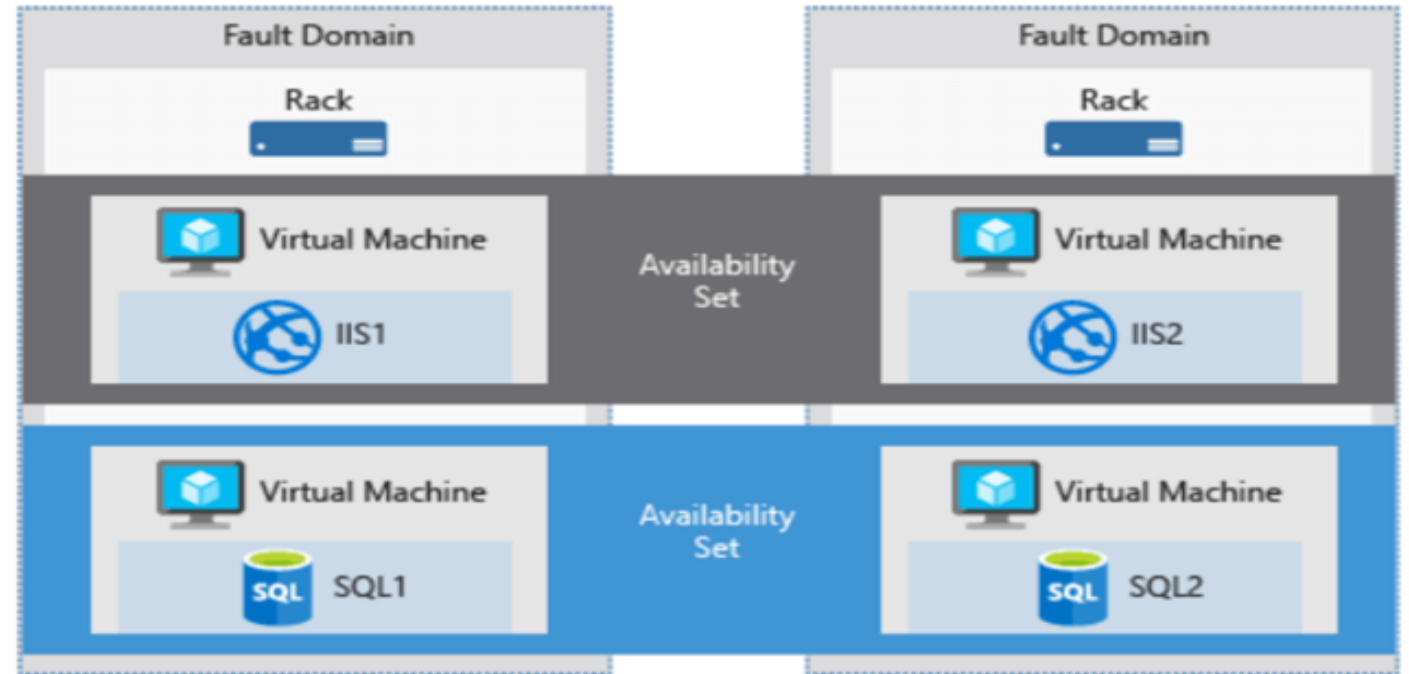
- Zonal services (Virtual Machines, Disks, etc.)
- Zone-redundant services (SQL, Storage, etc.)





Availability Set

- Availability Sets ensure that the Azure virtual machines are deployed across multiple isolated hardware nodes in a cluster.
- By deploying your VM's across multiple hardware nodes Azure ensures that if hardware or software failure happens within Azure, only a sub-set of your virtual machines are impacted and your overall solution is safe and in working condition.



- Availability set provides redundancy for your virtual machines.
- Availability set spreads your virtual machines across multiple fault domains and update domains.
- If you want to leverage Microsoft's 99.95% SLA from Microsoft you must place your VMs inside availability set except your VMs are having premium storage.

Fault Domains And Update Domains

Fault Domain

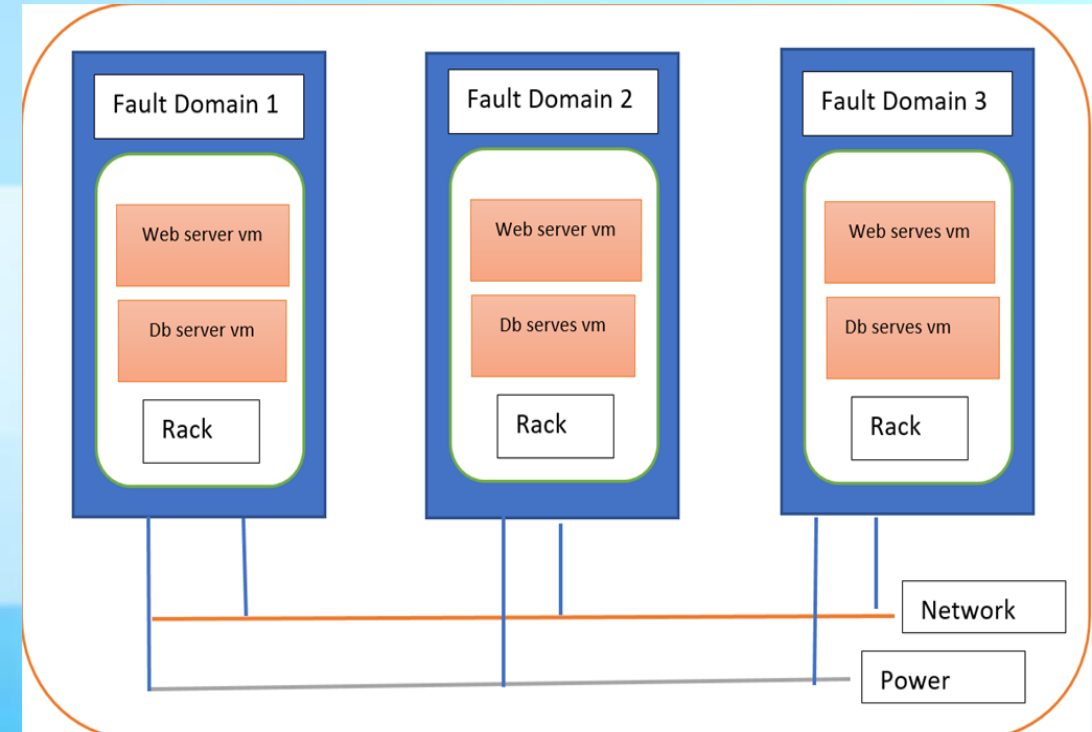
Fault domains define the group of virtual machines that share a common power source and network switch.

Each and every fault domain contains some racks and each rack contains virtual machine.

Each of these Fault domain shares a power supply and a network switch.

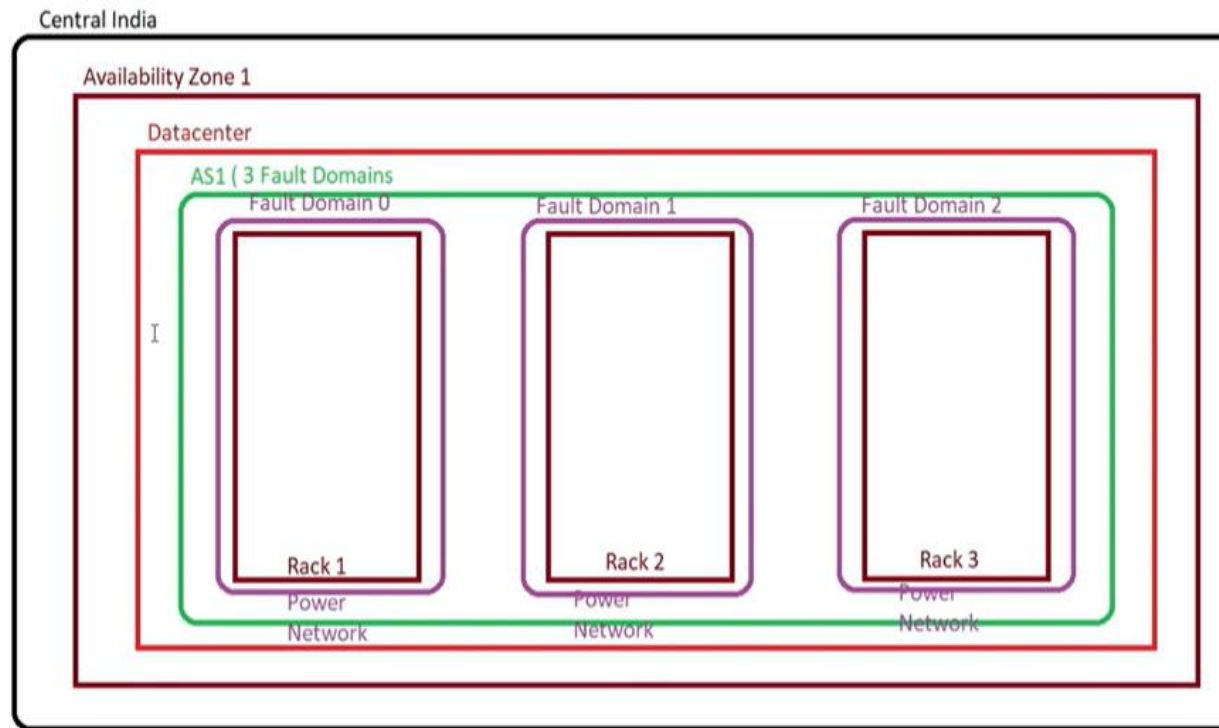
If there is a failure in the fault domain then all the resources in the fault domain become unavailable.

You should place your vms such a way that each fault domain get one web server, one database server and like that.



Availability Set

Fault Domain and Update Domain



Azure Replication Methods

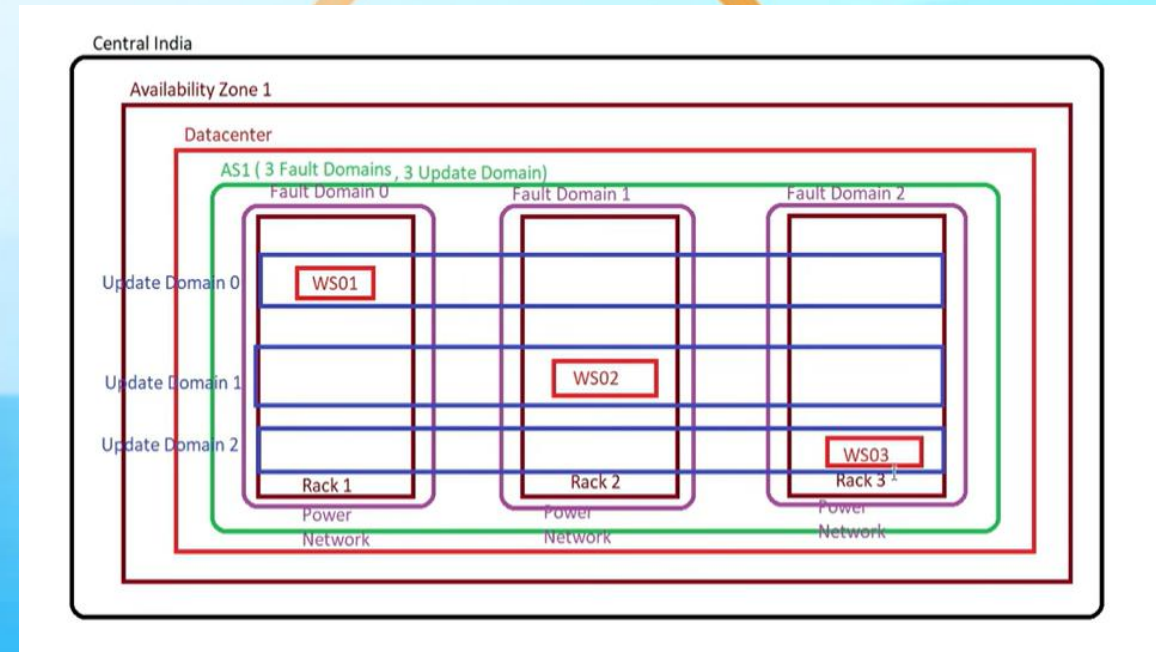
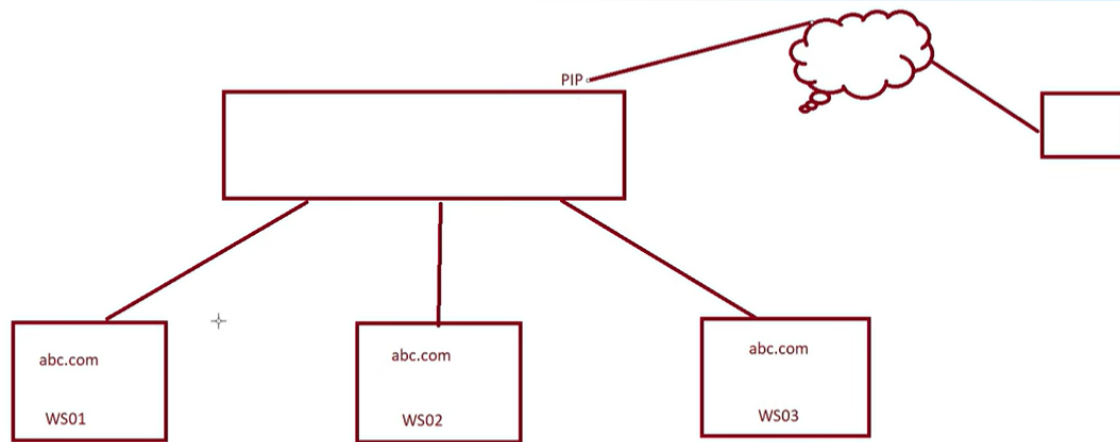
Update Domains

Virtual machines get update domains automatically once they are put inside availability set.

All virtual machines within that update domain will reboot together.

Update domains are used for patching of the virtual machines.

Only one update domain would be updated at the time



Azure accounts and subscriptions

Azure account gives access to the Azure resources and Azure subscription

You can create multiple subscription in Azure account to create separation

Eg: Billing and management purpose of Azure resources

Azure account is tied to a specific identity and holds information like:

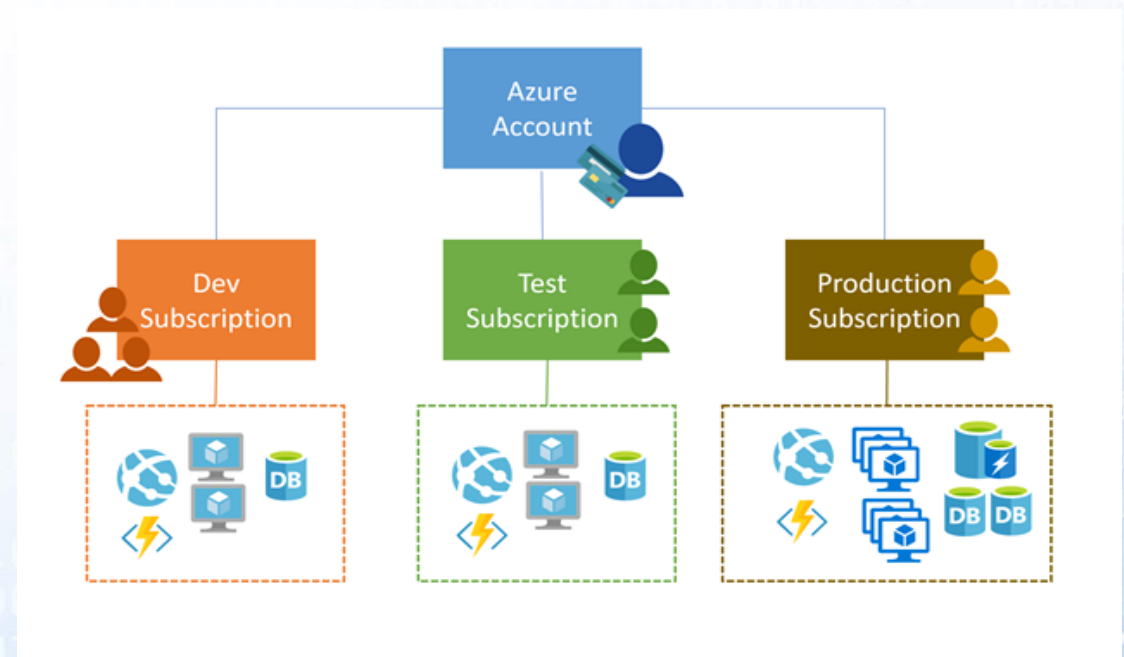
- Name, email, and contact preferences
- Billing information such as a credit card

An Azure account is what you use to sign in to the Azure website and administer or deploy services.

Every Azure account is associated with one or more subscriptions

An Azure subscription is a logical container used to provision resources in Microsoft Azure.

It holds the details of all your resources like virtual machines, databases, etc



Azure accounts and subscriptions

Subscription types

Subscription	Usage
Free	Includes a \$200 credit for the first 30 days, free limited access for 12 months
Pay-As-You-Go	Charges you monthly
Enterprise	One agreement, with discounts for new licenses and Software Assurance - targeted at enterprise-scale organizations.
Student	Includes \$100 for 12 months – must verify student access

Note: You must verify your student status through your organizational email address

Billing

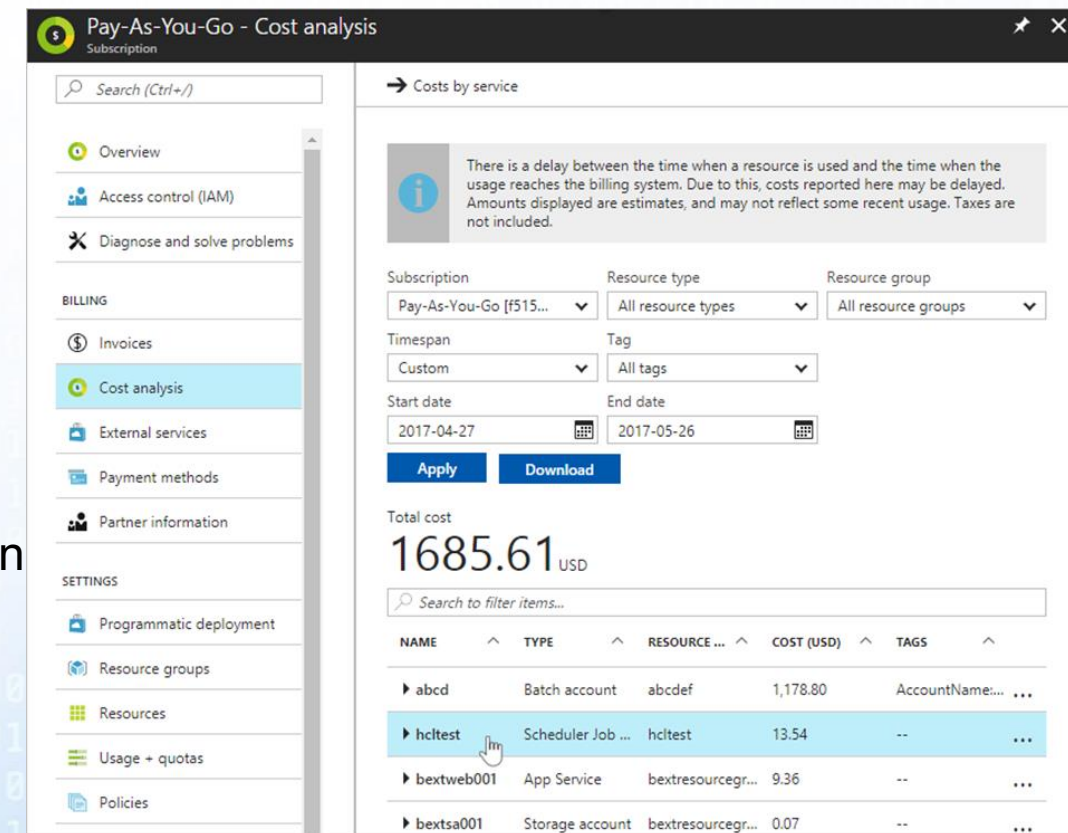
One bill is generated for every Azure subscription on a monthly basis. The payment is charged automatically to the associated account credit or debit card within 10 days after the billing period ends.

Note

Subscriptions are billed independently, but the account owner is responsible for payment. In the case of "Pay-as-you-go" subscriptions, the account credit card will be charged for all associated subscriptions

You can analyze your bill in the Azure portal - this will provide access to all your invoices, as well as a cost analysis breakdown of what got charged each month.

You can set spending limits on each subscription to ensure you aren't surprised at the end of the month.
Reports can be generated by subscriptions,



Pay-As-You-Go - Cost analysis

Subscription: Pay-As-You-Go [f515...]

Resource type: All resource types

Resource group: All resource groups

Timespan: Custom

Tag: All tags

Start date: 2017-04-27

End date: 2017-05-26

Apply **Download**

Total cost: **1685.61** USD

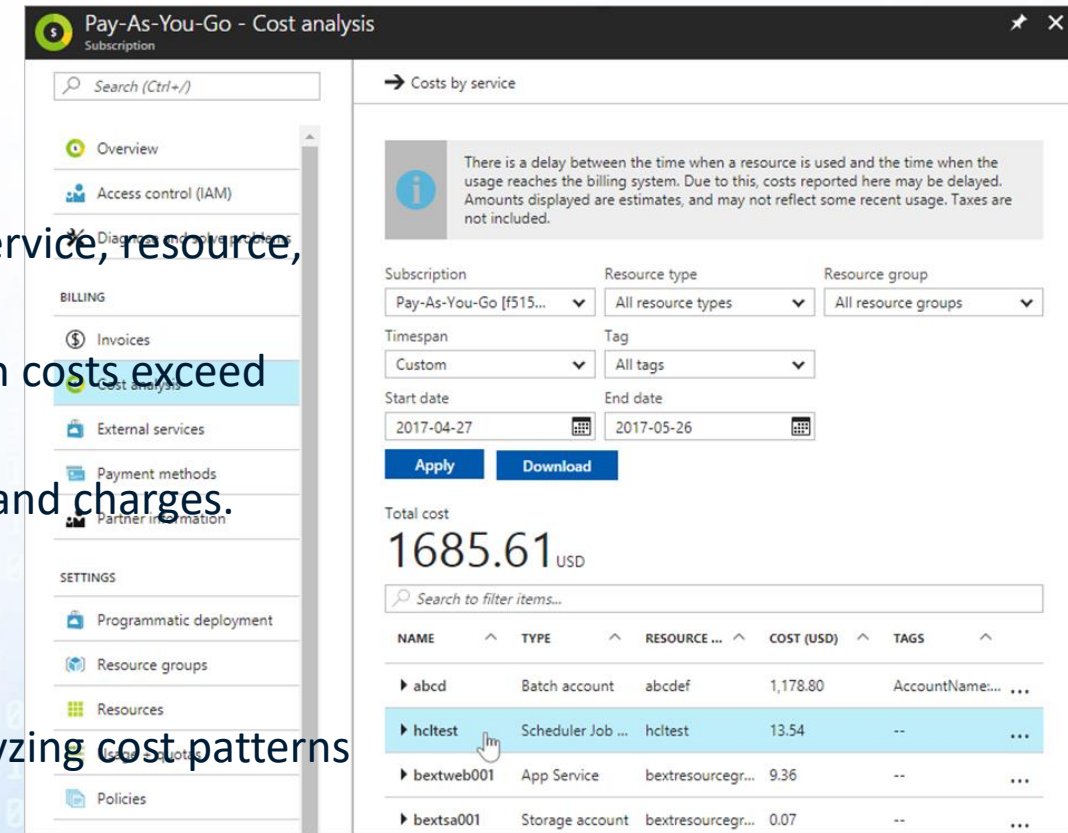
Search to filter items...

NAME	TYPE	RESOURCE ...	COST (USD)	TAGS
abcd	Batch account	abcdef	1,178.80	AccountName:...
hcltest	Scheduler Job ...	hcltest	13.54	--
bextweb001	App Service	bextresourcegr...	9.36	--
bextsa001	Storage account	bextresourcegr...	0.07	--

Azure's Cost Management + Billing service helps you understand, report on, and analyze your Azure costs. It provides tools like Cost Analysis, which allows you to explore costs at different granularities, including by service, resource group, or subscription. You can also set budgets and alerts to manage your spending, download usage details, and view invoice details.

Key features of Azure Cost Management + Billing:

- **Cost Analysis:** Explore costs across various dimensions, including by service, resource, and tags.
- **Budgets and Alerts:** Set spending limits and receive notifications when costs exceed those limits.
- **Usage and Charges Details:** Download detailed reports of your usage and charges.
- **Invoice Details:** View the charges as they appear on your invoice.
- **Forecasting:** Forecast future costs based on historical usage patterns.
- **Cost Optimization:** Identify and optimize your cloud spending by analyzing cost patterns and trends.
- **Integration with Power BI:** Create custom dashboards and reports using Power BI to analyze your costs.



Pay-As-You-Go - Cost analysis

Subscription: Pay-As-You-Go [f515...]

Resource type: All resource types

Resource group: All resource groups

Timespan: Custom

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bextsa001	Storage account	bextresourcegr...	0.07	--

Tags

- Tags are metadata elements that you apply to your Azure resources.
- They're key-value pairs that help you identify resources based on settings that are relevant to your organization.
- If you want to track the deployment environment for your resources, add a key named Environment.
- To identify the resources deployed to production, give them a value of Production. The full key-value pair is Environment = Production.
- You can use tags to group your billing data.



Resource Tags

You can apply tags to your Azure resources to logically organize them by categories.

Each tag consists of a name and a value. For example, you can apply the name “Environment” and the value “Production” or “Development” to your resources.

After creating your tags, you associate them with the appropriate resources.

With tags in place, you can retrieve all the resources in your subscription with that tag name and value.

Perhaps one of the best uses of tags is to group billing data.

When you download the usage CSV for services, the tags appear in the Tags column.

For example, you could group virtual machines by cost center and production environment.

Each resource or resource group can have a maximum of 15 tag name/value pairs.

Tags applied to the resource group are not inherited by the resources in that resource group



Billing

The Pricing Calculator provides estimates in all areas of Azure including compute, networking, storage, web, and databases.

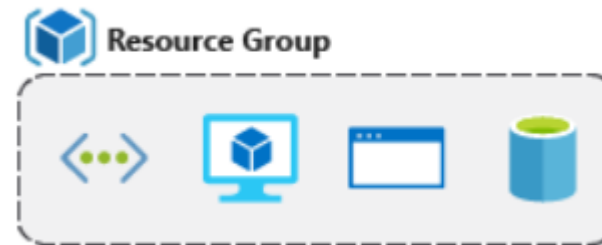
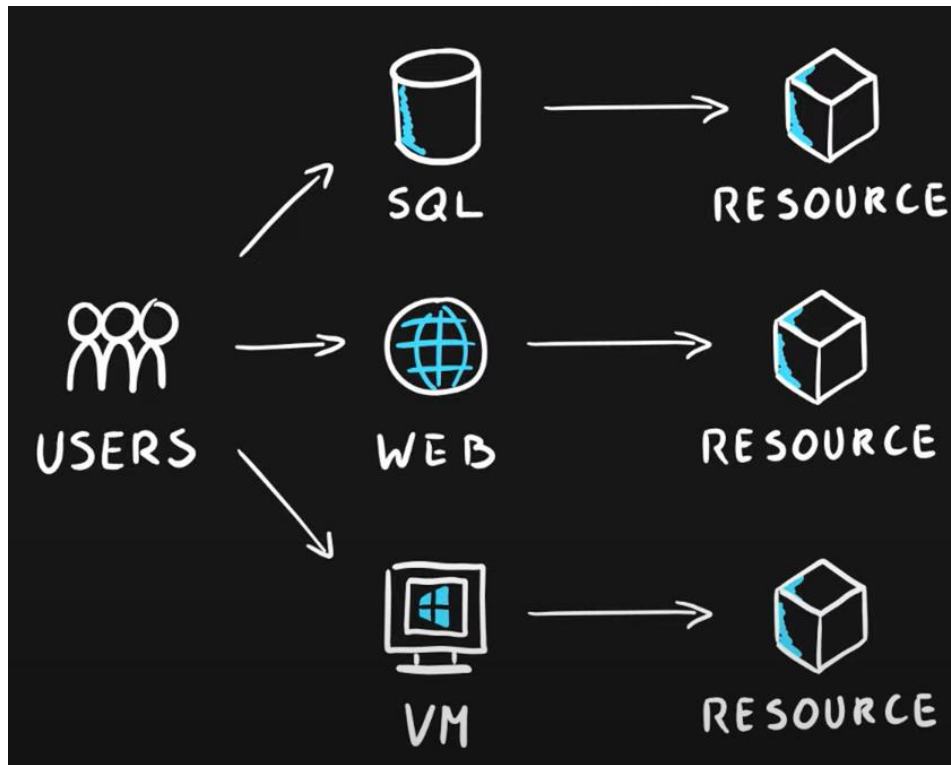
Billing Alerts help you monitor and manage billing activity for your Azure accounts. Billing alerts is available from the Account portal. You can set up a total of five billing alerts per subscription, with a different threshold and up to two email recipients for each alert

Reservations helps you save money by pre-paying for one-year or three-years of virtual machine, SQL Database compute capacity, Azure Cosmos DB throughput, or other Azure resources. Pre-paying allows you to get a discount on the resources you use

Reservations provide a billing discount and don't affect the runtime state of your resources

Budgets help you plan for and drive organizational accountability. With budgets, you can account for the Azure services you consume or subscribe to during a specific period. They help you inform others about their spending to proactively manage costs, and to monitor how spending progresses over time.





A resource is **an entity managed by Azure**.

For example, virtual machines, virtual networks, and storage accounts are all referred to as Azure resources.

A **resource group** is a container that holds related resources for an Azure solution..

Resource groups make it easier to apply access controls, monitor activity, and track the costs related to specific workloads

The resource group collects metadata from each individual resource to facilitate more granular management than at the subscription level

Resources can only exist in one resource group

Groups cannot be renamed

Groups can have resources of many different types (services)

Groups can have resources from many different regions

Deployments are incremental

Resources grouped
(Web + DB, VM, Storage) in one group



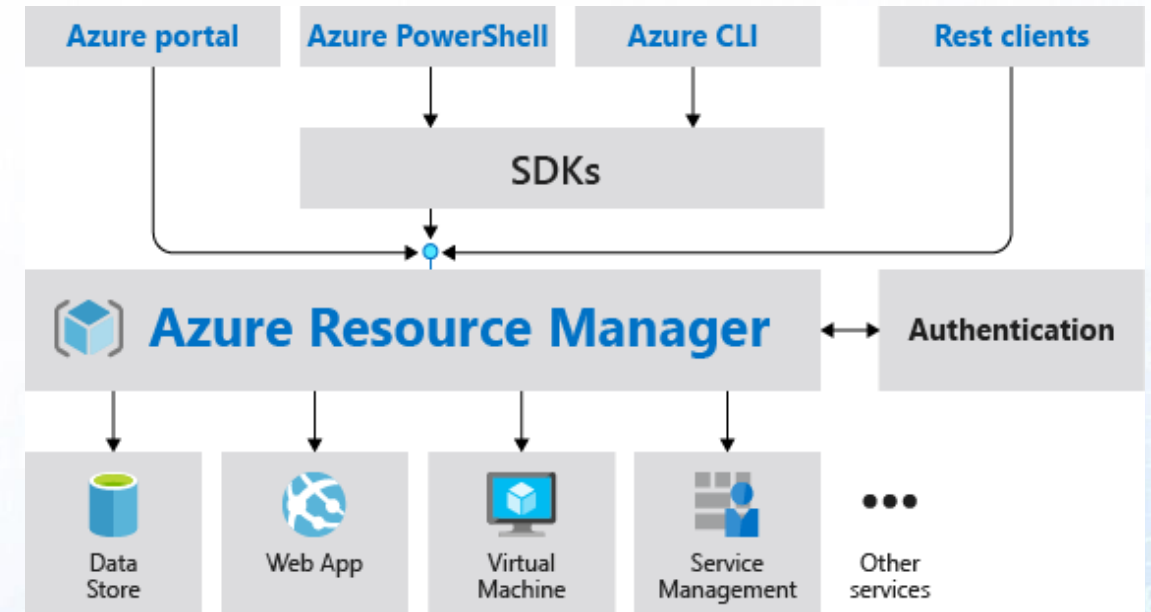
Provides a consistent management layer

Enables you to work with the resources in your solution as a group

Deploy, update, or delete in a single, coordinated operation

Provides security, auditing, and tagging features

Choose the tools and APIs that work best for you

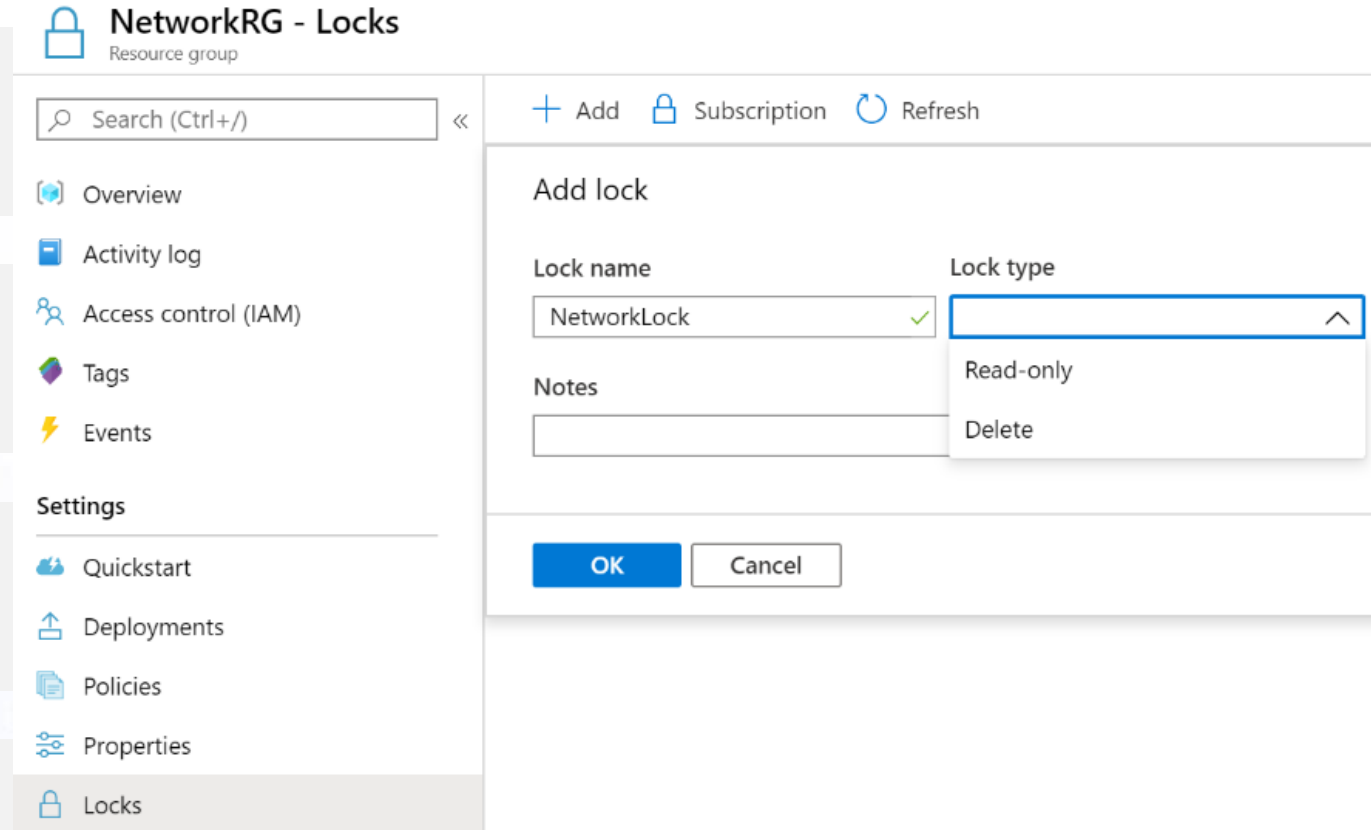


Associate the lock with a subscription, resource group, or resource

Locks are inherited by child resources

Read-Only locks prevent any changes to the resource

Delete locks prevent deletion



NetworkRG - Locks
Resource group

Search (Ctrl+ /)

Overview
Activity log
Access control (IAM)
Tags
Events

Settings

Quickstart
Deployments
Policies
Properties
Locks

+ Add Subscription Refresh

Add lock

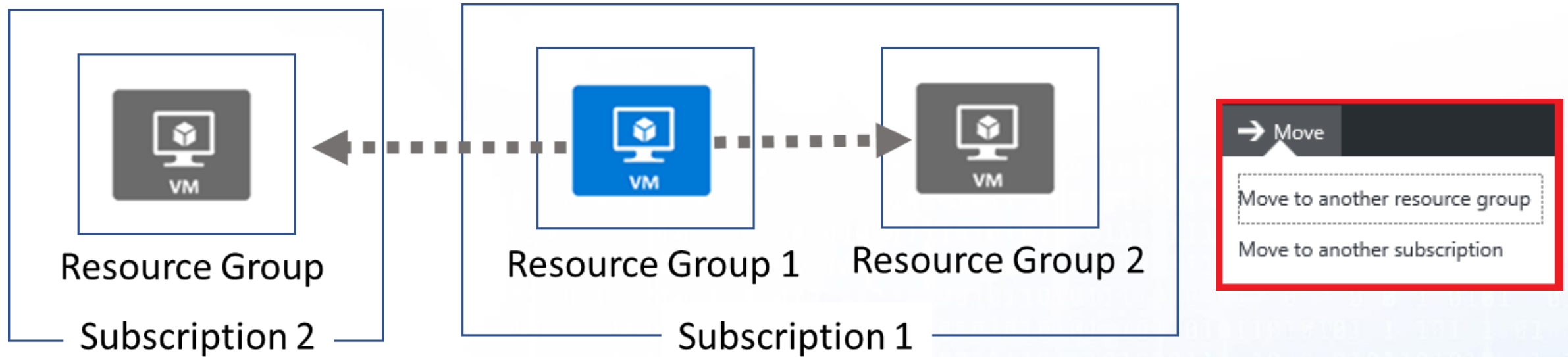
Lock name: NetworkLock ✓

Lock type: Read-only (selected)
Delete

Notes:

OK Cancel

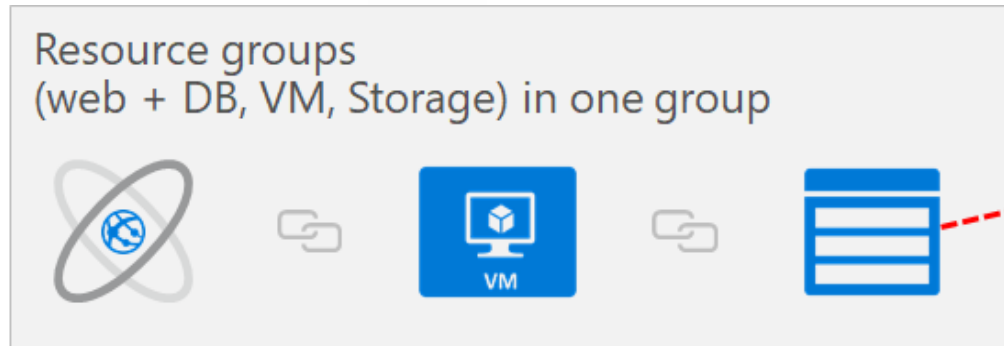




When moving resources, both the source group and the target group are locked during the operation

Services that cannot be moved: Azure AD Domain Services, ExpressRoute, and Site Recovery. Other restrictions apply





Refresh → Move **Delete**

Resource group (change) : ASH

Location : West Central US

Subscription (change) : Visual Studio Enterprise

```
Get-AzResourceGroup -Name 'az104-03*' | Remove-AzResourceGroup -Force -AsJob
```

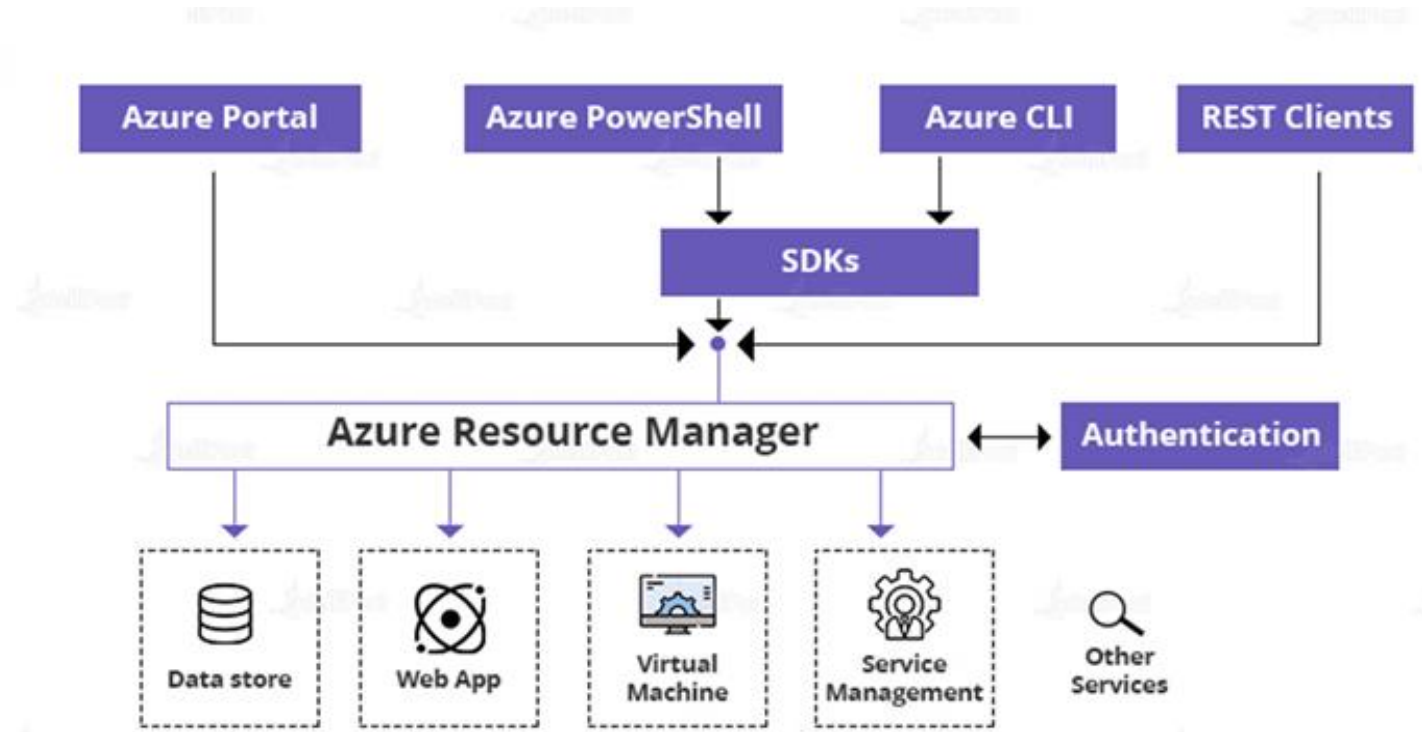
Remove Azure resources
that you no longer use

Ensures you will not see
unexpected charges

Remove individual resources
or remove the resource group



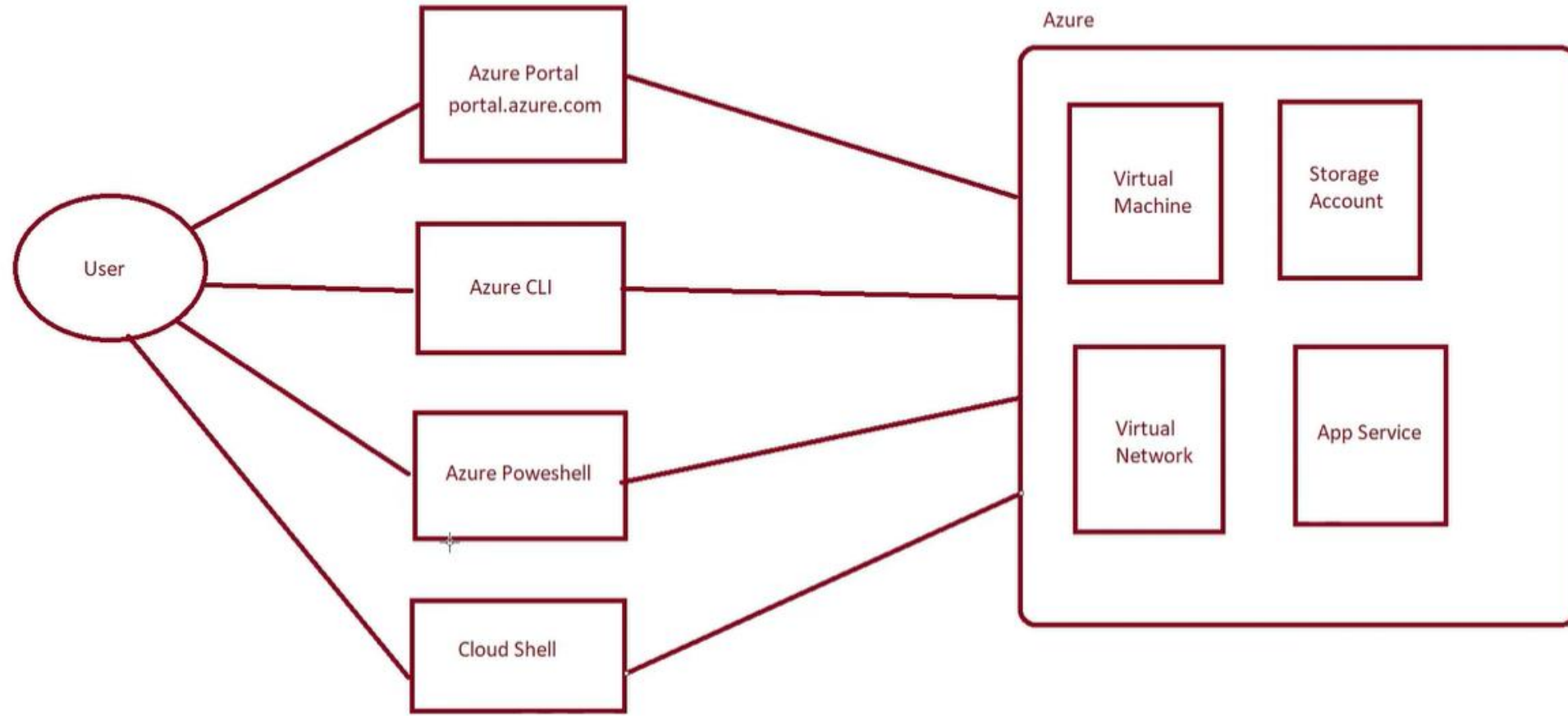
Microsoft
Azure



Azure Cloud shell require
Azure Storage Account,
Resource group and Azure file
share

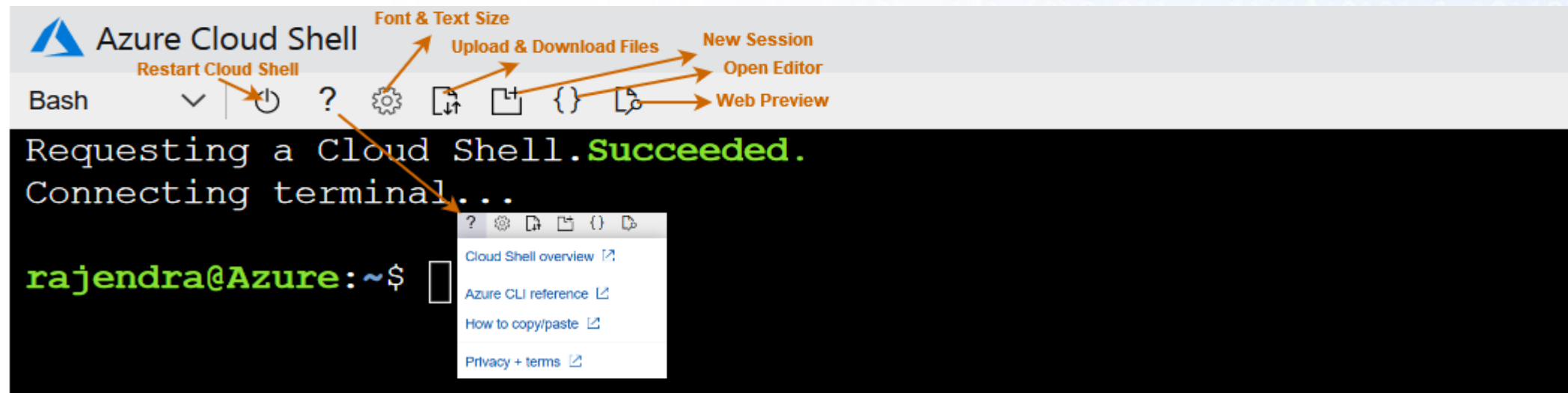
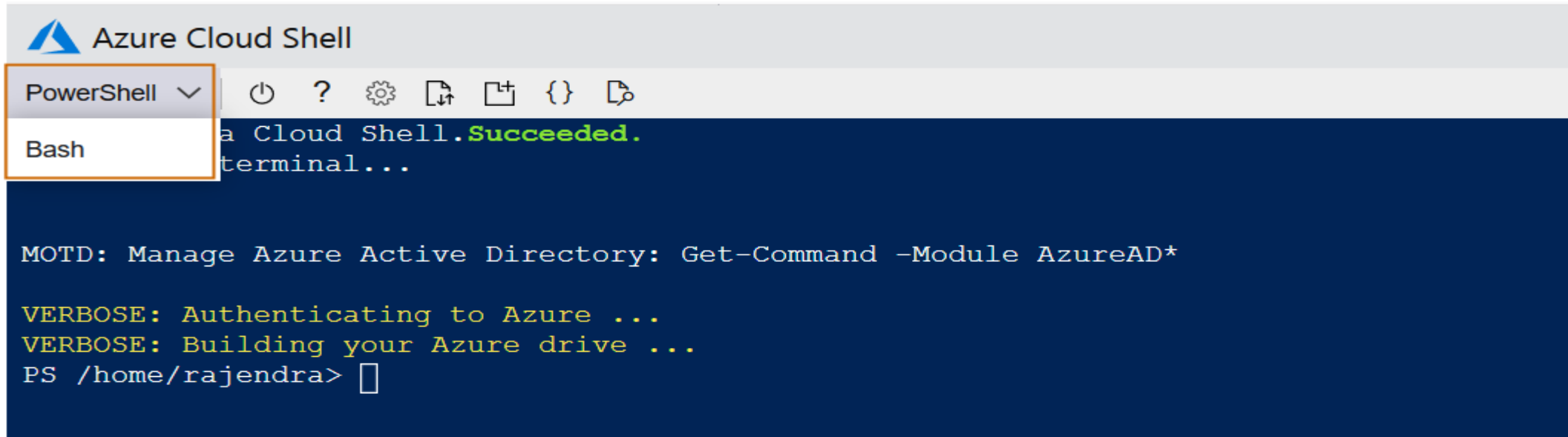
Cloud Shell logs out after 20
min without interactive activity

You can open a free support
case to increase limits to
published maximums



- Azure Cloud Shell is an interactive, authenticated, browser-accessible terminal for managing Azure resources.
-
- It provides the flexibility of choosing the shell experience that best suits the way you work, either Bash or PowerShell.
- Cloud Shell runs on a temporary host provided on a per-session, per-user basis. Your Cloud Shell session times out after 20 minutes without interactive activity.
- Cloud Shell persists your files in your \$HOME location using a 5-GB file share.
- **Shell.azure.com**





Managing Virtual Machines:

az vm stop	Stops a virtual machine.
az vm start	Starts a virtual machine.
az vm restart	Restarts a virtual machine.
az vm redeploy	Redeploys a virtual machine.
az vm delete	Deletes a virtual machine.
az vm list	Lists all virtual machines in a resource group.

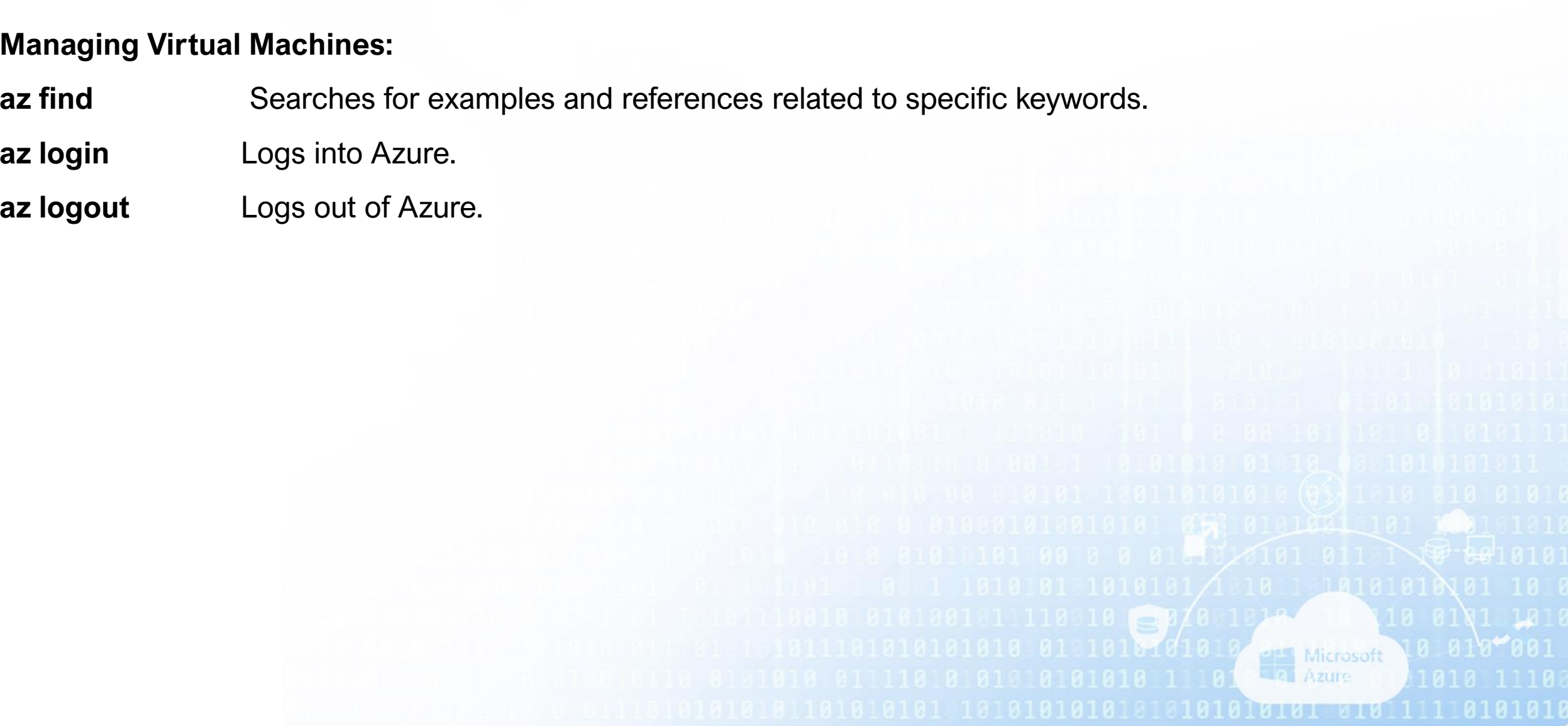
Managing Resource Groups:

az group create	Creates a new resource group.
az group list	Lists all resource groups.
az group delete	Deletes a resource group.



Managing Virtual Machines:

az find	Searches for examples and references related to specific keywords.
az login	Logs into Azure.
az logout	Logs out of Azure.



```
PS C:\> az vm stop -n Win2k16VM1 -g TESTVMRG --verbose
```

```
PS C:\> az vm stop --name win2k16vm1 --resource-group testvmrg
```

Alternatively, you can also use the full parameter name as shown below.

```
PS C:\> az vm stop --name win2k16vm1 --resource-group testvmrg
```

To stop the Azure VM without waiting for the VM to stop and move the execution to the next step, we can use `--no-wait` parameter.

```
PS C:\> az vm stop --name win2k16vm1 --resource-group testvmrg --no-wait
```



Create a resource group

```
az group create --name myResourceGroup --location eastus
```

Create a Linux VM

```
az vm create --resource-group myResourceGroup --name myVM --image LinuxImageName
```

```
az vm create --resource-group myResourceGroup --name myVM --image win2016datacenter
```



Manage VM state

Task	Azure CLI commands
Start a VM	<code>az vm start --resource-group myResourceGroup --name myVM</code>
Stop a VM	<code>az vm stop --resource-group myResourceGroup --name myVM</code>
Deallocate a VM	<code>az vm deallocate --resource-group myResourceGroup --name myVM</code>
Restart a VM	<code>az vm restart --resource-group myResourceGroup --name myVM</code>
Redeploy a VM	<code>az vm redeploy --resource-group myResourceGroup --name myVM</code>
Delete a VM	<code>az vm delete --resource-group myResourceGroup --name myVM</code>

Get VM information

Task	Azure CLI commands
List VMs	<code>az vm list</code>
Get information about a VM	<code>az vm show --resource-group myResourceGroup --name myVM</code>
Get usage of VM resources	<code>az vm list-usage --location eastus</code>
Get all available VM sizes	<code>az vm list-sizes --location eastus</code>

Disks and images

Task	Azure CLI commands
Add a data disk to a VM	<code>az vm disk attach --resource-group myResourceGroup --vm-name myVM --disk myDataDisk --size-gb 128 --new</code>
List attached disks to a VM	<code>az vm show --resource-group groupName --name vmName --query "storageProfile"</code>
Remove a data disk from a VM	<code>az vm disk detach --resource-group myResourceGroup --vm-name myVM --disk myDataDisk</code>
Resize a disk	<code>az disk update --resource-group myResourceGroup --name myDataDisk --size-gb 256</code>
Snapshot a disk	<code>az snapshot create --resource-group myResourceGroup --name mySnapshot --source myDataDisk</code>
Create image of a VM	<code>az image create --resource-group myResourceGroup --source myVM --name myImage</code>
Create VM from image	<code>az vm create --resource-group myResourceGroup --name myNewVM --image myImage</code>

- > **get-azsubscription**
- **get-AzResourceGroup**
- **Get-Azstorageaccount**

- **New-AZresourceGroup -name netlabs -location "East US"**
- **Get-AZresourceGroup | select resource Group**
- **Get-AZresourceGroup | select location**
- **Get-AZresourceGroup | select resource Group, location, resourceid**

- **Az group list -o table**

- **Az vm create -name Linuxserver --resourcegroup Netlabs --image 'ubuntu2204' --generate-ssh-keys --location "East US"**

- > **Az vm list -o table**



- > **get-azsubscription**
- **get-AzResourceGroup**
- **Get-Azstorageaccount**

- **New-AZresourceGroup -name netlabs -location "East US"**
- **Get-AZresourceGroup | select resource Group**
- **Get-AZresourceGroup | select location**
- **Get-AZresourceGroup | select resource Group, location, resourceid**

- **Az group list -o table**

- **Az vm create -name Linuxserver --resourcegroup Netlabs --image 'ubuntu2204' --generate-ssh-keys --location "East US"**

- > **Az vm list -o table**



Azure PowerShell

- It is a programming language as well as a command-line shell.
- It can communicate with a variety of technologies.
- Windows PowerShell provides complete access to all .NET framework types.
- PowerShell is object-oriented.
- Many of the graphical user interfaces (GUIs) built by Microsoft for its many products serve as front-end interfaces to PowerShell.
- It's safer than using VBScript or other scripting languages.
- It enables more effective execution of repeated operations by combining several instructions and building scripts.



Azure Resource Manager (ARM)

- Azure Resource Manager is a service that provides a way to organize resources in Azure.
- define and deploy Azure resources as a single logical unit called a resource group.
- This resource group can contain various Azure resources such as virtual machines, storage accounts, and network interfaces.

primary benefits of using ARM are:

Consistency: ARM provides a consistent way to create, deploy, and manage Azure resources.

Modularity: ARM enables you to define your infrastructure as a set of reusable templates, making it easier to manage and maintain.

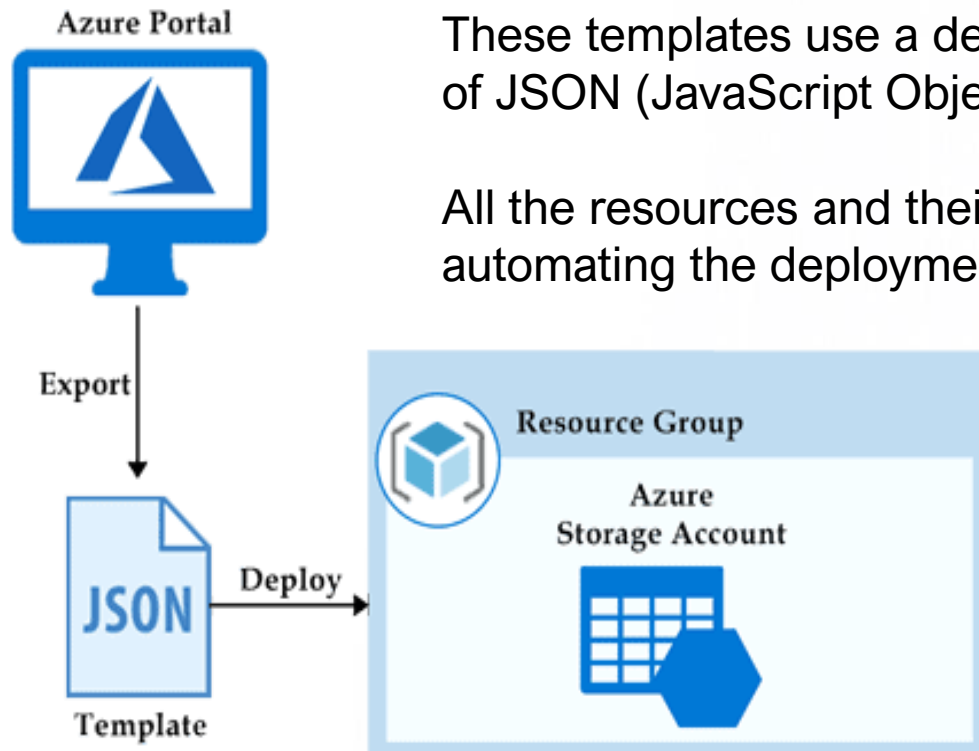
Automation: ARM enables you to automate the deployment and management of your infrastructure, saving time and reducing errors.

Tagging: ARM enables you to tag your resources, making it easier to organize, track, and manage them.

ARM (Azure Resource Manager) template is a block of code that defines the infrastructure and configuration for your project.

These templates use a declarative syntax to let you define your deployment in the form of JSON (JavaScript Object Notation) files.

All the resources and their properties are defined in this template. This helps in automating the deployment process in a constant flow.



Benefits of Using ARM Templates

Using ARM Templates, we can declare network infrastructure, storage, and any other resources.

Over the development lifecycle, ARM Templates allow the deployment of resources repeatedly in a consistent manner.

User can deploy templates parallelly, and only one command is sufficient to deploy all your resource settings.

Templates can be divided into different modules. In other words, templates can be broken into multiple templates so that a parent template can consist of small templates.



Understanding ARM Template

Template Format

The ARM Templates file contains various key-value pairs in JSON format. For example, below, you can see a format of ARM Templates.

```
{
"$schema": "https://schema.management.azure.com/schemas/2015-01-01/deploymentTemplate.json#",
"contentVersion": "1.0.0",
"parameters": {},
"variables": {},
"functions": [],
"resources": [],
"outputs": {}
}
```



Schema – This ‘schema’ defines the location of the JSON file and specifies the version of the template language that you want to use in this template.

This schema depends on the purpose of your deployment. Some schema types are listed below.

Resource Group Deployment – <https://schema.management.azure.com/schemas/2019-04-01/deploymentTemplate.json#>

• **Subscription Group Deployment** – <https://schema.management.azure.com/schemas/2018-05-01/subscriptionDeploymentTemplate.json#>



Parameters

Parameters in templates define the configuration.

These parameters are used in run time or during deployment. In a parameter, we need to define the name, type, values, and properties.

We can also set some allowed values and default values to the parameters, so when a value is not passed during deployment, then the default or allowed values will be used.

```
"parameters": {
  "adminUsername": {
    "type": "string",
    "defaultValue": "Admin",
    "metadata": {
      "description": "Username for the Virtual Machine."
    }
  },
```

```
  "adminPassword": {
    "type": "securestring",
    "defaultValue": "12345",
    "metadata": {
      "description": "Password for the Virtual Machine."
    }
  }
}
```

Variables

Variables define values used throughout the template.

Variables also become an advantage when you want to update all the values and references in a template.

Then you can update the variable and its value only.

```
"variables": {  
  "nicName": "myVMNic",  
  "addressPrefix": "10.0.0.0/16",  
  "subnetName": "Subnet",  
  "subnetPrefix": "10.0.0.0/24",  
  "publicIPAddressName": "myPublicIP",  
  "virtualNetworkName": "MyVNet"  
}
```

Select any Resource--- Overview ----- Deployment ----- AutomationScript



Day2: Identity and Access Management



Module Overview

- Lesson 01: Microsoft Entra ID
- Lesson 02: Users and Groups
- Lesson 03: Module 01 Lab and Review

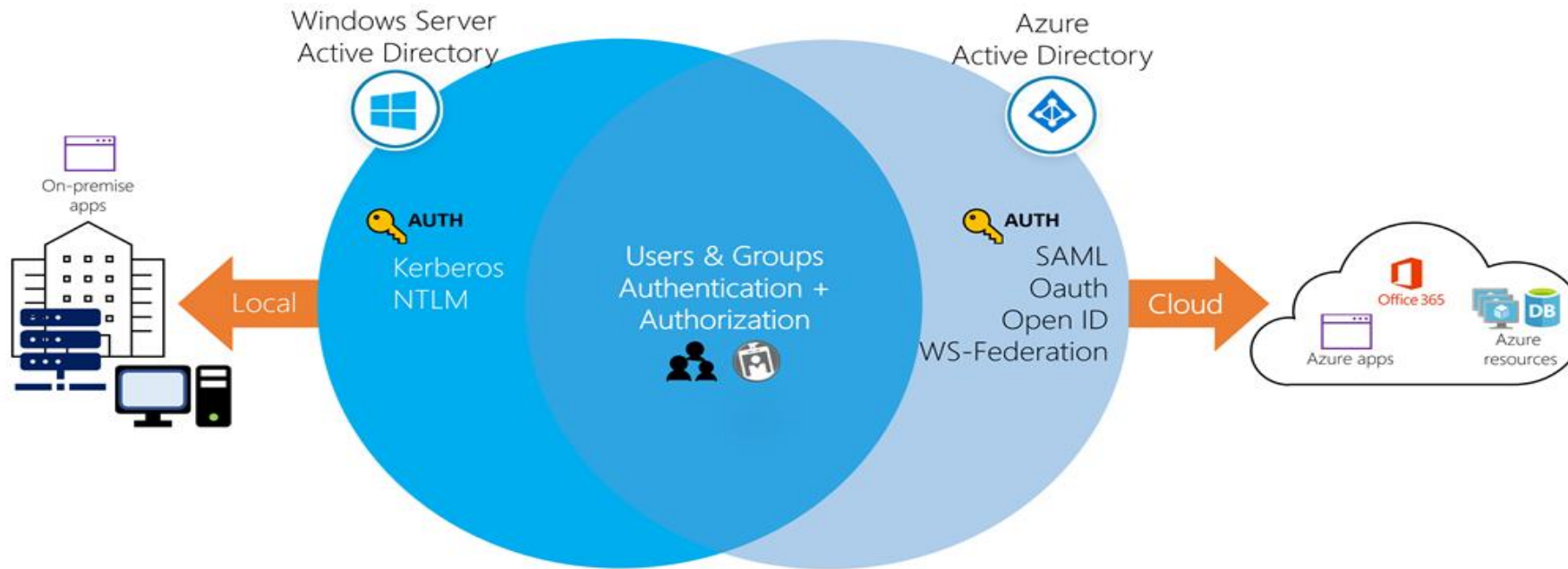
Lesson1: Identity and Governance



Azure Entra ID Overview

- Azure Entra ID
- Azure AD Concepts
- AD DS vs. Azure Active Directory
- Azure Active Directory Editions
- Azure AD Join
- Multi-Factor Authentication
- Self-Service Password Reset

Microsoft Entra ID (ME-ID)



- A cloud-based suite of identity management capabilities that enables you to securely manage access to Azure services and resources for your users
- Provides application management, authentication, device management, and hybrid identity

Microsoft Entra ID (ME-ID)

Azure AD product icons are replaced with the Microsoft Entra ID product icon.

Azure AD product icons	Microsoft Entra ID product icon
	

- A cloud-based suite of identity management capabilities that enables you to securely manage access to Azure services and resources for your users
- Provides application management, authentication, device management, and hybrid identity

Azure AD Concepts

Concept	Description
Identity	An object that can be authenticated.
Account	An identity that has data associated with it.
Azure AD Account	An identity created through Azure AD or another Microsoft cloud service.
Azure tenant	A dedicated and trusted instance of Azure AD that's automatically created when your organization signs up for a Microsoft cloud service subscription.
Azure AD directory	Each Azure tenant has a dedicated and trusted Azure AD directory.
User subscription	Used to pay for Azure cloud services.

AD DS vs Azure Active Directory

- Azure AD is primarily an identity solution, and designed for HTTP and HTTPS communications
- Queried using the REST API over HTTP and HTTPS. Instead of LDAP.
- Uses HTTP and HTTPS protocols such as SAML, WS-Federation, and OpenID Connect for authentication (and OAuth for authorization). Instead of Kerberos
- Includes federation services, and many third-party services (such as Facebook)
- Azure AD users and groups are created in a flat structure, and there are no Organizational Units (OUs) or Group Policy Objects (GPOs)

Azure Active Directory Editions/Licenses

Azure Active Directory free

User and group management, on-premises directory synchronization, basic reports, self-service password change for cloud users, and single sign-on.

Azure Active Directory Premium P1

All the free features of Azure AD, lets hybrid users to access both on-premises and cloud resources, dynamic groups management, self-service group management, Microsoft Identity Manager, and password write-back.

Azure Active Directory Premium P2

In addition to the free and Premium P1 features, P2 license includes Conditional Access and Privileged Identity Management.

"Pay as you go" feature licenses

Business-to-Customer (B2C) features.

Azure AD Free

Azure AD Premium P1

Also included in Microsoft 365 E3

Azure AD Premium P2

Also included in Microsoft 365 E5

Azure AD External identities



Microsoft Entra ID Free

Microsoft Entra ID P1

Also included in Microsoft 365 E3

Microsoft Entra ID P2

Also included in Microsoft 365 E5

Microsoft Entra External ID

Lesson 02: Users and Groups



Users and Groups Overview

- User Accounts
- Managing User Accounts
- Bulk User Accounts
- Group Accounts
- Azure AD Connect
- Azure AD Connect Health
- Azure AD B2B and B2C
- Demonstration – Users and Groups

User Accounts

**Users | All users**
Microsoft - Azure Active Directory

 All users
 Deleted users
 Password reset
 User settings
 Diagnose and solve problems

	Name	User name	User type	Source
☐	 Ziaulla	ziaulla@mac...	Guest	External Azure Active Directory
☐	 Retail Crisis Notificati	rscrisis@mic...	Member	Windows Server AD
☐	 "Planning & Launch Se	plsoem@mi...		Windows Server AD
☐	 'amckenziecec	'amckenziec...	Guest	Invited user
☐	 'Evento FY20 Colombia	kickcolo@mi...	Member	Windows Server AD

- All users must have an account
- The account is used for authentication and authorization
- Identity Sources: Cloud, Directory-synchronized, and Guest

Managing User Accounts

+ New user + New guest user ↑ Bulk create ↑ Bulk invite ↑ Bulk delete ↓ Download users ↻ Refresh 🔑 Reset password ↗ Multi-Factor Authentication ...

New user

Microsoft



Create user

Create a new user in your organization. This user will have a user name like `alice@Microsoft.onmicrosoft.com`.

[I want to create users in bulk](#)



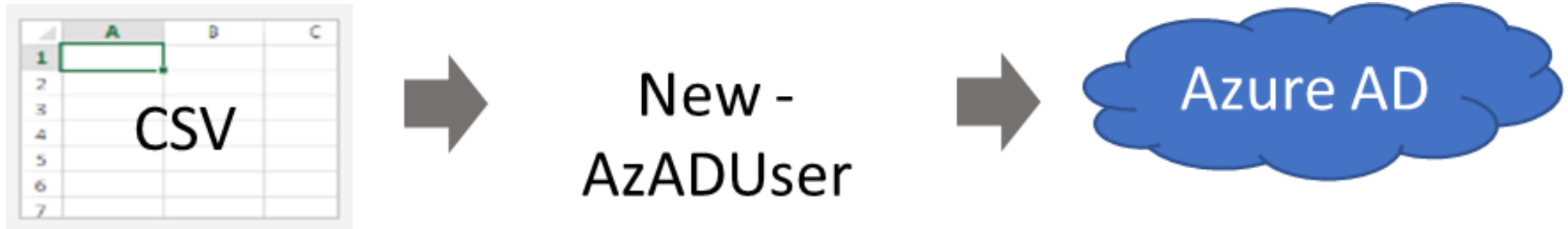
Invite user

Invite a new guest user to collaborate with your organization. The user will be emailed an invitation they can accept in order to begin collaborating.

[I want to invite guest users in bulk](#)

- Must be Global Administrator to manage users
- User profile (picture, job, contact info) is optional
- Deleted users can be restored for 30 days
- Sign in and audit log information is available

Bulk User Accounts



- Create the comma-separated values (CSV) file with the list of all the users and their properties
- Loop through the file processing each user
- Consider error handling, duplicate users, initial password settings, empty properties, and when the account is enabled

Group Accounts

Group Types

- Security groups
- Office 365 groups

Assignment Types

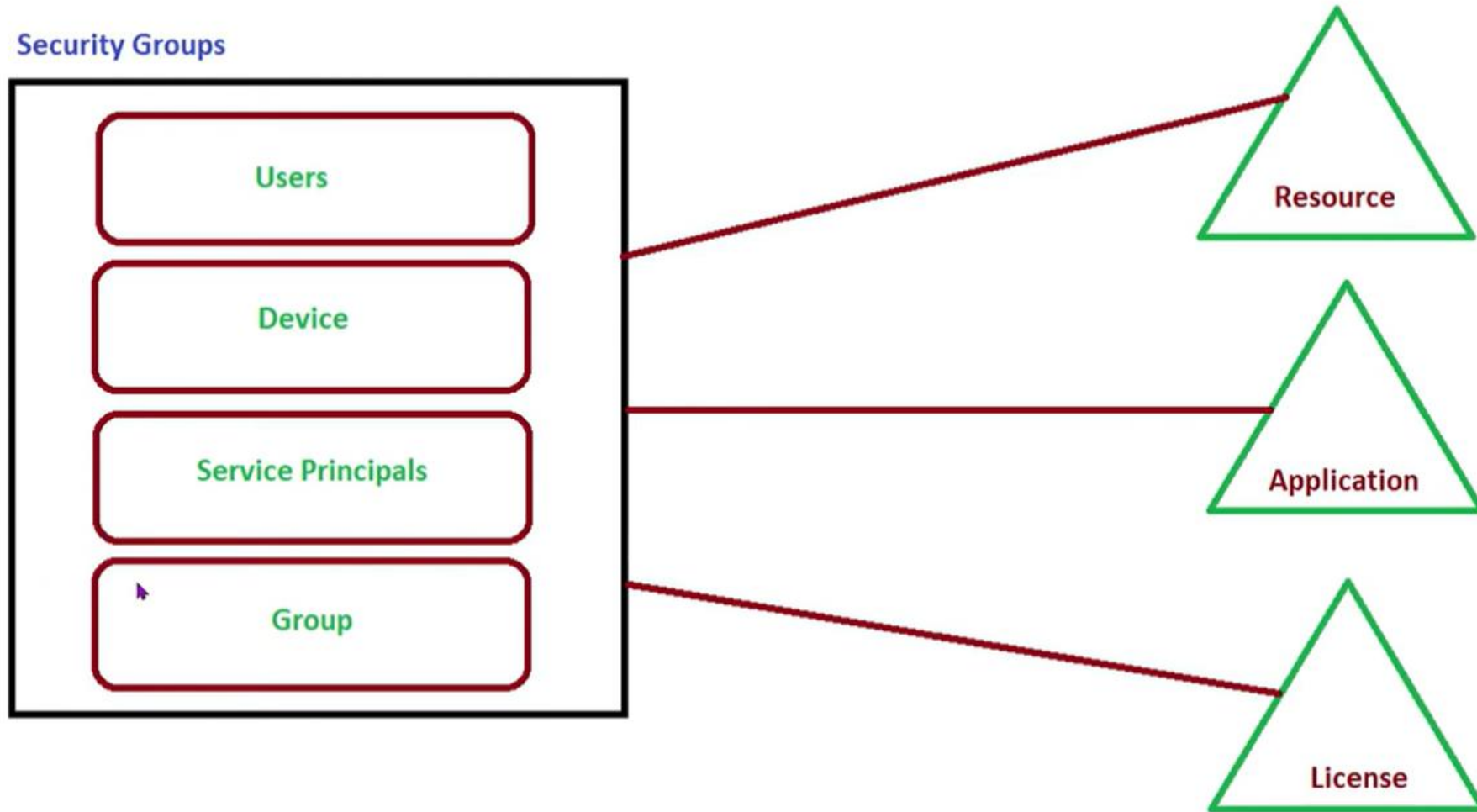
- Assigned
- Dynamic User
- Dynamic Device (Security groups only)

 Search groups

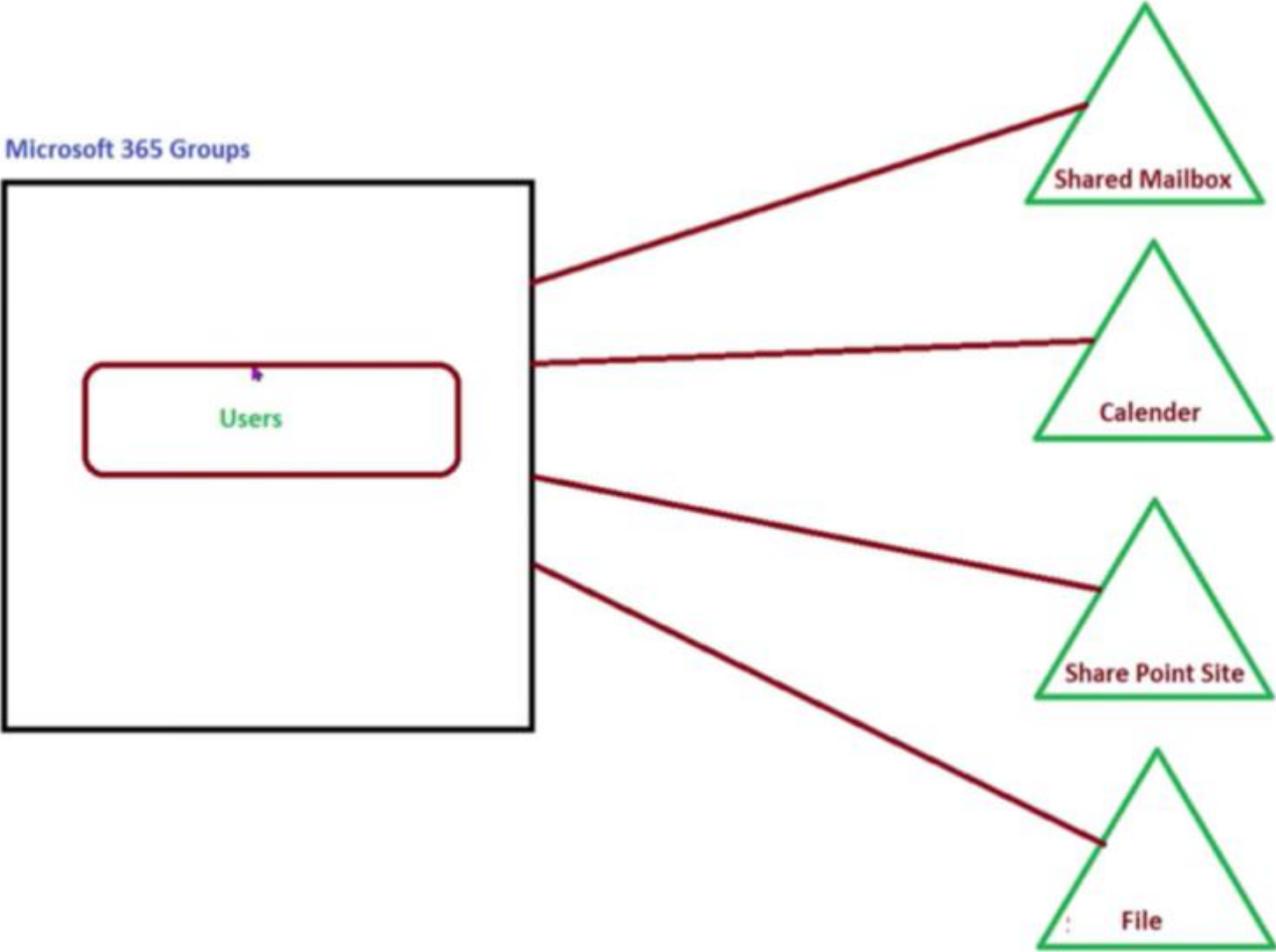
 Add filters

Name		↑↓	Group Type	Membership Type
☐	MA Managers		Security	Assigned
☐	VM Virtual Machine Administrators		Security	Assigned
☐	VN Virtual Network Administrators		Security	Assigned

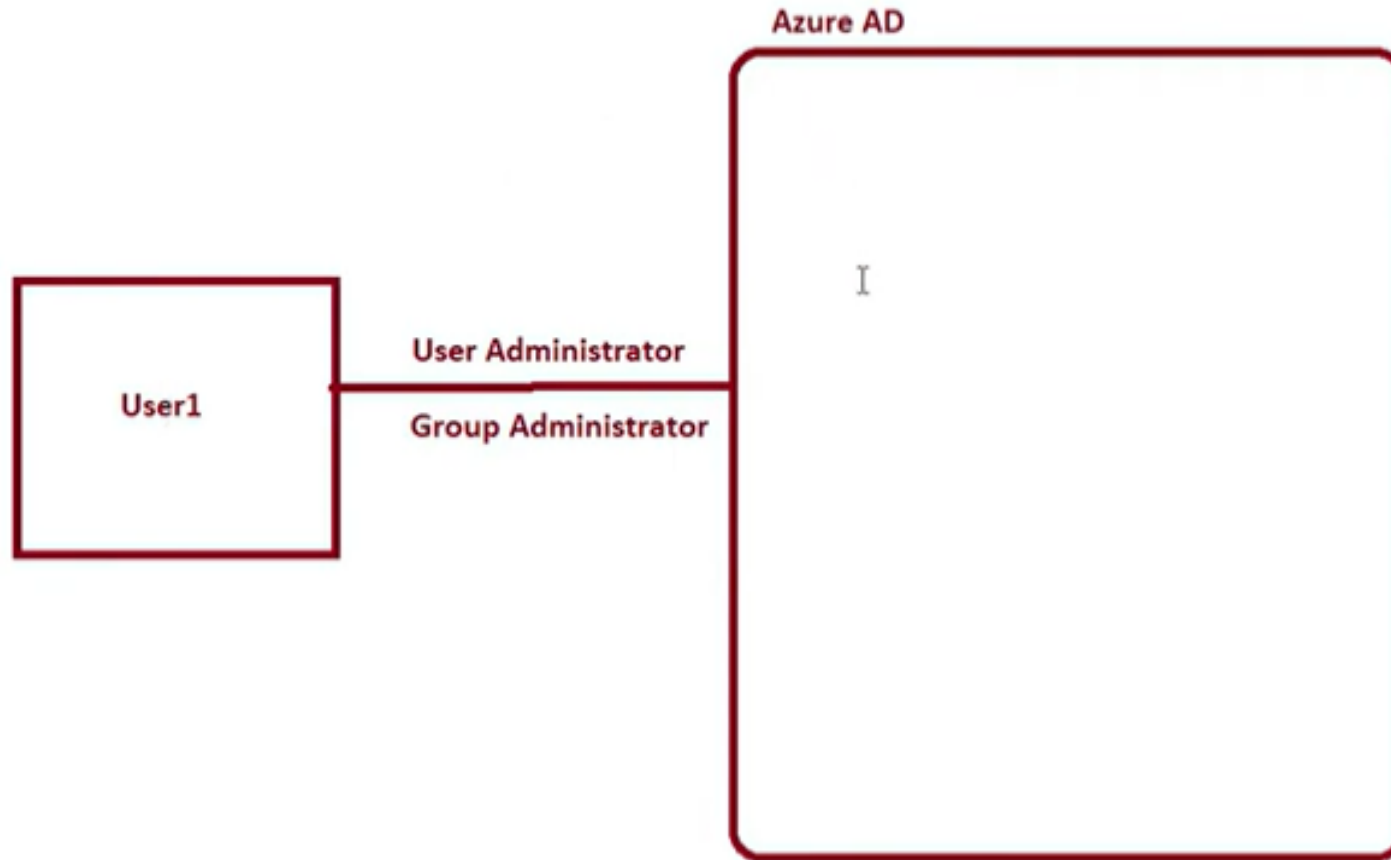
Security Groups



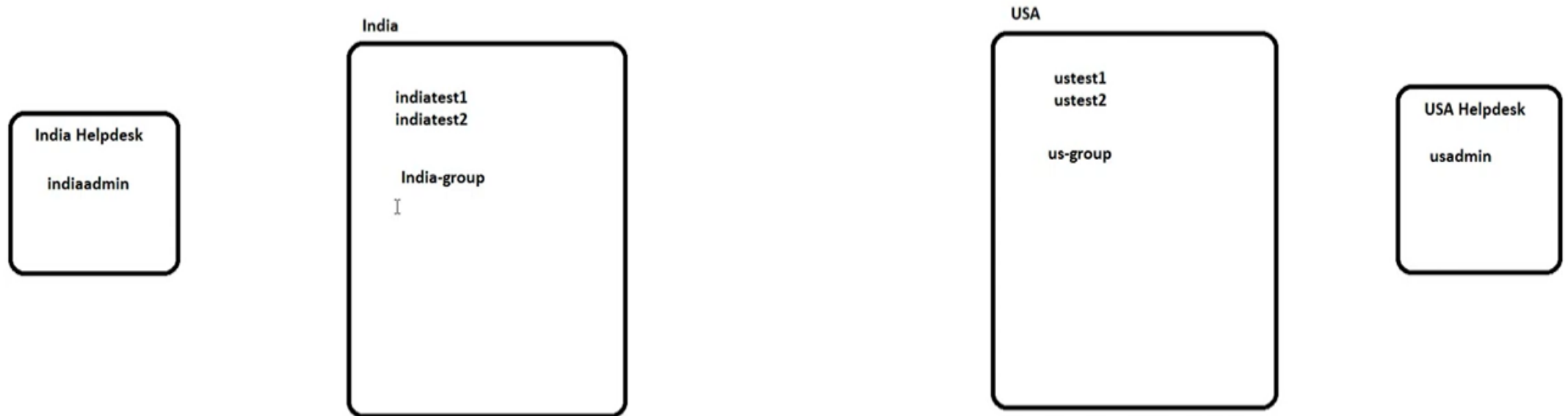
Microsoft 365



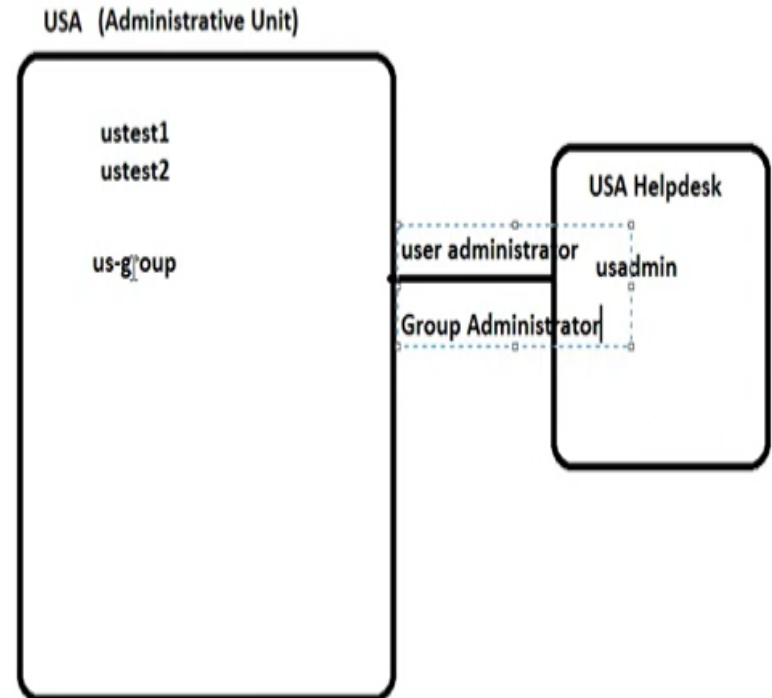
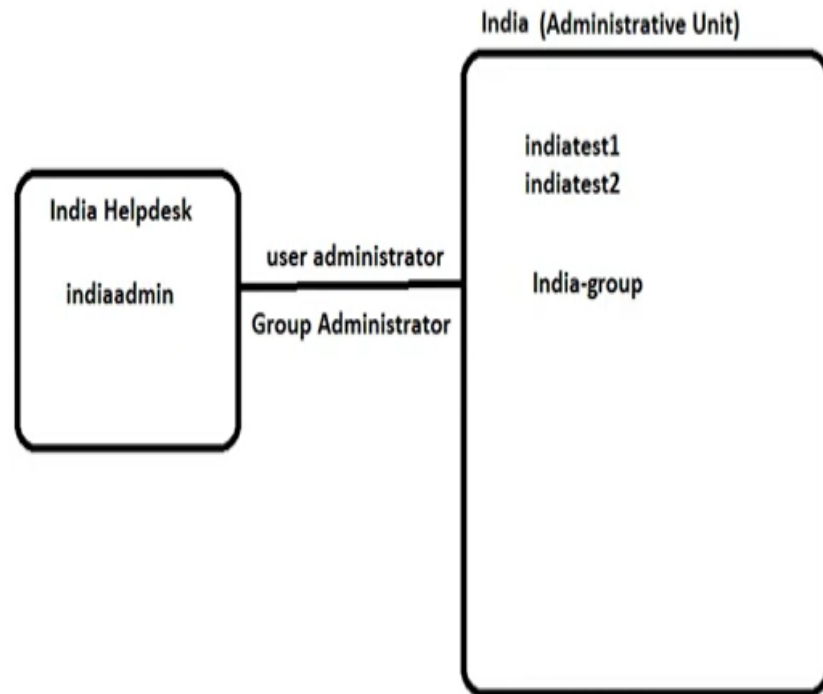
Azure AD Administrative Units



Azure AD Administrative Units



Azure AD Administrative Units



Lesson 01: Virtual Machine



Azure Compute service is a hosting model for the computing resources on which our application runs. Azure compute service can be divided broadly into three categories.

- ❖ Infrastructure as a service
- ❖ Platform as a service
- ❖ Serverless services

business scenarios of VMs

Test and development

Website hosting

Storage, backup, and recovery.

Web apps

High-performance computing

Big data analysis.

Extended Datacenter (Adding VMs to the existing DC instead of adding physical Hardware)



Location and Pricing

Decide the location for the VM

The location can limit your available options. Each region has different hardware available and configurations are not available in all regions.

There are price differences between locations. If your workload isn't bound to a specific location, it is cost effective to check your required configuration in multiple regions to find the lowest price.



There are two separate costs the subscription will be charged for every VM: compute and storage.

Compute costs - Compute expenses are priced on a per-hour basis but billed on a per-minute basis. For example, you are only charged for 55 minutes of usage if the VM is deployed for 55 minutes. You are not charged for compute capacity if you stop and deallocate the VM since this releases the hardware.

The cost for a VM includes the charge for the Windows operating system.

Linux-based instances are cheaper because there is no operating system license charge.



Location and Pricing

Storage costs - You are charged separately for the storage the VM uses.

The status of the VM has no relation to the storage charges that will be incurred; even if the VM is stopped/deallocated and you aren't billed for the running VM, you will be charged for the storage used by the disks.



Choose from two payment options for compute costs:

1. **Consumption-based** - With the consumption-based option, you pay for compute capacity by the second. You're able to increase or decrease compute capacity on demand as well as start or stop at any time. Prefer this option if you run applications with short-term or unpredictable workloads that cannot be interrupted.
2. **Reserved Virtual Machine Instances** - The Reserved Virtual Machine Instances (RI) option is an advance purchase of a virtual machine for one or three years in a specified region. The commitment is made up front, and in return, you get up to 72% price savings compared to pay-as-you-go pricing.



VM Type	Sizes	Purpose
General Purpose	B, Dsv3, Dv3, DSv2, Dv2, Av2, DC	Testing and development, small to medium databases, and low to medium traffic web servers.
Compute Optimized	Fsv2, Fs, F	Medium traffic web servers, network appliances, batch processes, and application servers.
Memory Optimized	Esv3, Ev3, M, GS, G, DSv2, Dv2	Relational database servers, medium to large caches, and in-memory analytics.
Storage Optimized	Lsv2, Ls	Ideal for VMs running databases.
GPU	NV, NVv2, NC, NCv2, NCv3, ND, NDv2 (Preview)	Ideal for model training and inferencing with deep learning.
High Performance Compute	H	Fastest and most powerful CPU virtual machines with optional high-throughput network interfaces.

What if my size needs change?

If you stop and deallocate the VM, you can then select any size available in your region since this removes your VM from the cluster it was running on.

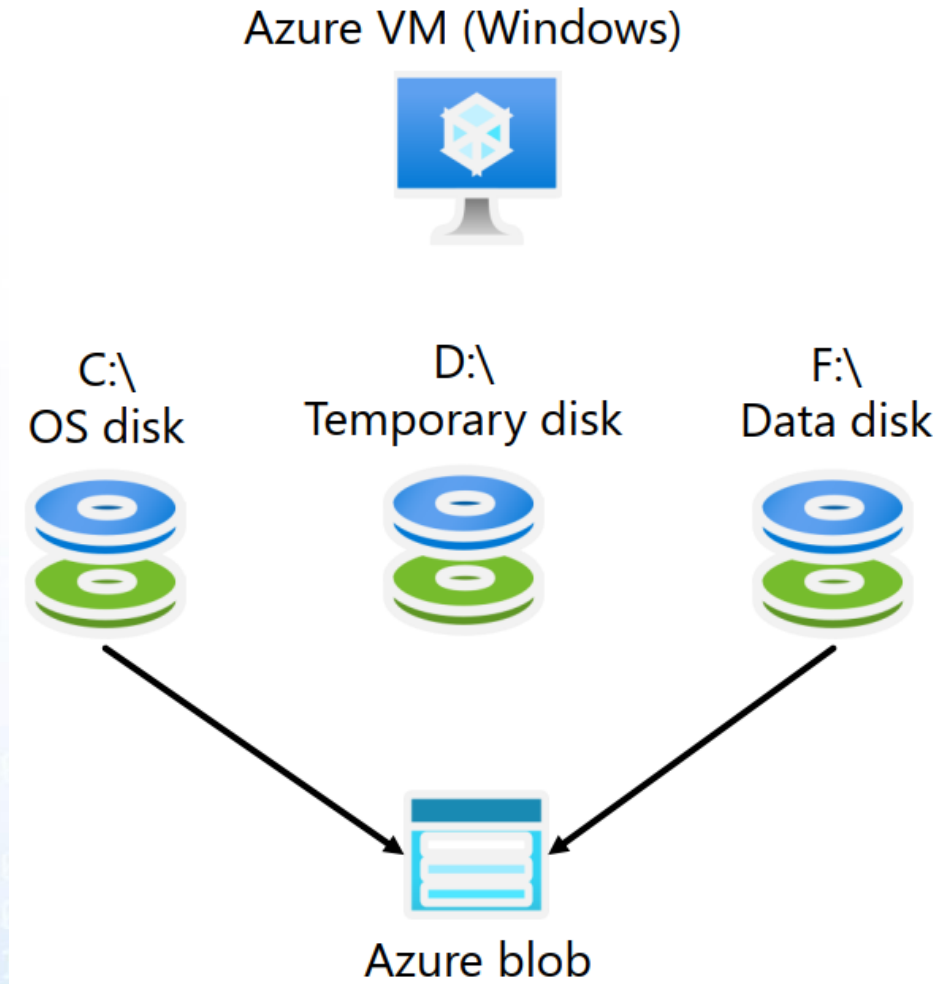
✓ Be cautious when resizing production VMs - they will be rebooted automatically which can cause a temporary outage and change some configuration settings such as the IP address.



Operating System Disks are SATA drives, labeled as C:

Temporary Disks provides short term storage

Data Disks are SCSI drives and depend on your virtual machine type



Disks used by VMs

- **Operating system disk:** In Azure, every virtual machine will have an operating system disk.
- **Temporary disk:** Each VM contains a temporary drive. The temporary disk provides short-term storage for applications and processes.
- **Data disk:** A data disk is a VHD that's attached to a virtual machine to store application data or other data we need to keep.

Performance tiers

- **Standard Storage:** It is backed by HDDs and deliver cost-effective storage while still being performant. It is ideal for development and testing, not-critical, and Infrequent access because the max throughput and IOPS per disk is 60MB/s and 500, respectively.
- **Premium Storage:** It is backed by SSDs, and deliver high-performance, low-latency disk support for VMs running I/O-intensive workloads. The maximum throughput and IOPS per disk are 250MB/s and 7500, respectively.



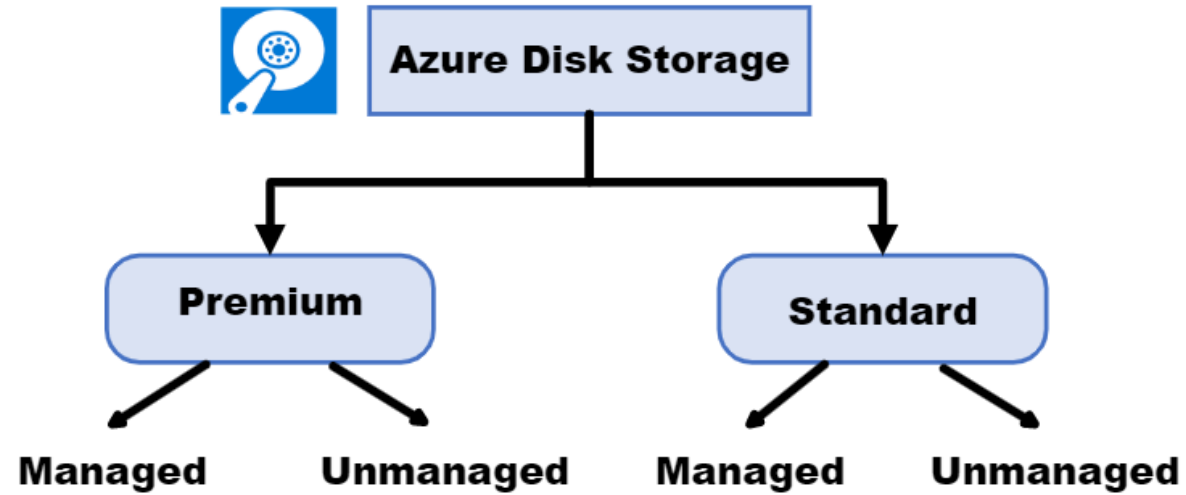
Types of the disk in Azure

- **Unmanaged disks:** It is the traditional type of disks that have been used by VMs. We can create our storage account and specify the storage account when you create the disk. The scalability targets of SA (20, 000 IOPS) are not exceeded.
- **Managed disks:** Managed disks handles the storage account creation/management. We do not have to care about scalability limits the storage account. Microsoft always recommends us to use Azure Managed Disks for new VMs.

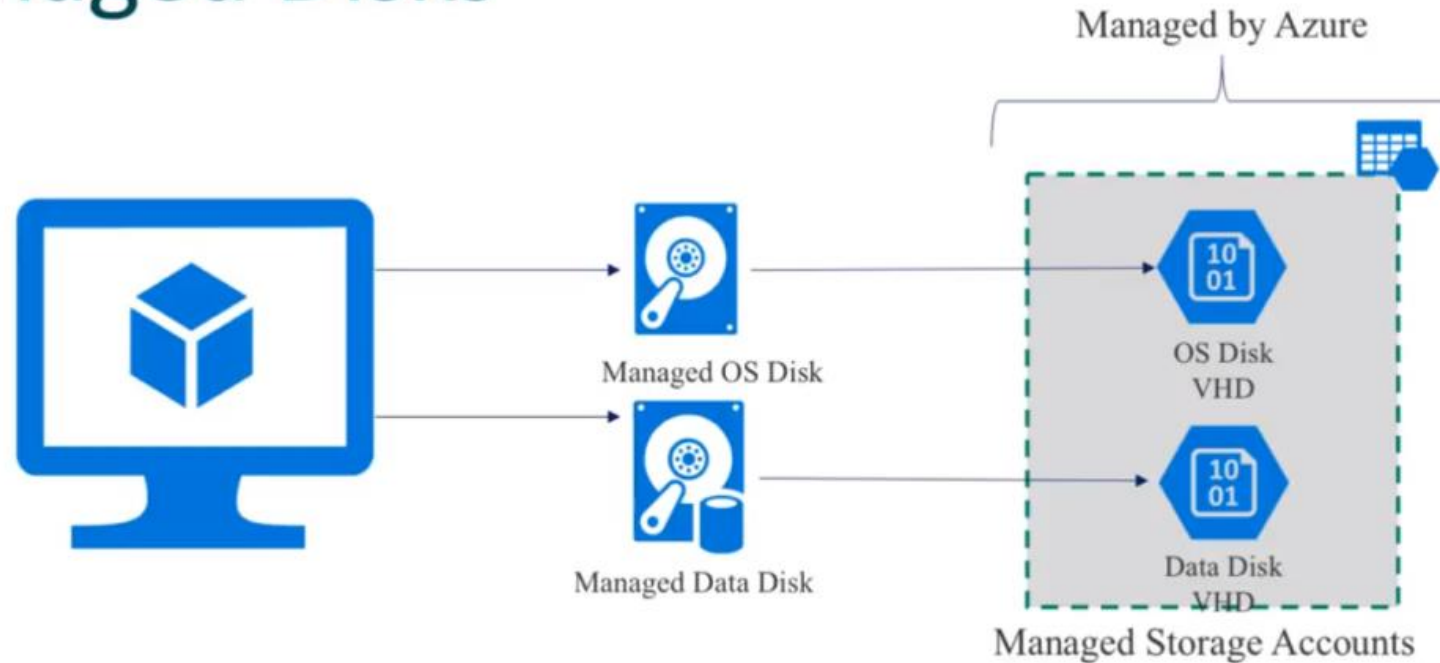
Disk encryption

- **Storage Service Encryption:** Azure Storage Service Encryption provides encryption-at-rest and safeguards our data to meet our organizational security and compliance commitments. It is enabled by default for all Managed Disks, Snapshots, and Images in every region where managed disks are available.
- **Azure Disk Encryption:** Azure Disk Encryption allows you to encrypt the OS and Data disks used by an IaaS Virtual Machine. For Windows, the drives are encrypted using industry-standard **BitLocker encryption technology**. For Linux, the disks are encrypted using the **DM-Crypt technology**.

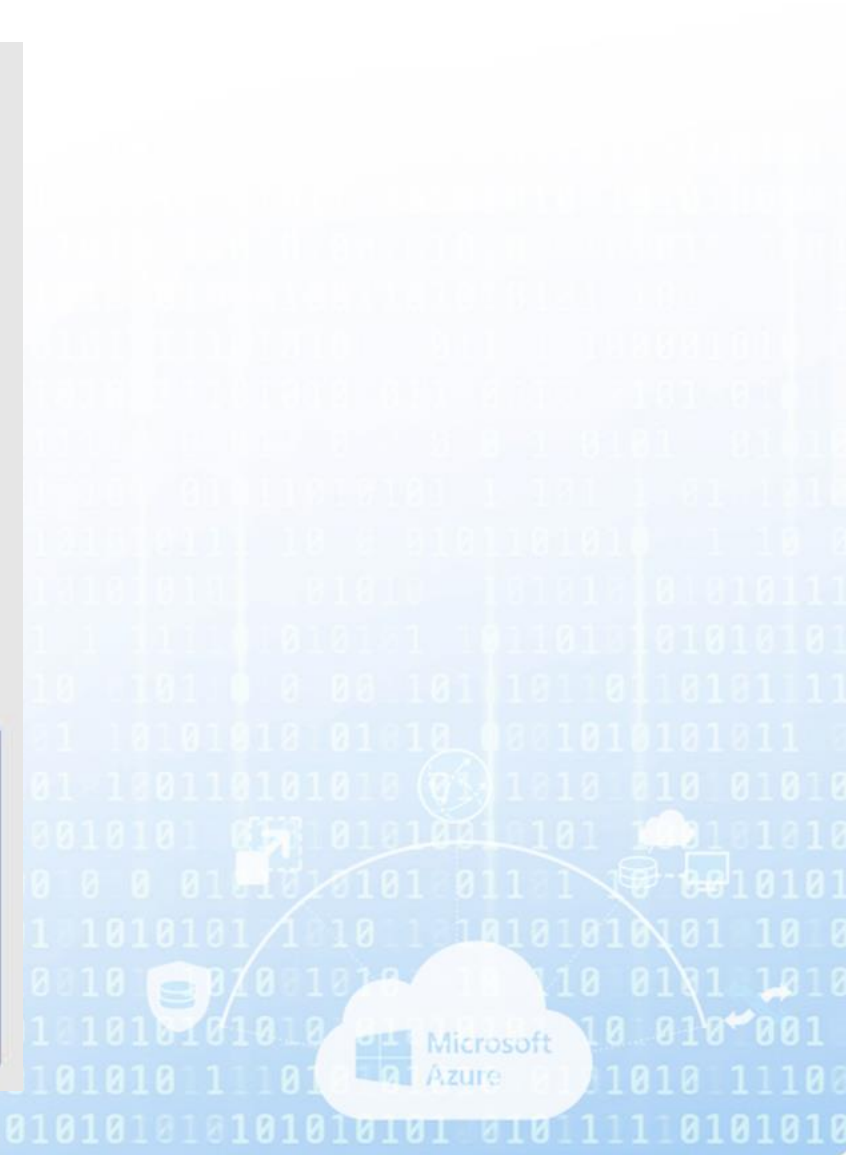
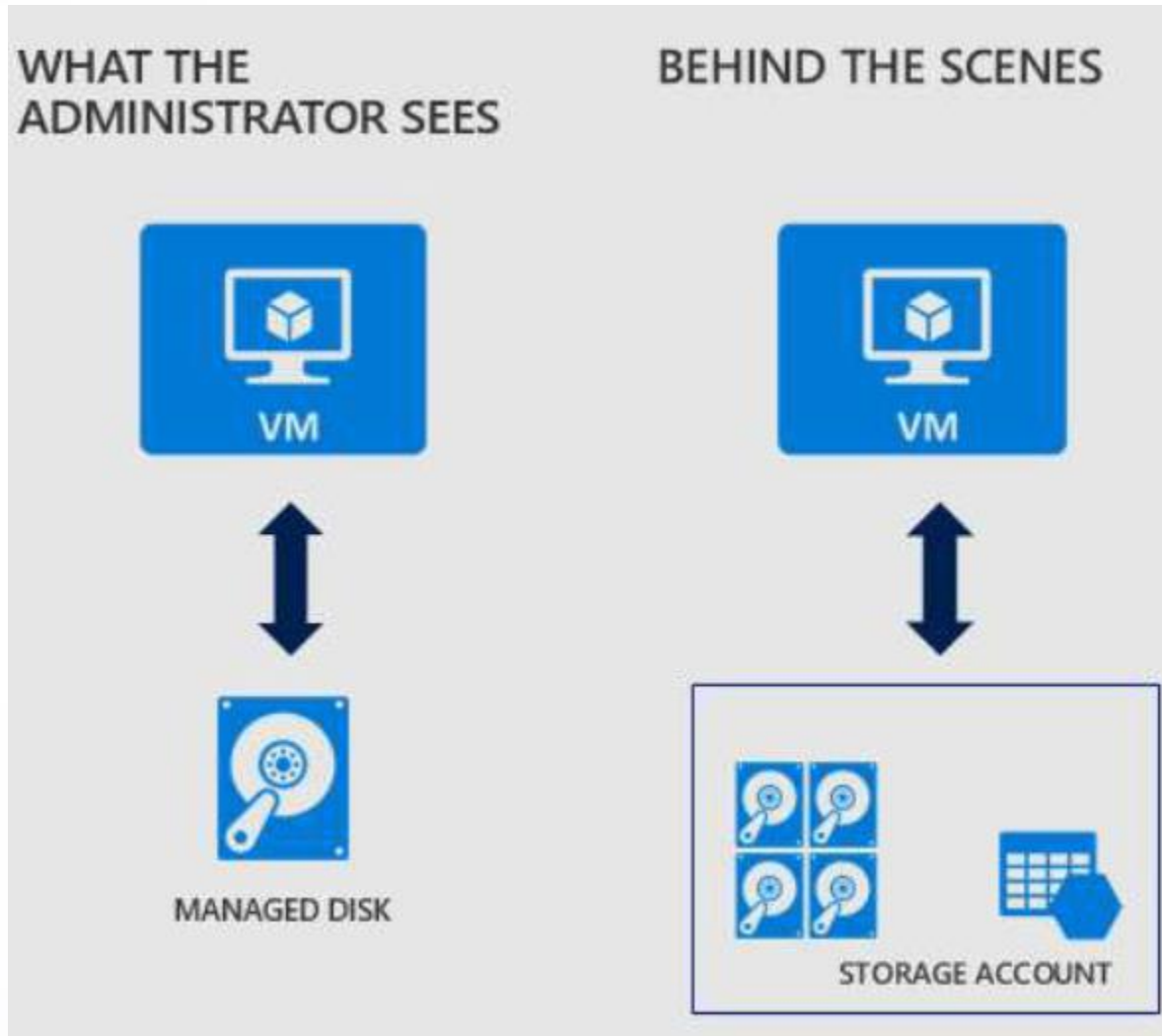




Managed Disks



- ✓ **Simple** - Abstracts storage accounts from customers
- ✓ **Granular access control** – Top level ARM resource, apply Azure RBAC
- ✓ **Storage account limits do not apply** – Enables scaling free of storage account limitations
- ✓ **Secure** – No public access to the underlying blob
- ✓ **Support VM level disk encryption** – secure data at rest
- ✓ **Better Storage Resiliency** - Prevents single points of failure due to storage



Disk encryption

- **Storage Service Encryption:** Azure Storage Service Encryption provides encryption-at-rest and safeguards our data to meet our organizational security and compliance commitments. It is enabled by default for all Managed Disks, Snapshots, and Images in every region where managed disks are available.
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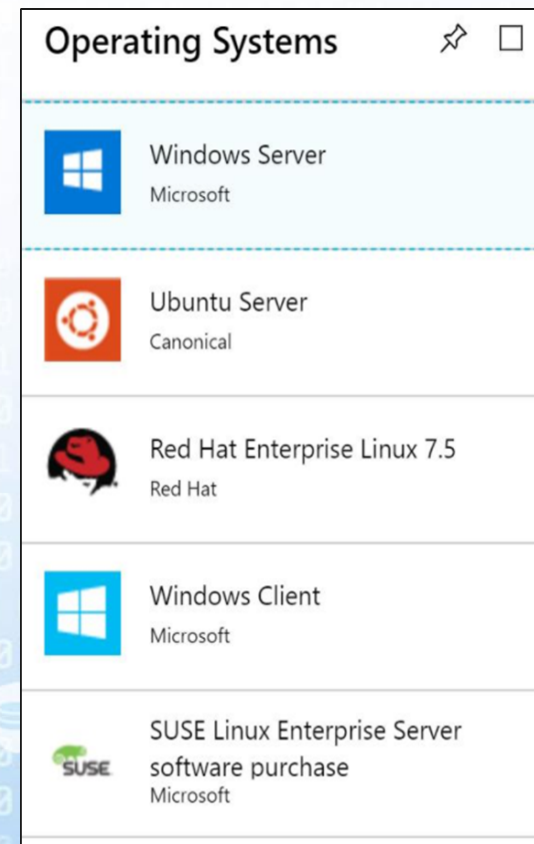
Azure provides a variety of OS images that you can install into the VM, including several versions of Windows and flavors of Linux

The choice of OS will influence your hourly compute pricing as Azure bundles the cost of the OS license into the price

Search the **Azure Marketplace** for more sophisticated install images that include the OS and popular software tools installed for specific scenarios.

you can create your disk image with what you need, upload it to Azure storage, and use it to create an Azure VM.

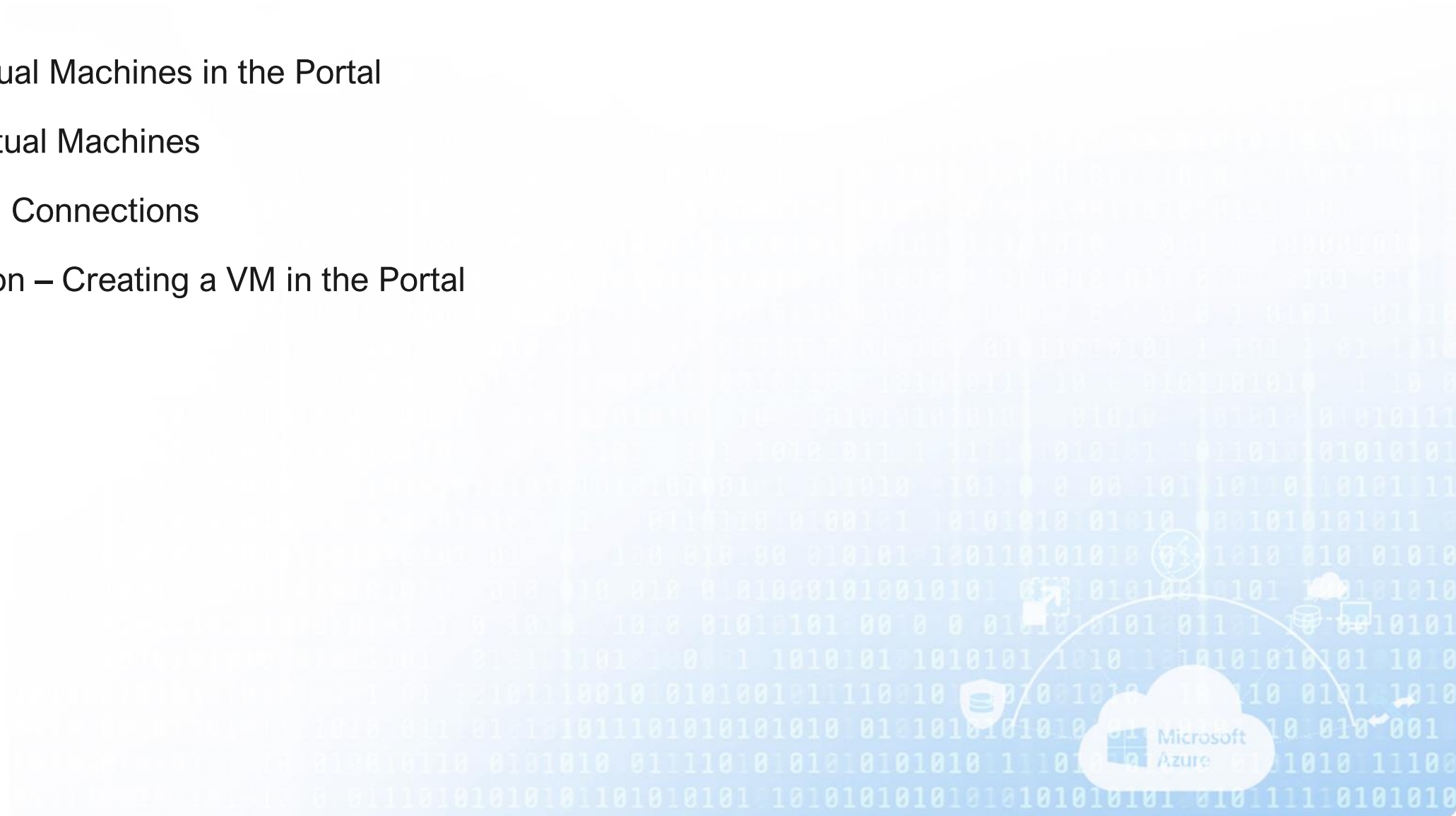
Keep in mind that Azure only supports 64-bit operating systems.



Lesson 02: Creating Virtual Machines



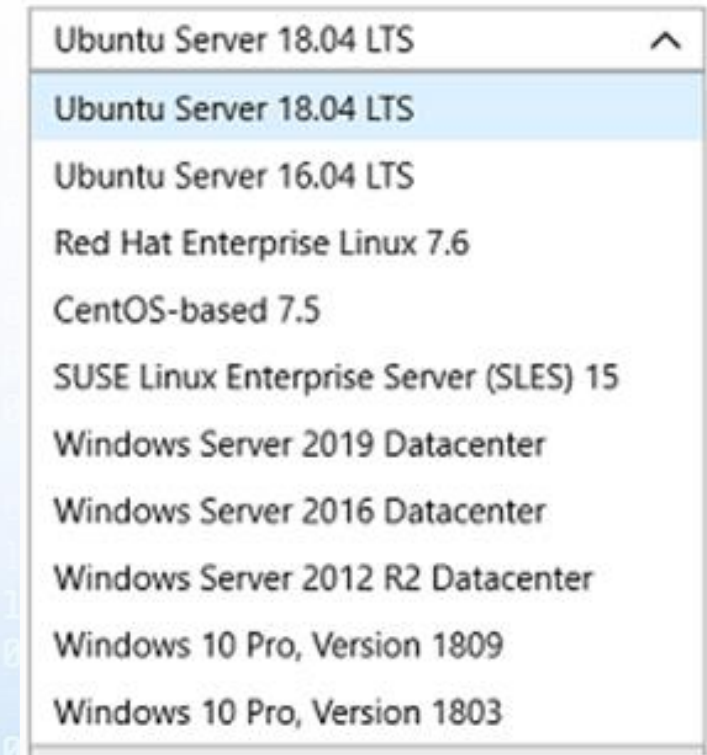
- ❖ Creating Virtual Machines in the Portal
- ❖ Windows Virtual Machines
- ❖ Windows VM Connections
- ❖ Demonstration – Creating a VM in the Portal



Create a virtual machine

Basics **Disks** Networking Management Guest config Tags Review + create

- **Basic** (required) - Project details, Administrator account, Inbound port rules
- **Disks** - OS disk type, data disks
- **Networking** - Virtual networks, load balancing
- **Management** - Monitoring, Auto-shutdown, Backup
- **Guest config** - Add additional configuration, agents, scripts or applications

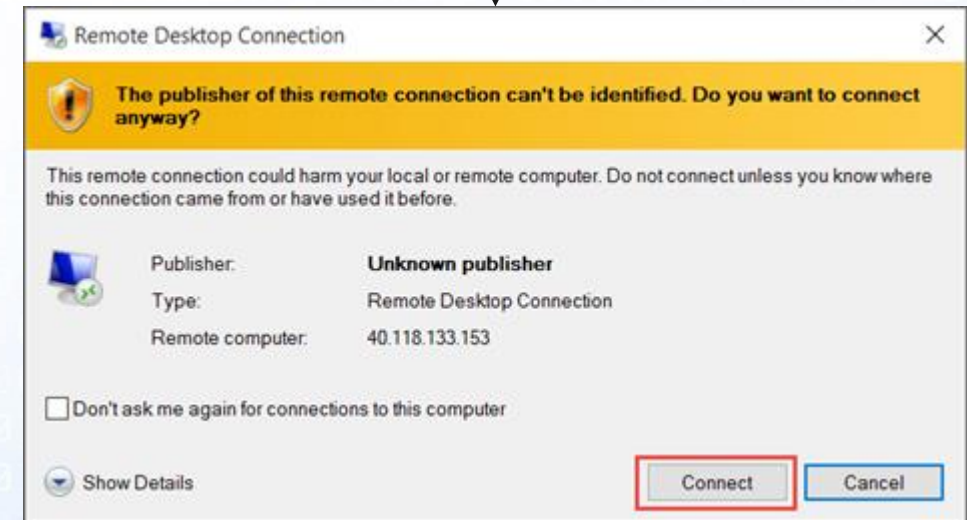




- Unique hybrid capabilities
- Advanced multi-layer security
- Faster innovation for applications
- Unprecedented hyper-converged infrastructure



- **Remote Desktop Protocol (RDP)** creates a GUI session and accepts inbound traffic on TCP port 3389



- Create the virtual machine
- Connect to the virtual machine
- Install the Web Server role
- View the IIS welcome page



Debian Linux

By credativ

Debian GNU/Linux for Microsoft Azure provided by credativ.

Software plans start at Free

Get it now



Clear Linux OS

By Clear Linux Project

A reference Linux distribution optimized for Intel Architecture.

Bring your own license

Get it now



SUSE Linux Enterprise Server

By SUSE

SUSE Linux Enterprise Server

Software plans start at Free

Get it now



Red Hat Enterprise Linux 7.4

By Red Hat

Red Hat Enterprise Linux 7 is the world's leading enterprise Linux platform built to meet the needs of today...

Get it now

- Hundreds of community-built images in the Azure Marketplace
- Linux has the same deployment options as for Windows VMs
- Manage Linux VMs with many popular open-source DevOps tools



* Authentication type

SSH public key Password

* SSH public key ⓘ

Provide an RSA public key in the single-line format (starting with "ssh-rsa") or the multi-line PEM format. You can generate SSH keys using ssh-keygen on Linux and OS X, or PuTTYGen on Windows.

- Authenticate with a SSH public key or password
- SSH is an encrypted connection protocol that allows secure logins over unsecured connections
- There are public and private keys



Demonstration – Connect to Linux VMs

- Create the SSH keys
- Create the Linux machine and assign the public SSH key
- Access the server using SSH



Virtual Machine Availability

Maintenance and Downtime



An Unplanned Hardware Maintenance event occurs when the Azure platform predicts that the hardware or any platform component associated to a physical machine, is about to fail.

When the platform predicts a failure, it will issue an unplanned hardware maintenance event. Azure uses Live Migration technology to migrate the Virtual Machines from the failing hardware to a healthy physical machine.

Unexpected Downtime is when the hardware or the physical infrastructure for the virtual machine fails unexpectedly.

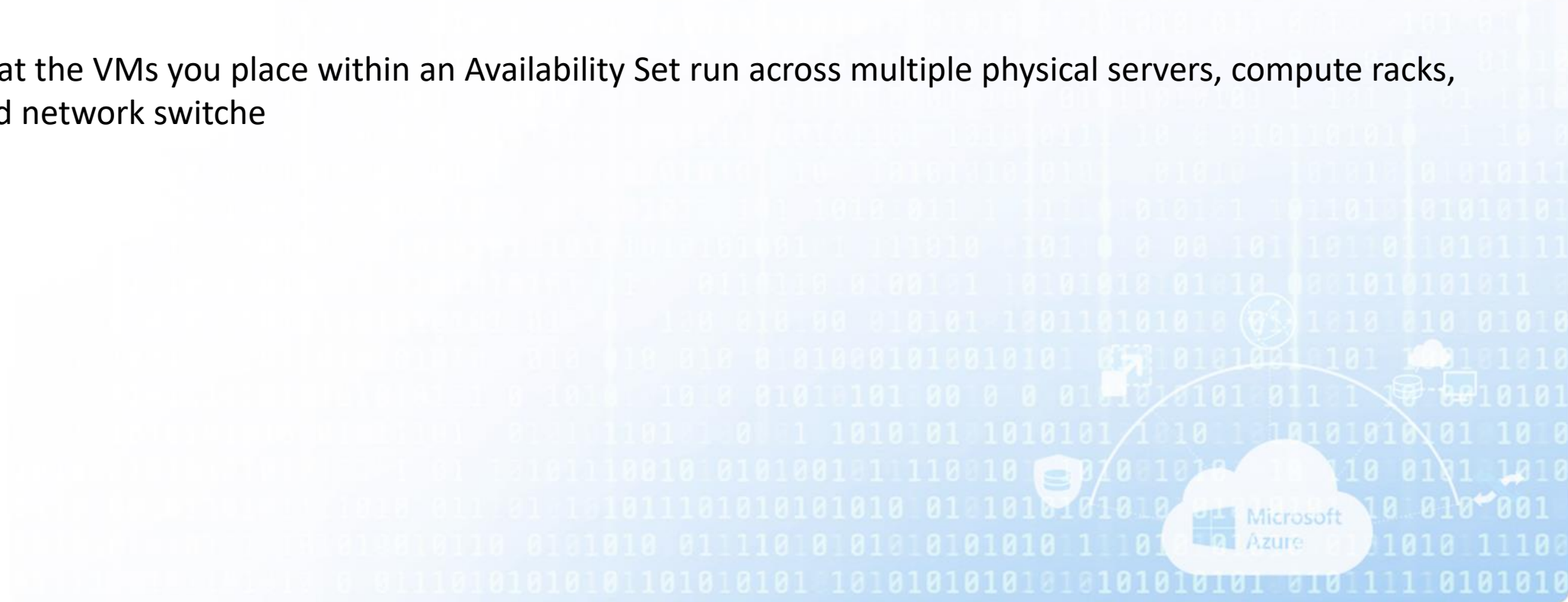
When detected, the Azure platform automatically migrates (heals) your virtual machine to a healthy physical machine in the same datacenter

Availability Sets

Availability Set is a logical feature used to ensure that a group of related VMs are deployed so that they aren't all subject to a single point of failure and not all upgraded at the same time during a host operating system upgrade in the datacenter.

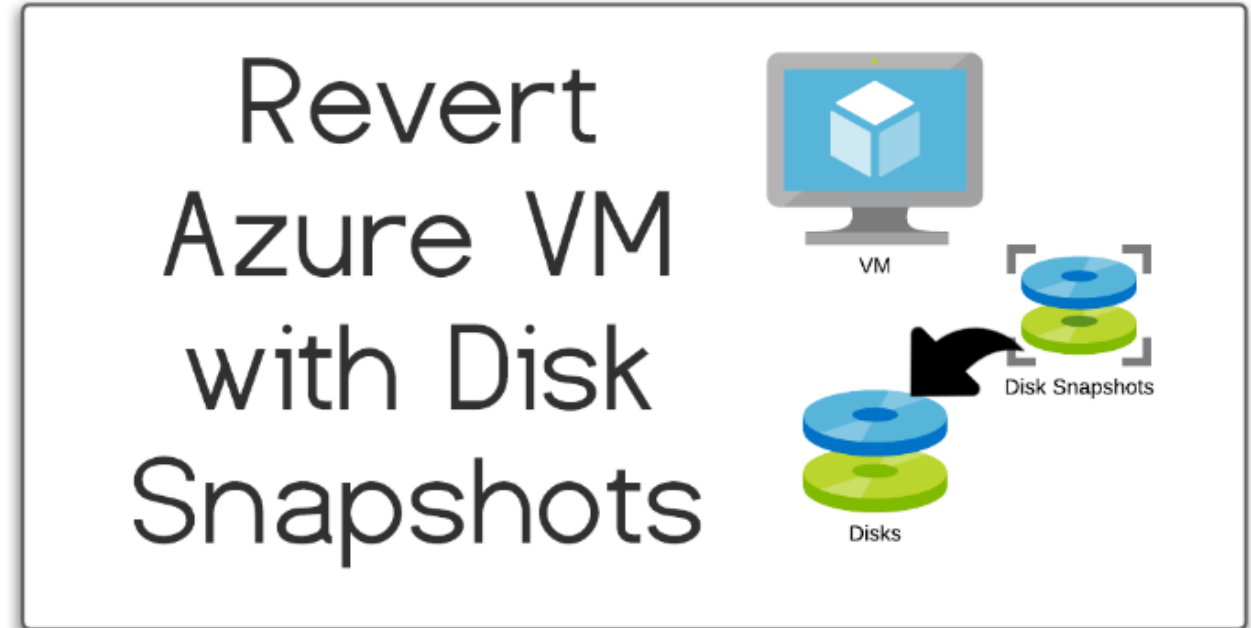
VMs placed in an availability set should perform an identical set of functionalities and have the same software installed.

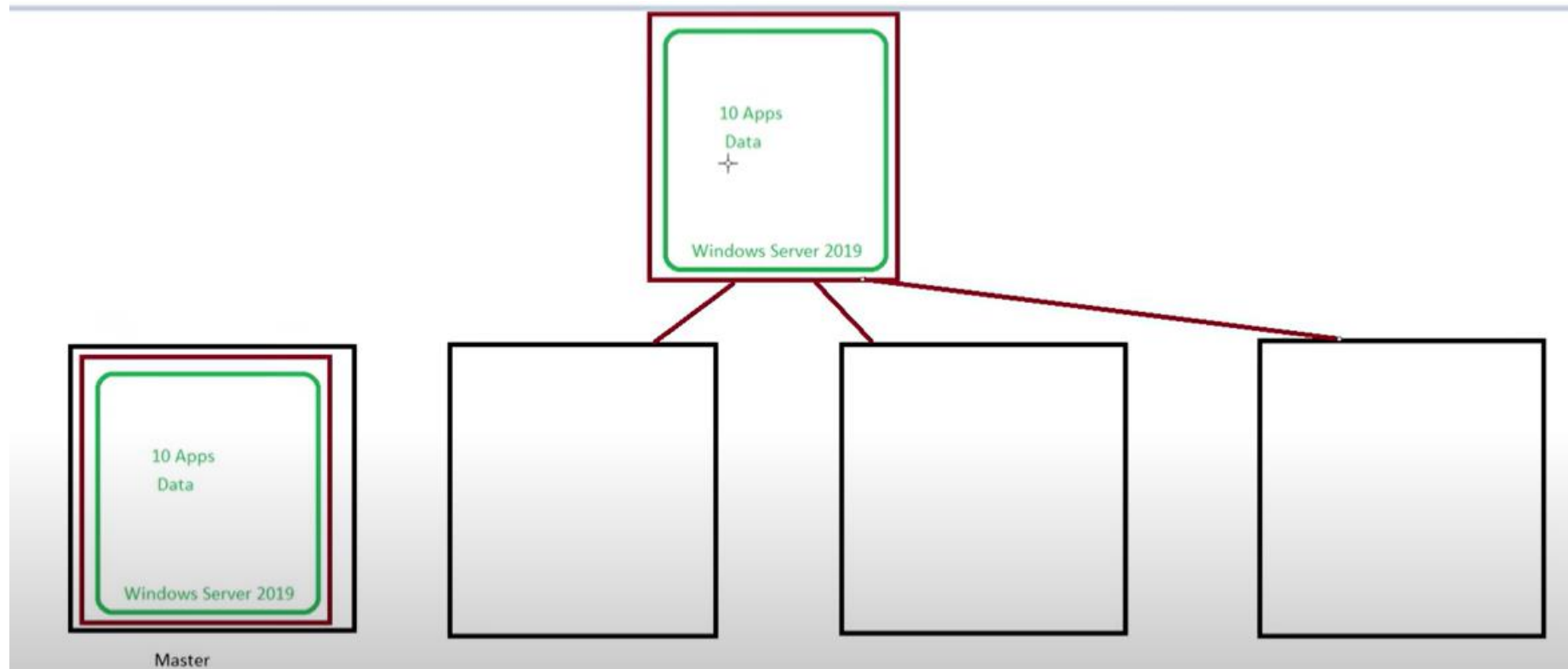
Azure ensures that the VMs you place within an Availability Set run across multiple physical servers, compute racks, storage units, and network switches



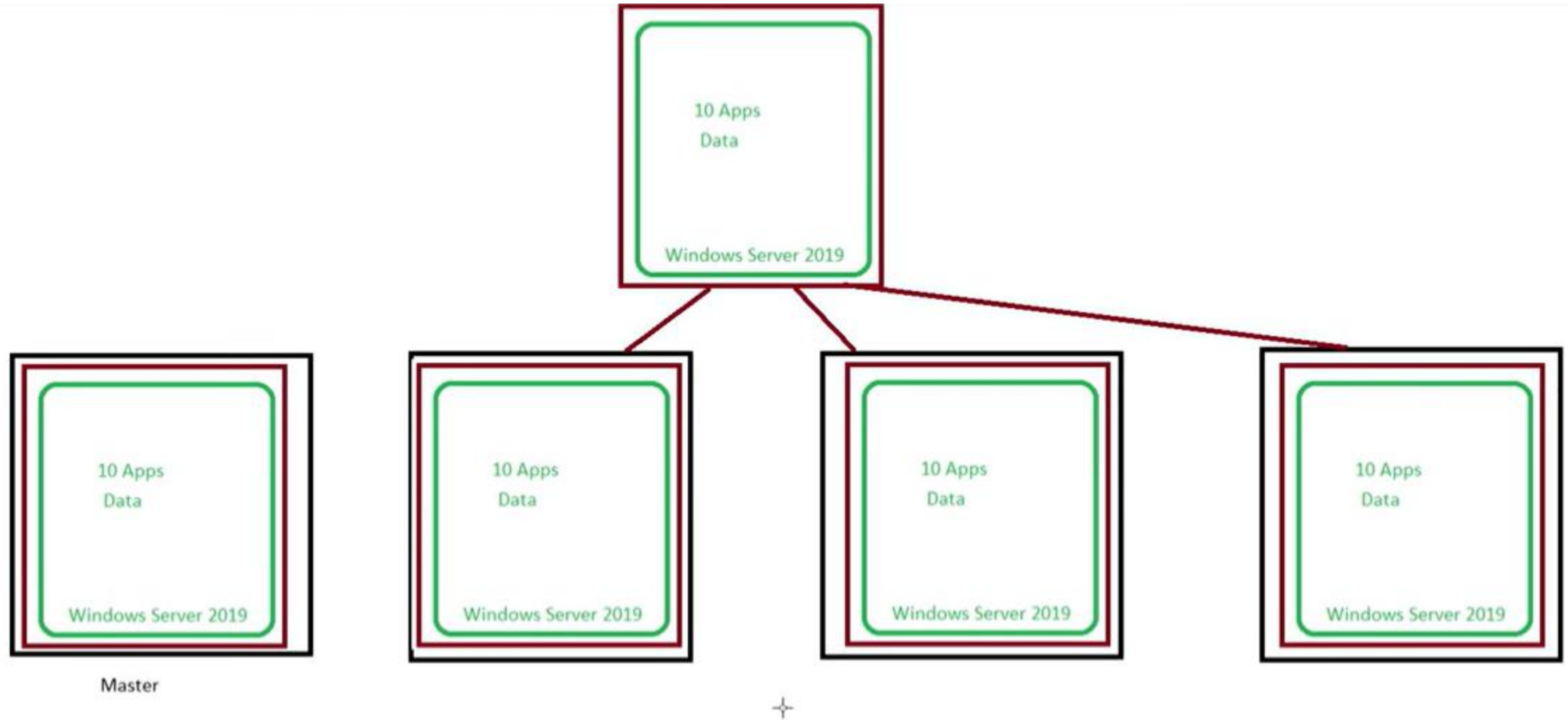
A snapshot is a point-in-time copy of data capturing its current state, often allowing for quick recovery or system rollback.

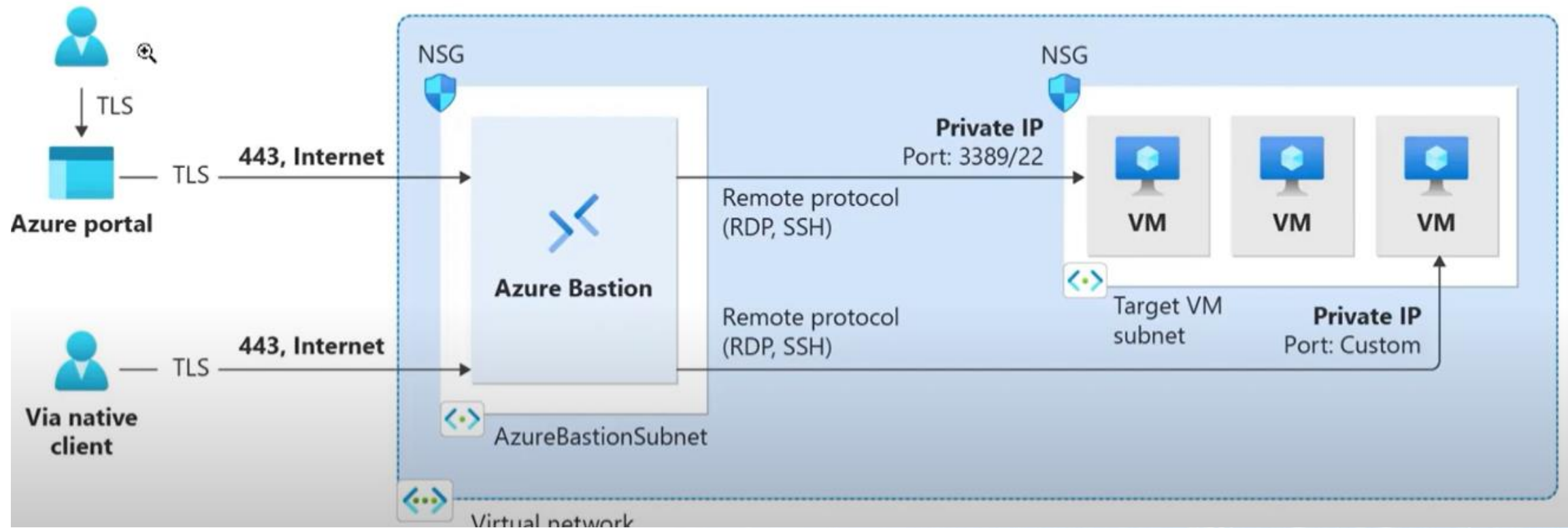
A Managed Snapshot is a full point-in-time copy of a VM Managed Disk.





Images stored in an Azure Compute Gallery (formerly known as Shared Image Gallery),
 You can use VMs that already have accounts created on them (specialized) or
 You can generalize the VM before creating the image to remove machine accounts and other machines specific information.





Azure's Stopped State

This means that any non-static public IPs will not be released, but you'll also be paying for the VM's Compute costs.

The image shows a Windows VM desktop on the left with the Start menu open, highlighting the 'Shut down' option. On the right, the Azure portal 'Virtual machines' page is displayed for the subscription 'MVSE05'. A table lists one VM with the status 'Stopped', highlighted by a red dashed box and a red arrow. The table columns are Name, Type, Subscription, Status, Operating system, Size, and Public IP address.

Name	Type	Subscription	Status	Operating system	Size	Public IP address
vm-1185303510	Virtual machine	MVSE05	Stopped	Windows		40.71.255.182

Azure's Deallocated State

Home >

Virtual machines

TSG Service and Trading Limited Company (tsgvietnam.onmicrosoft.com)

+ Create ▾ ⇄ Switch to classic ⌚ Reservations ▾ ⚙️ Manage view ▾ ↻ Refresh ⬇️ Export to CSV 🔗 Open query | 🏷️ Assign tags ▶ Start ⏪

Filter for any field... Subscription equals **MVSE05** Type equals **all** Resource group equals **all** × Location equals **all** × Status equals **Stopped**

Showing 1 to 2 of 2 records.

☐ Name ↑↓	Type ↑↓	Subscription ↑↓	Status ↑↓	Operating system ↑↓	Size ↑↓	Public IP address ↑
☐  vm-210251813	Virtual machine	MVSE05	Stopped (deallocated)	Windows	Standard_DS2_v2	40.71.252.173
☐  vm-253623263	Virtual machine	MVSE05	Stopped (deallocated)	Windows	Standard_DS2_v2	52.224.109.239

This means that any non-static public IPs will be released, but you'll also stop paying for the VM's Hard disk costs.

Azure virtual machine extensions

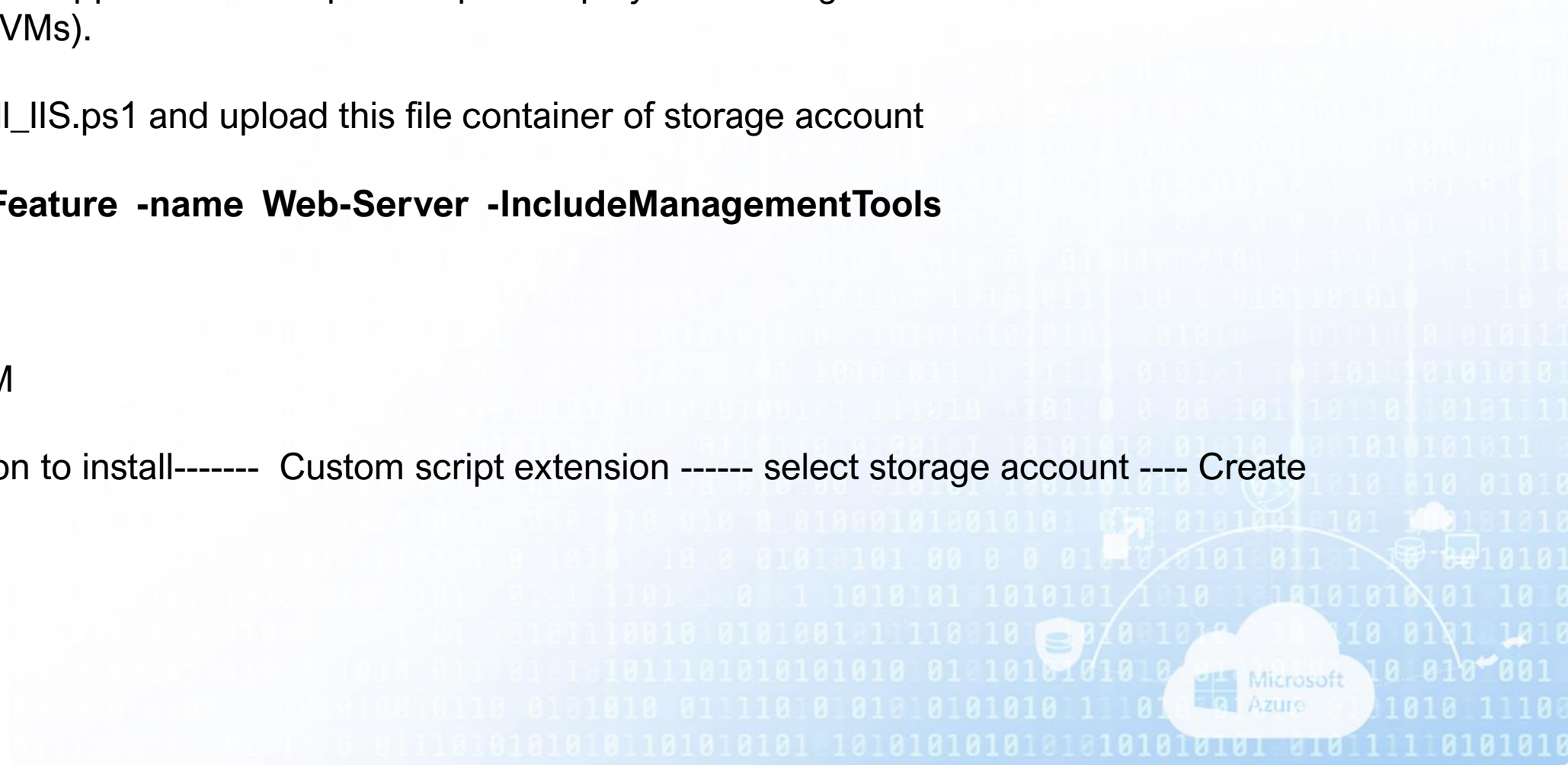
Extensions are small applications that provide post-deployment configuration and automation on Azure virtual machines (VMs).

Create a file Install_IIS.ps1 and upload this file container of storage account

Install-WindowsFeature -name Web-Server -IncludeManagementTools

While Creating VM

Select an extension to install----- Custom script extension ----- select storage account ---- Create



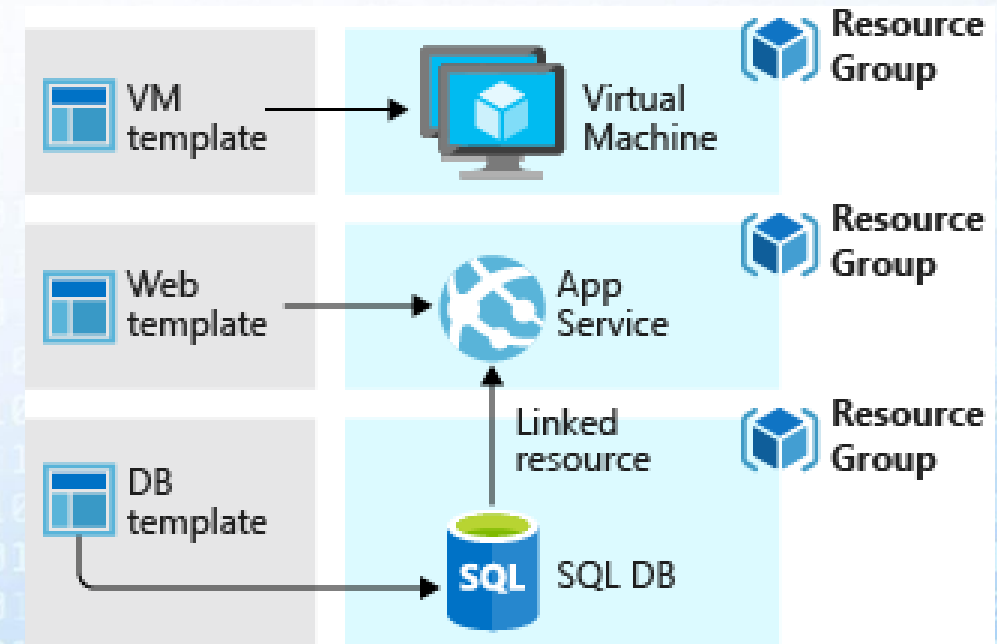
What is ARM?

When you deploy resources in Azure, whether through the Azure Portal, Azure CLI, or other tools, you are likely interacting with Azure Resource Manager (ARM)

ARM provides a consistent way to deploy and manage resources, making it easier to automate deployments, enforce policies, and maintain Infrastructure as Code practices

ARM templates describe the desired state of resources in a declarative manner, allowing you to specify what resources should exist and their configurations

ARM templates are written in JSON (JavaScript Object Notation) and are used to define and deploy resources into the Microsoft Azure cloud platform.



Using ARM templates to deploy resources into Azure comes with all the generic benefits of using IaC.

These include consistent deployments, reusability, modularity, versioning, ability to rollback, automation, and orchestration, Built in Validation

Note: While creating a Resource , you will get an option to download template

or

Go to resource ----Automation ----- Export Template --- download

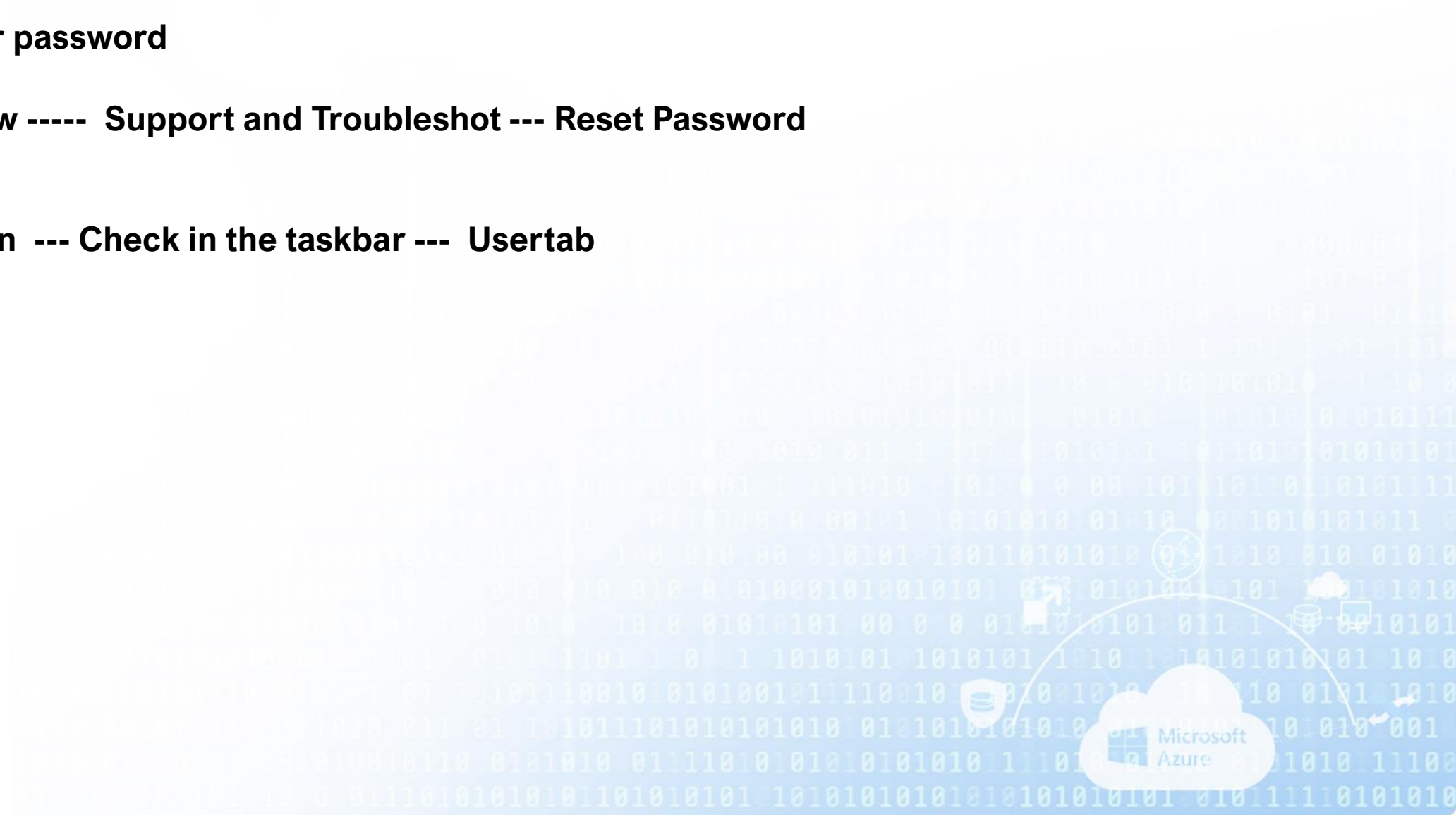
or

Manually create script using either PowerShell or CLI

Resetting User password

VM--- Overview ----- Support and Troubleshoot --- Reset Password

Multiuser Login --- Check in the taskbar --- Usertab



Azure Backup

On-premises - Back up files, folders, system state using the Microsoft Azure Recovery Services (MARS) agent. Or use the DPM or Azure Backup Server (MABS) agent to protect

on-premises VMs (Hyper-V and VMware) and other on-premises workloads

Azure VMs - Back up entire Windows/Linux VMs (using backup extensions) or back up files, folders, and system state using the MARS agent.

Azure Managed Disks - Back up Azure Managed Disks

Azure Files shares - Back up Azure File shares to a storage account

SQL Server in Azure VMs - Back up SQL Server databases running on Azure VMs

SAP HANA databases in Azure VMs - Backup SAP HANA databases running on Azure VMs

Azure Database for PostgreSQL servers - Back up Azure PostgreSQL databases and retain the backups for up to 10 years

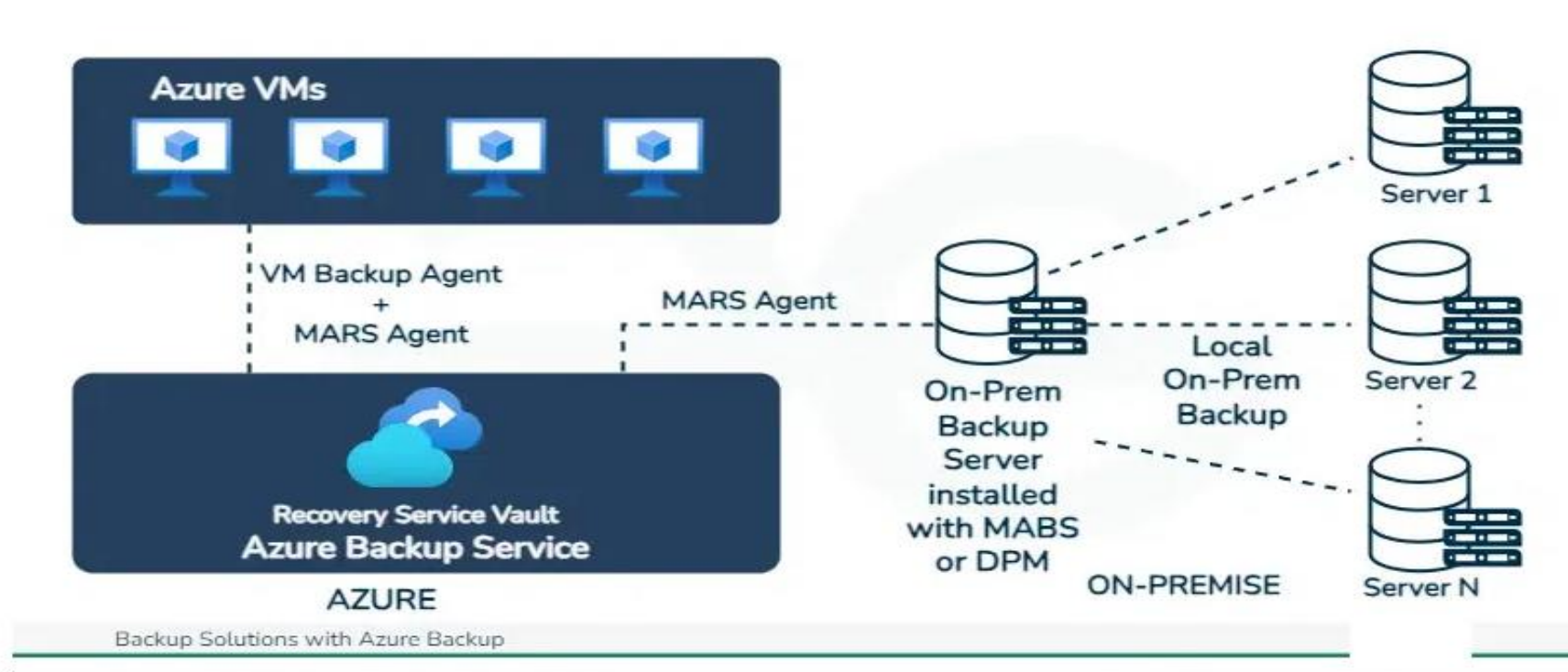
Azure Blobs - Overview of operational backup for Azure Blobs

Azure Kubernetes service - Overview of AKS backup

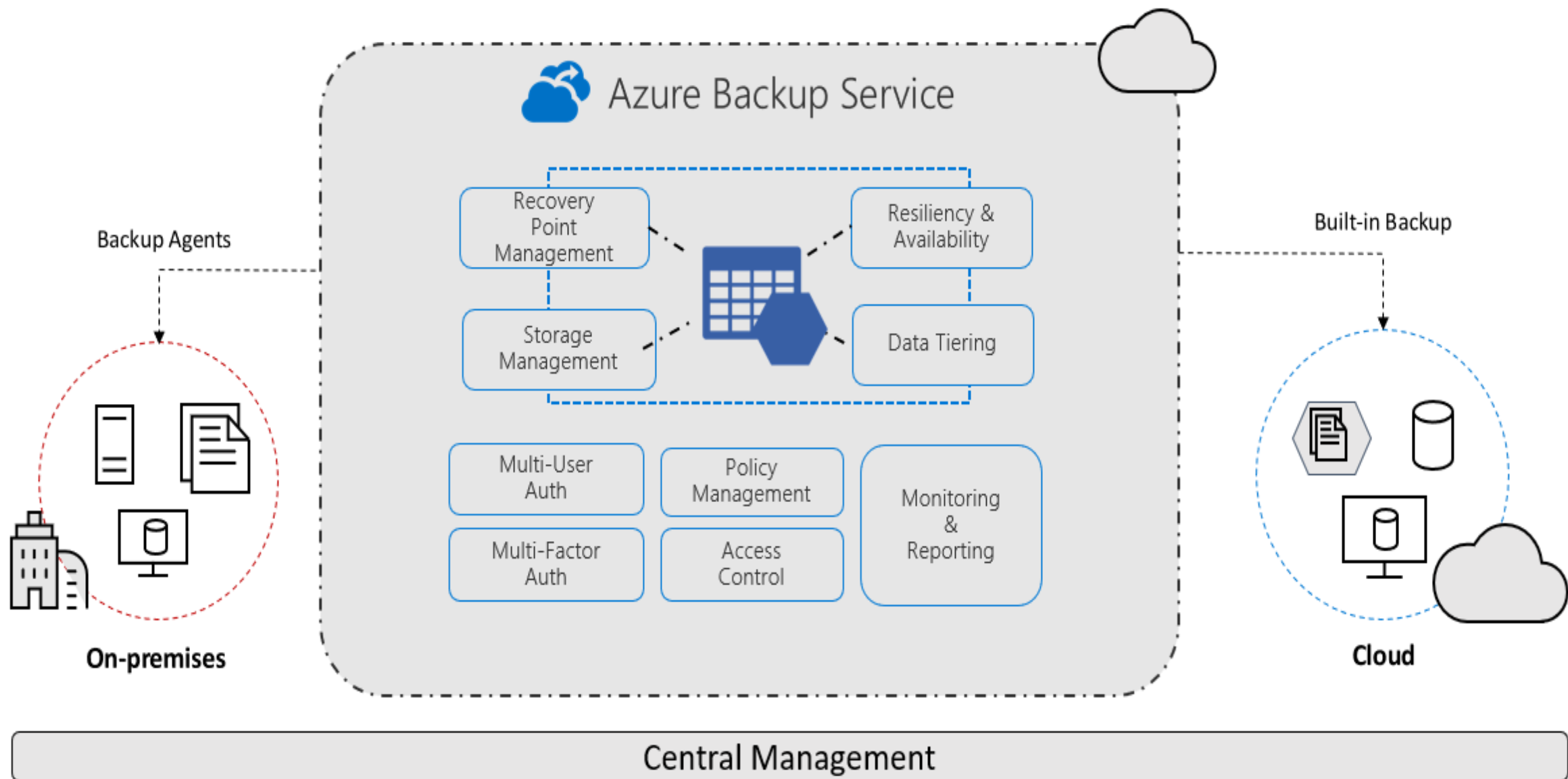
Azure Backup Service

Azure Backup is a service that allows you to back up your data to the Microsoft Azure Cloud.

Azure Backup is a service that can supplement existing backup solutions or replace existing solutions altogether.



On-premises - Back up files, folders, system state using the Microsoft Azure Recovery Services (MARS) agent. Or use the DPM or Azure Backup Server (MABS) agent to protect on-premises VMs (Hyper-V and VMware) and other on-premises workloads



What can I back up?

Azure VMs - Back up entire Windows/Linux VMs (using backup extensions) or back up files, folders, and system state using the MARS agent.

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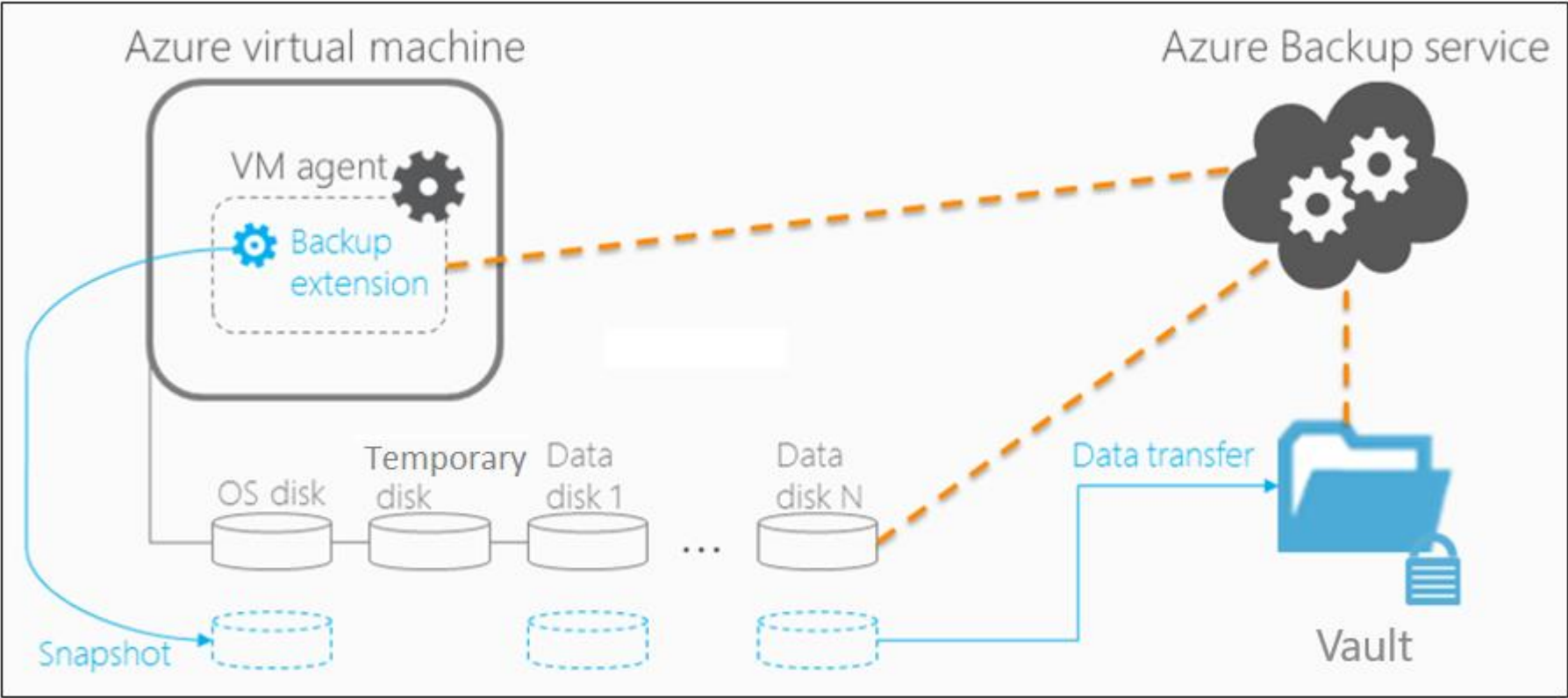
SAP HANA databases in Azure VMs - Backup SAP HANA databases running on Azure VMs

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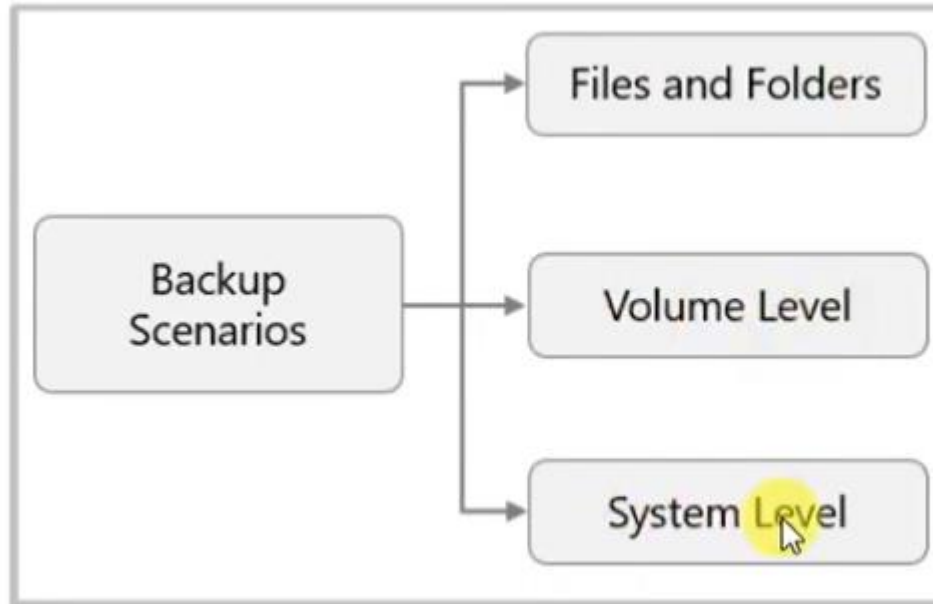
Azure Kubernetes service - Overview of AKS backup

Azure Backup stores backups in a Recovery Services vault and manages recovery points automatically. You can configure and scale backups easily, restore data quickly, and optimize storage. Use one of the following methods to back up your Azure VMs



Backup scenarios

The MARS agent supports the following backup scenarios:



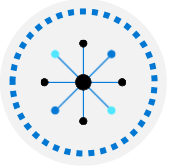
- **Files and Folders:** Selectively protect Windows files and folders.
- **Volume Level:** Protect an entire Windows volume of your machine.
- **System Level:** Protect an entire Windows system state.

1. For Azure VMs that are selected for backup, Azure Backup starts a backup job according to the backup schedule you specify.
2. If you have opted for application or file-system consistent backups, the VM needs to have a backup extension installed to coordinate for the snapshot process.
3. If you have opted for [crash-consistent backups](#), no agents are required in the VMs.
4. During the first backup, a backup extension is installed on the VM if the VM is running.
 1. For Windows VMs, the [VMSnapshot extension](#) is installed.
 2. For Linux VMs, the [VMSnapshotLinux extension](#) is installed.

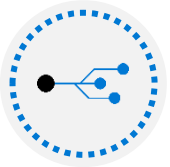


Day3: Virtual Networking





Lesson 01: Virtual Networks



Lesson 02: IP Addressing



Lesson 03: Network Security Groups



Lesson 04: Azure Express Route

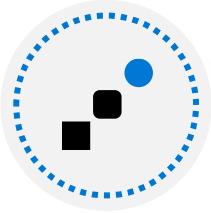


Lesson 06: Module 04 Lab and Review



Lesson 01: Virtual Networks





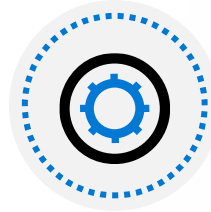
Azure Networking Components



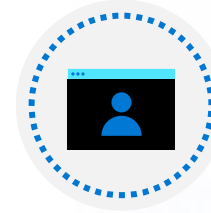
Virtual Networks



Subnets



Implementing Virtual Networks



Demonstration – Creating Virtual Networks



Isolation and segmentation:

To deploy resources such as virtual machines into virtual networks, they will be isolated from other resources.

Communication with the Internet: All resources in a virtual network can communicate outbound to the Internet by default. But it needs to establish an inbound connection from the Internet. We can either use public IP or load balancers

Communication between resources: Communication between the number of resources inside the virtual network

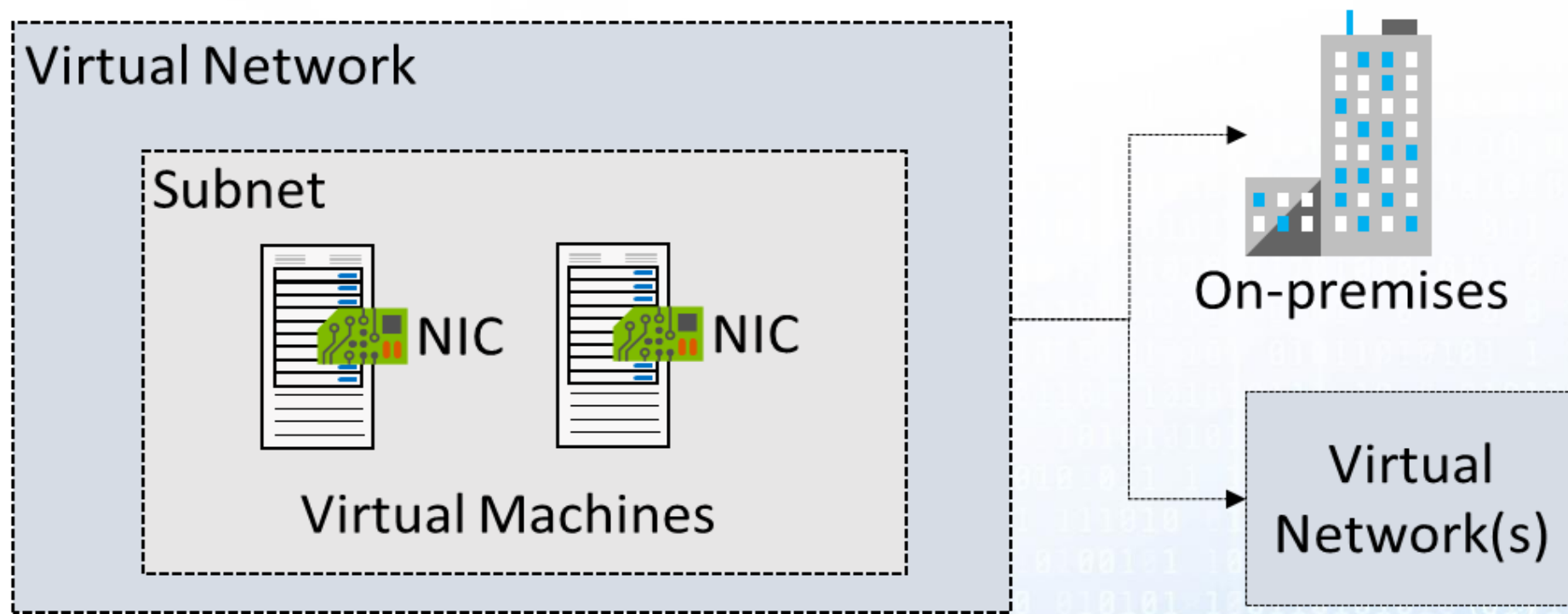
Communication with on-premises resources: By establishing either point to site VPN or site to site VPN or Express route, your workloads within Azure virtual network can seamlessly communicate with workloads within our on-premises data center.

Filter network traffic: We can use Network Security Groups, Application Security Group, Azure firewall, or third-party network virtual appliance to filter the traffic coming to the resources in the virtual network.

Route network traffic: We can route the network traffic using the routing tables, we can configure user-defined routes to route all the outbound traffic, let's say via a firewall.

Monitor network traffic: By network security groups and traffic analytics monitoring solution, you'll be able to carry out extensive monitoring on both inbound and outbound communications.





Logical representation
of your own network

Create a dedicated
private cloud-only
virtual network

Securely extend
your datacenter with
virtual networks

Enable hybrid
cloud scenarios

Adopting cloud solutions can save time and simplify operations

Azure requires the same types of networking functionality as on-premises infrastructure

Azure networking offers a wide range of services and products



Virtual Network

Microsoft

Create a logically isolated section in Microsoft Azure and securely connect it outward.



Load Balancer

Microsoft

A load balancer that distributes incoming traffic among backend virtual machine instances.



Application Gateway

Microsoft

Scalable layer-7 load balancer offering various traffic routing rules and SSL termination for backend



Traffic Manager profile

Microsoft

Create a Microsoft Azure Traffic Manager Profile that allows you to control the distribution of user



Virtual network gateway

Microsoft

The VPN device in your Azure virtual network and used with site-to-site and VNet-to-VNet VPN



Virtual WAN

Microsoft

Azure Virtual WAN is a networking service that provides optimized and automated branch-to-branch



[+ Subnet](#)
[+ Gateway subnet](#)
[↻ Refresh](#)

🔍 Search subnets

Name	↑↓	Address range	↑↓	IPv4 available addresses	↑↓	Delegated to	↑↓	Security group
subnet0		10.1.0.0/24		251		-		nsg0
subnet1		10.1.1.0/24		251		-		-
subnet2		10.1.2.0/24		251		-		nsg2
GatewaySubnet		10.1.255.0/24		251		-		-

A virtual network can be segmented into one or more subnets

Subnets provide logical divisions within your network

Subnets can help improve security, increase performance, and make it easier to manage the network

Each subnet must have a unique address range cannot overlap with other subnets in the virtual network in the subscription

Create new virtual networks at any time

Add virtual networks when you create a virtual machine

Need to define the address space, and at least one subnet

Be careful with overlapping address spaces

Create virtual network

Basics IP Addresses Security Tags Review + create

Project details

Subscription * ⓘ

Visual Studio Enterprise



Resource group * ⓘ

Lab04

[Create new](#)

Instance details

Name *

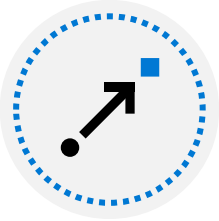
VNet2

Region *

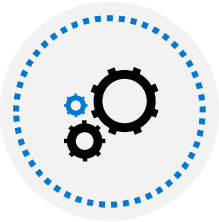
(US) East US 2

Lesson 02: IP Addressing

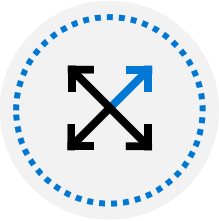




IP Addressing



Creating IP Addresses

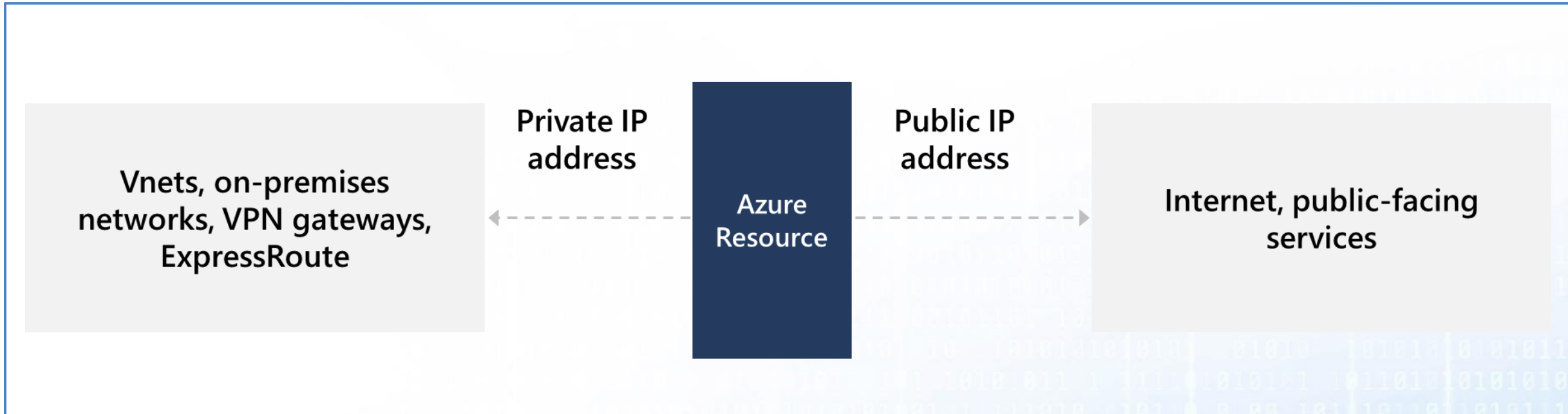


Public IP Addresses



Private IP Addresses





Private IP addresses - used within an Azure virtual network (VNet), and your on-premises network, when you use a VPN gateway or ExpressRoute circuit to extend your network to Azure

Public IP addresses - used for communication with the Internet, including Azure public-facing services

Available in IPv4 or IPv6 or both

Basic vs Standard SKU

Available in Dynamic, Static or both
(depending on SKU):

Zone redundant

Not mixable or immutable

Range of contiguous addresses available as a prefix

Create public IP address

IP Version * ⓘ

☒ IPv4 ☐ IPv6 ☐ Both

SKU * ⓘ

☒ Basic ☐ Standard

IPv4 IP Address Configuration

Name *

IP address assignment *

☒ Dynamic ☐ Static

Azure Network Security

A network security group consists of security rules that allow or deny inbound/Outbound network traffic to or from different types of Azure resources that we will host in our Azure virtual network.

Network security group at different levels, Subnet and VM level

Security rule properties:

Name: The name of the network should be unique within the network security group.

Priority: Security rules are processed in priority order with a lower number has the highest priority.

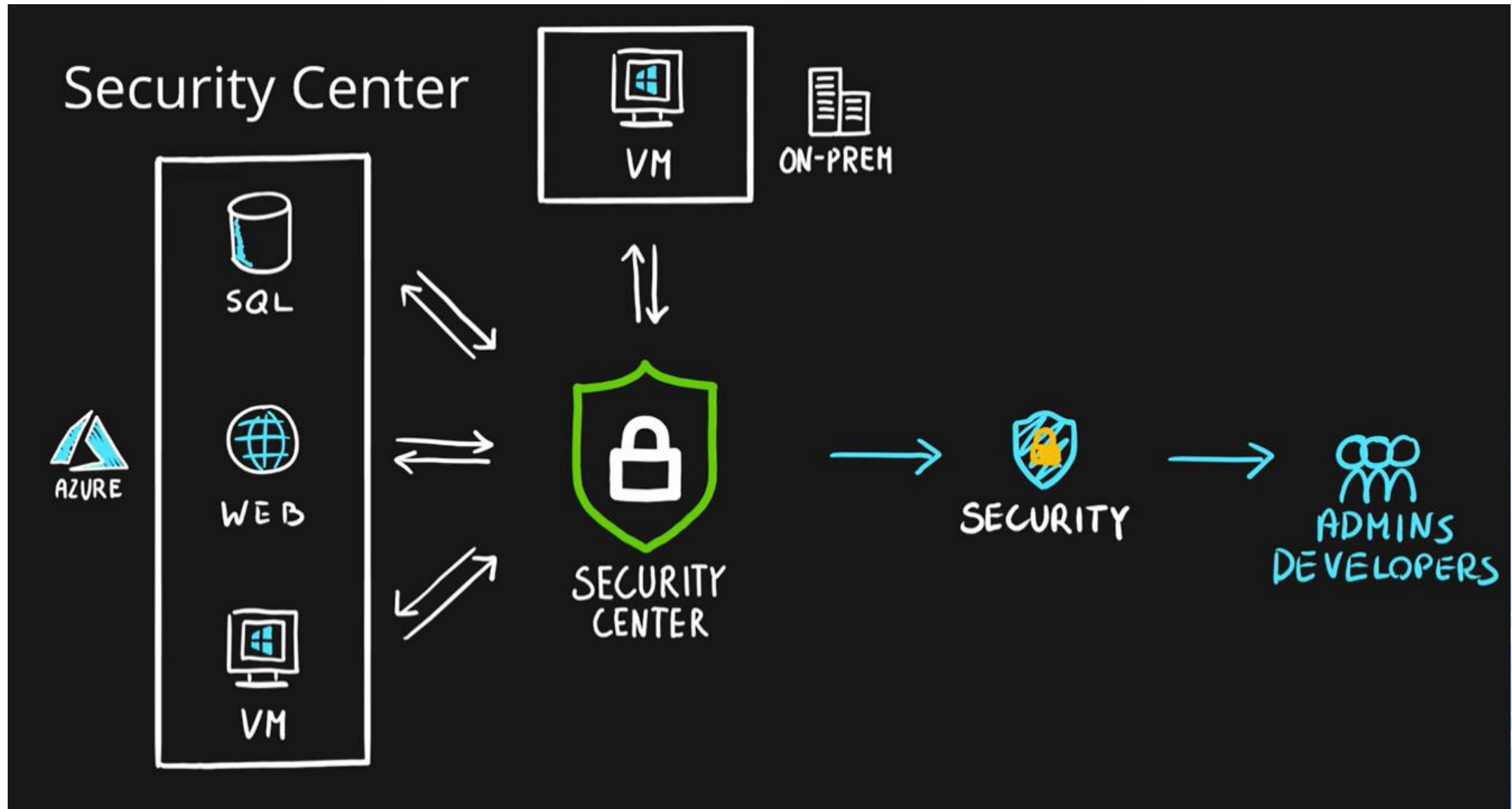
Source or Destination: (The IP address, CIDR (Classless inter-domain routing) block, service tag, or application security group) The ability to specify multiple individual IP addresses and ranges in a rule is referred to as augmented security rules.

Protocol: TCP, UDP, etc.

Port range: we can specify an individual or range of ports

Action: Allow or Deny

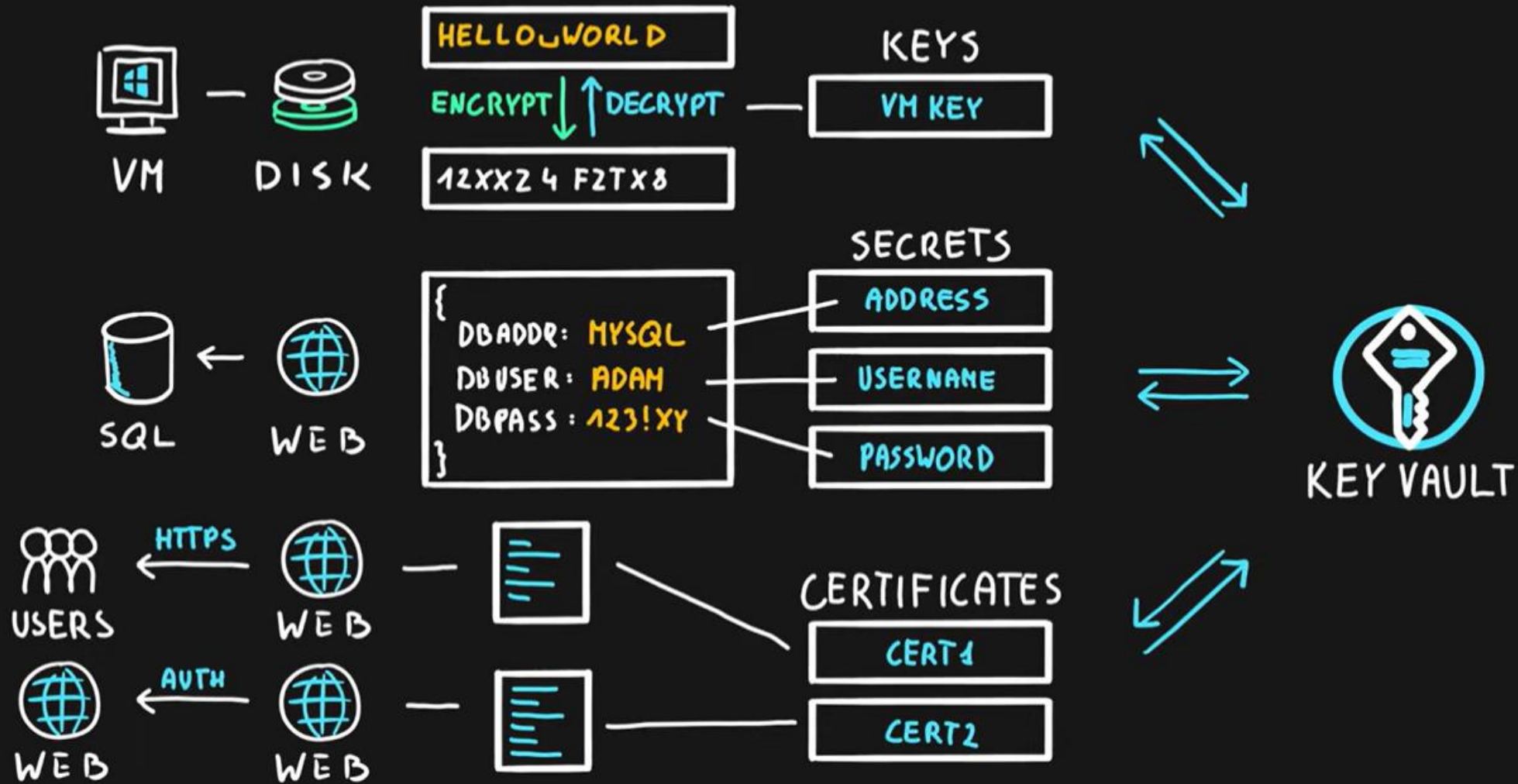


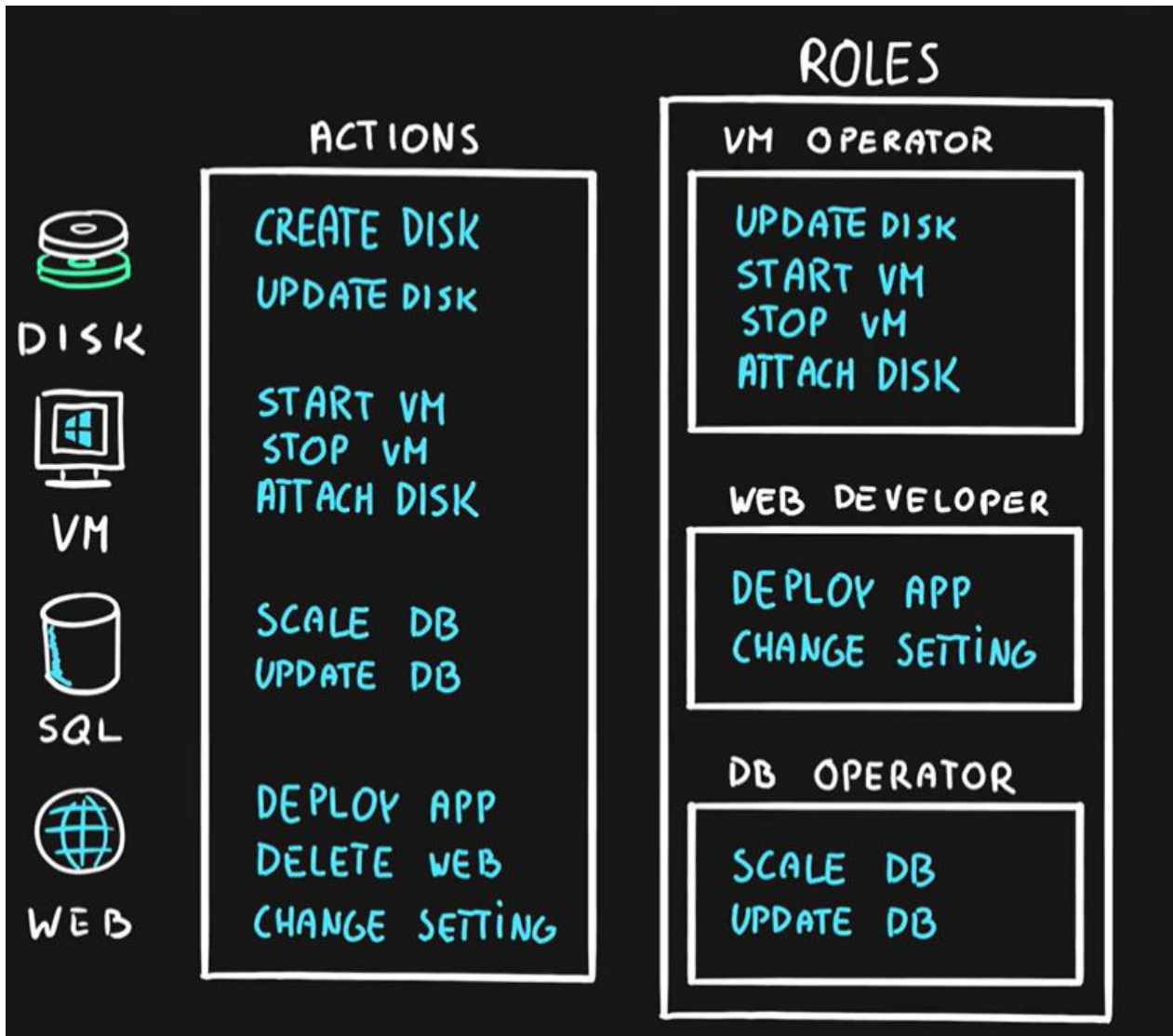


Security Center



Azure Key Vault

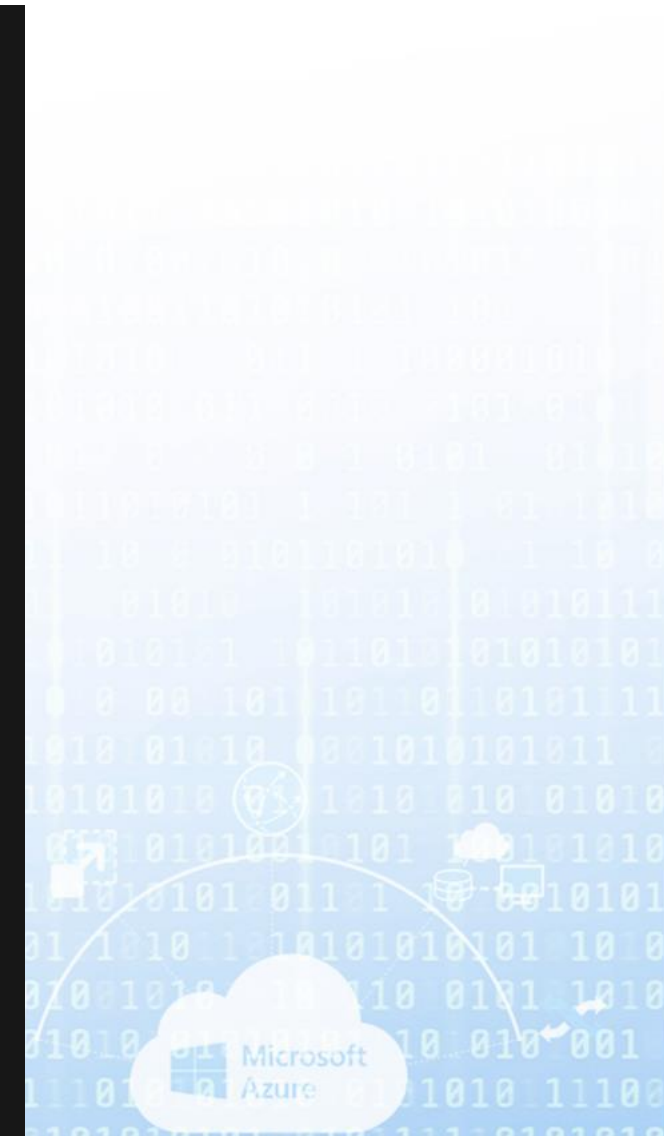




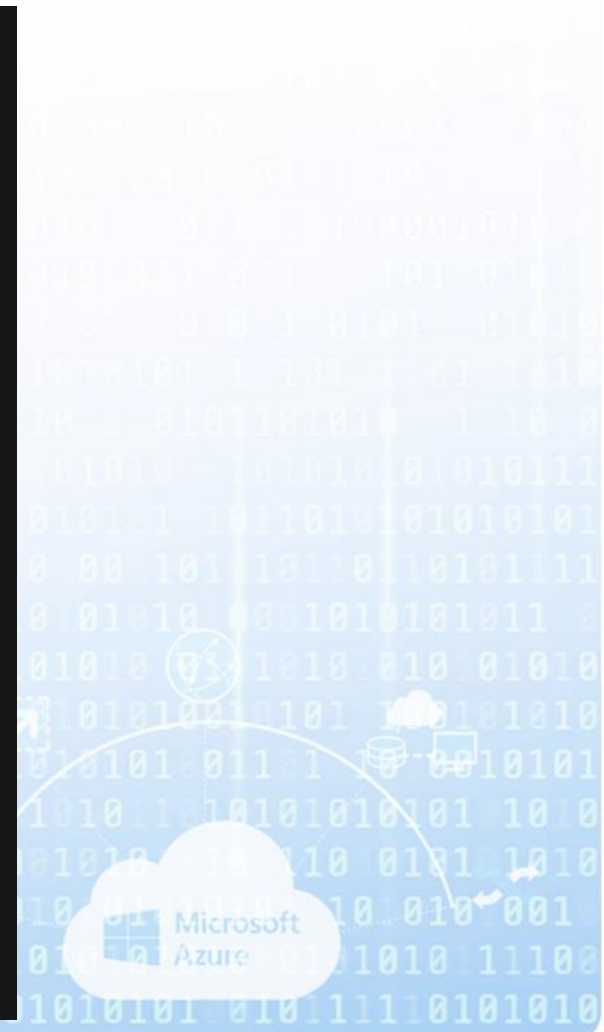
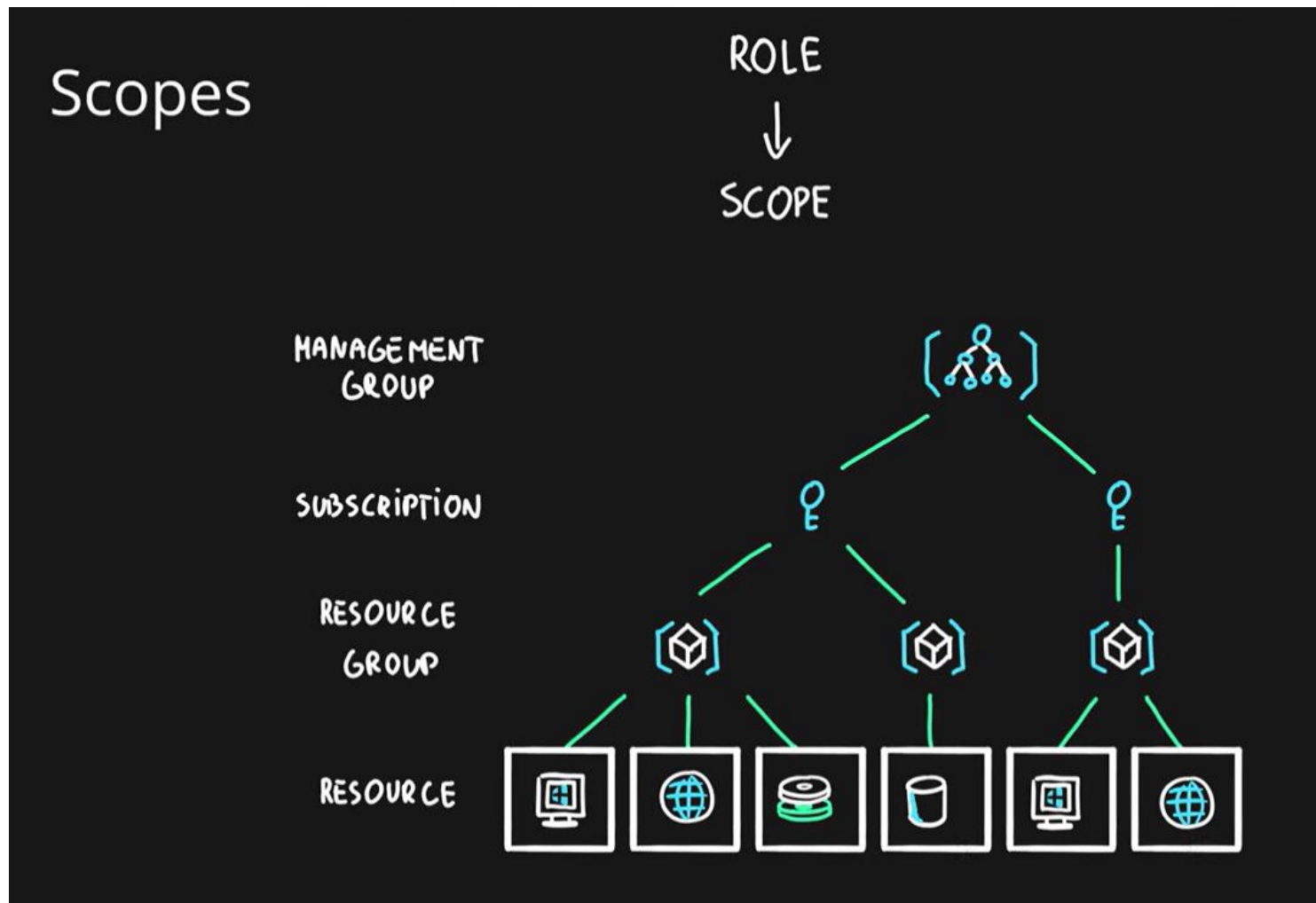
Role (role definition) is a collection of actions that the assigned identity will be able to perform.

*Role definition is an answer to a question
"What can be done?"*





Role-Based Access Control (RBAC)



Key Characteristics

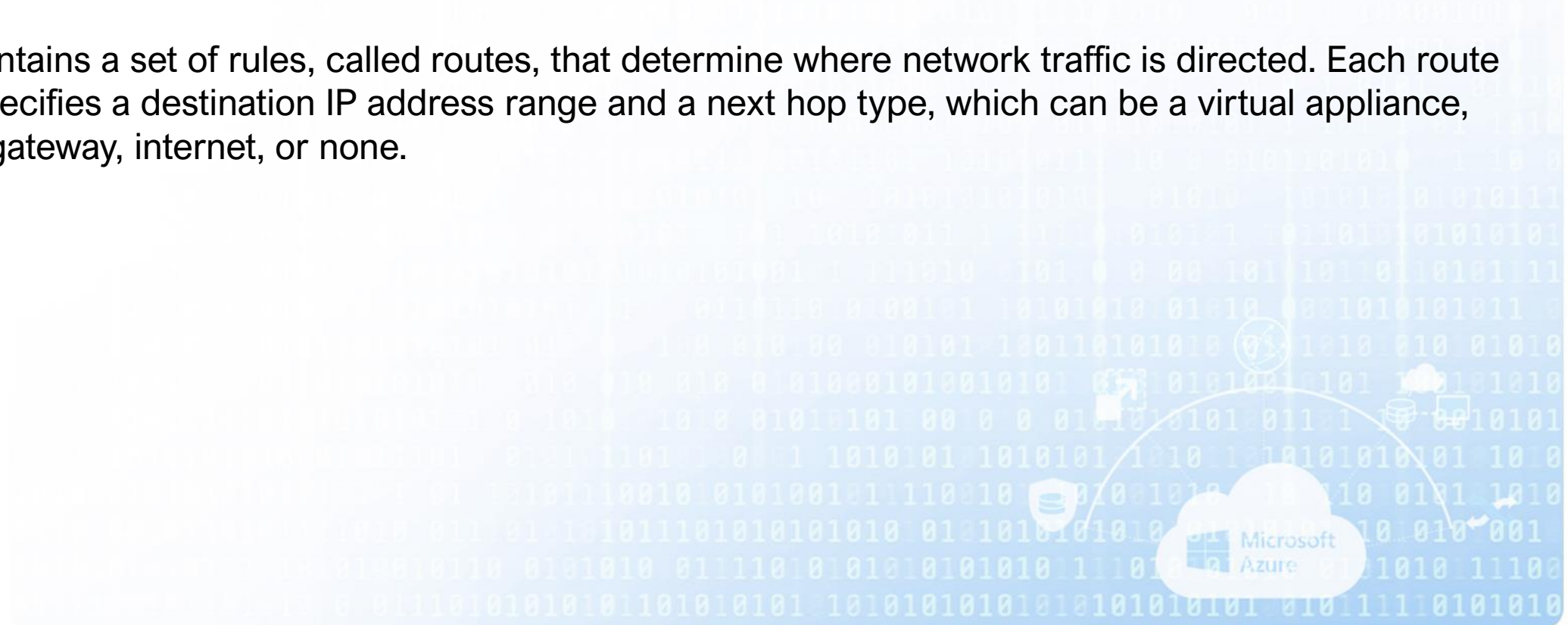
- Authorization system built on Azure Resource Manager (ARM)
- Designed for fine-grained access management of Azure Resources
- Role assignment is combination of
 - Role definition – list of permissions like create VM, delete SQL, assign permissions, etc.
 - Security principal – user, group, service principal and managed identity and
 - Scope – resource, resource groups, subscription, management group
- Scopes are hierarchical
 - Management Groups > Subscriptions > Resource Groups > Resources
- Built-in and Custom roles are supported

Key Characteristics

- Managed service for securing sensitive information (application/platform) (PaaS)
- Secure storage service for
 - Keys,
 - Secrets and
 - Certificates
- Highly integrated with other Azure services (VMs, Logic Apps, Data Factory, Web Apps, etc.)
- Centralization
- Access monitoring and logging



- Microsoft Azure's Route Table is a powerful tool that allows for the control and direction of network traffic within a virtual network.
- An Azure Route Table is a set of custom routes that dictate how network traffic should move within a virtual network (VNet). It offers a way to control the flow of data, ensuring it reaches the correct endpoint.
- A route table contains a set of rules, called routes, that determine where network traffic is directed. Each route within a table specifies a destination IP address range and a next hop type, which can be a virtual appliance, virtual network gateway, internet, or none.



Key Use Cases and Scenarios

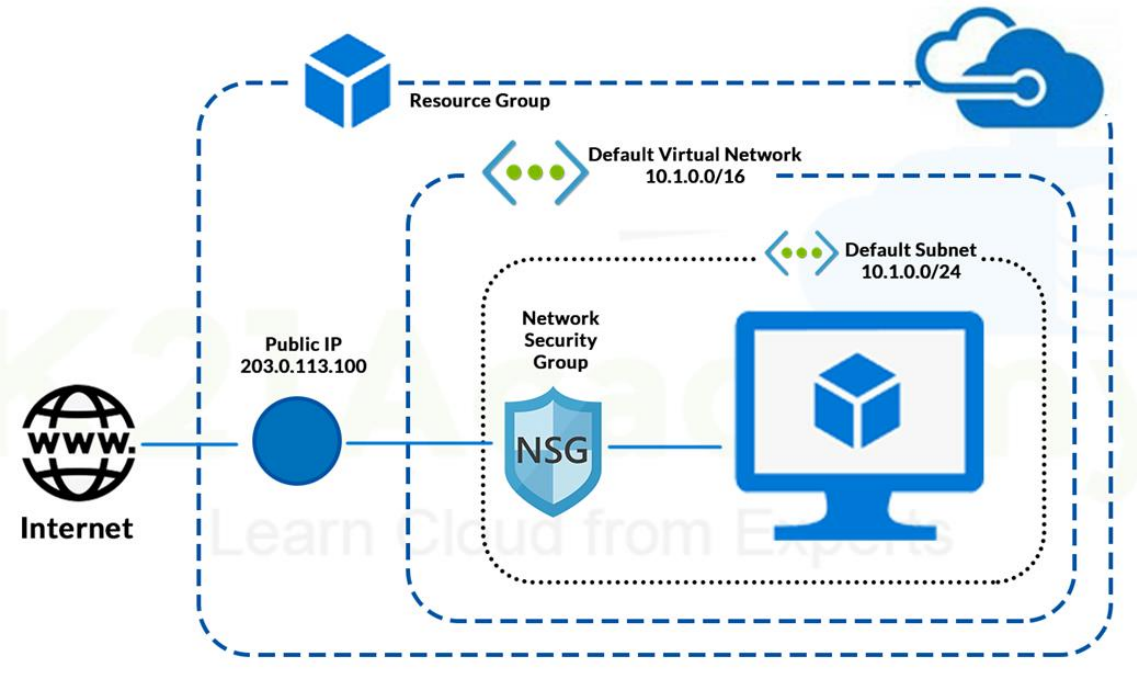
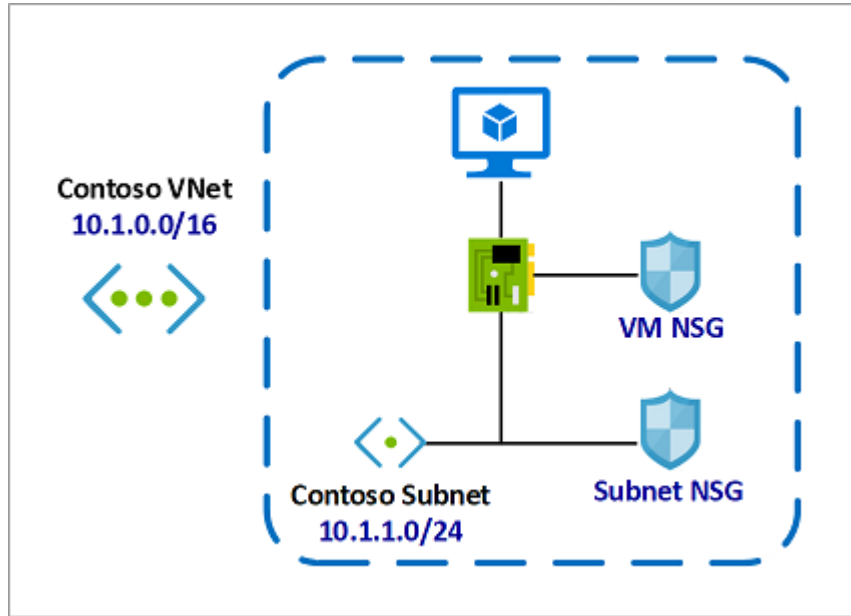
- **Inter-VNet Communication:** Route tables are essential for directing traffic between different virtual networks (VNets) in Azure. This is particularly useful in complex architectures where multiple VNets need to communicate with each other.
- **Forced Tunneling:** If you want to route all outbound traffic to your on-premises network for inspection or audit purposes, you can use route tables to configure forced tunneling.
- **Integrating with Virtual Appliances:** Route tables allow you to route traffic through network virtual appliances (NVAs) for purposes like firewalling, intrusion detection, or data analysis.
- **Bypassing Azure's Default System Routes:** Azure creates default system routes. However, custom routes in route tables can override these for specific traffic flow requirements.



System routes

- Azure automatically creates system routes and assigns the routes to each subnet in a virtual network. You can't create system routes, and you can't remove system routes, but you can override some system routes with custom routes.
- **User-defined routes**
 - To customize your traffic routes, you shouldn't modify the default routes. You should create custom or user-defined (static) routes, which override the Azure default system routes.
 - In Azure, you create a route table and then associate the route table to zero or more virtual network subnets. Each subnet can have zero or one route table associated to it.





Default Rules

Some default rules are created by default when we create NSG. There are two types of default rules.

Inbound Security rules

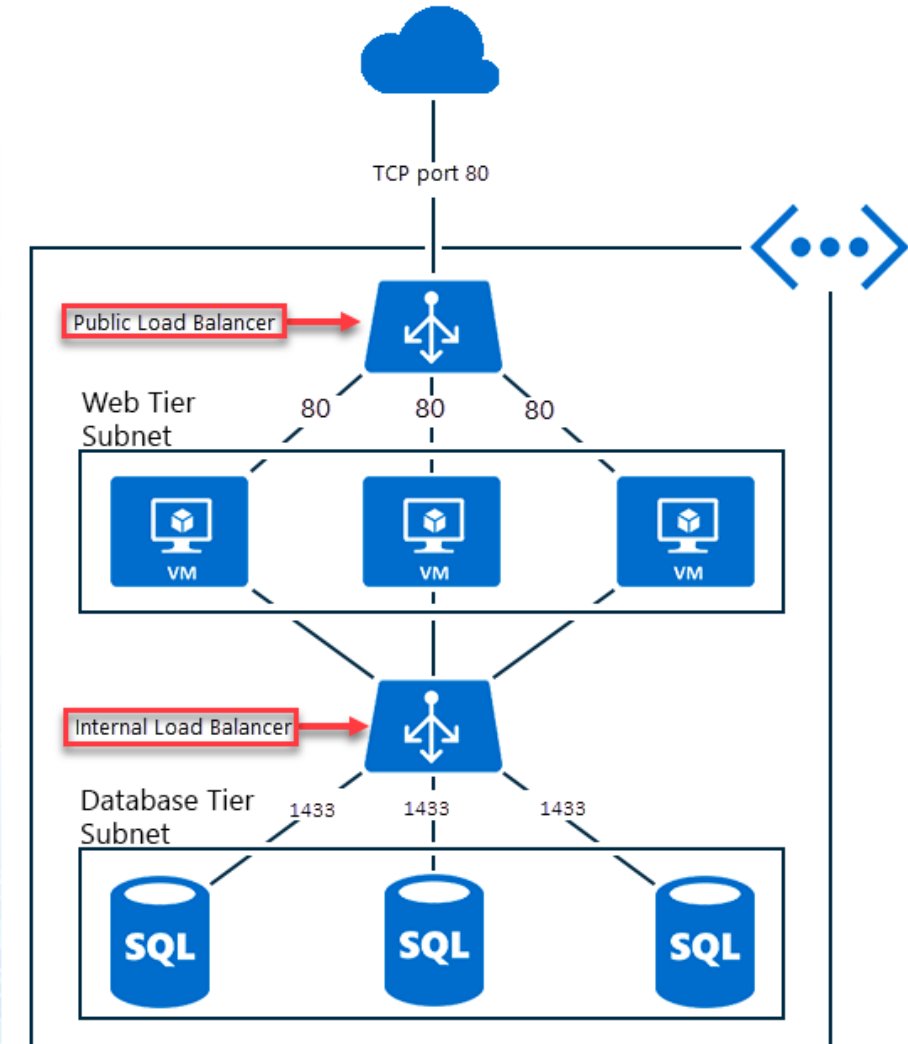
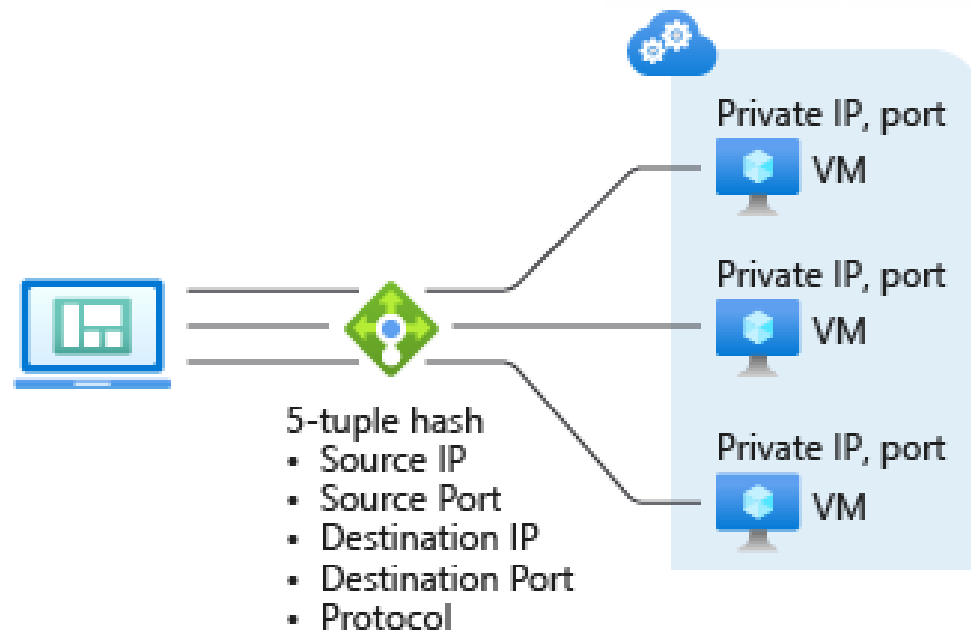
- **AllowVNetInbound:** Traffic is allowed from any resources within the VNet
- **AllowAzureLoadBalancerInbound:** Any traffic originating from Azure load-balancer to any of the virtual machines within the network is permitted.
- **DenyAllInbound:** By default, virtual machines in the virtual network can communicate with each other, and also Azure load balancer can communicate with virtual machines within the virtual network.

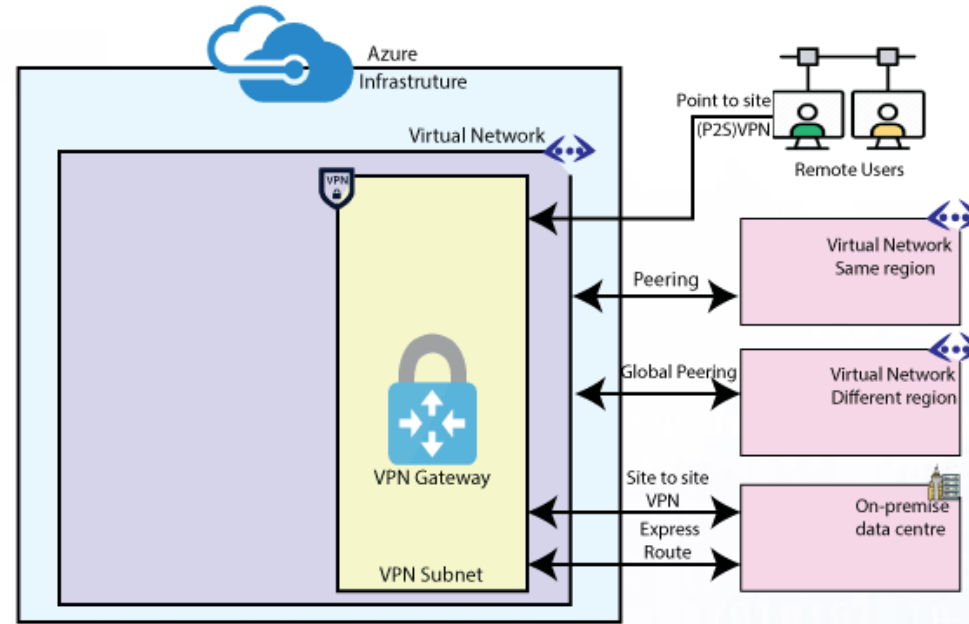
Outbound Security rules

- **AllowVNetOutBound: Traffic** is allowed through any resources within the VNet
- **AllowInternetOutBound:** Traffic originating from any resources in the VNet to the Internet is allowed.
- **DenyAllOutBound:** By default, virtual machines in a virtual network can communicate with each other, and also Azure load balancer can interact with the virtual machine within the virtual network.



The load balancer is used to distribute the incoming traffic to the pool of virtual machines. It stops routing the traffic to a failed virtual machine in the pool.





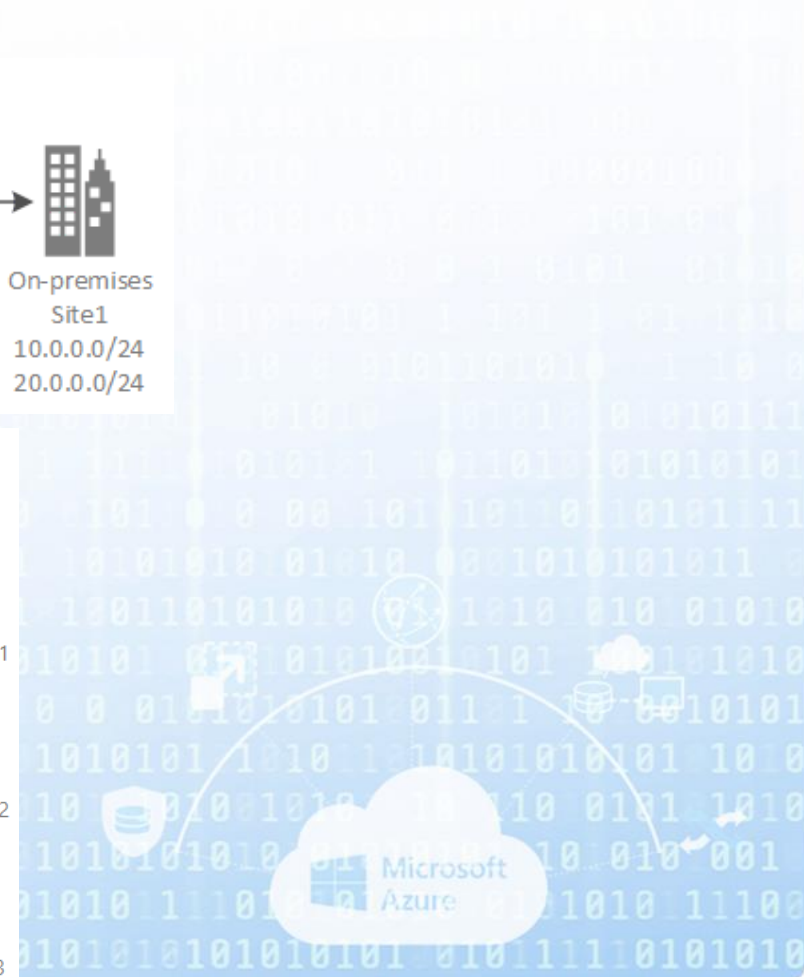
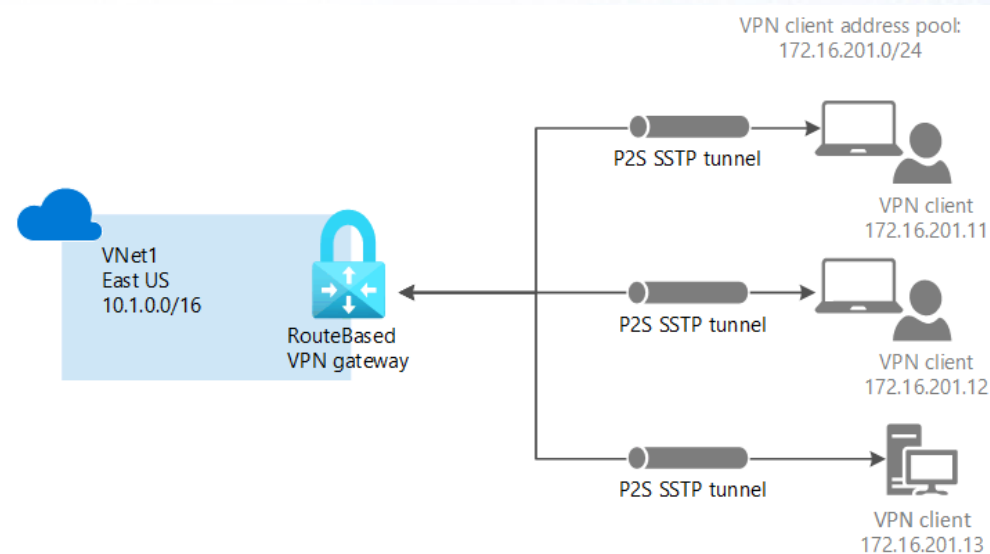
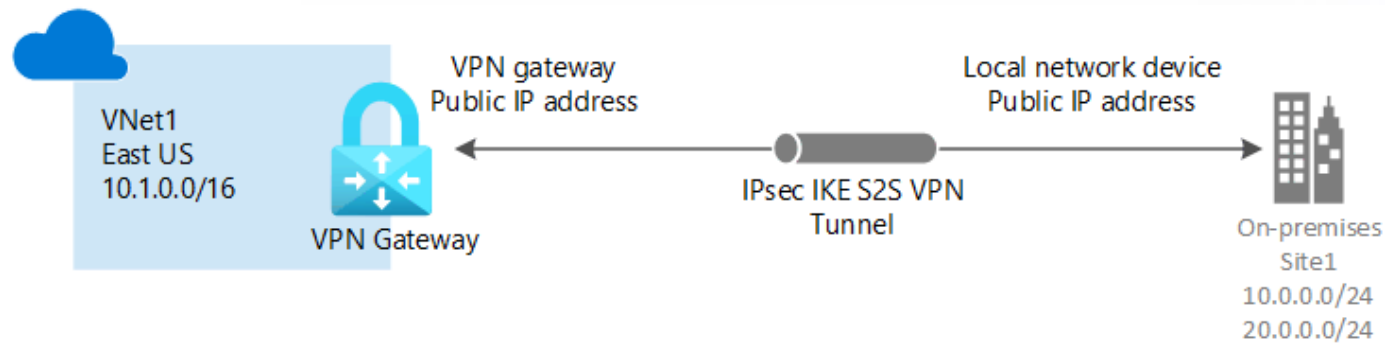
Virtual network peering enables us to connect two VNet in the same or across regions. If both of the virtual networks are in Azure and also within the same region, then you can use peering.

The traffic between different virtual machines in peered virtual networks is routed directly from the Microsoft backbone infrastructure, not through a gateway or over the public Internet.

Regional Peering and Global peering

If we have a virtual network in azure that exists in different regions, then we can use Global peering. Gateway transit is supported for both VNet peering and Global VNet Peering.

If we have an on-premises virtual network, and we may have other virtual networks existing in other cloud providers. To connect to our virtual network in Azure with the network that is an on-premises data center, we can use Site to site VPN.

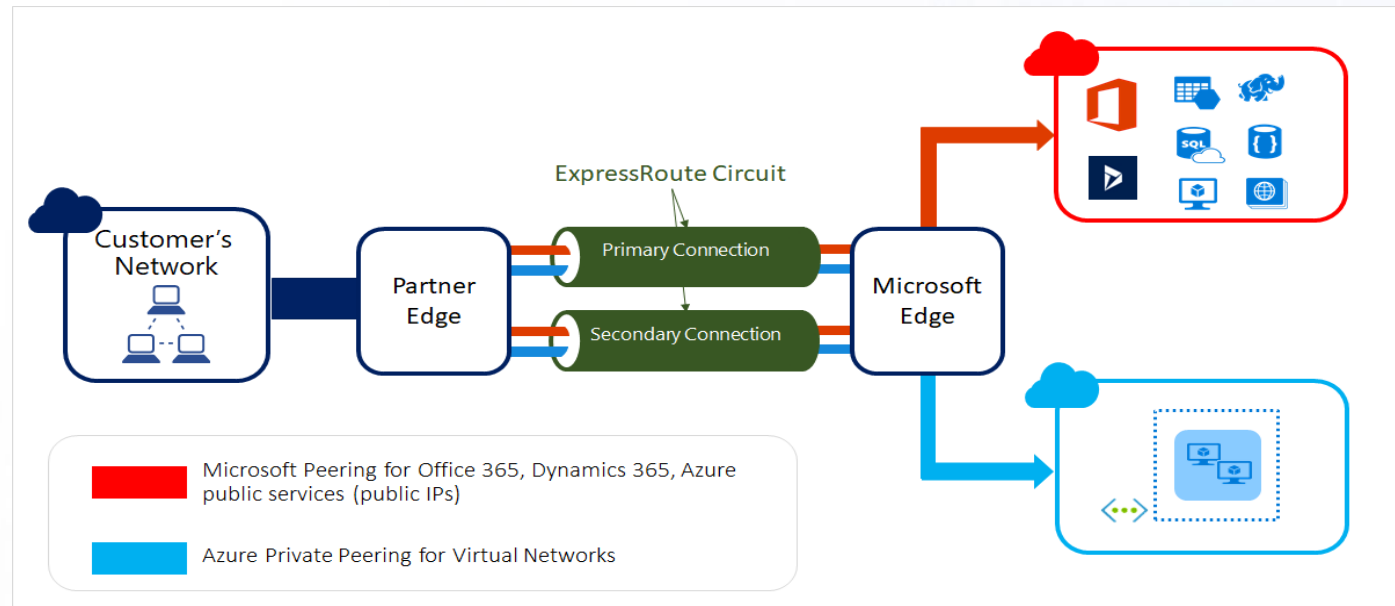


ExpressRoute lets you extend your on-premises networks into the Microsoft cloud over a private connection with the help of a connectivity provider.

Connectivity can be from an any-to-any (IP VPN) network, a point-to-point Ethernet network,

ExpressRoute connections don't go over the public Internet.

This allows ExpressRoute connections to offer more reliability, faster speeds, consistent latencies, and higher security than typical connections over the Internet.



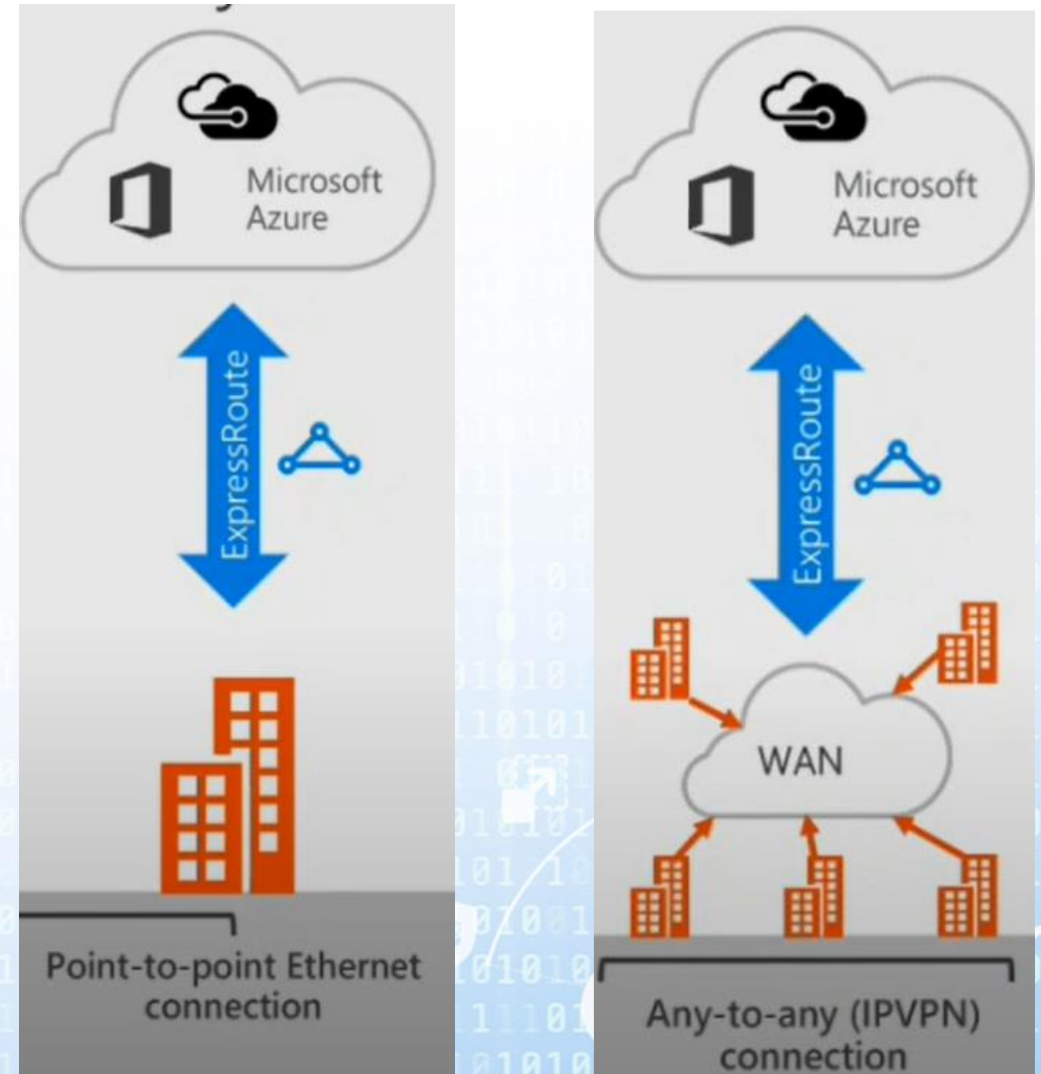
Bandwidth options

You can purchase ExpressRoute circuits for a wide range of bandwidths. The supported bandwidths are listed as followed. Be sure to check with your connectivity provider to determine the bandwidths they support.

- 50 Mbps
- 100 Mbps
- 200 Mbps
- 500 Mbps
- 1 Gbps
- 2 Gbps
- 5 Gbps
- 10 Gbps

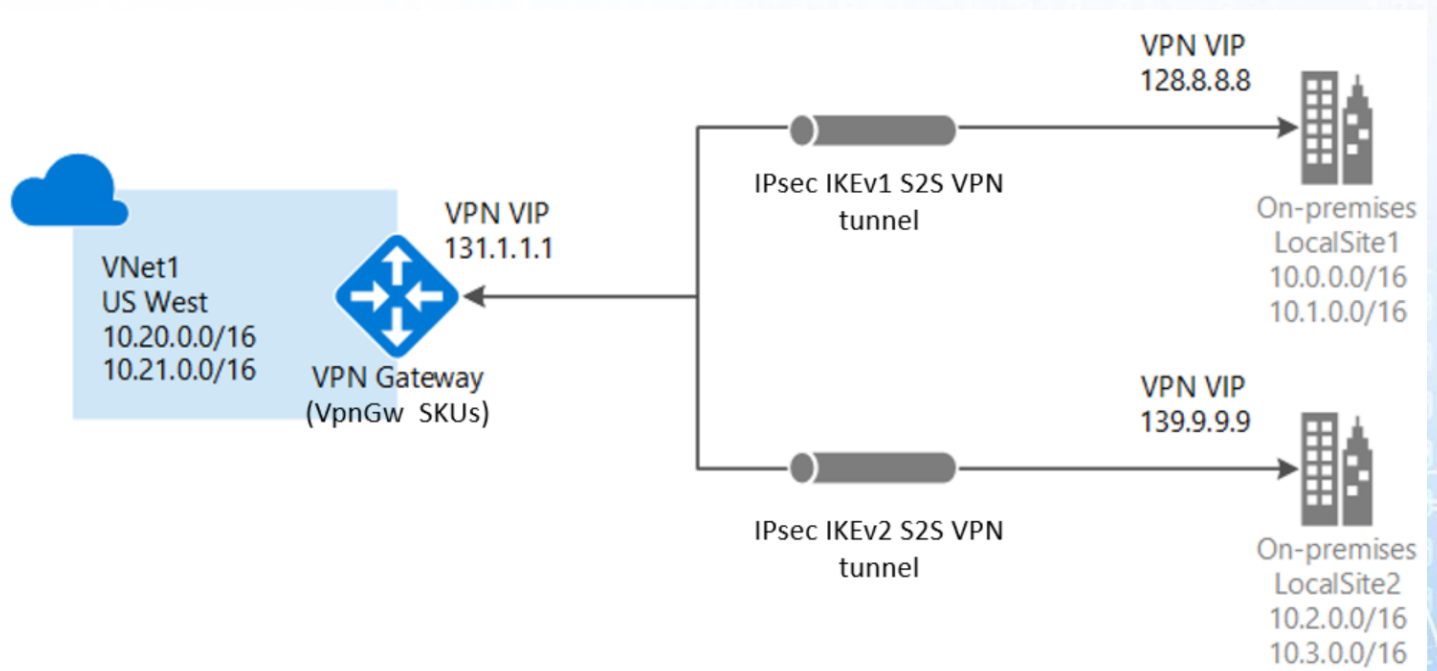
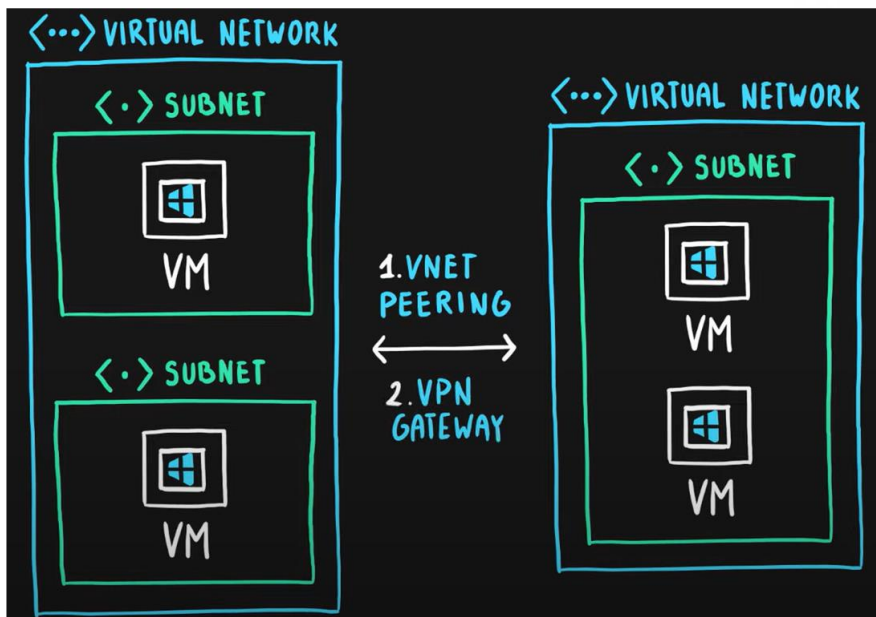
Dynamic scaling of bandwidth

You can increase the ExpressRoute circuit bandwidth (on a best effort basis) without having to tear down your connections.



A VPN gateway is a specific type of virtual network gateway, which is used to send encrypted traffic between an Azure virtual network and an on-premises location over the public internet.

VPN gateway act as a middle man on both sides of the virtual networks. And if the workloads in those virtual networks need to communicate with each other, they will communicate via this encrypted channel of communication between the VPN gateways of both virtual networks.



Azure CDN caches web content at a strategically placed location to provide maximum throughput for delivering content to users.



Features of Content Delivery Network (CDN)

- ❖ **Dynamic site acceleration:** It is the capability to deliver dynamic web content with minimum latency. It is achieved by using different techniques such as route optimization to avoid congestion points, TCP optimization, etc.
- ❖ **HTTPS support:** It provides us the HTTPS support of secure web content.
- ❖ **Query string caching:** Based on query string caching, we can cache the content also within CDN location.
- ❖ **Geo-Filtering:** We can apply some geo-filtering if we want certain content filtered for a particular geographical region.
- ❖ **Azure diagnostics logs:** It provides the facility of records of diagnosis.



Module 05: Identity and Access Management



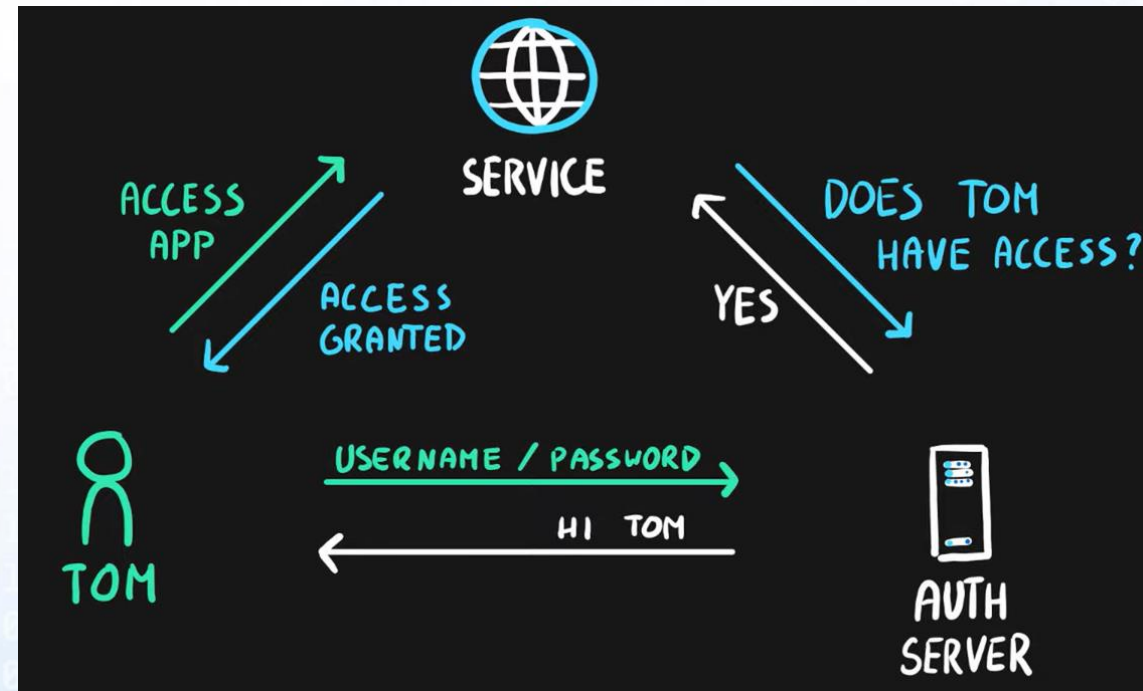
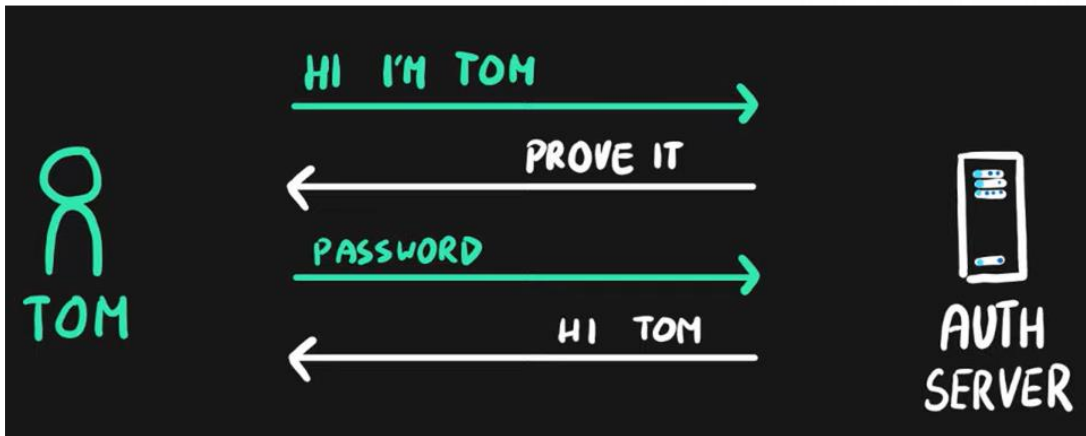
Lesson 01: Identity and Access Management



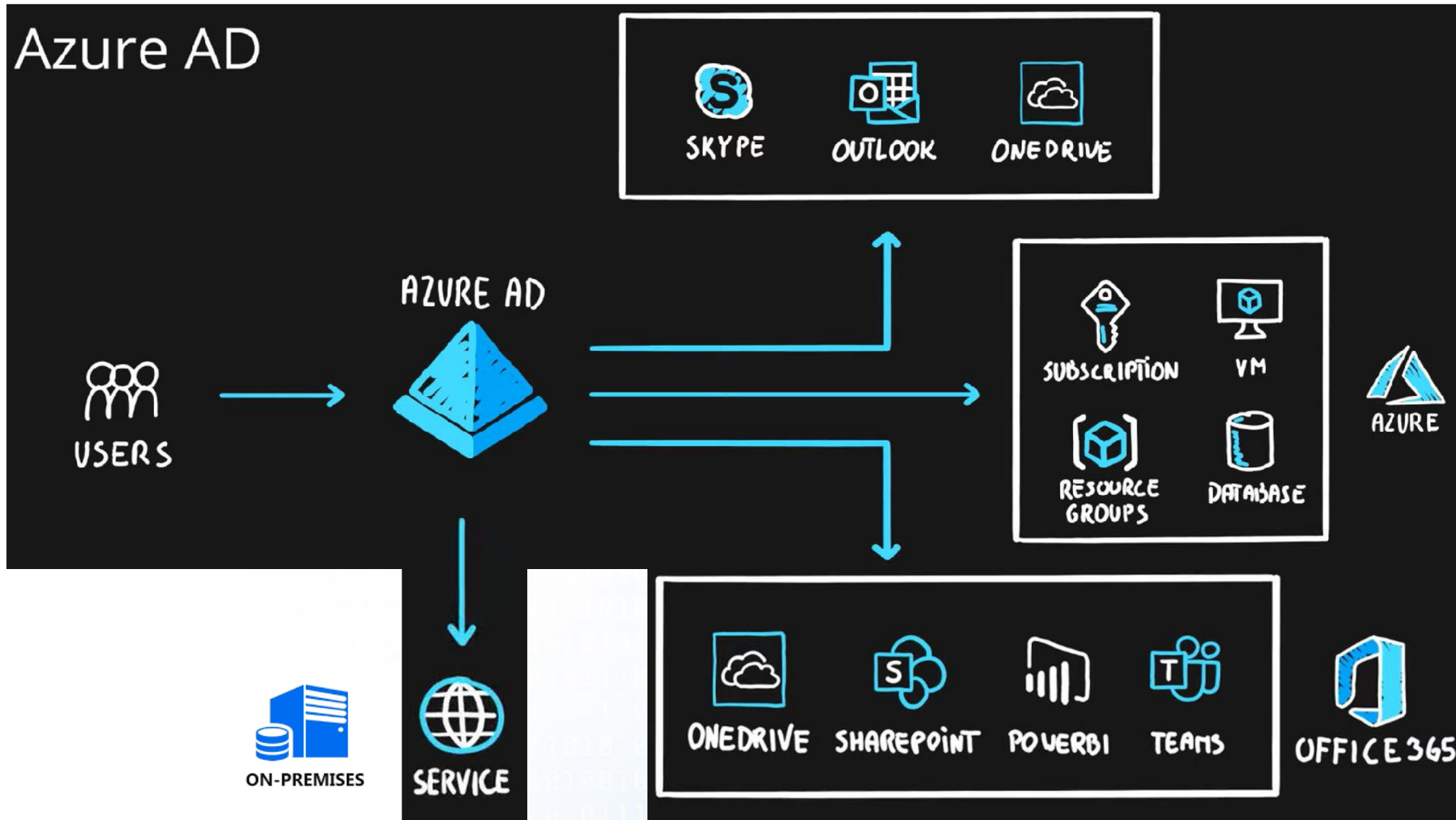
Identity : User with User Name and Password, Application, server with token or certificate



Authentication and Authorization



aa



Azure Active Directory is Microsoft's multi-tenant, cloud-based directory and identity management service. For an organization, Azure AD helps employees sign up to multiple services and access them anywhere over the cloud with a single set of login credentials.

Who uses Azure AD

- IT admins.
- App developers.
- Microsoft 365, Office 365, Azure, or Dynamics CRM Online subscribers

❖ Sync services to synchronize on premises active directory users and groups with Azure AD

❖ Multifactor Authentication

Azure AD Multi-Factor Authentication works by requiring two or more of the following authentication methods:

- Something you know, typically a password.
- Something you have, such as a trusted device that's not easily duplicated, like a phone or hardware key.
- Something you are - biometrics like a fingerprint or face scan.

❖ Single sign-on (SSO)

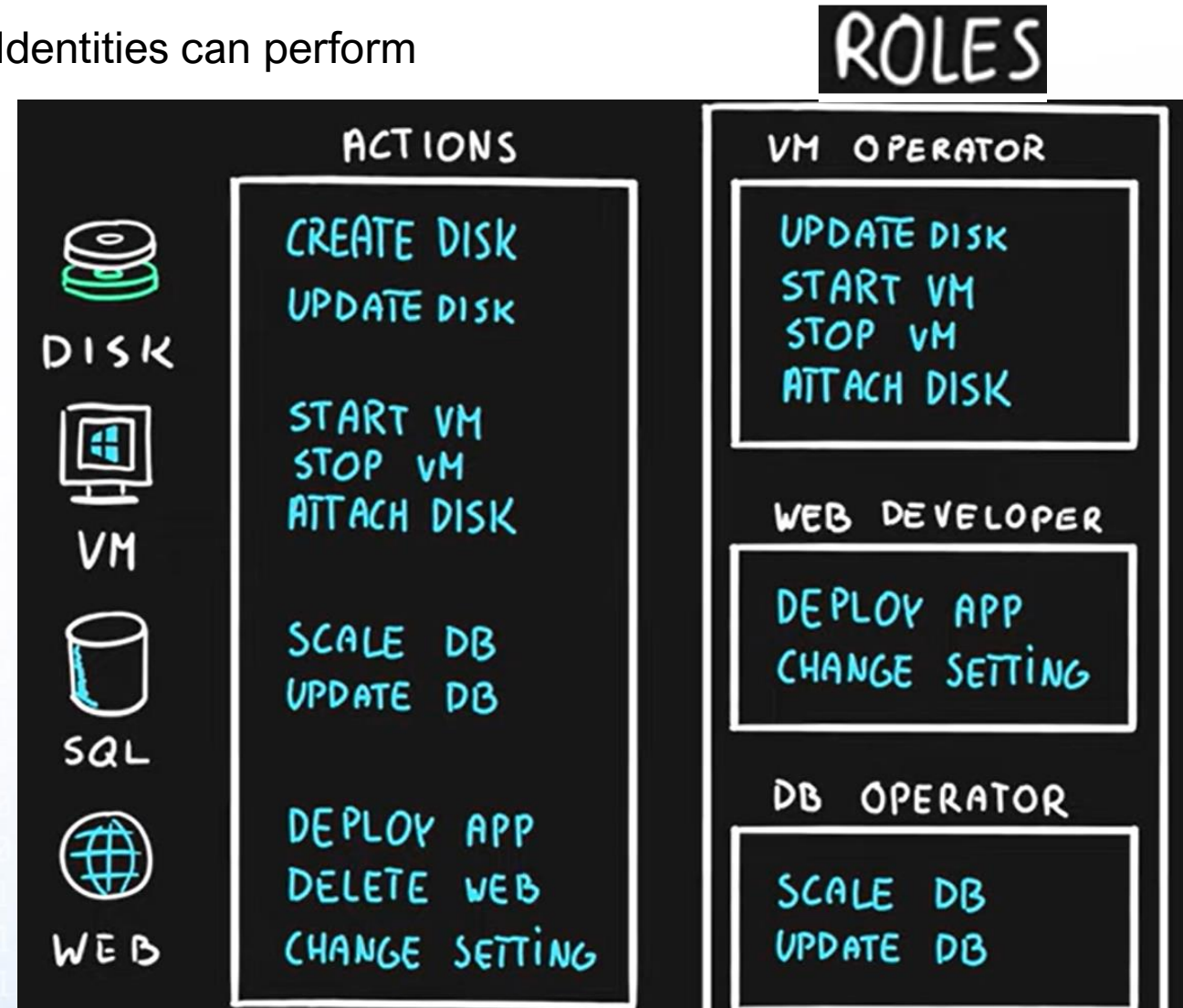
- ❖ Azure role-based access control (Azure RBAC) helps you manage who has access to Azure resources, what they can do with those resources, and what areas they have access to.
- ❖ Azure RBAC is an authorization system built on Azure Resource Manager that provides fine-grained access management of Azure resources.
 - Allow one user to manage virtual machines in a subscription and another user to manage virtual networks
 - Allow a DBA group to manage SQL databases in a subscription
 - Allow a user to manage all resources in a resource group, such as virtual machines, websites, and subnets
 - Allow an application to access all resources in a resource group

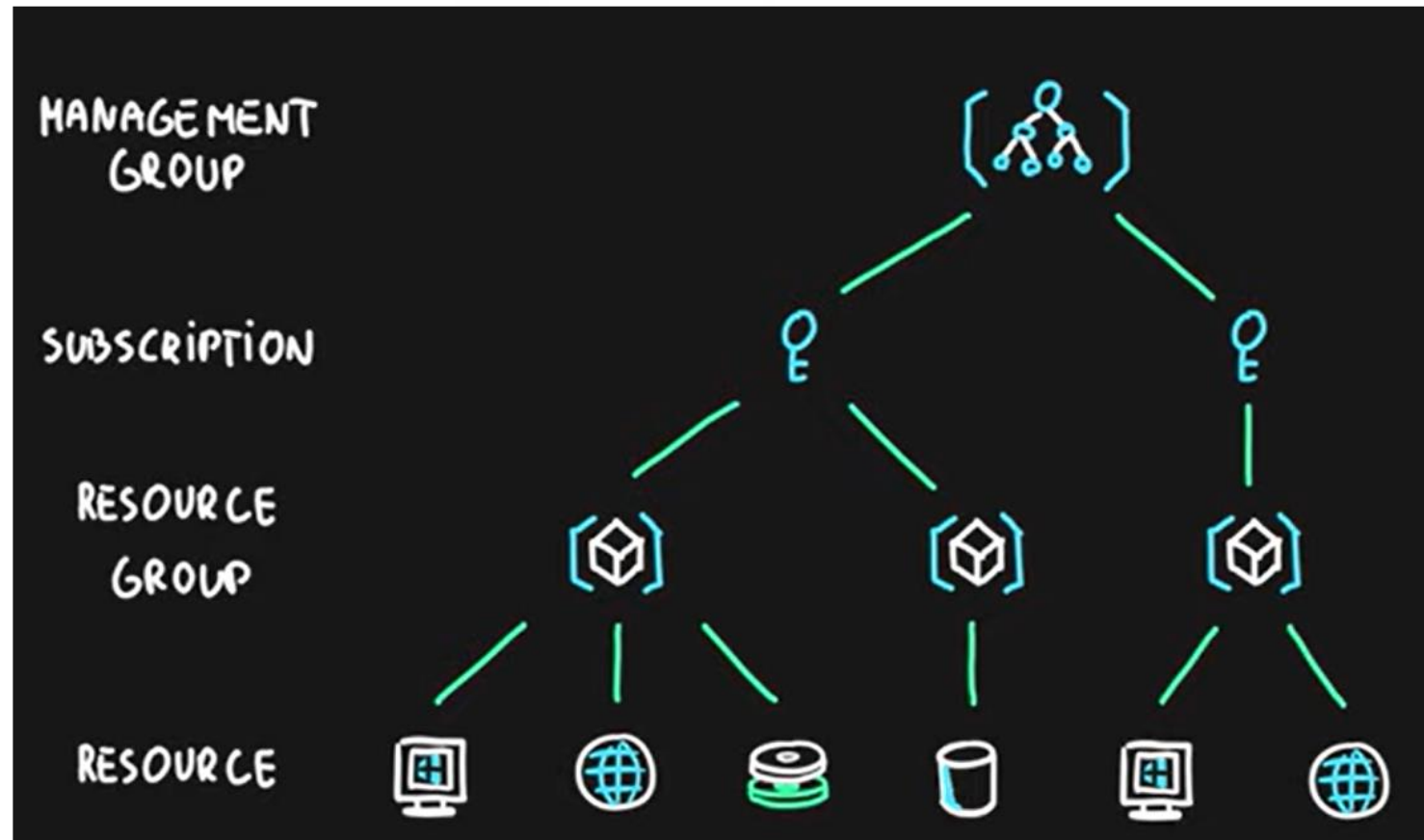


Role(Role Definition) is a collection of actions assigned Identities can perform

Identities : Security principals :

- users,
- groups,
- service principals (Application Accounts in Azure)
- Managed Identity (Application Accounts that are tied to Certain services)





Role assignment is a combination of the role definition, security principal and scope

"What can be done?"

OWNER - EVERYTHING



"Who can do it?"

USER - ADAM



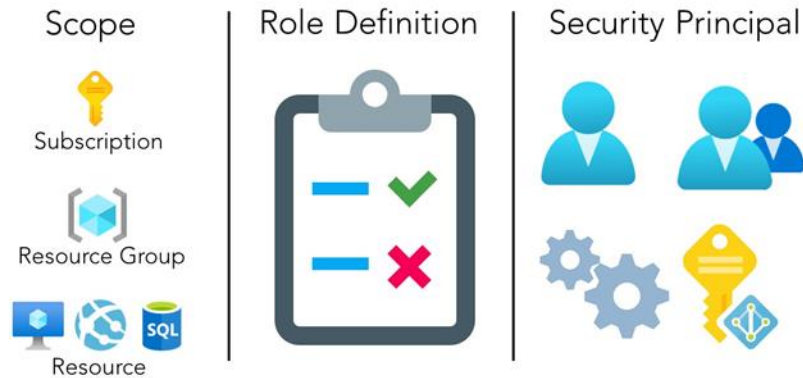
"Where can it be done?"

VM RESOURCE - DEV-VM



How Azure RBAC works (how permissions are enforced)

A role assignment consists of three elements: security principal, role definition, and scope.



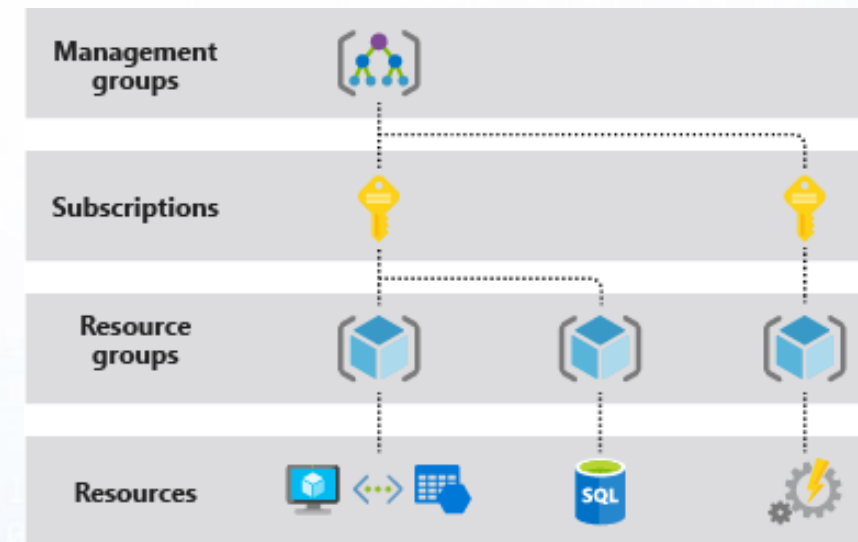
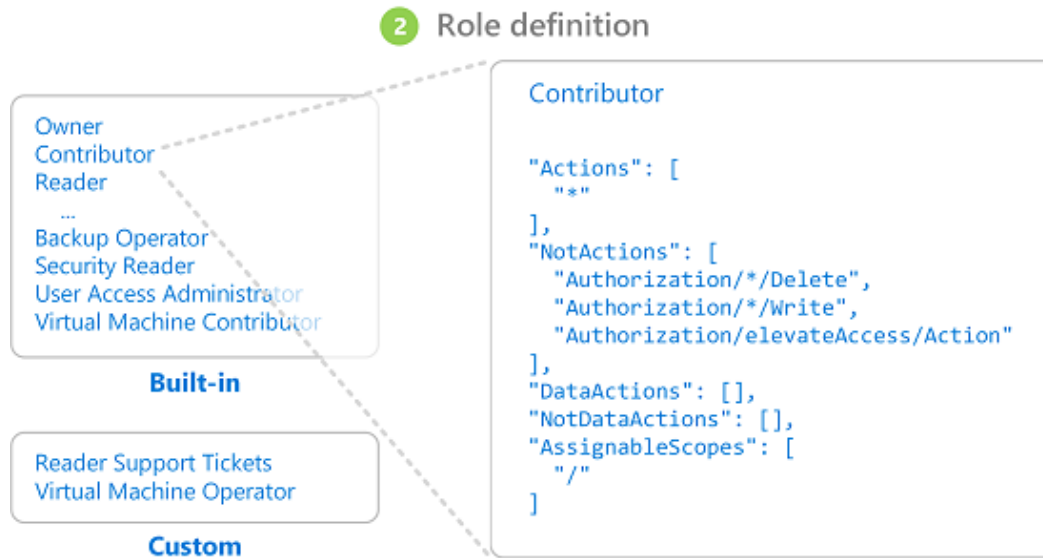
Security principal

A security principal is an object that represents a user, group, service principal, or managed identity



Role definition

A role definition is a collection of permissions. It's typically just called a role. A role definition lists the actions that can be performed, such as read, write, and delete. Roles can be high-level, like owner, or specific, like virtual machine reader.



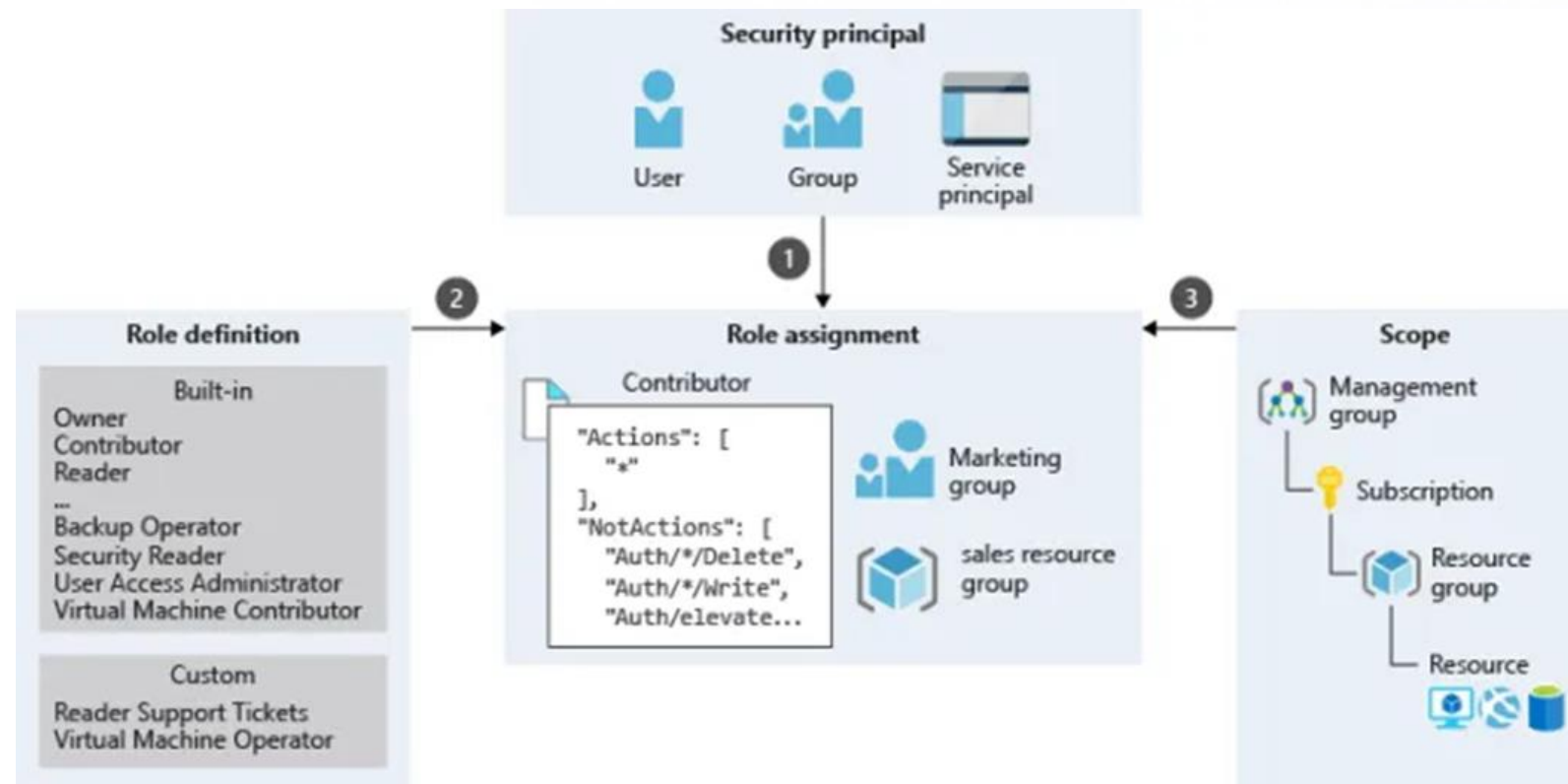
Scope

Scope is the set of resources that the access applies to. When you assign a role, you can further limit the actions allowed by defining a scope. This is helpful if you want to make someone a Website Contributor, but only for one resource group.



Role assignments

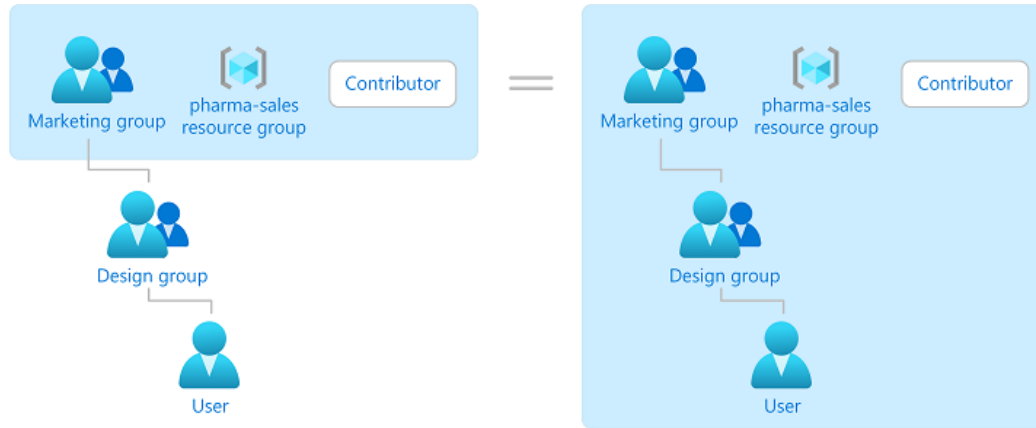
A role assignment is the process of attaching a role definition to a user, group, service principal, or managed identity at a particular scope for the purpose of granting access. Access is granted by creating a role assignment, and access is revoked by removing a role assignment.



You can assign roles using the Azure portal, Azure CLI, Azure PowerShell, Azure SDKs, or REST APIs.

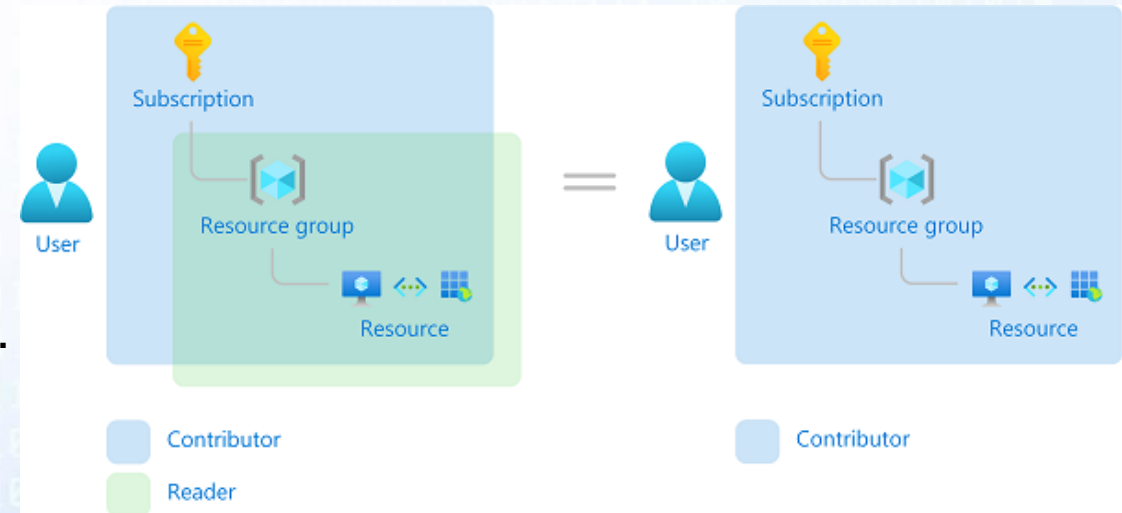
Groups

Role assignments are transitive for groups which means that if a user is a member of a group and that group is a member of another group that has a role assignment, the user will have the permissions in the role assignment.



Multiple role assignments

Azure RBAC is an additive model, so your effective permissions are the sum of your role assignments.



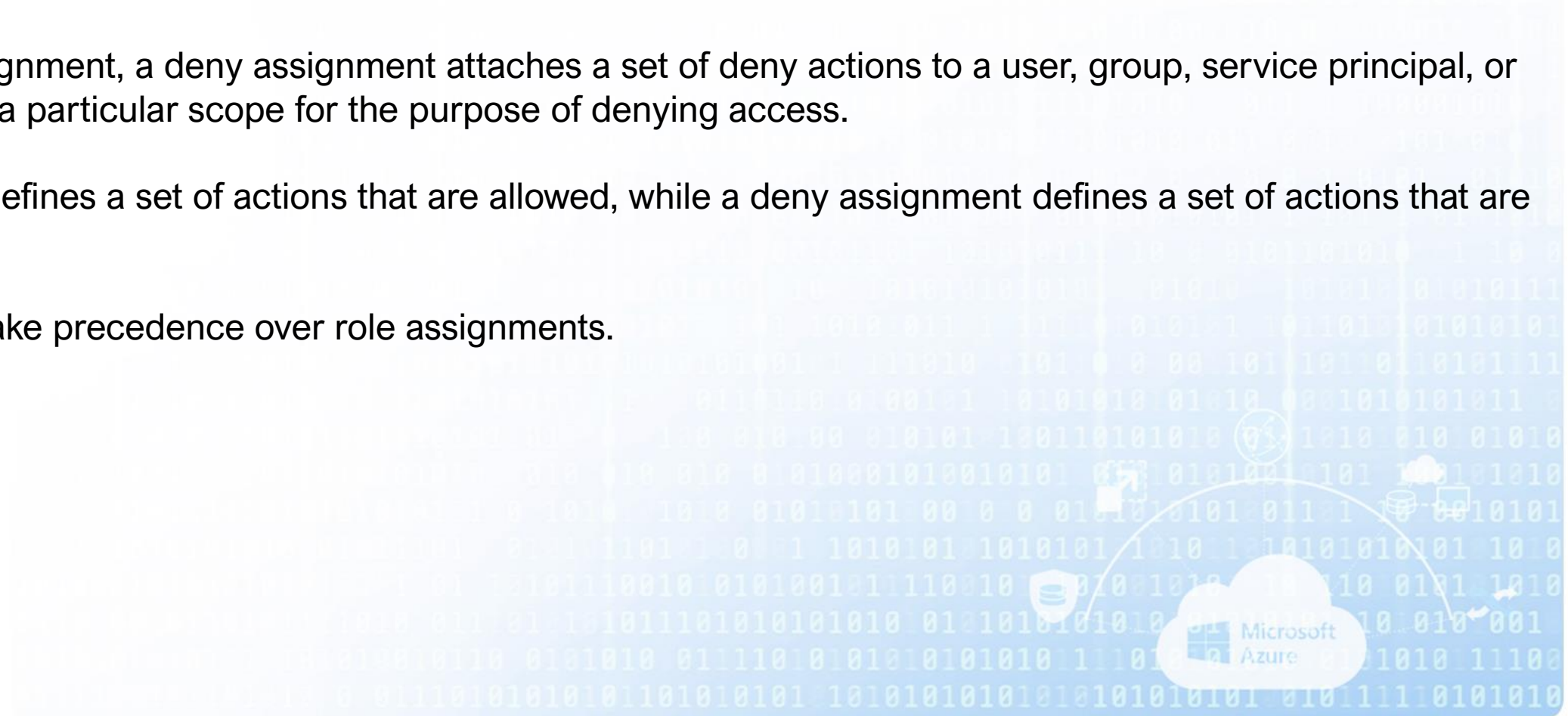
Deny assignments

Previously, Azure RBAC was an allow-only model with no deny, but now Azure RBAC supports deny assignments in a limited way.

Similar to a role assignment, a deny assignment attaches a set of deny actions to a user, group, service principal, or managed identity at a particular scope for the purpose of denying access.

A role assignment defines a set of actions that are allowed, while a deny assignment defines a set of actions that are not allowed.

Deny assignments take precedence over role assignments.



Azure Policy evaluates resources in Azure by comparing the properties of those resources to business rules.

These business rules, described in JSON format, are known as policy definitions.

To simplify management, several business rules can be grouped together to form a policy initiative (sometimes called a policy Set).

the policy definition or initiative is assigned to any scope of resources that Azure supports, such as management groups, subscriptions, resource groups, or individual resources.

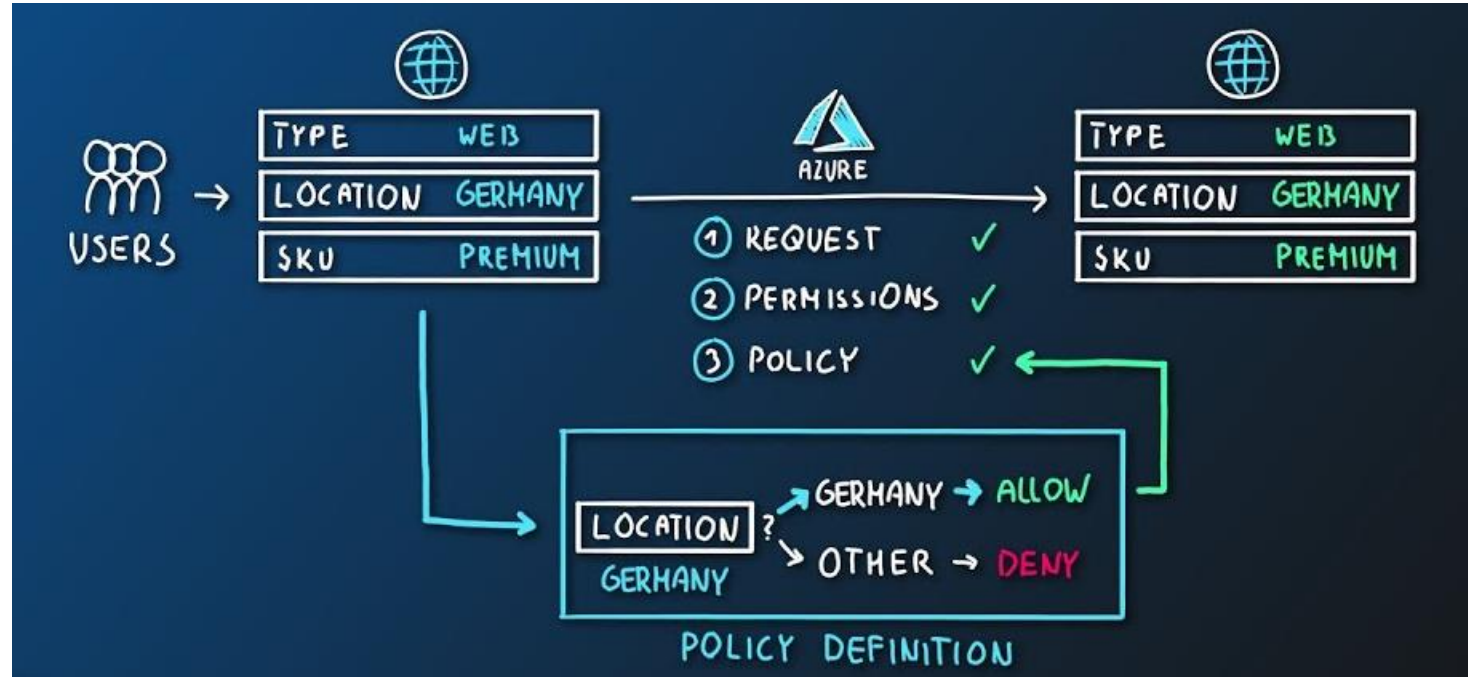
A resource is created or updated in a scope with a policy assignment.

A policy or initiative is newly assigned to a scope.

A policy or initiative already assigned to a scope is updated.

During the standard compliance evaluation cycle, which occurs once every 24 hours





Resources are evaluated at specific times during the resource lifecycle, the policy assignment lifecycle, and for regular ongoing compliance evaluation

- A resource is created or updated in a scope with a policy assignment.
- A policy or initiative is newly assigned to a scope.
- A policy or initiative already assigned to a scope is updated.
- During the standard compliance evaluation cycle, which occurs once every 24 hours.

Control the response to an evaluation

- Deny the resource change
- Log the change to the resource
- Alter the resource before the change
- Alter the resource after the change
- Deploy related compliant resources

Remediate non-compliant resources



Lesson 01: Storage Accounts



Azure Storage Account is a secure account, which provides you access to services in Azure Storage.

The storage account is like an administrative container, and within that, we can have several services like blobs, files, queues, tables, disks, etc.

To use any storage type in azure, you first have to create an account in Azure.

Create a storage account to store up to 500 TB of data in the cloud

Types of performance tiers

Standard performance: This tier is backed by magnetic drives and provides low cost/GB. They are best for applications that are best for bulk storage or infrequently accessed data.

Premium storage performance:

This tier is backed by solid-state drives and offers consistency and low latency performance.

They can only be used with Azure virtual machine disks, and are best for I/O intensive workload such as the database.



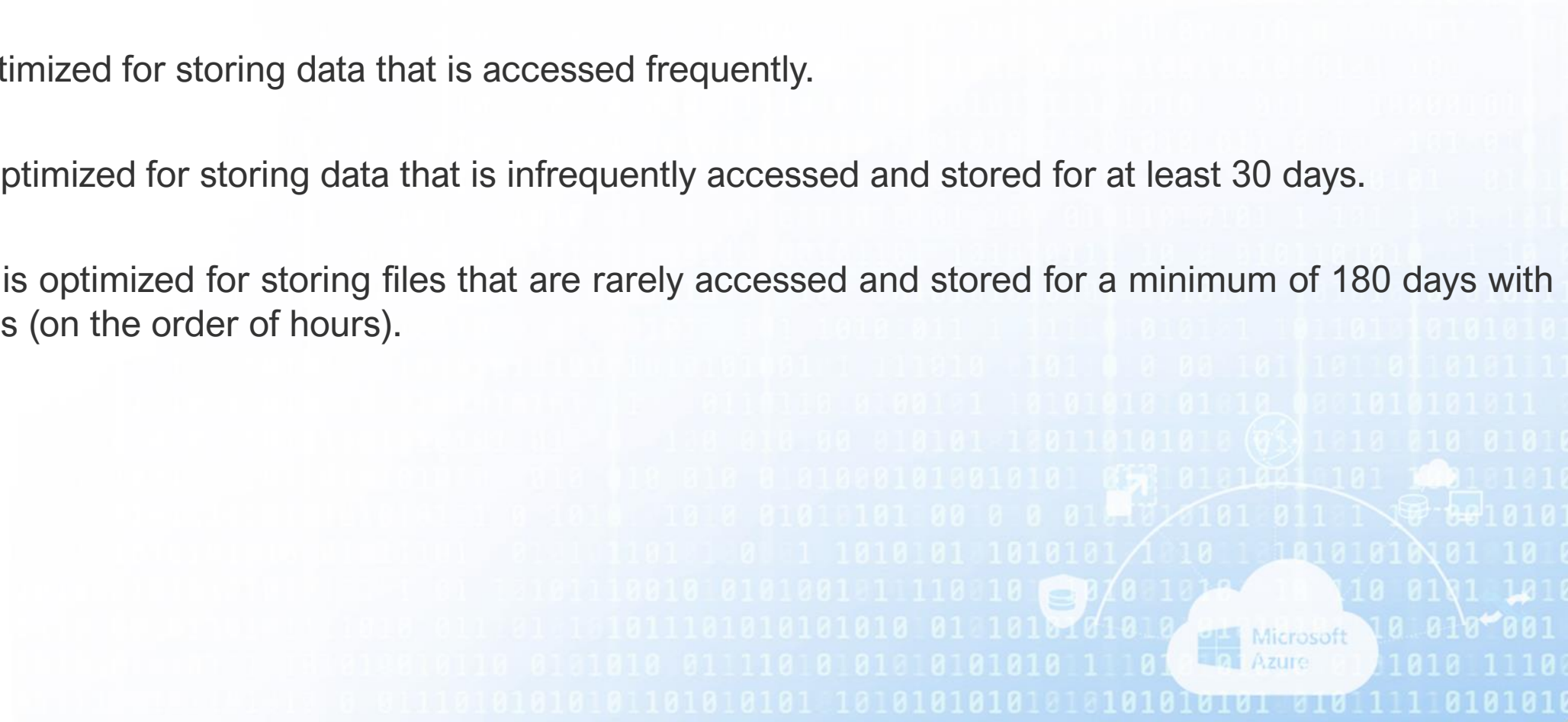
There are four types of access tiers available:

Premium Storage (preview): It provides high-performance hardware for data that is accessed frequently.

Hot storage: It is optimized for storing data that is accessed frequently.

Cool Storage: It is optimized for storing data that is infrequently accessed and stored for at least 30 days.

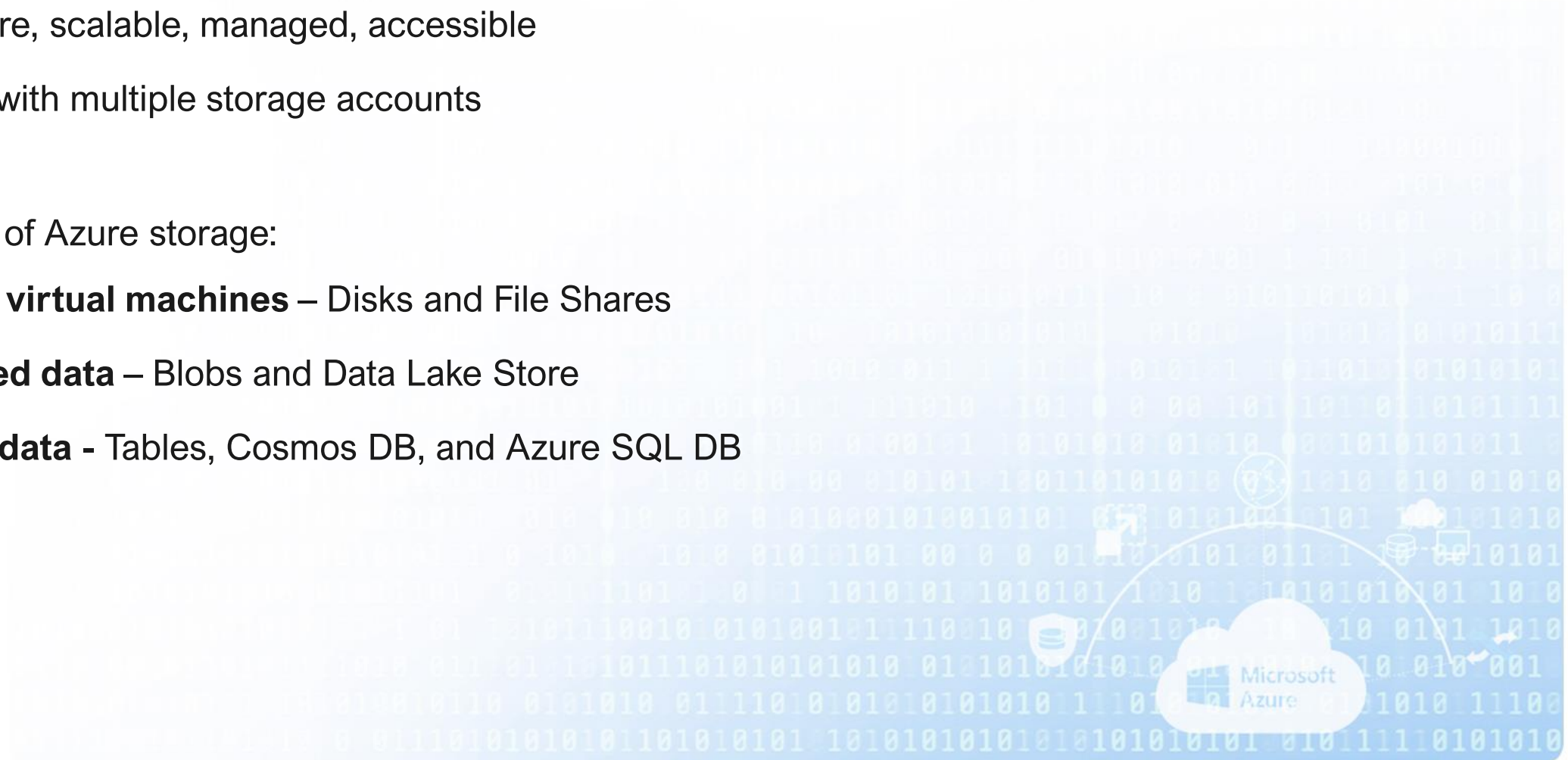
Archive Storage: It is optimized for storing files that are rarely accessed and stored for a minimum of 180 days with flexible latency needs (on the order of hours).



- A service that you can use to store files, messages, tables, and other types of information
- Durable, secure, scalable, managed, accessible
- Manage data with multiple storage accounts

Three categories of Azure storage:

- **Storage for virtual machines** – Disks and File Shares
- **Unstructured data** – Blobs and Data Lake Store
- **Structured data** - Tables, Cosmos DB, and Azure SQL DB



There are two types of access tier:

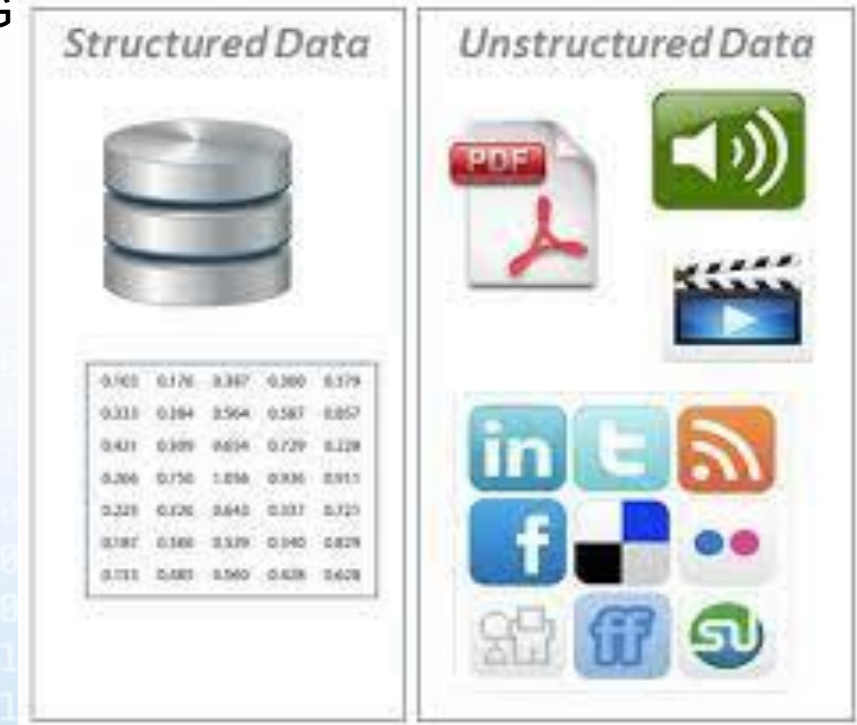
- **Hot:** This access tier grants us the lowest latency possible. Hence, it should be used with data which is frequently accessed. Naturally, since it offers low latency it is more expensive

Ex: Building Web Application which servers images to customers

- **Cold:** This access tier is less in performance than the “Hot” access tier i.e offers higher latency than the former access tier. That being said, it comes with a lesser price tag and hence can be used for data which is less frequently accessed.
- Ex: storing older version of data and backups
- **Archive :** Rarely accessed data, data life cycle, lowest pricing service



- **Structured Data:** Relational Data, there will be proper schema defined for it
- **Semi Structured Data :** still have table structure but no proper schema defined for it. here each row have its own property
- **Unstructured Data:** Images. Movies, txt files, binary files, exe files, PNG



Azure Storage includes these data services, each of which is accessed through a storage account

There are 4 types of storage types in azure:

- Tables
- Blobs
- Queues
- File Storage



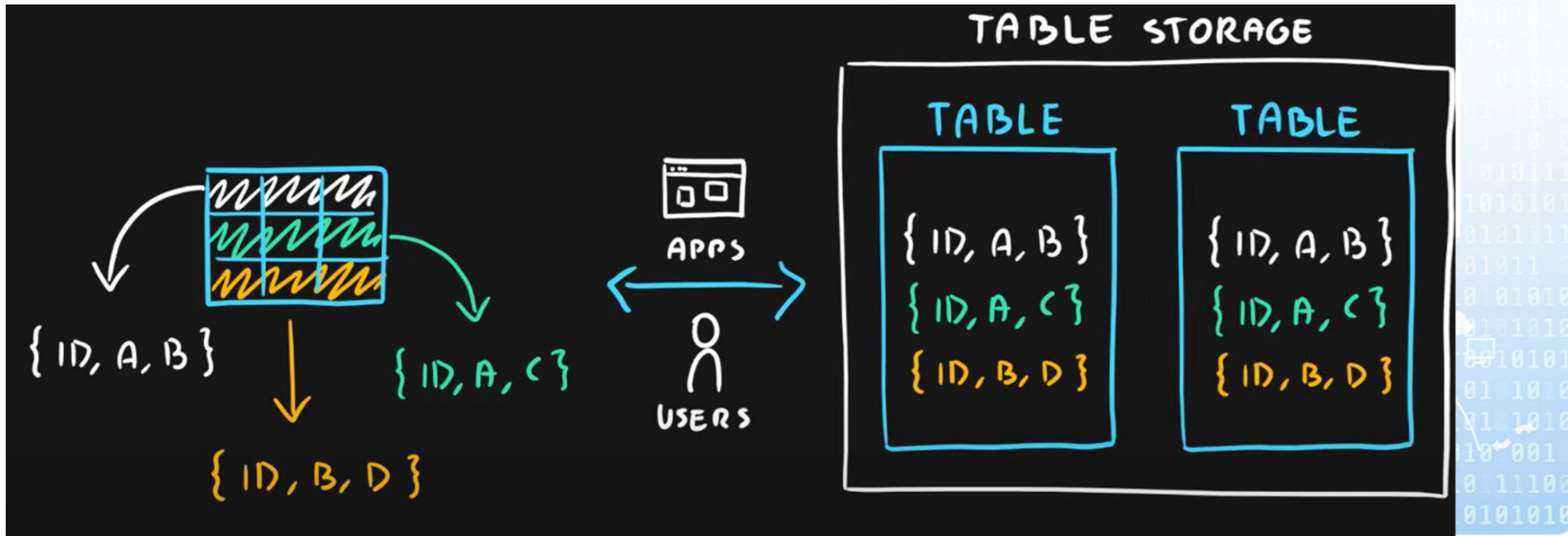
Azure Tables: A NoSQL store for schema less storage of structured data

The Azure Table storage service stores large amounts of structured data.

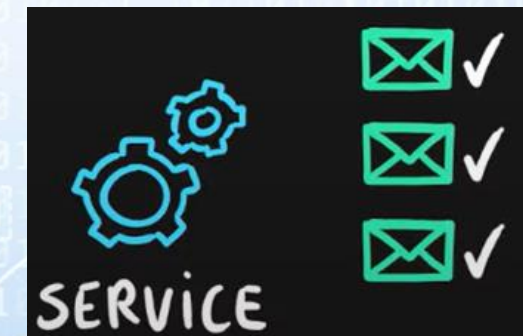
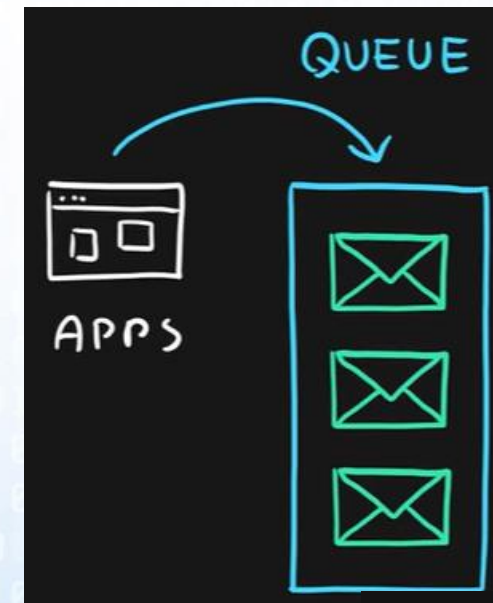
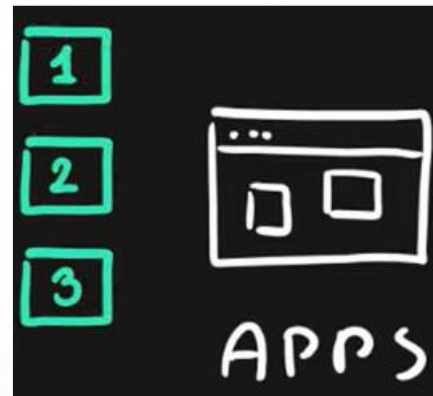
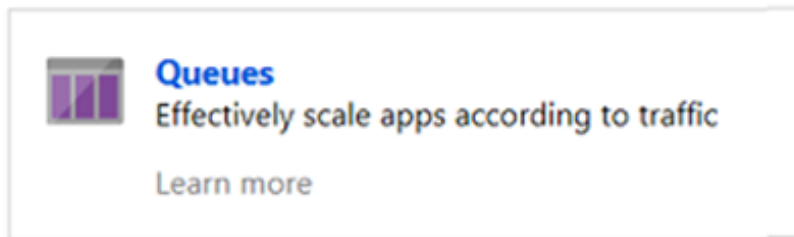
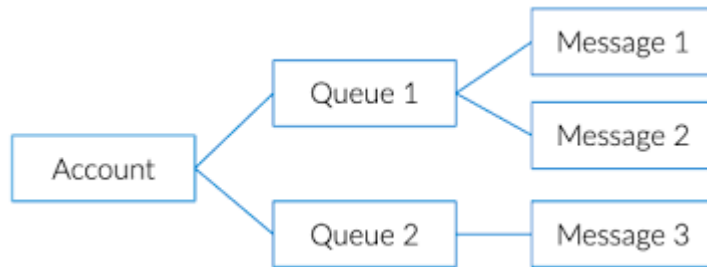
The service is a NoSQL datastore which accepts authenticated calls from inside and outside the Azure cloud.

Azure tables are ideal for storing structured, non-relational data.

Designed for faster access (access petabyte of data in milisec)

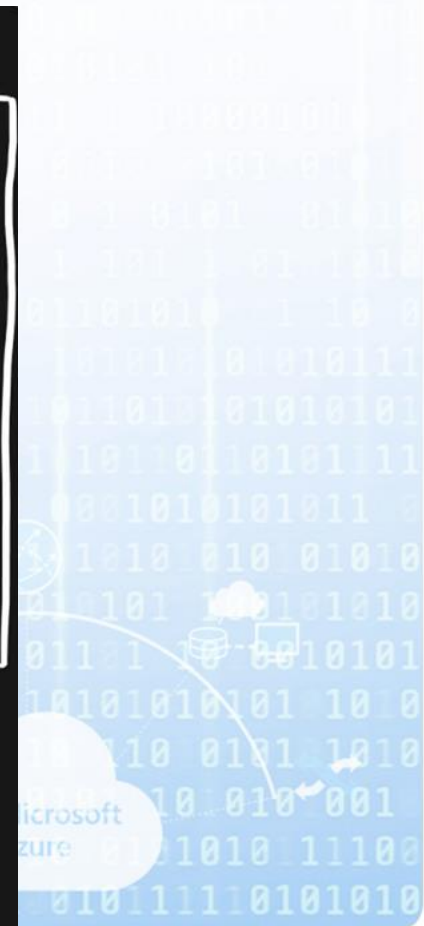
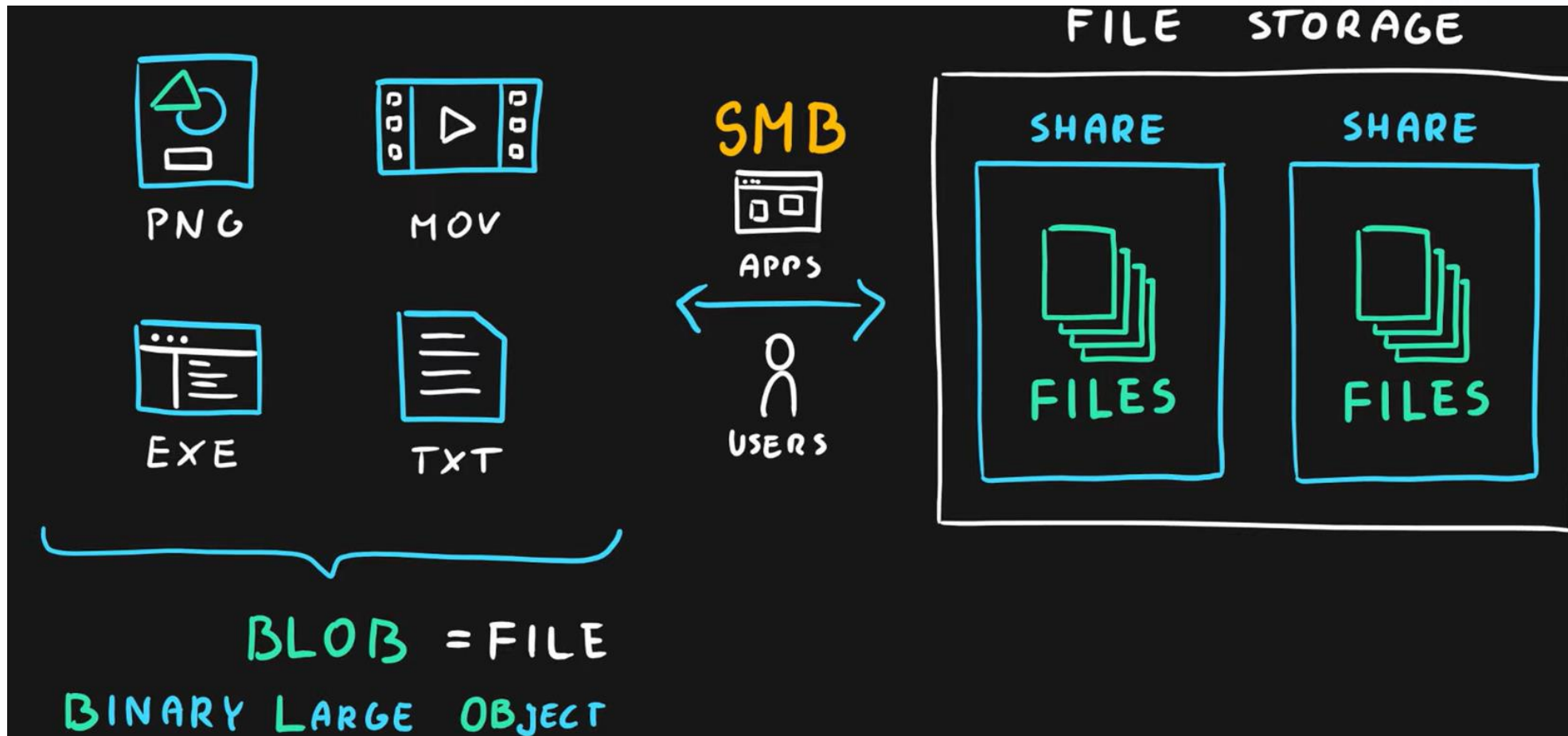


- **Azure Queues:** A messaging store for reliable messaging between Application components
- Azure Queue storage is a service for storing large numbers of messages that can be accessed from anywhere in the world via authenticated calls using HTTP or HTTPS. A single queue message can be up to 64 KB in size, and a queue can contain millions of messages, up to the total capacity limit of a storage account



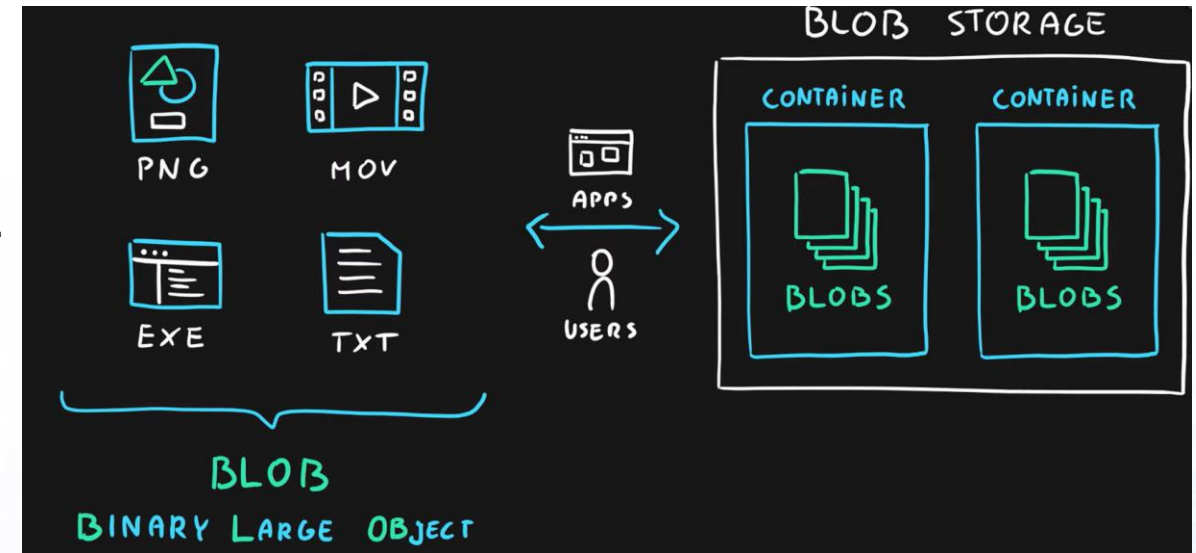
- **Azure Files:** Managed file shares for cloud or on-premises deployments

A File Storage share is an SMB file share in Azure. All directories and files must be created in a parent share. An account can contain an unlimited number of shares, and a share can store an unlimited number of files, up to the 5 TB total capacity of the file share.



Azure Blobs: storage is designed for:

- ❖ Serving images or documents directly to a browser.
- ❖ Storing files for distributed access.
- ❖ Streaming video and audio.
- ❖ Writing to log files.
- ❖ Storing data for backup and restore, disaster recovery, and archiving.
- ❖ Storing data for analysis by an on-premises or Azure-hosted service

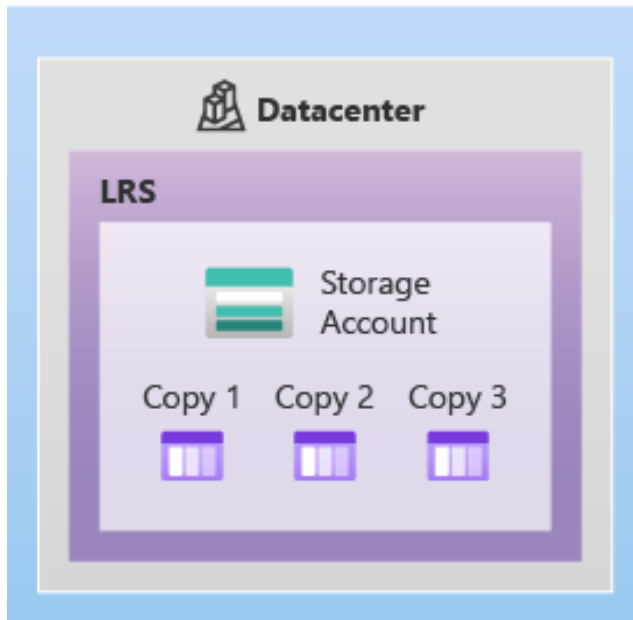


Users or client applications can access objects in Blob storage via HTTP/HTTPS, from anywhere in the world. Objects in Blob storage are accessible via the Azure Storage REST API, Azure PowerShell, Azure CLI,

Locally redundant storage (LRS)

Data is replicated three times within a single facility in a single region.

Primary region



Primary region

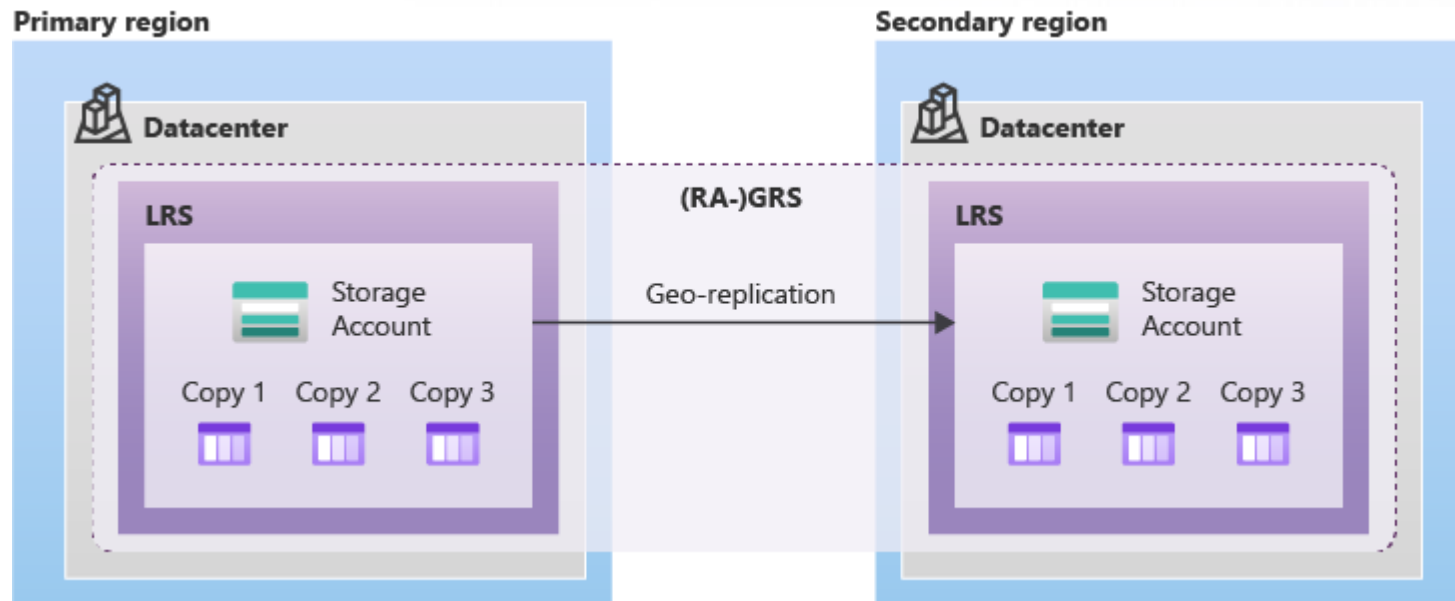


Zone-redundant storage (ZRS)

Data is replicated three times across two to three facilities, either within a single region or across two regions.

Geo-redundant storage (GRS)

Data is replicated three times within the primary region and replicated three times to the regions pair.



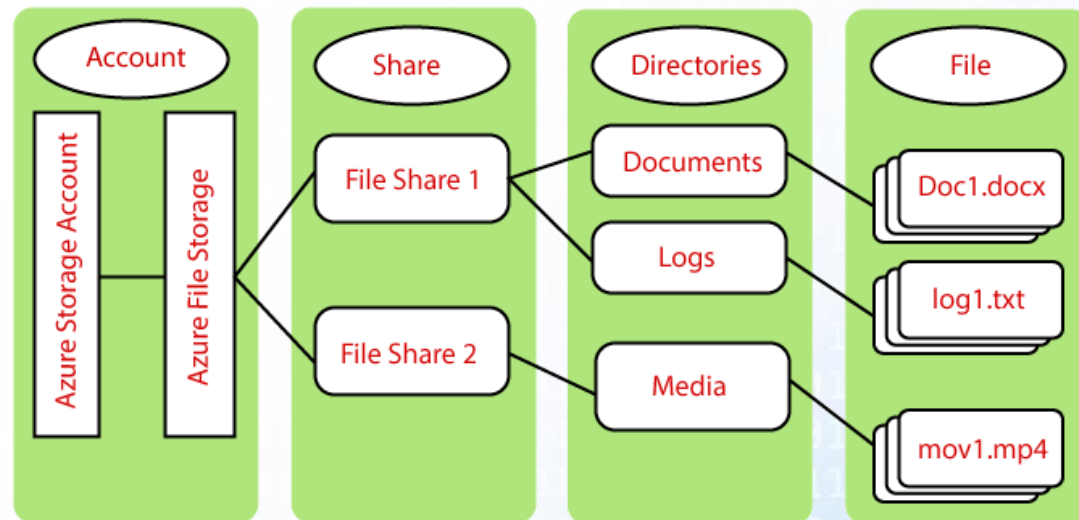
Read access geo-redundant storage (RA-GRS)

Data is replicated three times within the primary region and replicated with read-access to the region pair.

Azure File Storage Service

- ❖ Azure file storage mainly can be used if we want to have a shared drive between two servers or across users In the Azure file storage structure, the first thing we need to have is an Azure storage account
- ❖ We can create an unlimited number of file shares within a storage account.
- ❖ Once we create a file share, then we can create directories, just like folders, and then we can upload files into it.
- ❖ Once we create a file share, we can mount that on any virtual machine, whether it is in Azure or outside.

Azure file storage



Share: A file storage Share is an SMB file share in Azure. The directories and files must be created in a parent's Share, so we can't create a directory directly in a storage account. We need file storage share, first created, and then we can create directories to upload. An account may contain an unlimited number of shares, and a Share can store an unlimited number of files, up to 5 TB total capacity of the file share. But, in case we need more than the full capacity, then we can create another file share.

Directory: It is an optional hierarchy of directories.

File: A file in the share. A file may be up to 1 TB in size.

URL format: For a request to an Azure file share made with the file REST protocol

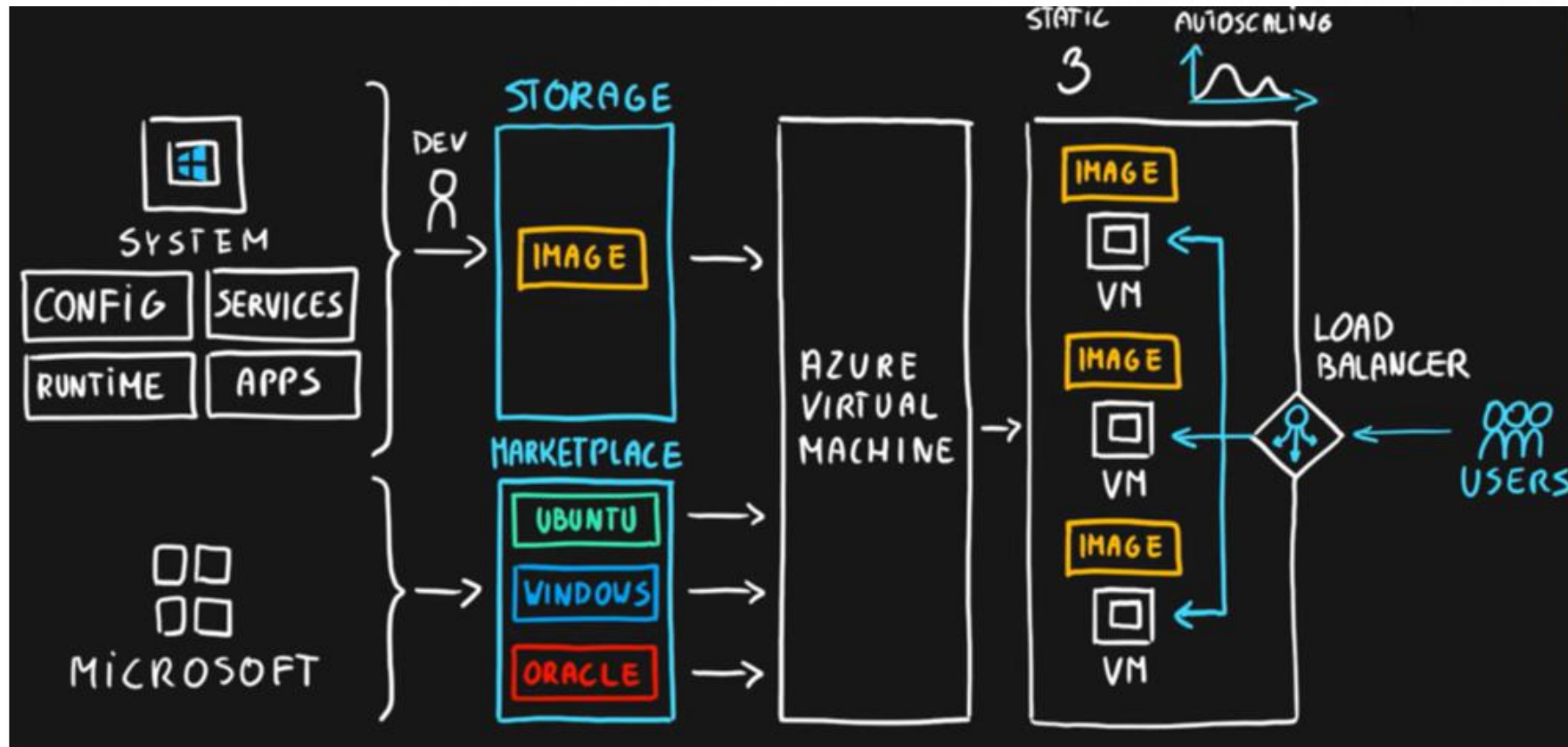
<https://www.javatpoint.com/azure-file-storage-service>

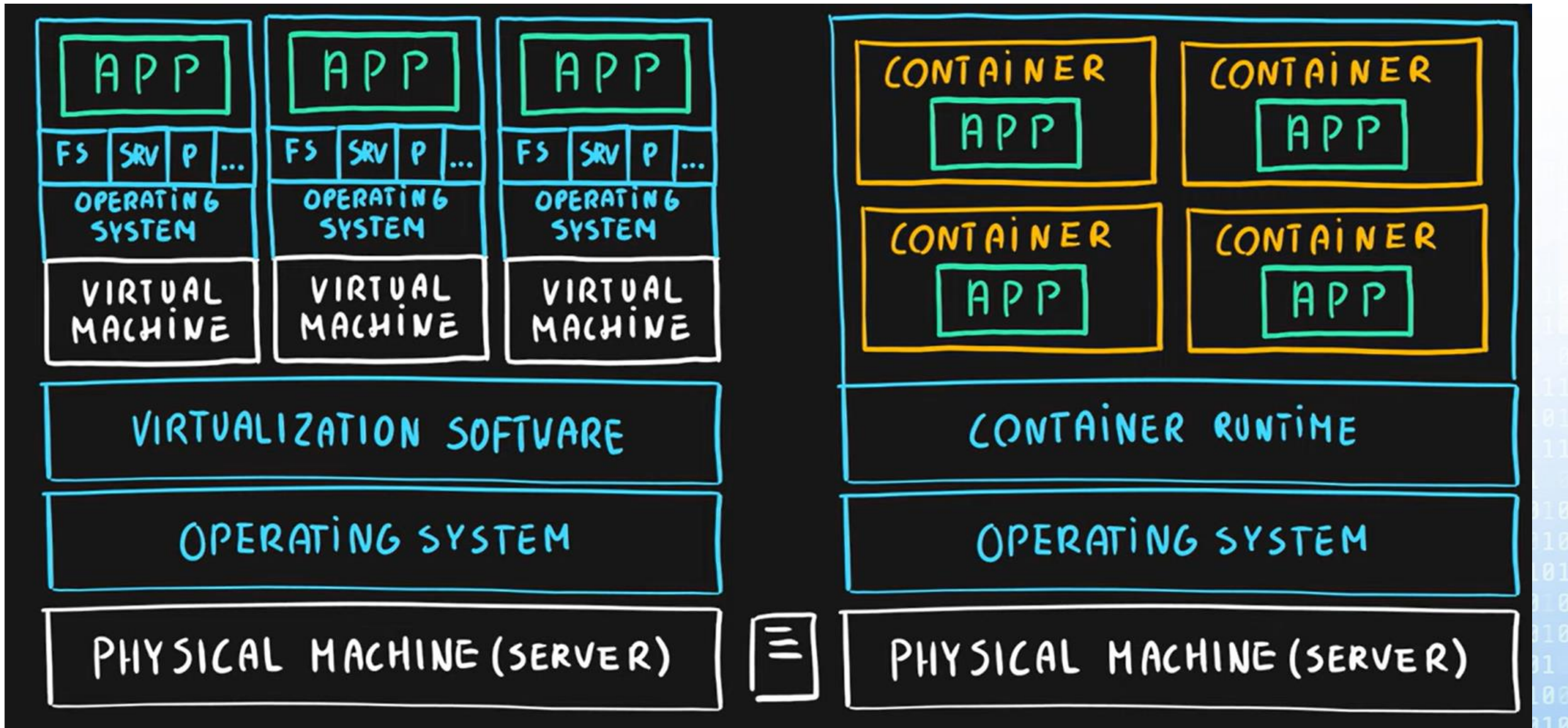


Module10: Azure Compute Services

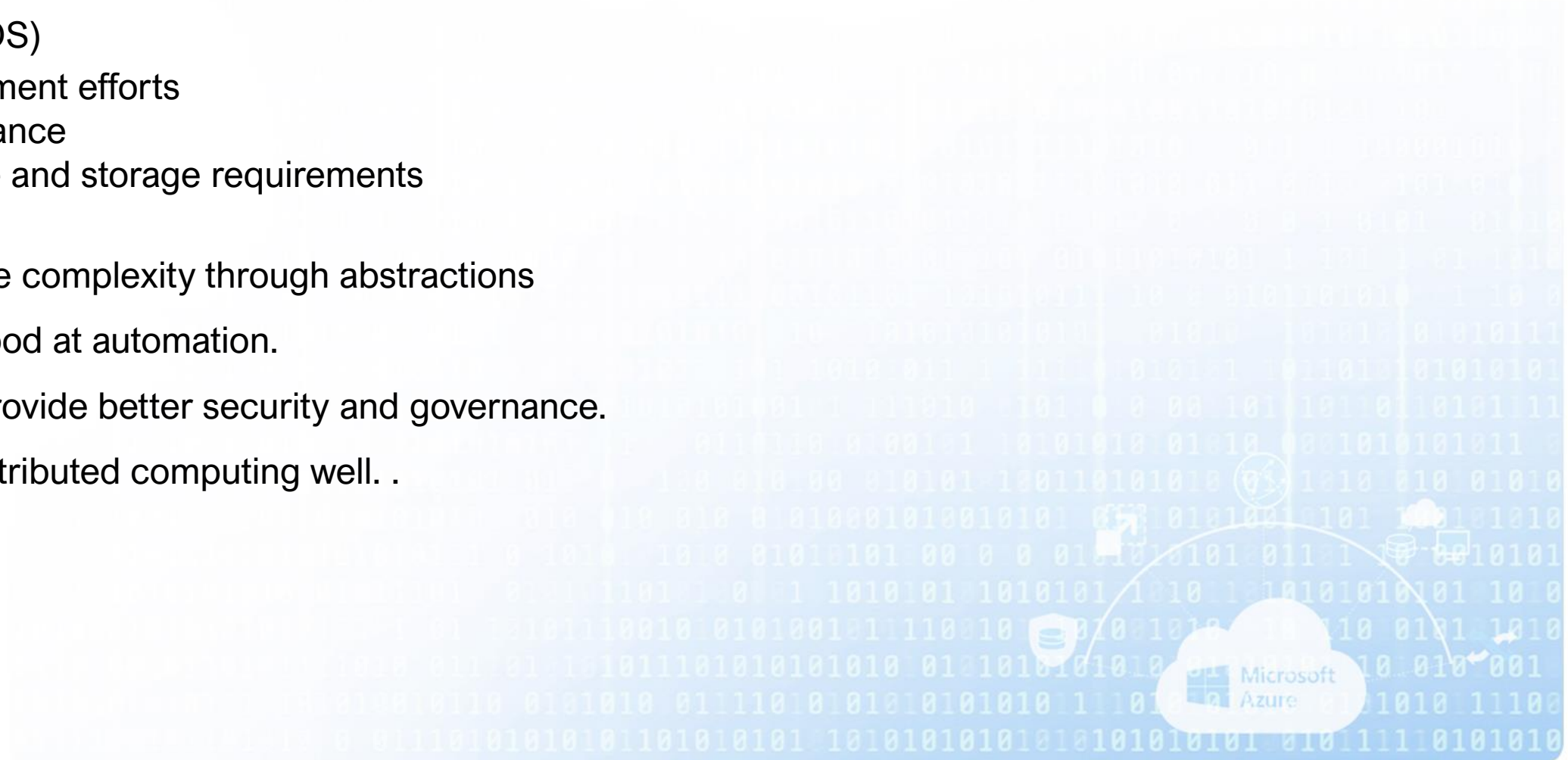


- Offered as Infrastructure as a service(IaaS)
- Set of identical Virtual Machines
- Built in Autoscaling feature
- Designed for manual and auto scaled workload like web servers, SQL Servers etc
- Auto scale up to 600 to 100 VM's





- ❖ Containers does not contain any operating system, use host operating system
- ❖ Containers emulated operating system (Virtual Machines emulate hardware)
- ❖ Lightweight (No OS)
 - development efforts
 - maintenance
 - compute and storage requirements
 -
- ❖ Containers reduce complexity through abstractions
- ❖ Containers are good at automation.
- ❖ Containers can provide better security and governance.
- ❖ Containers do distributed computing well. .



Containers are becoming the preferred way to package, deploy, and manage cloud applications.

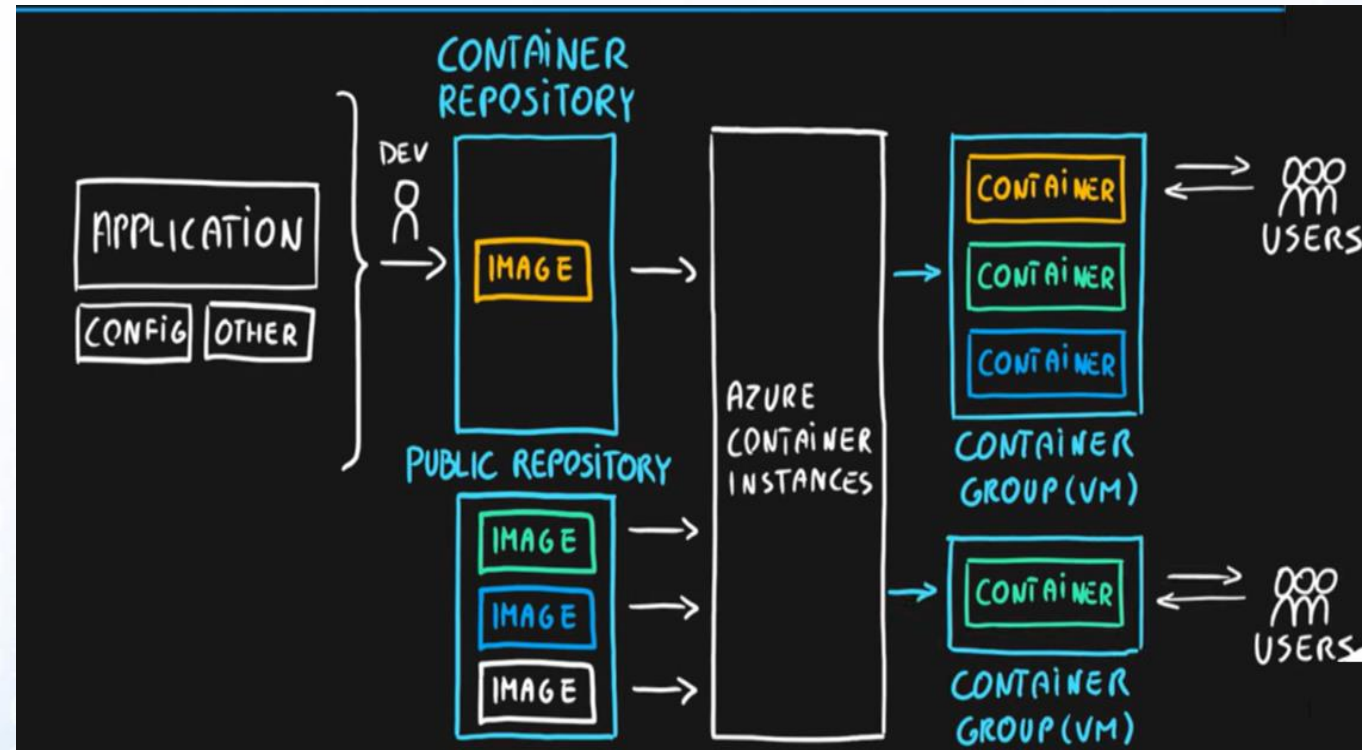
Azure Container Instances offers the fastest and simplest way to run a container in Azure, without having to manage any virtual machines and without having to adopt a higher-level service.

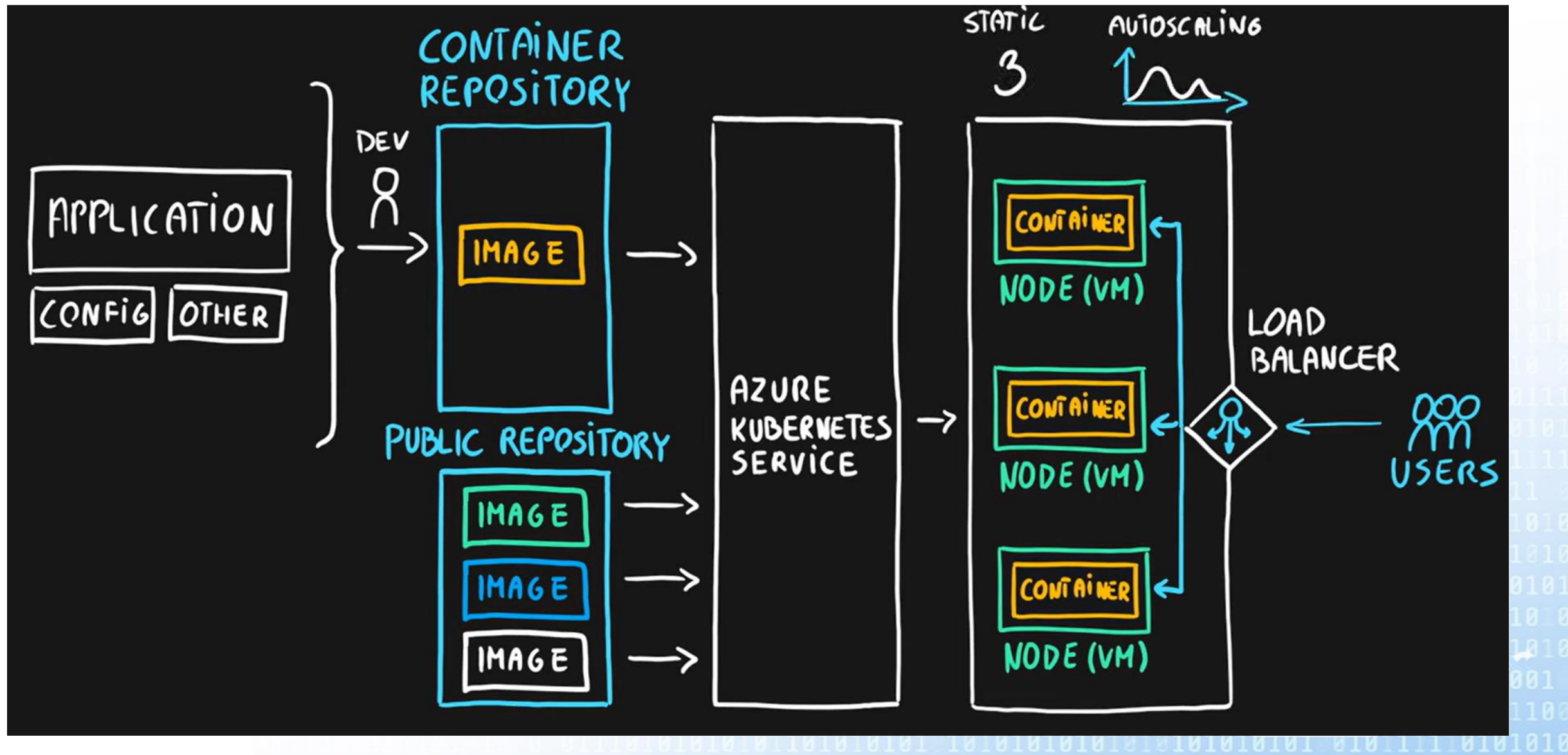
- ❖ Containers offer significant startup benefits over virtual machines (VMs).
- ❖ Azure Container Instances can start containers in seconds,
- ❖ Available PaaS
- ❖ It is called as a Serverless containers

Designed for

- small and simple web applications/Services
- Background Jobs
- schedule scripts

It will not support Auto scaling capabilities





Open-source container orchestration platform (available in all cloud provider services like AWS, GCP etc)

Available as PaaS

Highly scalable and customizable

Designed for highly scale container deployment

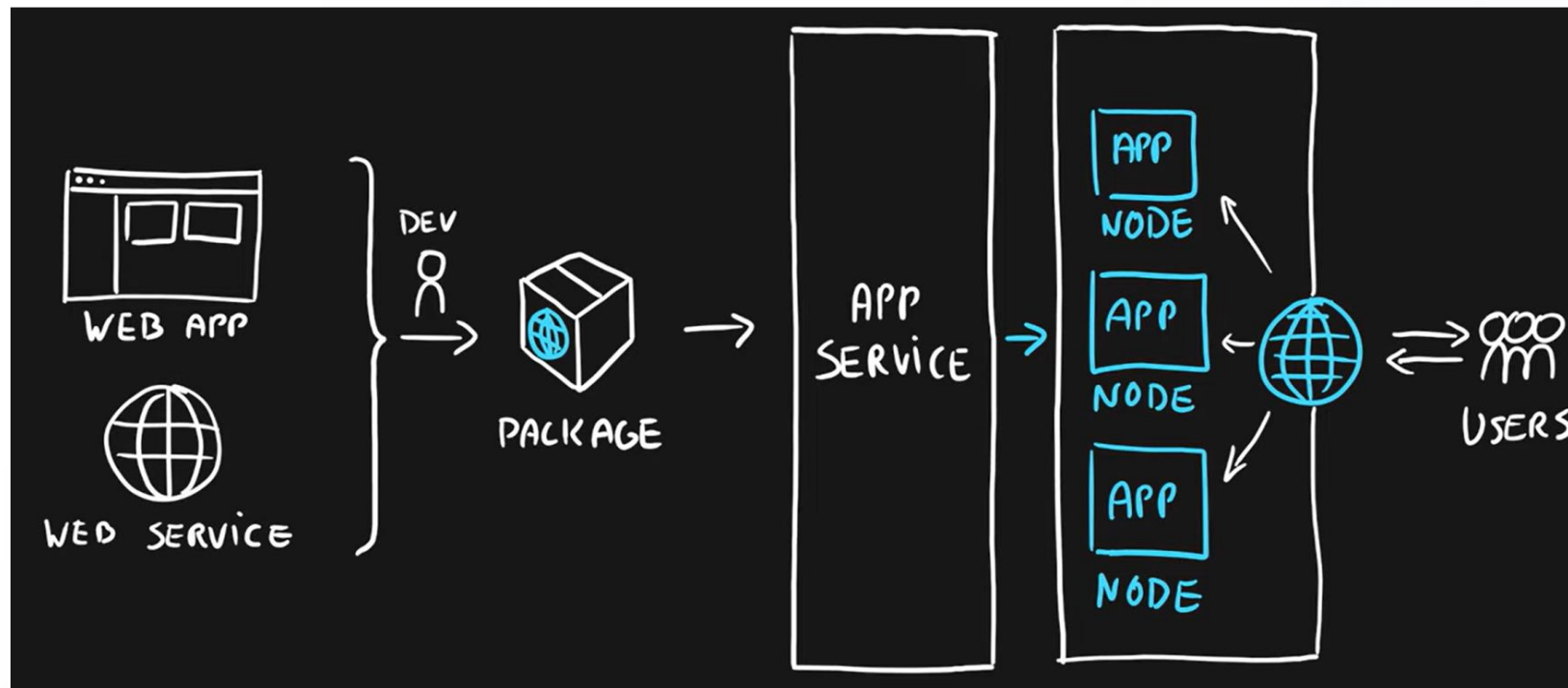
Service	Configuration Control / Maintenance	Autoscaling	Min Nodes	Max Nodes	Scalability
Virtual Machines	☆☆☆☆☆	No	1	1	☆
VM Scale Sets	☆☆☆☆☆	Yes	1	1000/600	☆☆☆☆☆
Container Instances	☆☆☆	No	0	20	☆☆
Kubernetes Service	☆☆☆☆	Yes	3	100	☆☆☆☆



Designed as enterprise Web application service

PaaS offering

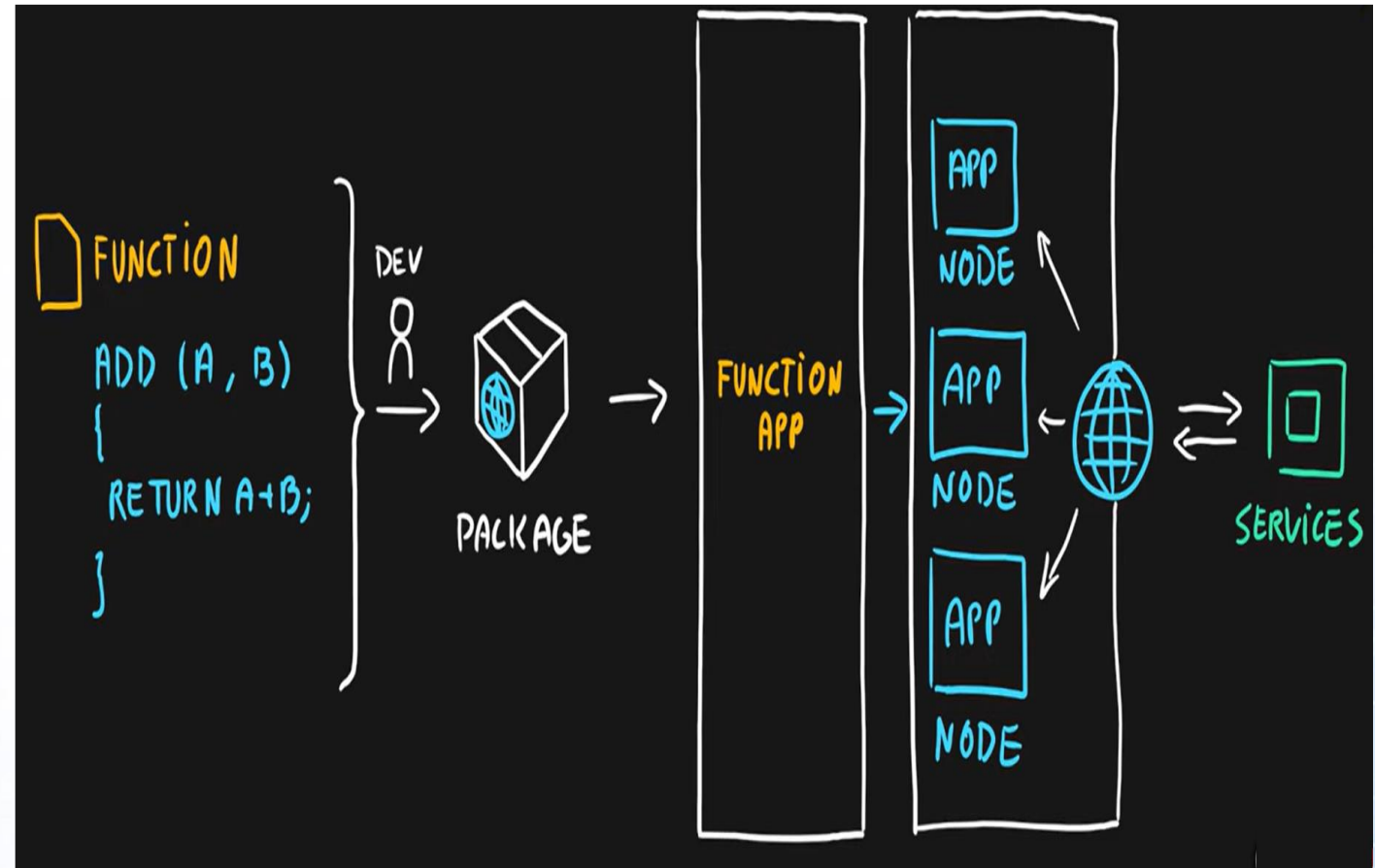
Supports multiple programming languages and containers



Serverless computing
PaaS offering

Two hosting/pricing Model
consumption based-plan
Dedicated Plan

This service is build for Micro or
Nano service applications
Least maintenance from development team

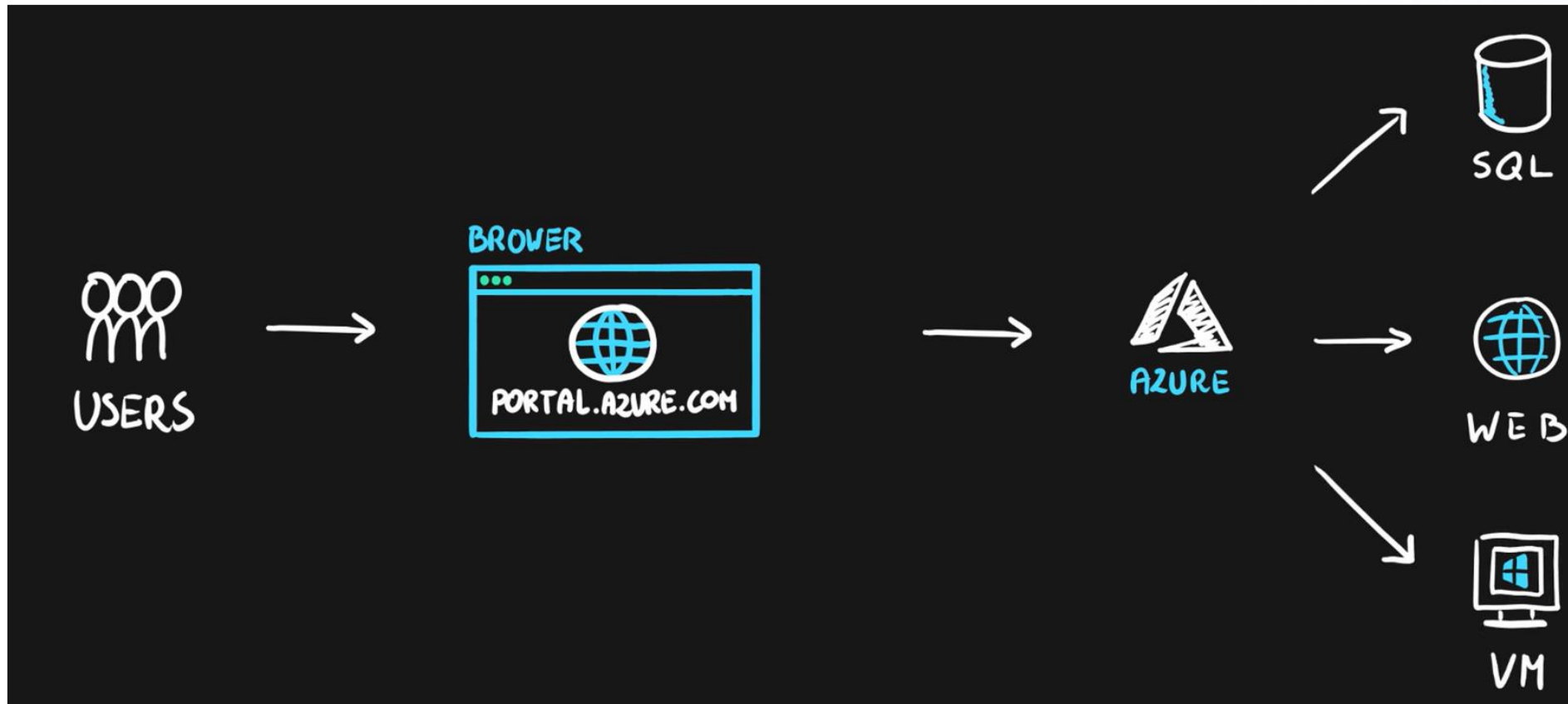


Module11: Azure Management Tools



Azure Portal : public web based interface for management of Azure platform

- Designed for Self service and easy to use
- Custom portal : create your custom Dashboard
- Possible to do Simple tasks, no complex deployments and automation tasks



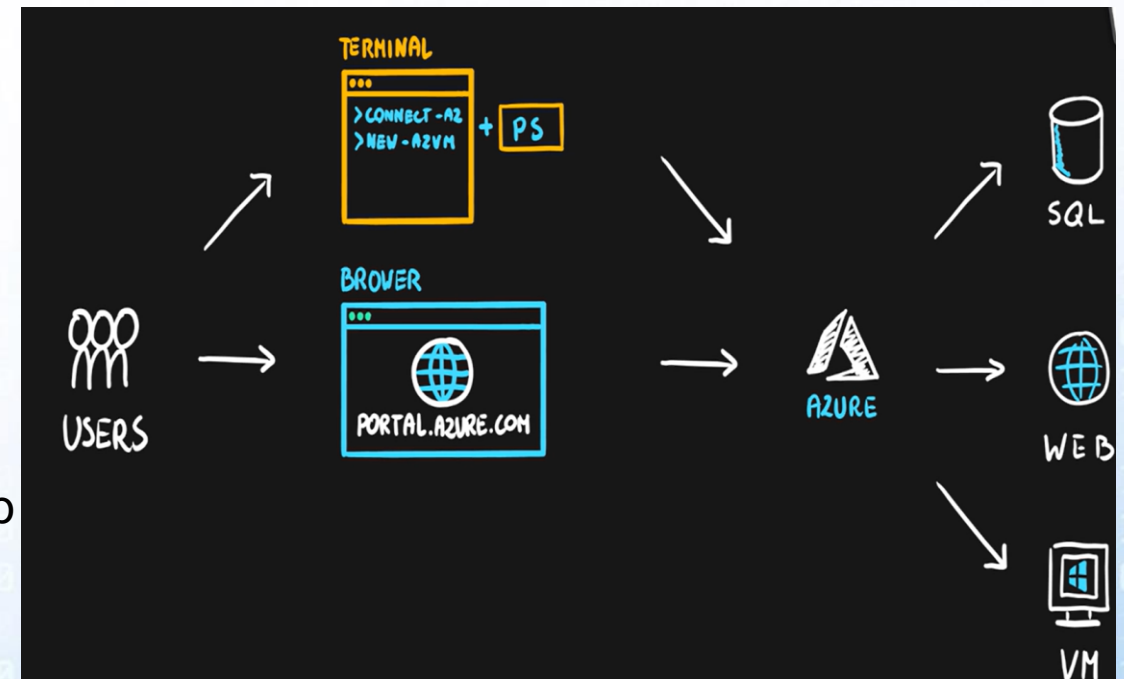
Azure PowerShell is a set of cmdlets for managing Azure resources directly from the PowerShell command line. Azure PowerShell is designed to make it easy to learn and get started with, but provides powerful features for automation.

- Written in .NET Standard, Azure PowerShell works with PowerShell 5.1 on Windows
- Multi platform with PowerShell core
- It support windows, Linux and Mac OS

install PowerShell module to work with Azure resources

Simple to use

Ex: connect-AzAccount --- to login into Azure
 Get-AzResourceGroup --- list resource groups
 New-AzResourceGroup -- To Create a new resource group
 New-AzVM --- To Create VM



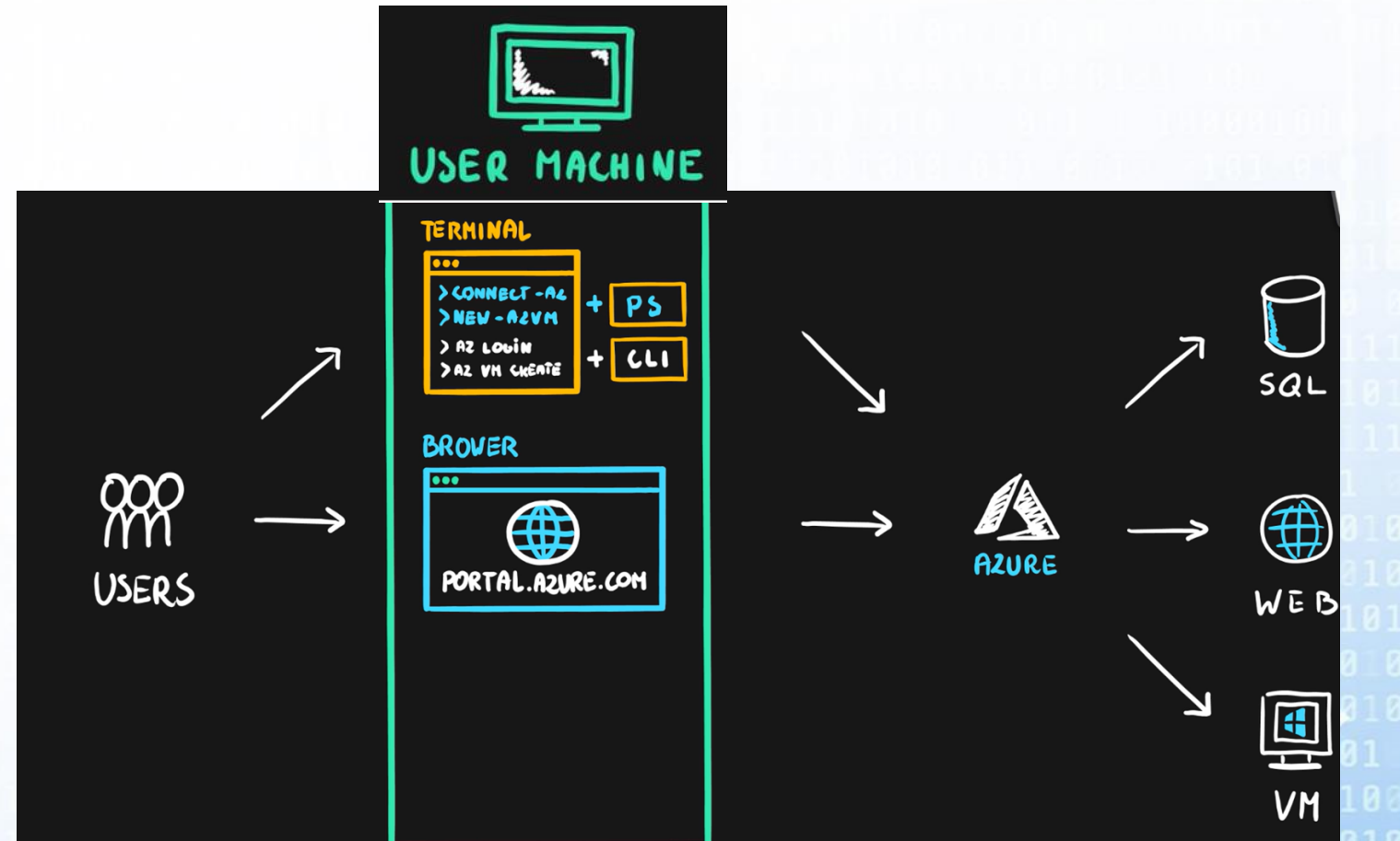
AzureCLI

The Azure command-line interface (Azure CLI) is a set of commands used to create and manage Azure resources. The Azure CLI is available across Azure services and is designed to get you working quickly with Azure, with an emphasis on automation.

Multiplatform and built on Python

```
az login  login in to Azure
az group list
az group create
az vm create
```

Native OS terminal scripting

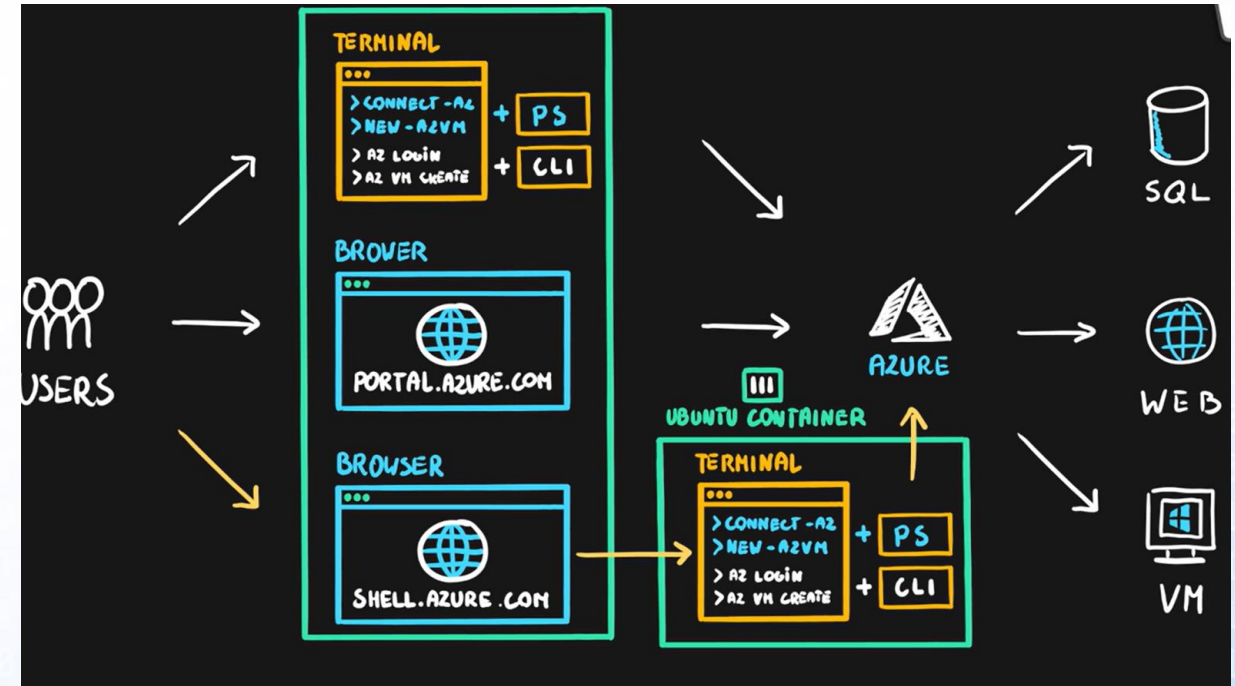


Azure Cloudshell

- Natively built-in the portal, no need to install any module
- Even you can access through mobile App
- Supports both PowerShell and azurecli

az group list -o table

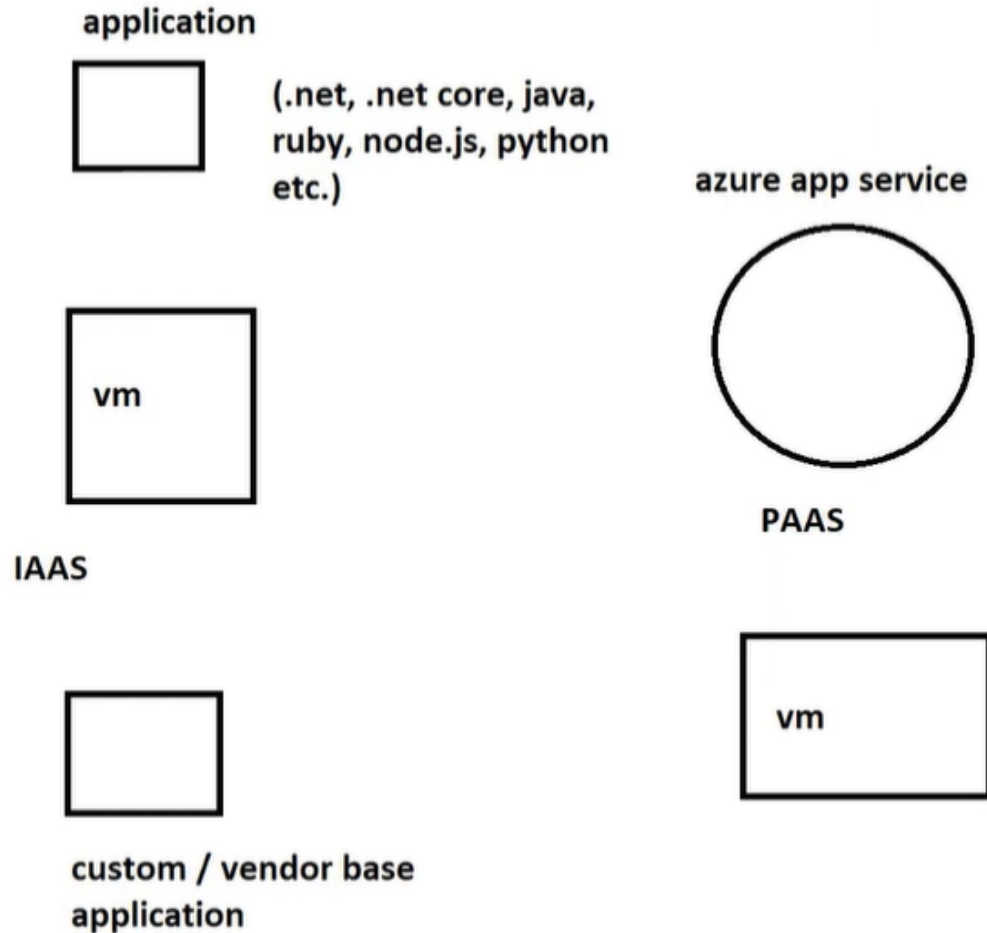
az group create --name netlab1 --location southindia



az vm create --name ubuntuserver --resource-group netlab1 --image ubuntuLTS --generate-ssh-keys --location southindia

Azure App Service





Azure App Service is a fully managed, cloud-based Platform-as-a-Service (PaaS) that allows developers to build, deploy, and scale web applications, RESTful APIs, and mobile backends without managing the underlying infrastructure.

Use Cases

Web Applications: Host static or dynamic websites.

Mobile Backends: Provide backend services for mobile applications.

RESTful APIs: Create and deploy RESTful APIs for various applications.

Azure Functions: Integrates with Azure Functions for serverless compute scenarios.

App Service is a Platform as a Service (PaaS) offering from Microsoft. use it to host web applications, REST API's and backend services for mobile applications.

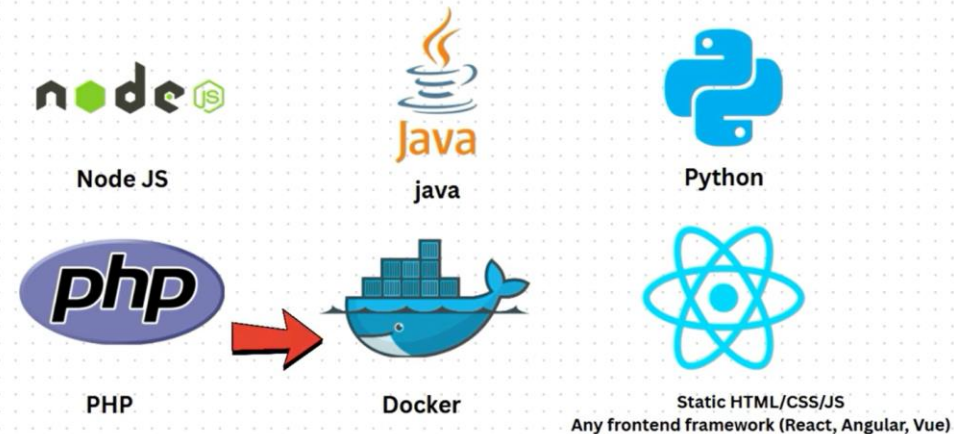
Web applications and services that are developed using any of the following programming languages

- NET
- .NET Core
- Java
- Ruby
- Node.js
- PHP
- Python

Benefits of using azure app service

- Fully managed environment
- Multiple programming languages and frameworks are supported
- Scalability
- Compliance
- Security
- Support for Containerization and Docker and DevOps optimization

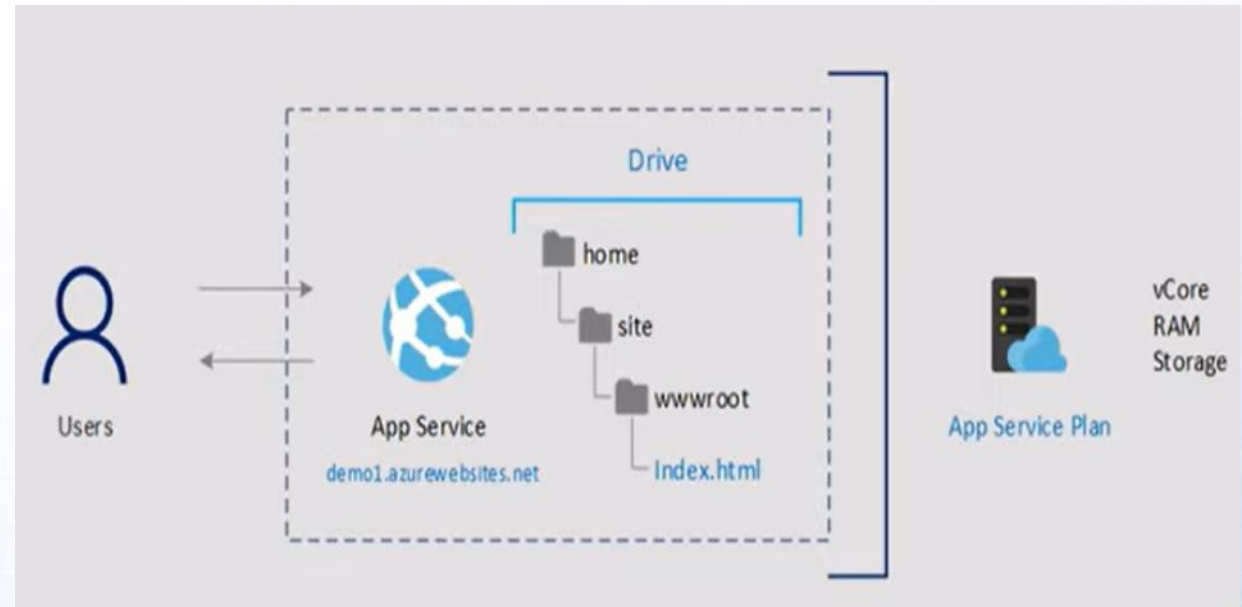
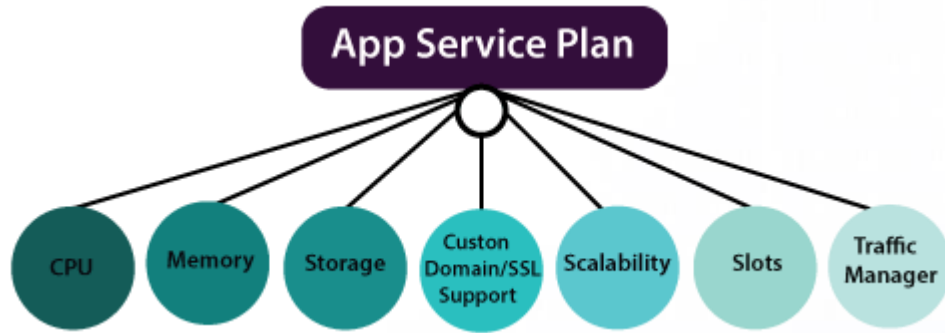
Supported Languages in App Service

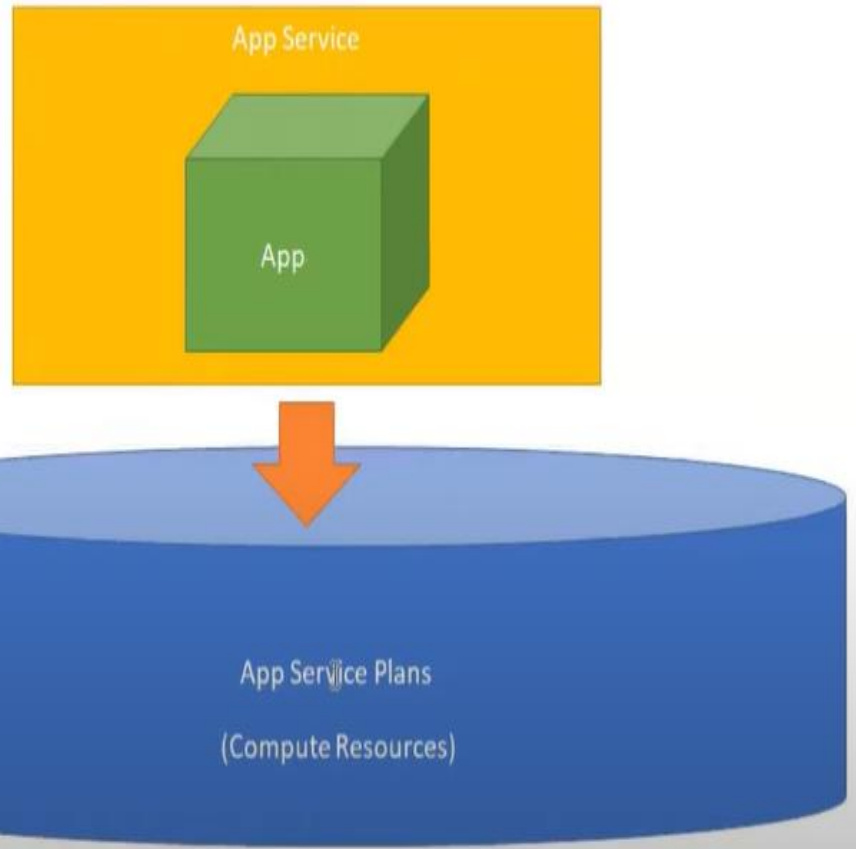


There are two types of hosting environments within App Service. App Service plan and App Service environment.

App Service plan

An app service plan denotes a set of features and capacity that we can share across multiple apps in the same subscription and geographical region. A single or dual app can be configured to run on the same computing resources.



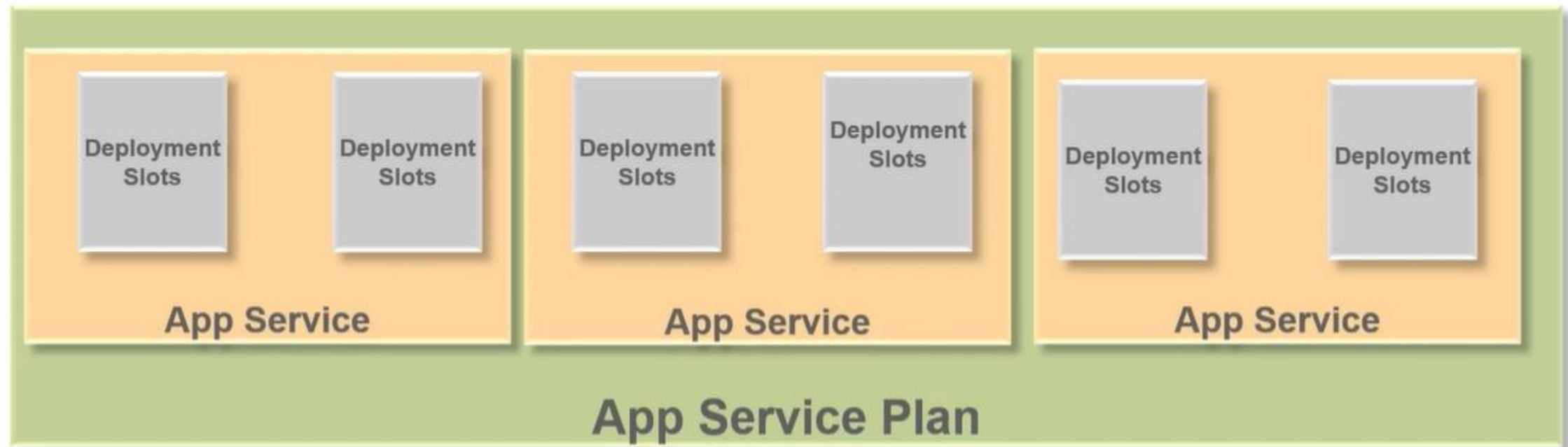


- Azure App Service plan defines a set of compute resources for a web app to run. An app service always runs in an App Service plan.

Each App Service plan defines:

- **Operating system** (Windows, Linux)
- **Region** (West US, East US, and so on)
- **Number of virtual machine (VM) instances**
- **Size of VM instances** (small, medium, large)
- **Pricing tier** (Free, Shared, Basic, Standard, Premium, PremiumV2, PremiumV3, PremiumV4 IsolatedV2)





Azure App Service pricing tiers include Free, Shared, Basic, Standard, Premium, and Isolated.

Tier	Use Case	Key Features
Free	Development, testing, small-scale applications	Limited features and performance
Shared	Development, testing, small-scale applications	Shared resources, limited performance
Basic	Development and testing environments	Dedicated Azure VMs
Standard	Production workloads	Higher performance, dedicated VMs, more memory and storage than Basic
Premium	Mission-critical production workloads	Faster processors, more memory and storage, additional features
Isolated	Mission-critical workloads requiring physical isolation	Dedicated, private network with faster processing and SSD storage



- **App Service Plan:** The App Service Plan is the core of pricing, as it defines the compute resources (CPU, memory, storage) and features available.
- **Tier and Instance Count:** The total cost depends on the selected pricing tier and the number of instances you scale to.
- **Pay-as-you-go:** Higher tiers are generally priced on a pay-as-you-go model, meaning you pay for the resources you consume.
- **Scalability:** You can start with a lower tier and scale up to a higher tier or scale out more instances as your traffic grows, allowing you to adjust costs based on your project needs



Monitoring Azure Resources



Azure Monitor



Azure Monitor is a powerful reporting and analytics tool.

Azure Monitor helps you maximize the availability and performance of your applications and services.

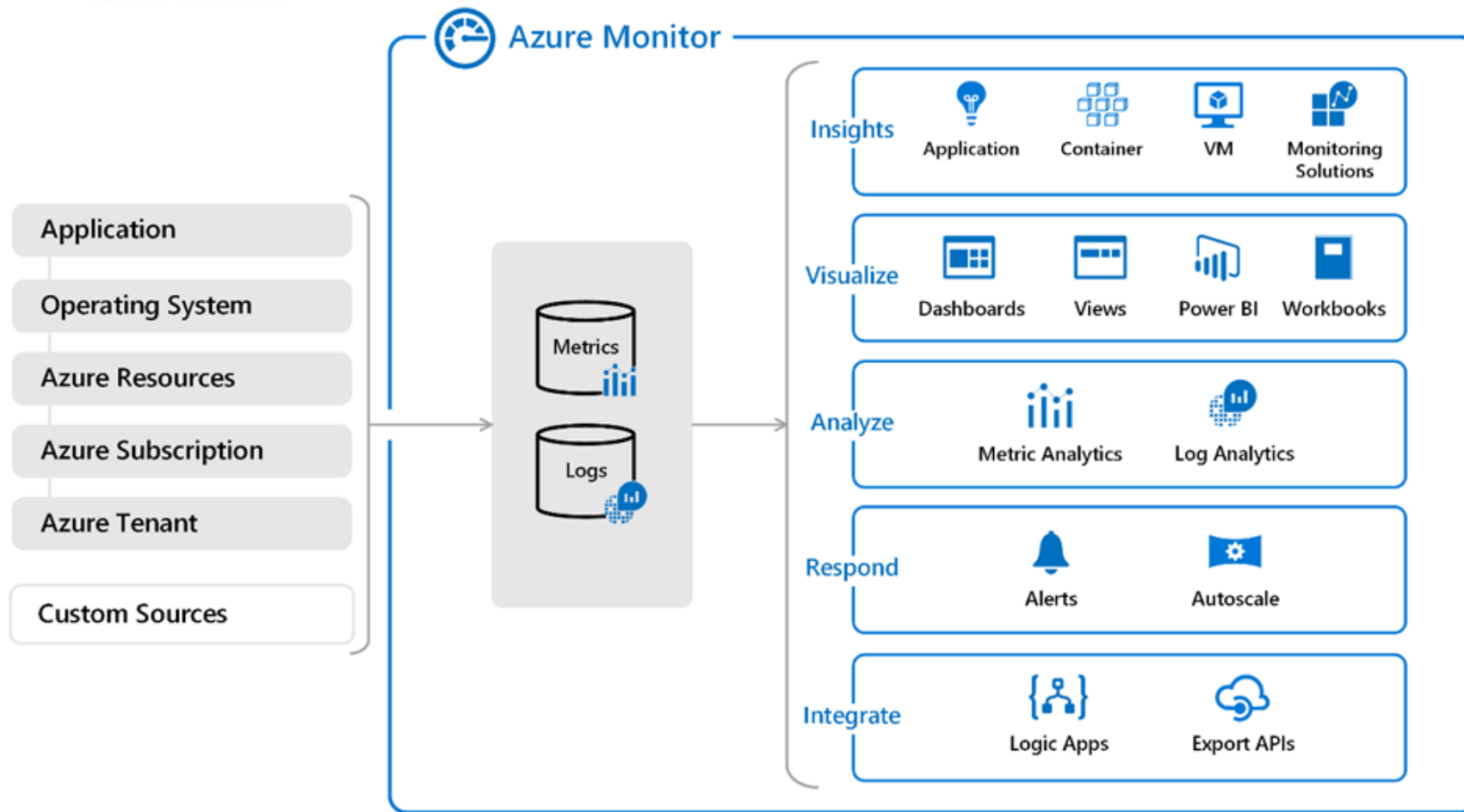
It delivers a comprehensive solution for collecting, analyzing, and acting on telemetry from your cloud and on-premises environments.

This information helps you understand how your applications are performing and proactively identify issues affecting them and the resources they depend on.

Azure Monitor include:

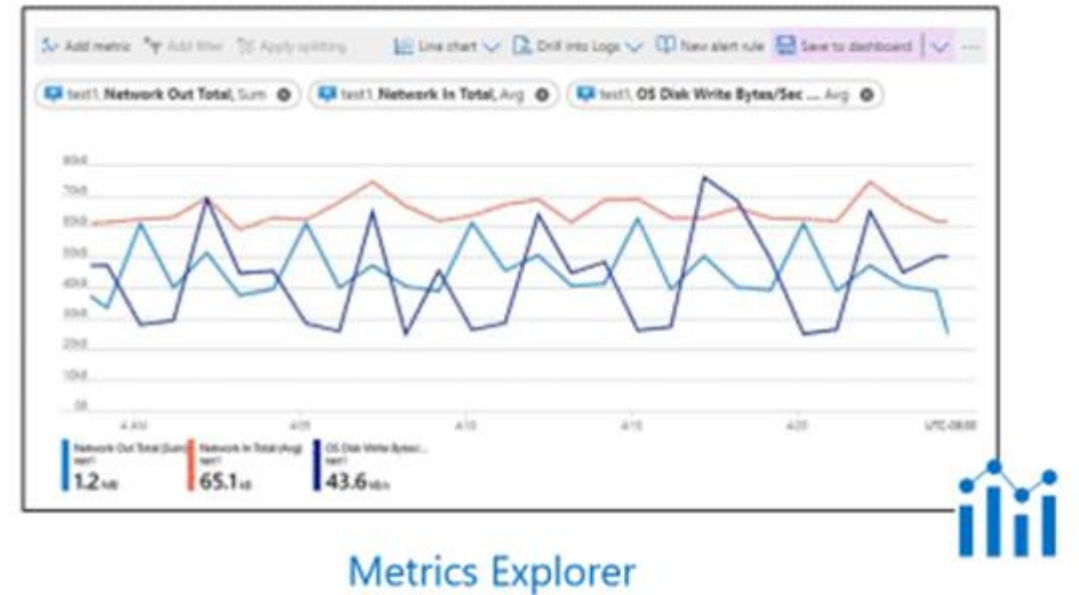
- Detect and diagnose issues across applications and dependencies with Application Insights.
- Correlate infrastructure issues with VM insights and Container insights.
- Drill into your monitoring data with Log Analytics for troubleshooting and deep diagnostics.
- Support operations at scale with smart alerts and automated actions.
- Create visualizations with Azure dashboards and workbooks.
- Collect data from monitored resources using Azure Monitor Metrics





Metrics

- ❖ Metrics are measures of a resource's certain characteristics over a given period.
- ❖ For instance, CPU utilization, disk IOPS, number of connections, etc.
- ❖ These are typically real-time, and since they are stored as values with a standard collection interval, they are ideally suitable for viewing as graphs to help you view results over time.
- ❖ Metrics are numerical values that describe some aspect of a system at a point in time. Azure Monitor can capture metrics in near real-time.



Logs

- Logs contain time-stamped information about changes made to resources.
- The type of information recorded varies by log source.
- The log data is organized into records, with different sets of properties for each type of record.
- The logs can include numeric values such as Azure Monitor metrics, but most include text data rather than numeric values.



Show legacy language converter

Event | where EventLevelName == "Error" | project TimeGenerated, Computer, EventLevelName, Source, EventID

4K Results List Table Advanced Analytics 00:00:00.339

Drag a column header and drop it here to group by that column.

TimeGenerated	Computer	EventLevelName	Source	EventID
7/17/2017 11:39:02 AM	srv01.contoso.com	Error	Microsoft-Windows-L...	5873
7/17/2017 11:39:12 AM	srv01.contoso.com	Error	HealthService	4502
7/17/2017 11:39:12 AM	srv02.contoso.com	Error	HealthService	4502
7/17/2017 11:39:12 AM	srv01.contoso.com	Error	HealthService	4502
7/17/2017 11:39:12 AM	srv03.contoso.com	Error	HealthService	4502
7/17/2017 11:39:26 AM	srv03.contoso.com	Error	NPMO Agent	100
7/17/2017 11:39:26 AM	srv02.contoso.com	Error	NPMO Agent	100

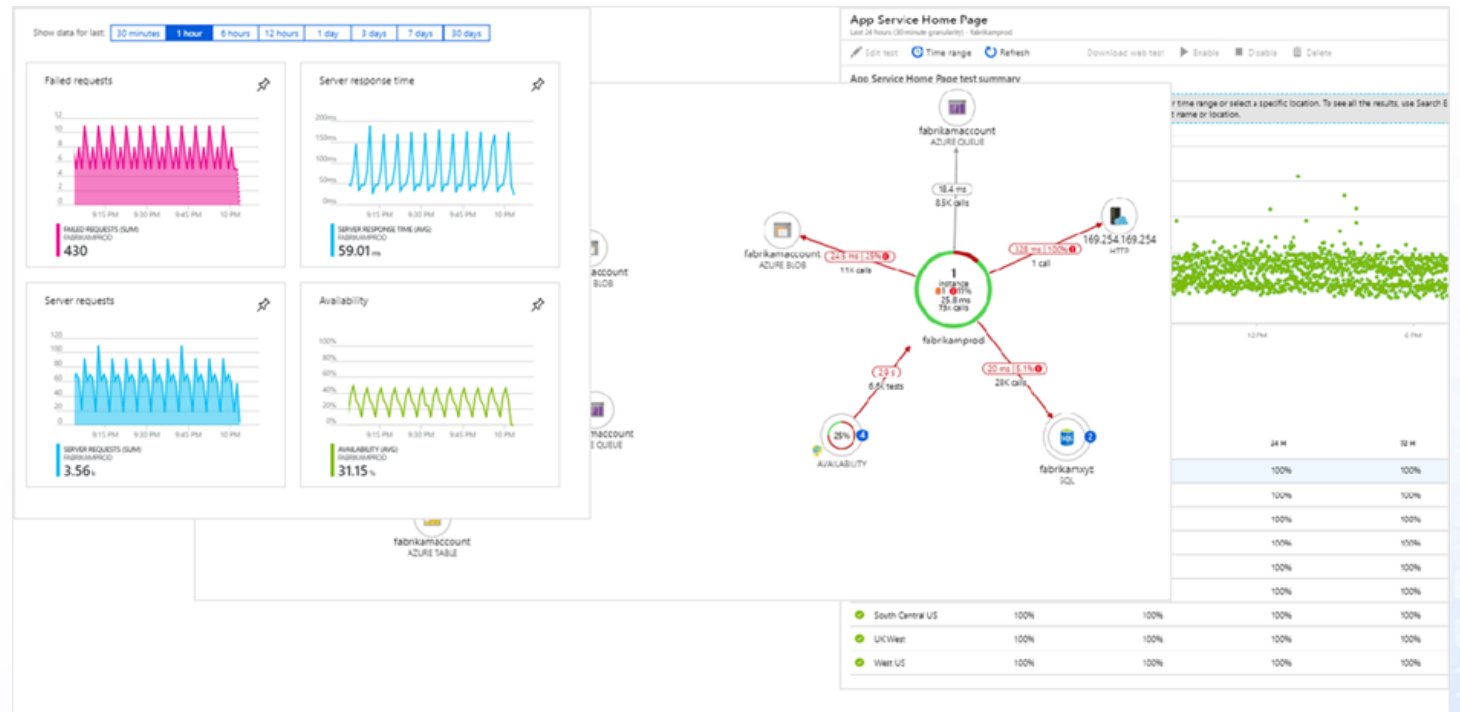
Log Analytics



Azure

What data does Azure Monitor collect?

- ❖ Application monitoring data
- ❖ Guest OS monitoring data
- ❖ Azure subscription monitoring
- ❖ Azure tenant monitoring data



Application Insights

Application Insights monitors the availability, performance, and usage of your web applications whether they're hosted in the **cloud or on-premises**

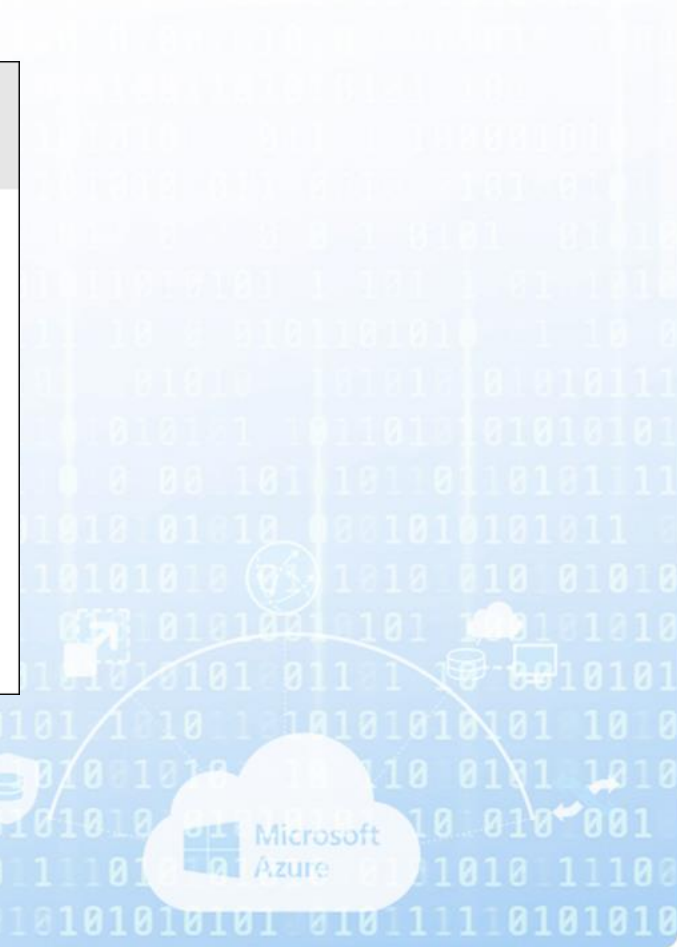
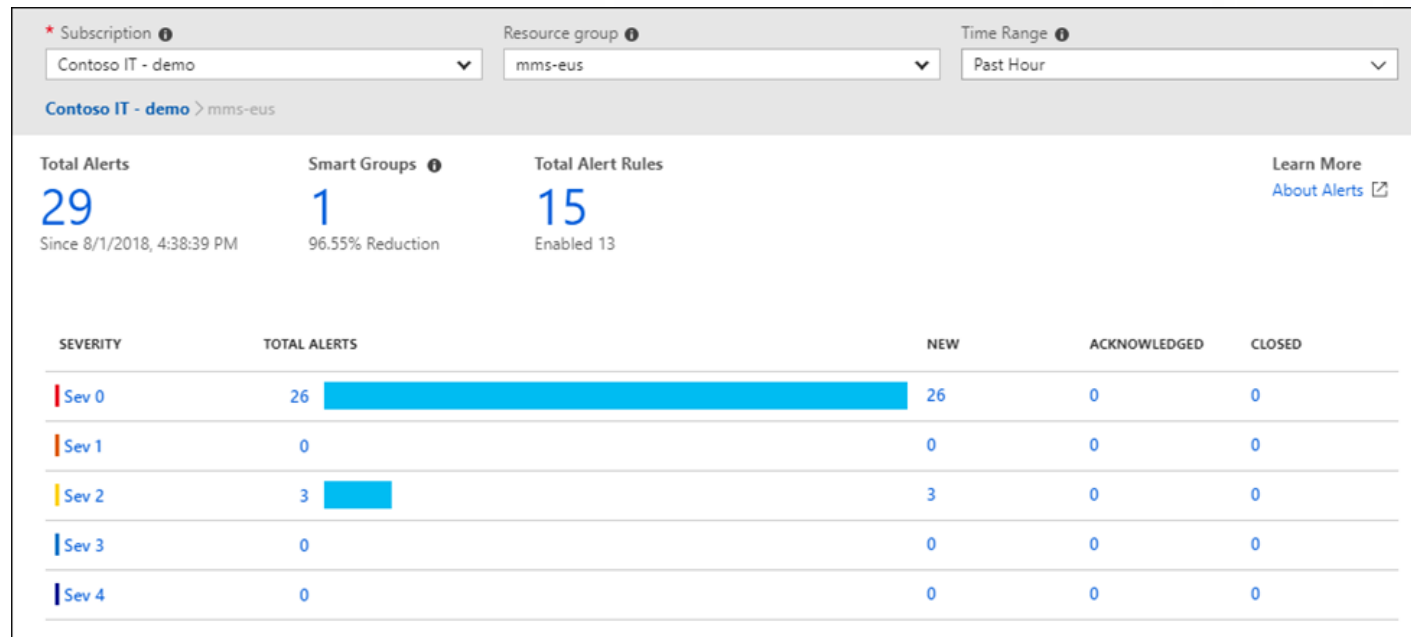
VM insights

VM insights monitors your Azure virtual machines (VM) at scale. It analyzes the performance and health of your Windows and Linux VMs and identifies their different processes and interconnected dependencies on external processes

Responding to critical situations

Alerts

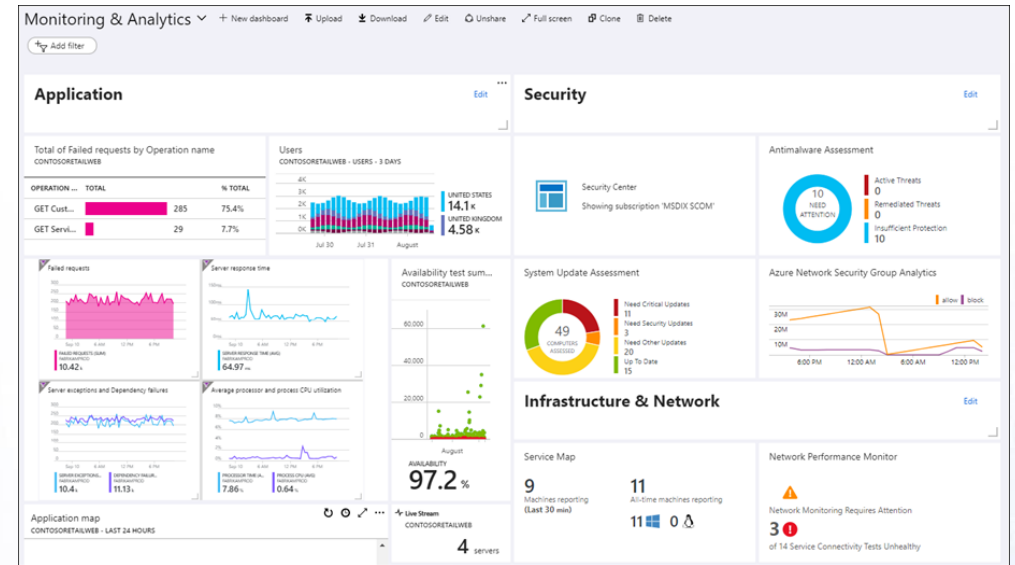
Alerts in Azure Monitor proactively notify you of critical conditions and potentially attempt to take corrective action.



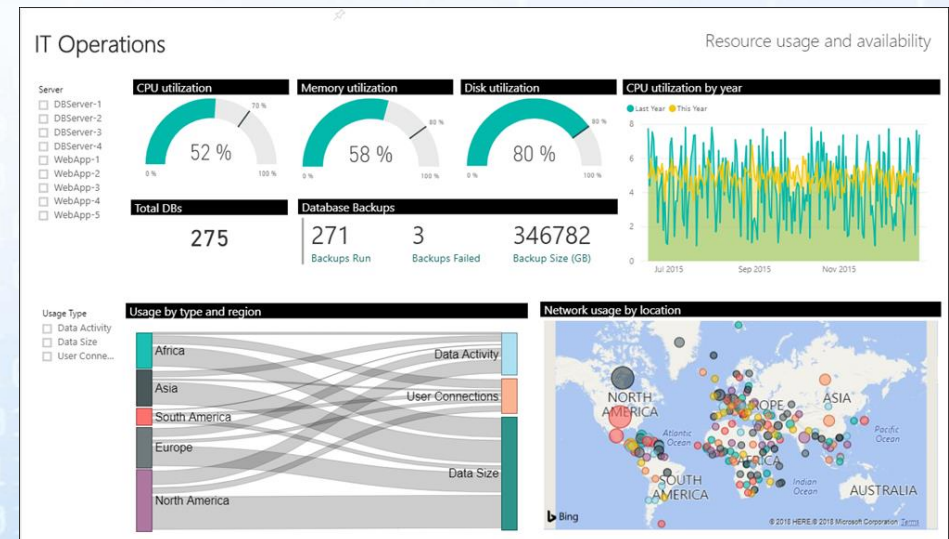
Visualizing monitoring data

Dashboards

Workbooks



Power BI



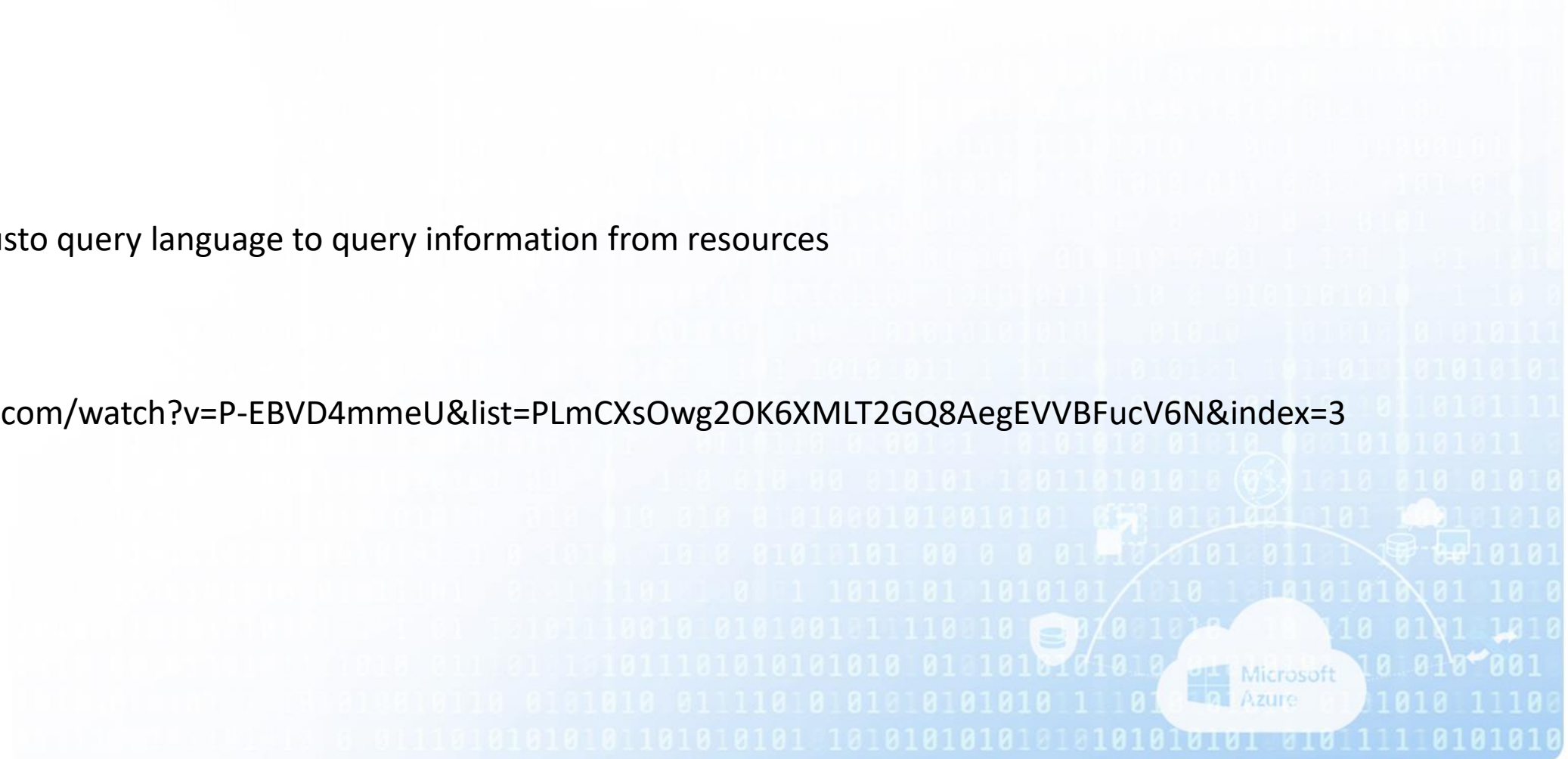
Integrate and export data

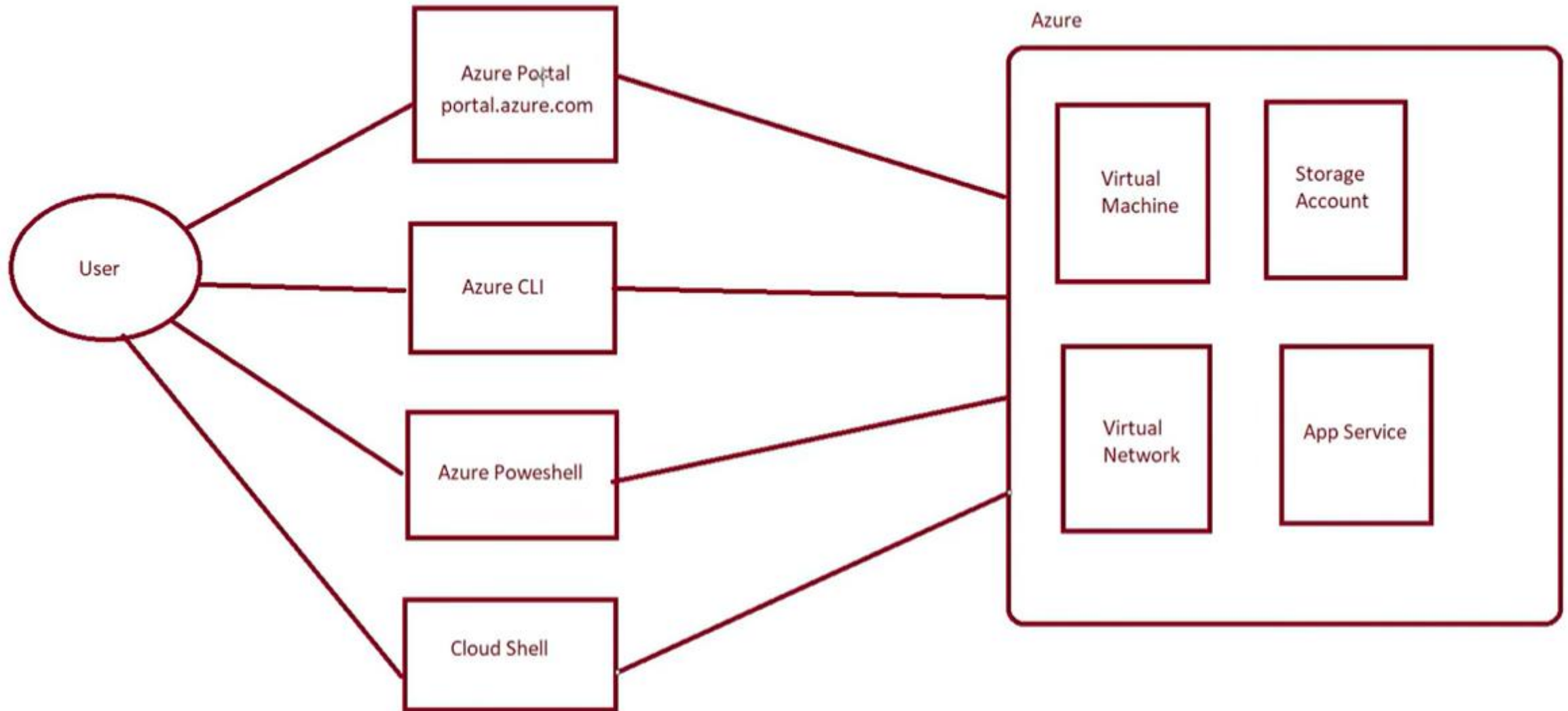
integrate Azure Monitor with other systems and to build custom solutions that use your monitoring data

- Event Hub
- Logic Apps
- API

Azure monitor uses Kusto query language to query information from resources

<https://www.youtube.com/watch?v=P-EBVD4mmeU&list=PLmCXsOwg2OK6XMLT2GQ8AegEVVBFucV6N&index=3>





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Installation on Windows PowerShell

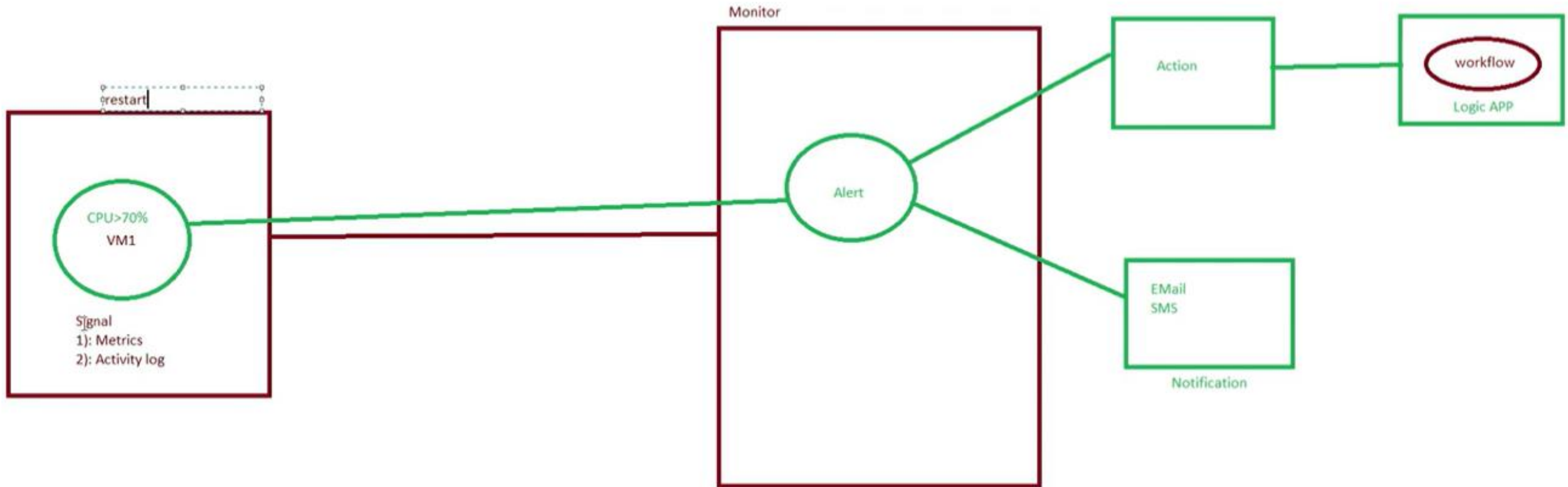
The Azure Az PowerShell module is also supported for use with **PowerShell 5.1** on Windows.

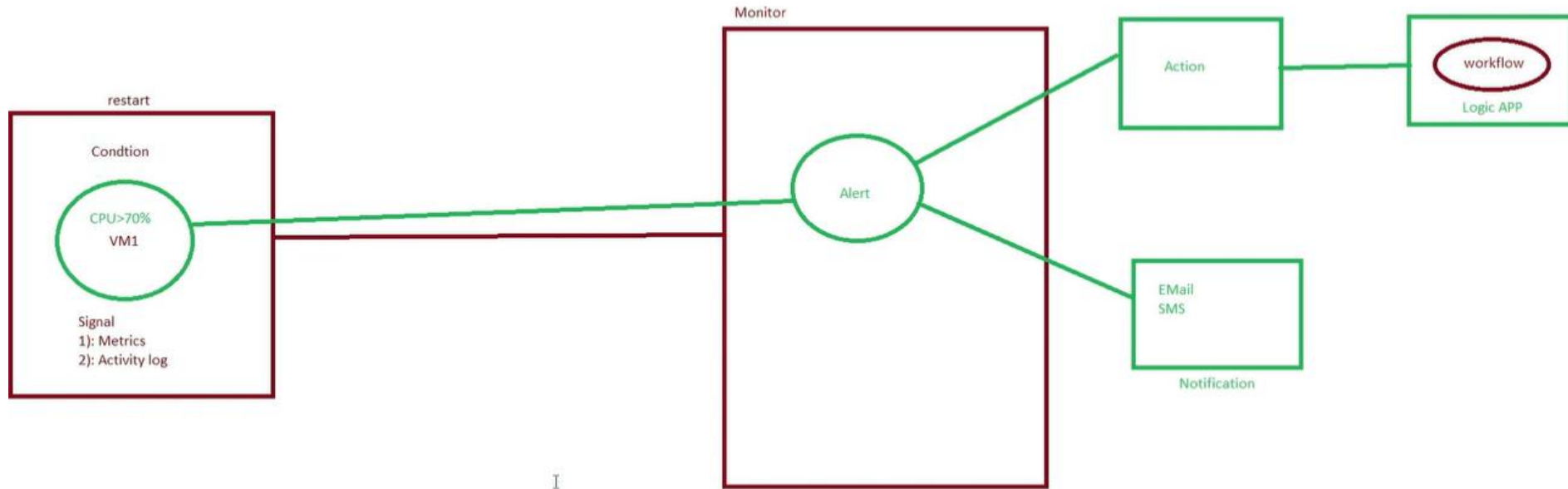
To use the Azure Az PowerShell module in PowerShell 5.1 on Windows:

- ✓ Update to Windows PowerShell 5.1 (If you're on Windows 10 version 1607 or higher, you already have PowerShell 5.1 installed)
- ✓ Install **.NET Framework 4.7.2** or later.
- ✓ Make sure you have the latest version of PowerShellGet.
- ✓ Run **Install-Module -Name PowerShellGet -Force**.



<https://www.youtube.com/watch?v=-kOI7BaFu-w&list=PLUGuCqrhcwZzht4r2sbByidApmrvEjL9m&index=14>





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